Falchook_NoWriting

Jack Dumanski

Summary

We analyzed data within residents and directors, and then we analyzed data between them. Out of 81 comparisons for residents, we found 7 to be significant at a 0.05 level, but 0 after adjusting for the amount of comparions. Out of 14 for directors, we found 0 to be true before and after adjustment. We had a total of 119 residents and 24 directors in our data, which for some questions with many categorical responses, made it difficult to find significance. However, when cross-comparing the groups on each common question, we found 20/51 comparisons to be significant, 18 after adjustment.

Methods

For our methodology, we conducted chi-squared tests for independence to see if there were differences in how the responses were distributed for each comparison. In doing so, most of the expected counts were lower than the requirements necessary for chi-squared tests due to a low sample size and large amount of potential responses for each question, so instead we used Fisher exact tests, as all of the comparisons violated the chi-square assumptions.

The Fisher exact test can calculate exact probabilities to see if there's a non-random association between groups, in which a low p-value (after adjustment) translates to having significant evidence for there existing some association between questions.

For questions that had multiple parts, we had to split them into individual questions (e.g. 13a, 13b, 13c…) to see if there were specific responses that were influenced by the grouping variable in question, which usually was the role (being a resident/director) in the case of cross-comparison, or other information like program, experience, or other variables in question. In practice, this saw us, for example, analyzing whether directors perceive residence having negative impact on marriage versus whether residents feel their residency having a negative impact on marriage. The same question was asked for personal health care, control over one's life, and work/life balance. All of that example was from data from question 13 on the director's survey.

Then, to visualize the difference, we created side-by-side bar charts which plot the distribution of question responses versus the grouping variable. For resident/director comparison, this shows side-by-side plots of the residents' response to a question versus the directors'. Noticeable/interesting plots and findings were placed in the "significant findings" section of this writeup and are also in a list of all comparison results and plots in the appendix.

Significant Findings

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.1     v readr     2.1.6
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.1     v tibble    3.3.0
v lubridate 1.9.4     v tidyr     1.3.1
v purrr     1.2.0
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becom

Attaching package: 'zoo'


The following objects are masked from 'package:base':

    as.Date, as.Date.numeric
```

Question1	Question2	P_value	Significant	test	Adj_P	Sig_Adj
Q13d	Q12	0.0095	TRUE	fisher	0.3240	FALSE
Q13h	Q2	0.0065	TRUE	fisher	0.3240	FALSE
Q18	Q11	0.012	TRUE	fisher	0.3240	FALSE
Q13a	Q1	0.0405	TRUE	fisher	0.4891	FALSE
Q13d	Q10	0.0725	FALSE	fisher	0.4891	FALSE
Q13d	Q5	0.0645	FALSE	fisher	0.4891	FALSE
Q13f	Q12	0.073	FALSE	fisher	0.4891	FALSE
Q14a	Q5	0.057	FALSE	fisher	0.4891	FALSE
Q14c	Q3	0.0545	FALSE	fisher	0.4891	FALSE
Q15a	Q3	0.0715	FALSE	fisher	0.4891	FALSE
Q16d	Q3	0.027	TRUE	fisher	0.4891	FALSE
Q17	Q10	0.043	TRUE	fisher	0.4891	FALSE
Q18	Q10	0.0785	FALSE	fisher	0.4891	FALSE
Q13b	Q2	0.086	FALSE	fisher	0.4976	FALSE
Q13e	Q10	0.1074	FALSE	fisher	0.5437	FALSE
Q16c	Q3	0.1069	FALSE	fisher	0.5437	FALSE
Q13c	Q1	0.1949	FALSE	fisher	0.6106	FALSE
Q13d	Q11	0.2174	FALSE	fisher	0.6106	FALSE
Q13e	Q12	0.2674	FALSE	fisher	0.6106	FALSE
Q13f	Q10	0.1769	FALSE	fisher	0.6106	FALSE
Q13g	Q1	0.2479	FALSE	fisher	0.6106	FALSE
Q13g	Q5	0.2319	FALSE	fisher	0.6106	FALSE
Q13h	Q12	0.2579	FALSE	fisher	0.6106	FALSE
Q13h	Q1	0.2049	FALSE	fisher	0.6106	FALSE
Q13h	Q5	0.1809	FALSE	fisher	0.6106	FALSE
Q13i	Q10	0.2714	FALSE	fisher	0.6106	FALSE
Q13i	Q11	0.2584	FALSE	fisher	0.6106	FALSE
Q13i	Q5	0.1419	FALSE	fisher	0.6106	FALSE
Q14b	Q5	0.2709	FALSE	fisher	0.6106	FALSE
Q14b	Q3	0.2319	FALSE	fisher	0.6106	FALSE
Q14c	Q5	0.1529	FALSE	fisher	0.6106	FALSE
Q14d	Q5	0.2329	FALSE	fisher	0.6106	FALSE
Q16a	Q3	0.2124	FALSE	fisher	0.6106	FALSE
Q16a	Q1	0.1899	FALSE	fisher	0.6106	FALSE
Q16c	Q1	0.2614	FALSE	fisher	0.6106	FALSE
Q17	Q11	0.1979	FALSE	fisher	0.6106	FALSE
Q13c	Q5	0.3233	FALSE	fisher	0.6547	FALSE
Q13i	Q1	0.3173	FALSE	fisher	0.6547	FALSE
Q14d	Q3	0.3203	FALSE	fisher	0.6547	FALSE
Q16b	Q3	0.3043	FALSE	fisher	0.6547	FALSE
Q13b	Q11	0.3848	FALSE	fisher	0.6873	FALSE
Q13c	Q10	0.3903	FALSE	fisher	0.6873	FALSE
Q13c	Q11	0.3553	FALSE	fisher	0.6873	FALSE
Q13e	Q11	0.3568	FALSE	fisher	0.6873	FALSE
Q13e	Q5	0.3713	FALSE	fisher	0.6873	FALSE
Q18	Q1	0.3798	FALSE	fisher	0.6873	FALSE
Q13f	Q5	0.4238	FALSE	fisher	0.7160	FALSE
Q13i	Q12	0.4243	FALSE	fisher	0.7160	FALSE
Q17	Q1	0.4468	FALSE	fisher	0.7386	FALSE
Q13g	Q12	0.4853	FALSE	fisher	0.7862	FALSE

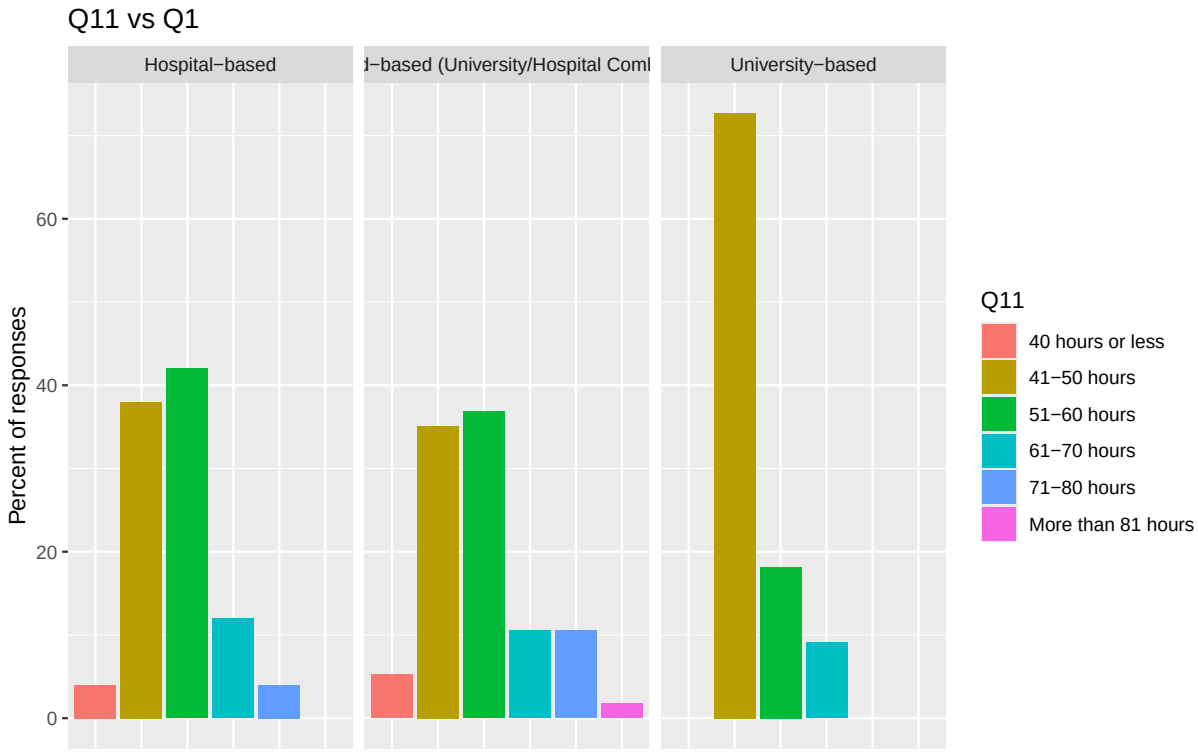
Question1	Question2	P_value	Significant	test	Adj_P	Sig_Adj
Q12h	Q2	0.1354	FALSE	fisher	0.9478	FALSE
Q21	Q2	0.1279	FALSE	fisher	0.9478	FALSE
Q10	Q2	0.2079	FALSE	fisher	0.9520	FALSE
Q12a	Q2	0.7866	FALSE	fisher	0.9520	FALSE
Q12b	Q2	0.8646	FALSE	fisher	0.9520	FALSE
Q12c	Q2	0.952	FALSE	fisher	0.9520	FALSE
Q12d	Q2	0.8016	FALSE	fisher	0.9520	FALSE
Q12e	Q2	0.951	FALSE	fisher	0.9520	FALSE
Q12f	Q2	0.8481	FALSE	fisher	0.9520	FALSE
Q12g	Q2	0.4648	FALSE	fisher	0.9520	FALSE
Q12i	Q2	0.8306	FALSE	fisher	0.9520	FALSE
Q22a	Q2	0.6712	FALSE	fisher	0.9520	FALSE
Q22b	Q2	0.6572	FALSE	fisher	0.9520	FALSE
Q22c	Q2	0.3718	FALSE	fisher	0.9520	FALSE

Question1	Question2	P_value	Significant	test	Adj_P	Sig_Adj
Q26	role	1e-04	TRUE	fisher	0.0051	TRUE
Q12	role	5e-04	TRUE	fisher	0.0064	TRUE
Q13f	role	5e-04	TRUE	fisher	0.0064	TRUE
Q21	role	5e-04	TRUE	fisher	0.0064	TRUE
Q13a	role	0.001	TRUE	fisher	0.0085	TRUE
Q22b	role	0.001	TRUE	fisher	0.0085	TRUE
Q13b	role	0.0015	TRUE	fisher	0.0096	TRUE
Q14b	role	0.0015	TRUE	fisher	0.0096	TRUE
Q20a	role	0.002	TRUE	fisher	0.0102	TRUE
Q20b	role	0.002	TRUE	fisher	0.0102	TRUE
Q13c	role	0.003	TRUE	fisher	0.0139	TRUE
Q3	role	0.004	TRUE	fisher	0.0157	TRUE
Q14a	role	0.004	TRUE	fisher	0.0157	TRUE
Q19	role	0.0045	TRUE	fisher	0.0164	TRUE
Q22a	role	0.0055	TRUE	fisher	0.0187	TRUE
Q20d	role	0.006	TRUE	fisher	0.0191	TRUE
Q22c	role	0.015	TRUE	fisher	0.0450	TRUE
Q15a	role	0.0165	TRUE	fisher	0.0468	TRUE
Q14c	role	0.0255	TRUE	fisher	0.0684	FALSE
Q16b	role	0.034	TRUE	fisher	0.0867	FALSE
Q13d	role	0.0525	FALSE	fisher	0.1175	FALSE
Q13h	role	0.0495	TRUE	fisher	0.1175	FALSE
Q20c	role	0.053	FALSE	fisher	0.1175	FALSE
Q9	role	0.063	FALSE	fisher	0.1199	FALSE
Q13e	role	0.06	FALSE	fisher	0.1199	FALSE
Q14d	role	0.0635	FALSE	fisher	0.1199	FALSE
Q24d	role	0.0625	FALSE	fisher	0.1199	FALSE
Q23a	role	0.08	FALSE	fisher	0.1457	FALSE
Q17	role	0.093	FALSE	fisher	0.1636	FALSE
Q13g	role	0.0995	FALSE	fisher	0.1640	FALSE
Q24b	role	0.1029	FALSE	fisher	0.1640	FALSE
Q24f	role	0.1004	FALSE	fisher	0.1640	FALSE
Q27	role	0.1164	FALSE	fisher	0.1799	FALSE
Q15b	role	0.1264	FALSE	fisher	0.1896	FALSE
Q23d	role	0.1369	FALSE	fisher	0.1995	FALSE
Q23c	role	0.1629	FALSE	fisher	0.2308	FALSE
Q8	role	0.2224	FALSE	fisher	0.3066	FALSE
Q13i	role	0.2294	FALSE	fisher	0.3079	FALSE
Q23b	role	0.2374	FALSE	fisher	0.3104	FALSE
Q18	role	0.2524	FALSE	fisher	0.3218	FALSE
Q1	role	0.3288	FALSE	fisher	0.4090	FALSE
Q15c	role	0.3473	FALSE	fisher	0.4217	FALSE
Q11	role	0.3923	FALSE	fisher	0.4653	FALSE
Q24c	role	0.4463	FALSE	fisher	0.5173	FALSE
Q16a	role	0.4883	FALSE	fisher	0.5534	FALSE
Q10	role	0.5172	FALSE	fisher	0.5734	FALSE
Q24e	role	0.7441	FALSE	fisher	0.8074	FALSE
Q25	role	0.934	FALSE	fisher	0.9924	FALSE
Q16c	role	1	FALSE	fisher	1.0000	FALSE
Q16d	role	1	FALSE	fisher	1.0000	FALSE

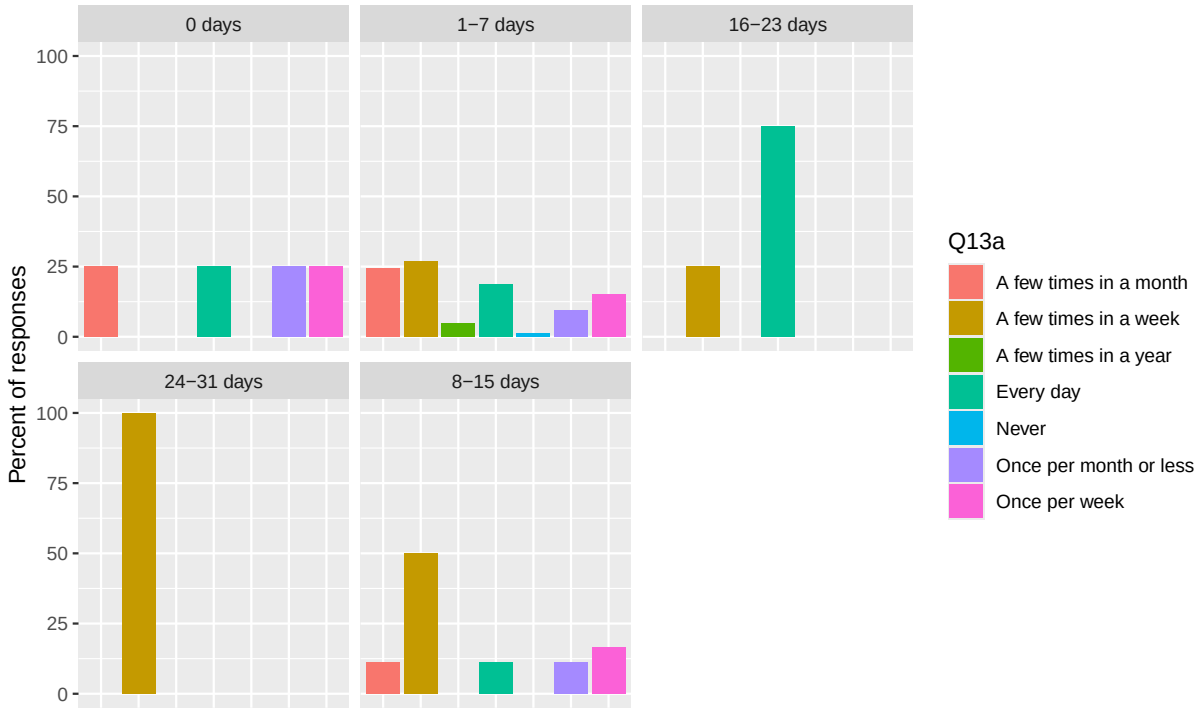
Residents Results Table

Directors Results Table

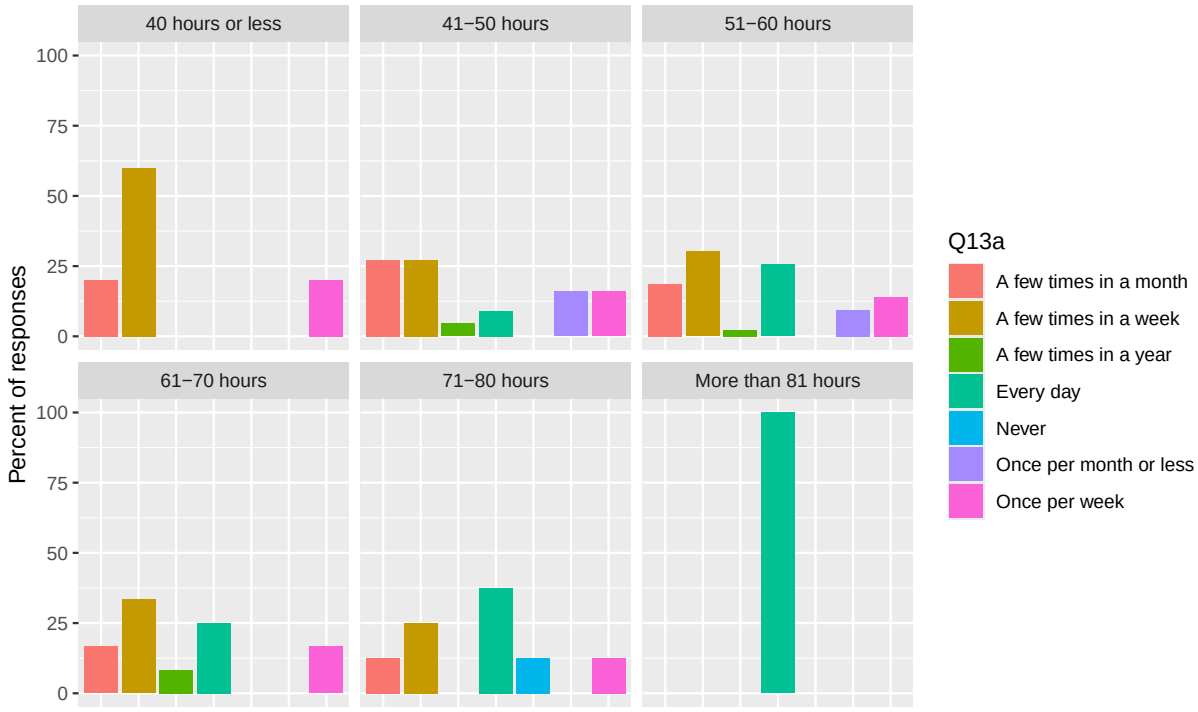
Appendix A: Plots



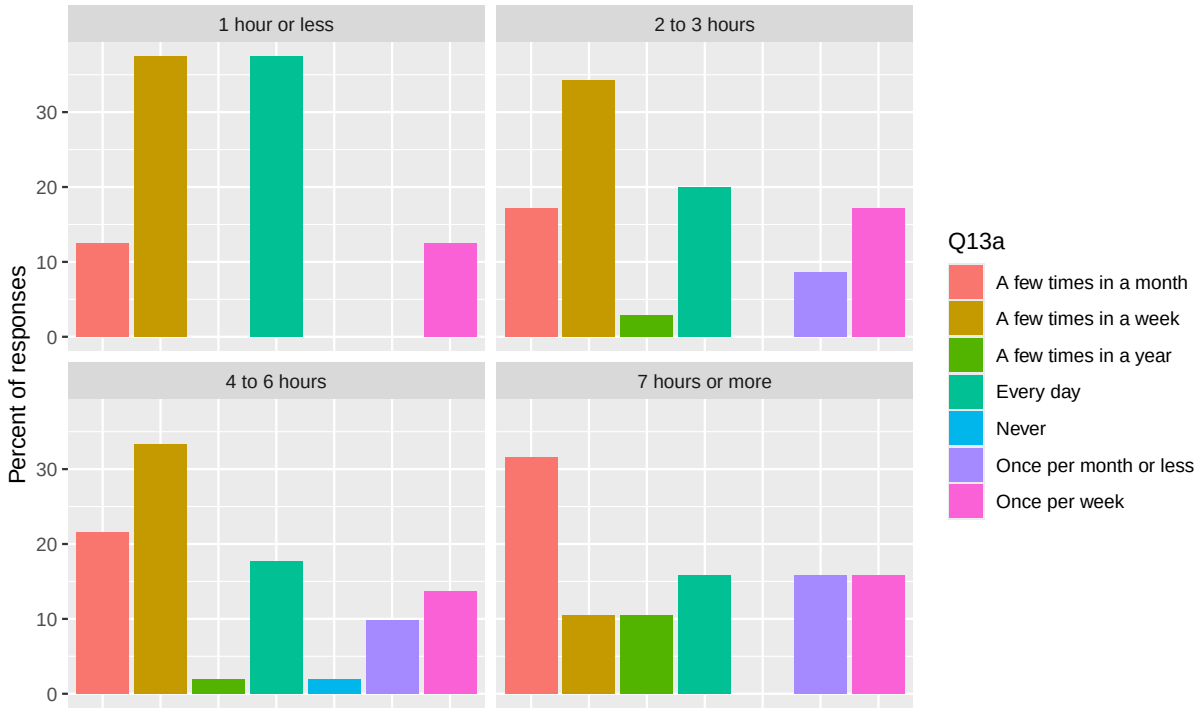
Q13a vs Q10



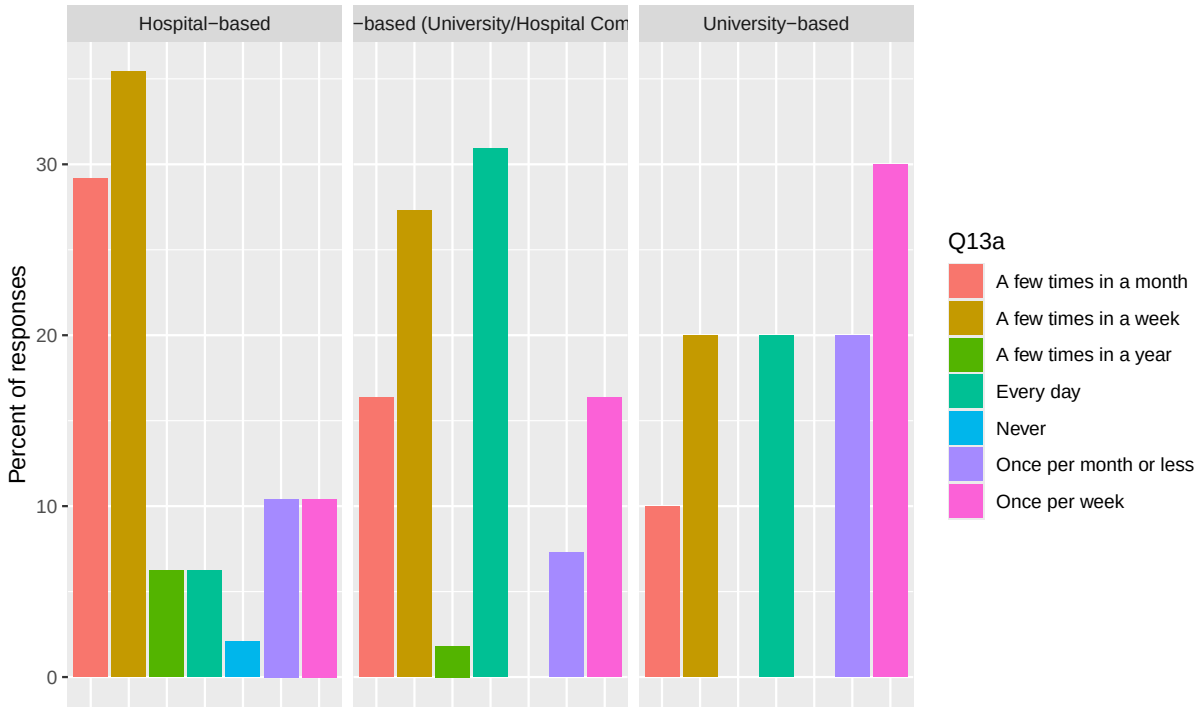
Q13a vs Q11



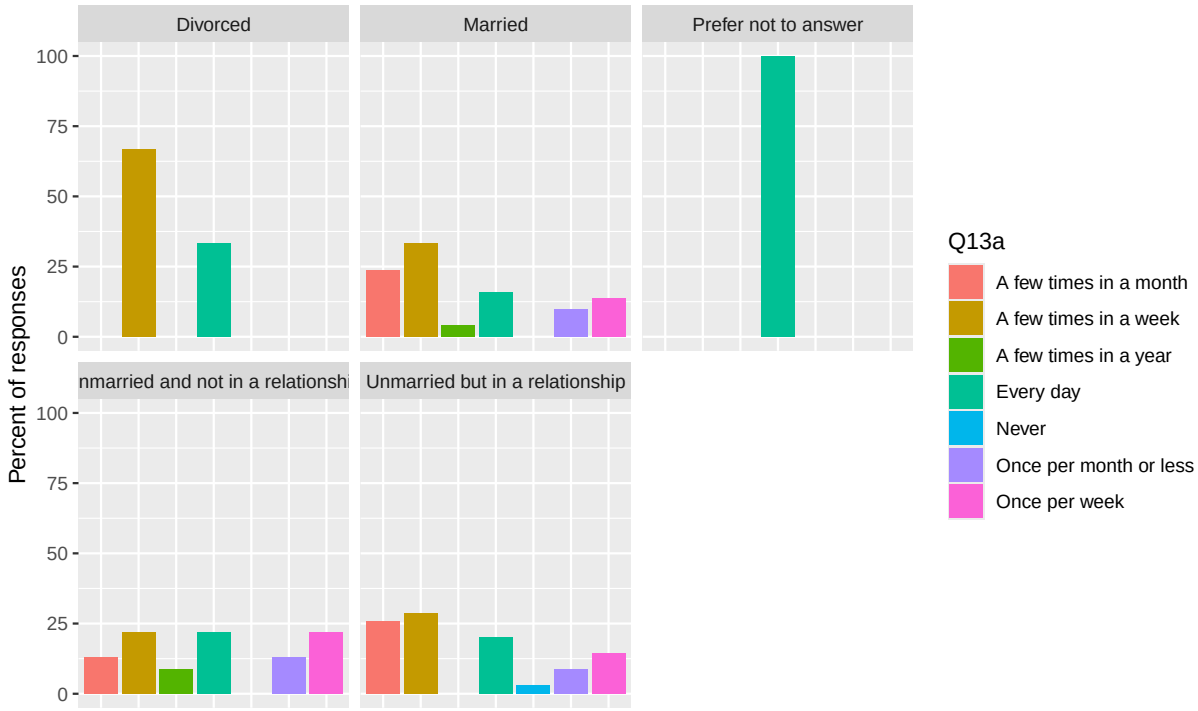
Q13a vs Q12



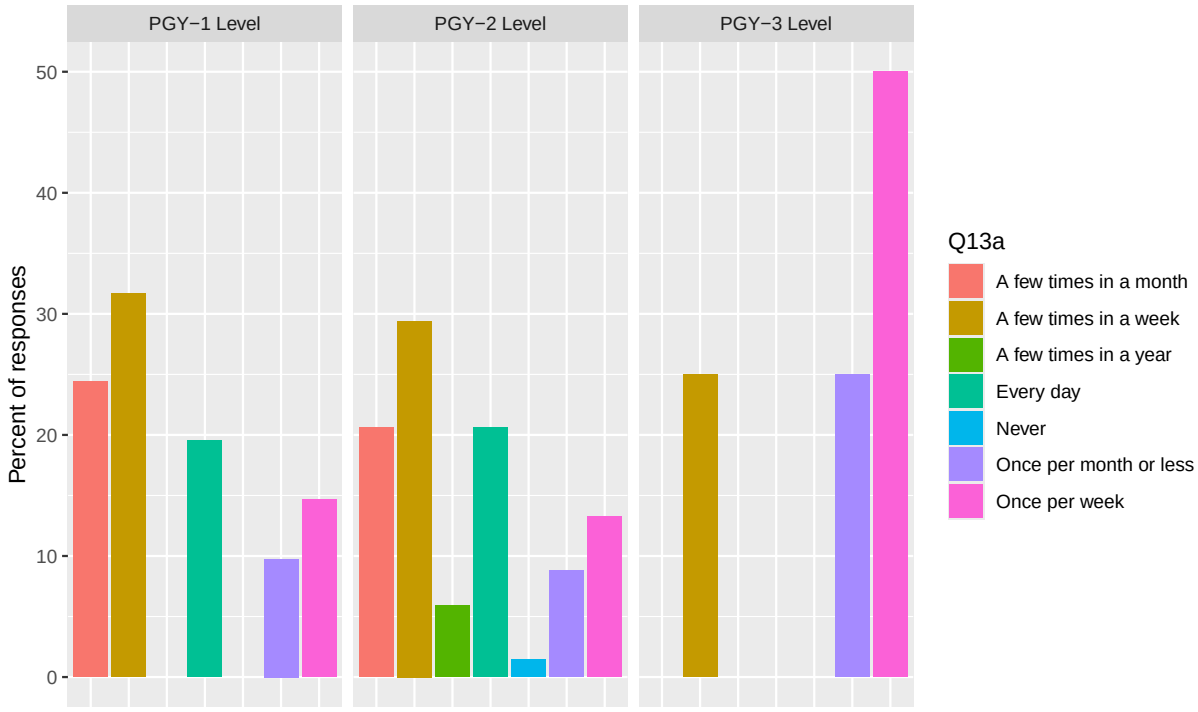
Q13a vs Q1

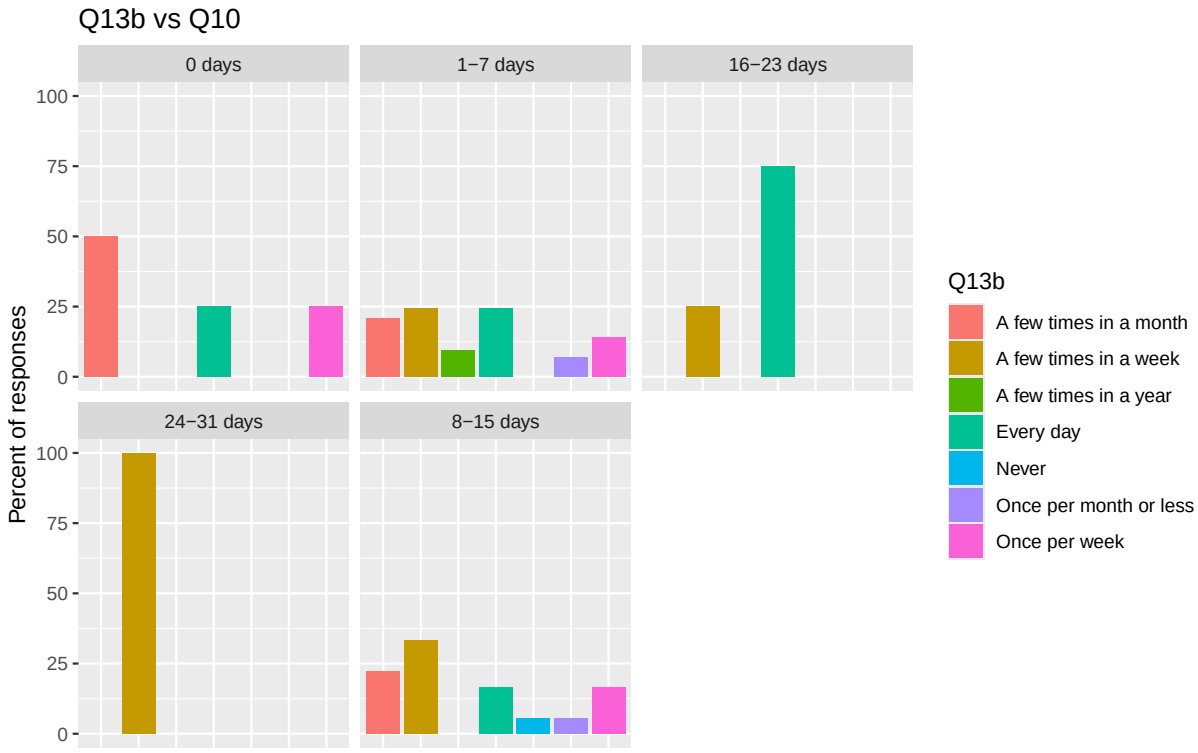


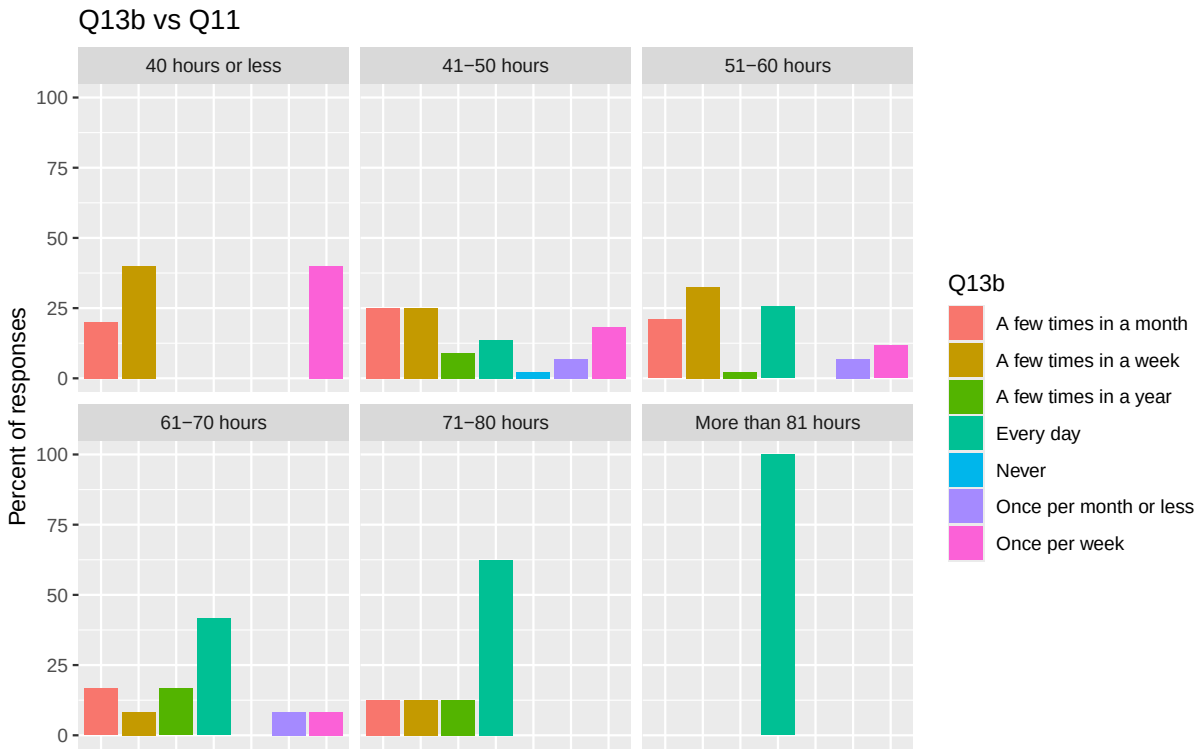
Q13a vs Q5

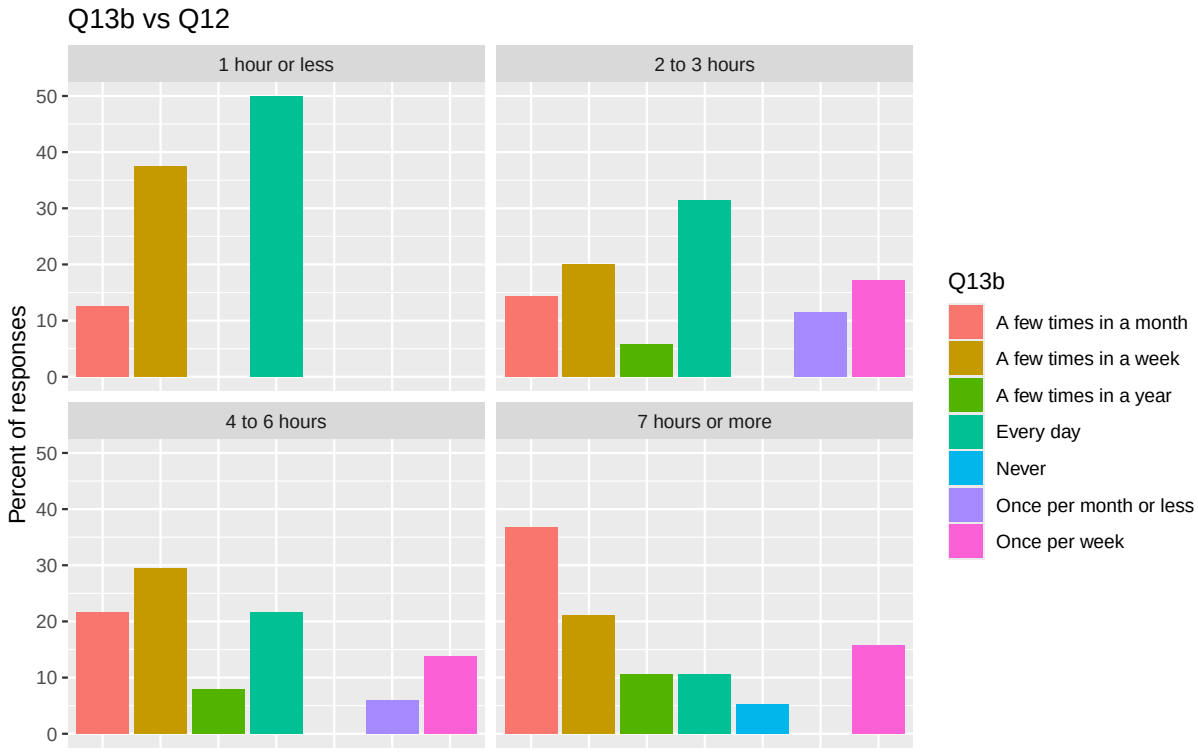


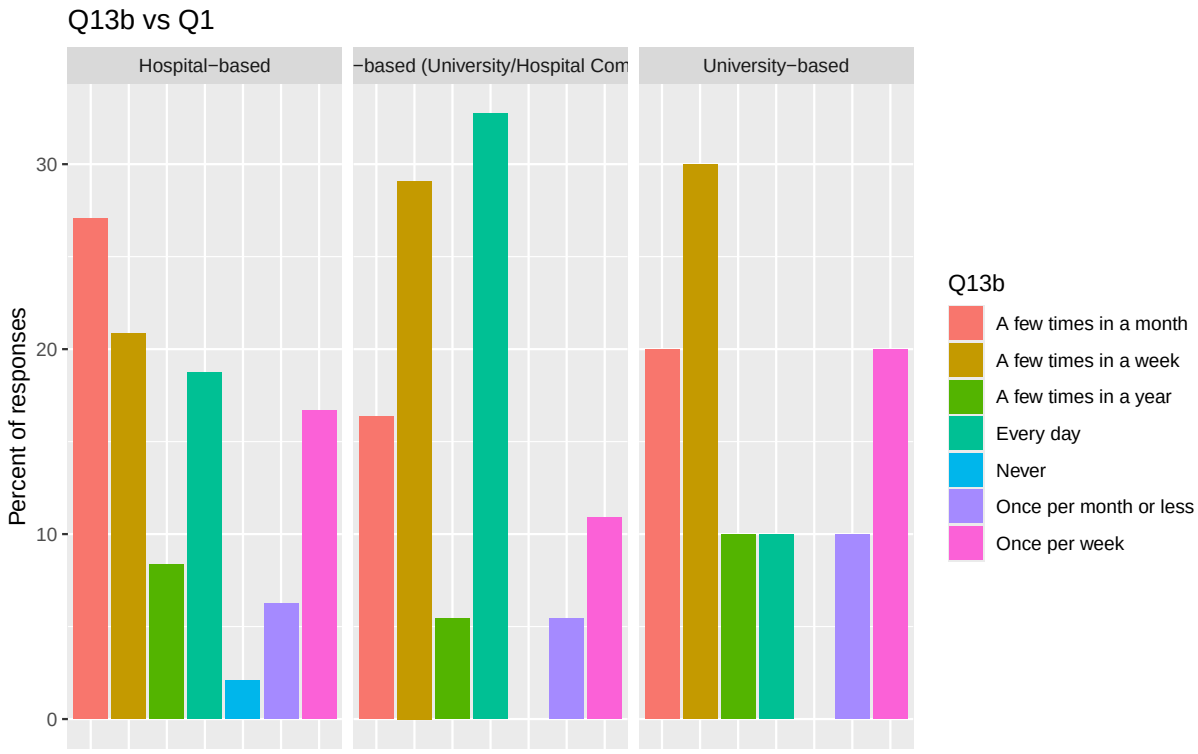
Q13a vs Q2

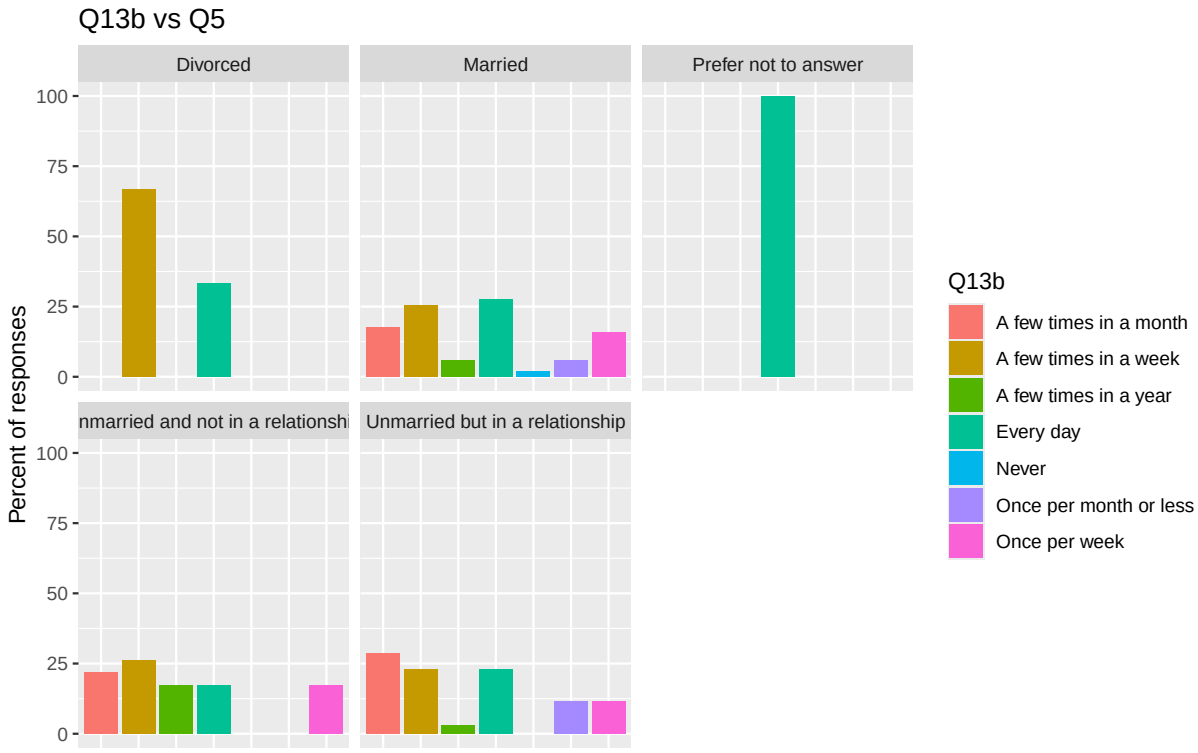


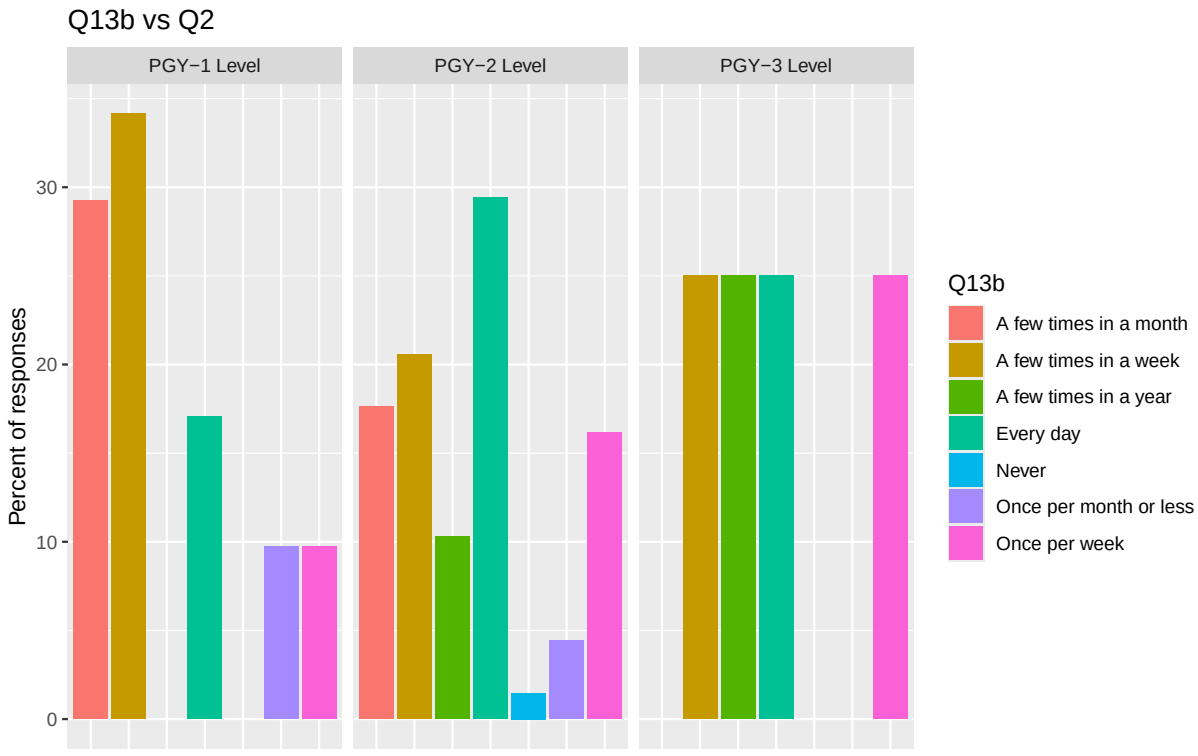

Q13b vs Q10
0 days
1−7 days
16−23 days
24−31 days
8−15 days
Percent of responses
100
75
50
25
0
Q13b
A few times in a month
A few times in a week
A few times in a year
Every day
Never
Once per month or less
Once per week

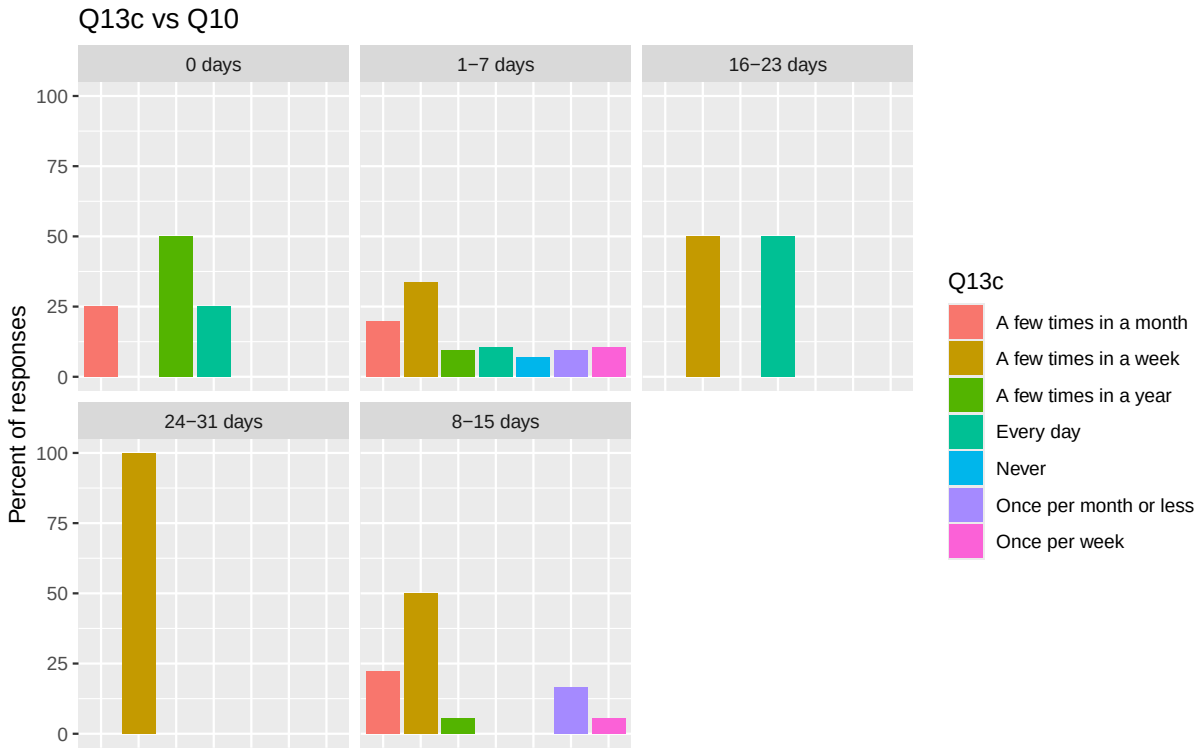

Q13b vs Q11
40 hours or less
41−50 hours
51−60 hours
61−70 hours
71−80 hours
More than 81 hours
Percent of responses
100
75
50
25
0
Q13b
A few times in a month
A few times in a week
A few times in a year
Every day
Never
Once per month or less
Once per week

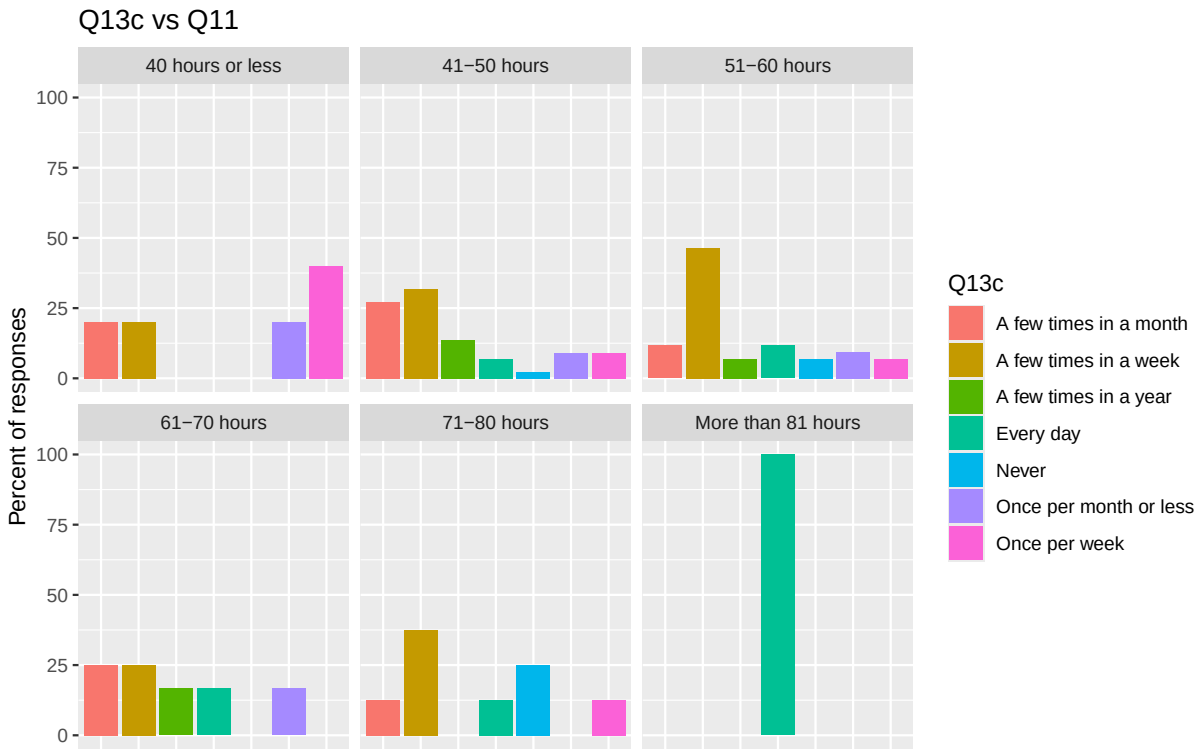

Q13b vs Q12
1 hour or less
2 to 3 hours
4 to 6 hours
7 hours or more
Percent of responses
0
10
20
30
40
50
Q13b
A few times in a month
A few times in a week
A few times in a year
Every day
Never
Once per month or less
Once per week

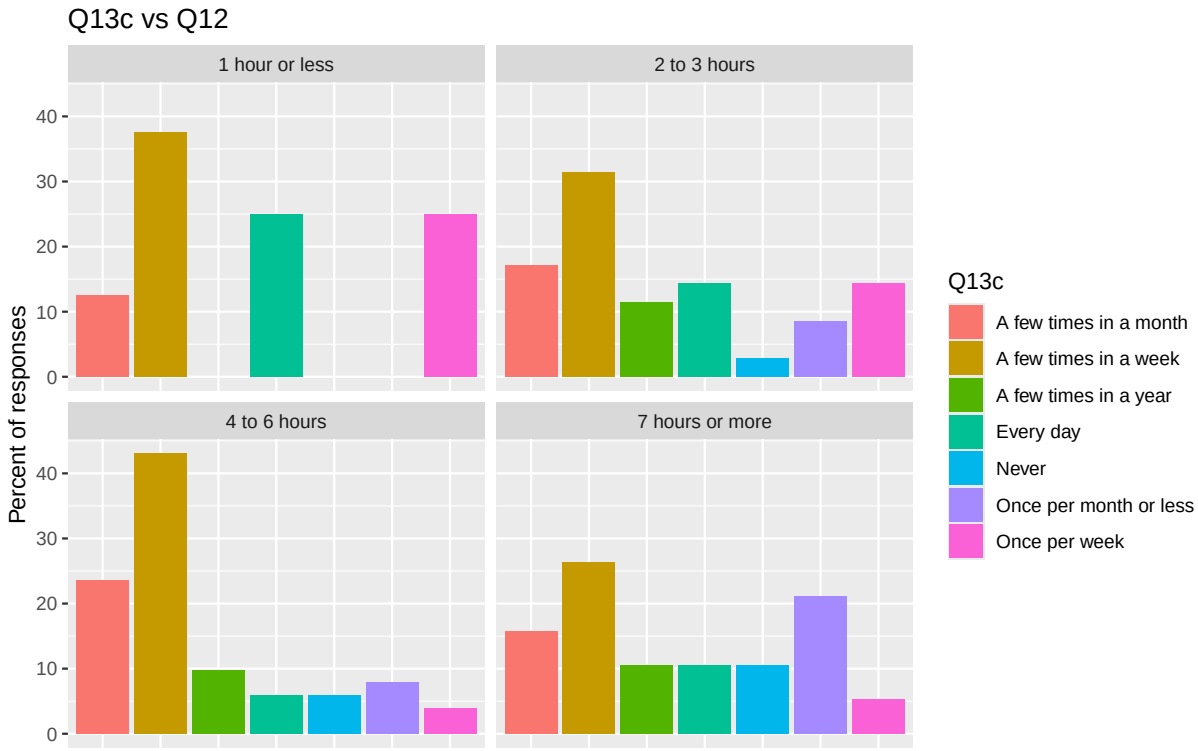

Q13b vs Q1
Hospital−based
−based (University/Hospital Com
University−based
Percent of responses
0
10
20
30
Q13b
A few times in a month
A few times in a week
A few times in a year
Every day
Never
Once per month or less
Once per week

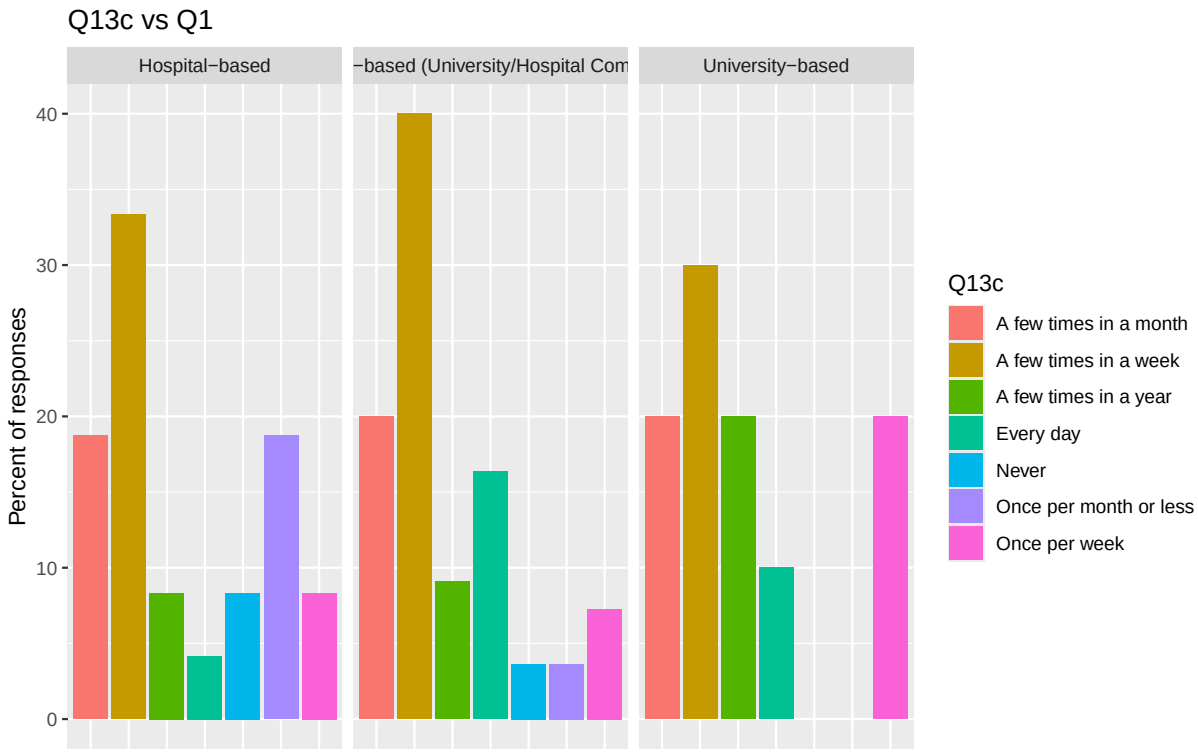

Q13b vs Q5
Divorced
Married
Prefer not to answer
nmarried and not in a relationshi
Unmarried but in a relationship
Percent of responses
100
75
50
25
0
Q13b
A few times in a month
A few times in a week
A few times in a year
Every day
Never
Once per month or less
Once per week


Q13b vs Q2
PGY−1 Level
PGY−2 Level
PGY−3 Level
Percent of responses
30
20
10
0
Q13b
A few times in a month
A few times in a week
A few times in a year
Every day
Never
Once per month or less
Once per week

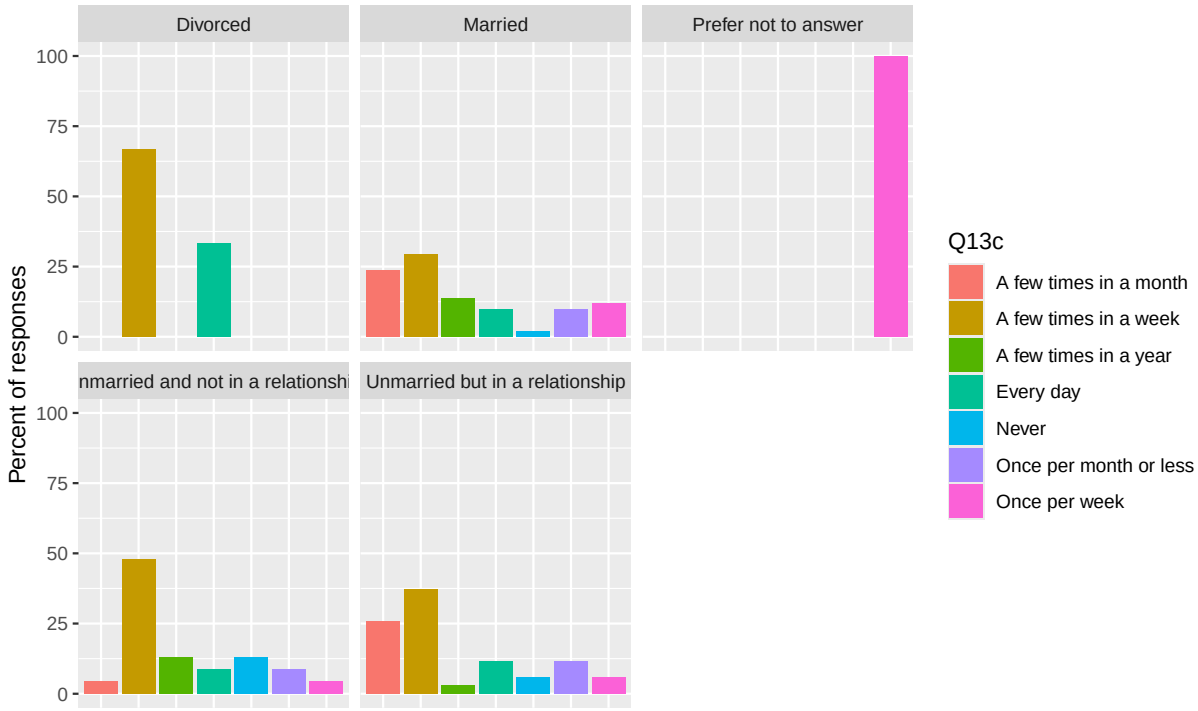

Q13c vs Q10
0 days
1−7 days
16−23 days
24−31 days
8−15 days
Percent of responses
100
75
50
25
0
Q13c
A few times in a month
A few times in a week
A few times in a year
Every day
Never
Once per month or less
Once per week


Q13c vs Q11
40 hours or less
41−50 hours
51−60 hours
61−70 hours
71−80 hours
More than 81 hours
Percent of responses
100
75
50
25
0
Q13c
A few times in a month
A few times in a week
A few times in a year
Every day
Never
Once per month or less
Once per week

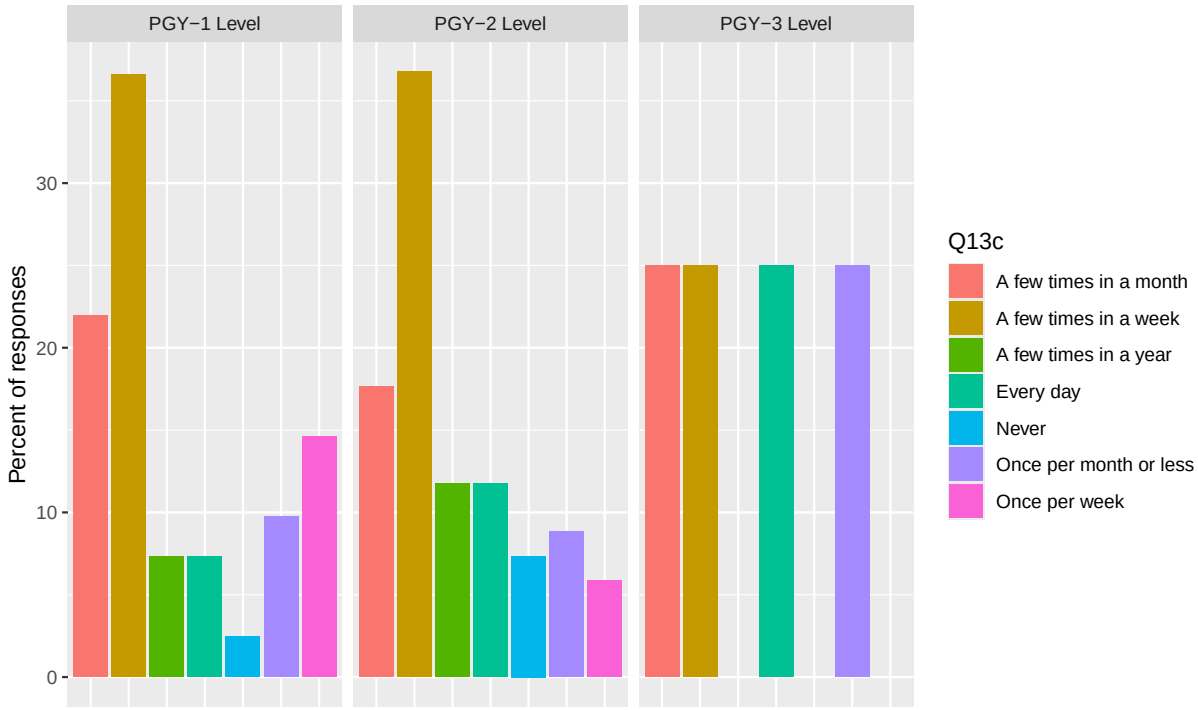

Q13c vs Q12
1 hour or less
2 to 3 hours
4 to 6 hours
7 hours or more
Percent of responses
0
10
20
30
40
Q13c
A few times in a month
A few times in a week
A few times in a year
Every day
Never
Once per month or less
Once per week

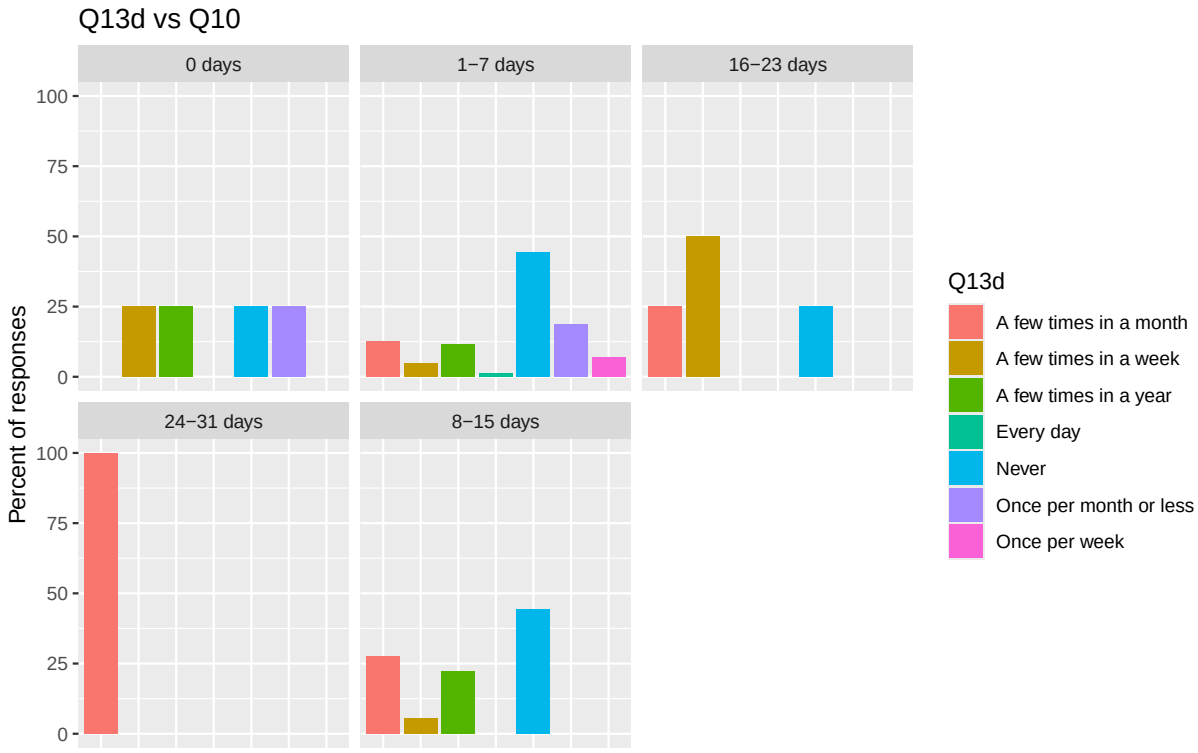

Q13c vs Q1
Hospital−based
−based (University/Hospital Com
University−based
Percent of responses
0
10
20
30
40
Q13c
A few times in a month
A few times in a week
A few times in a year
Every day
Never
Once per month or less
Once per week

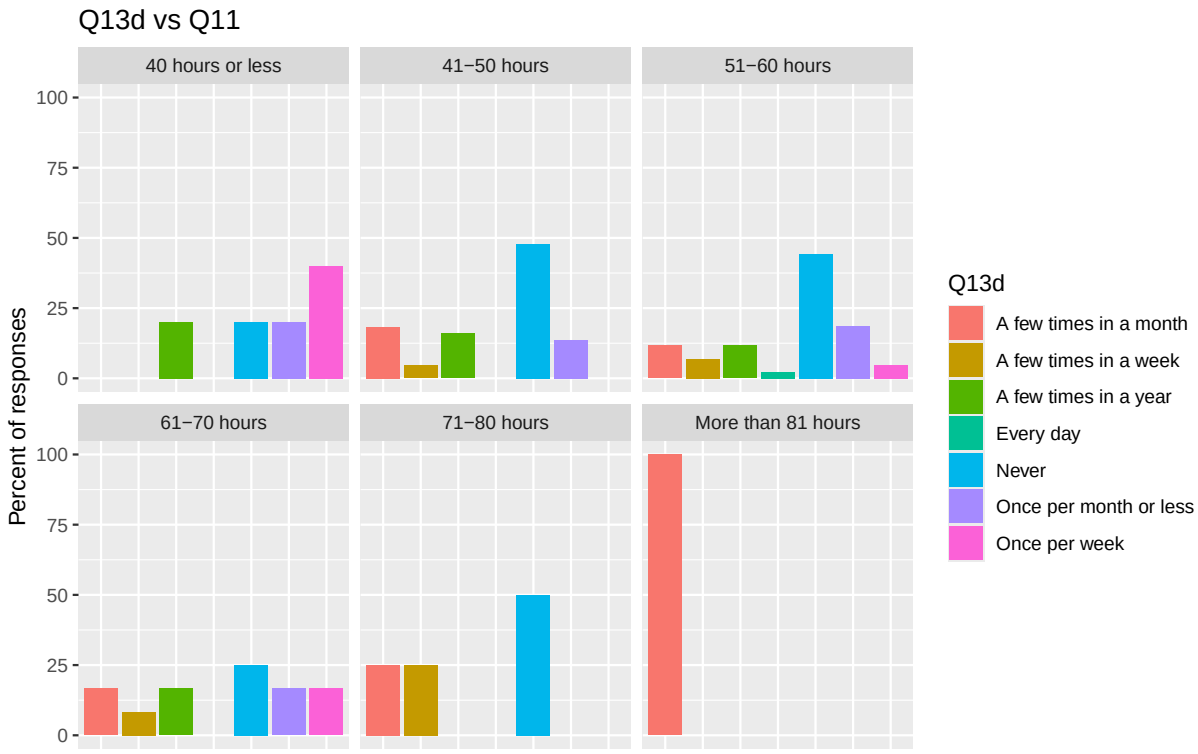
Q13c vs Q5

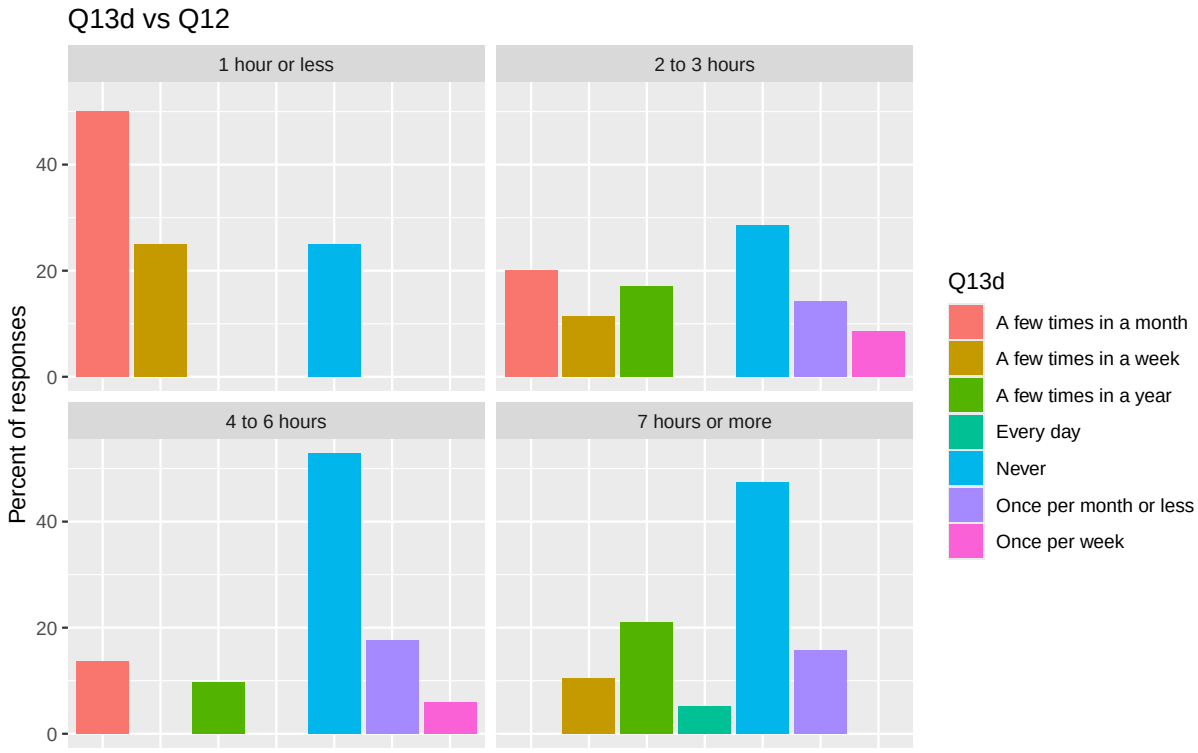


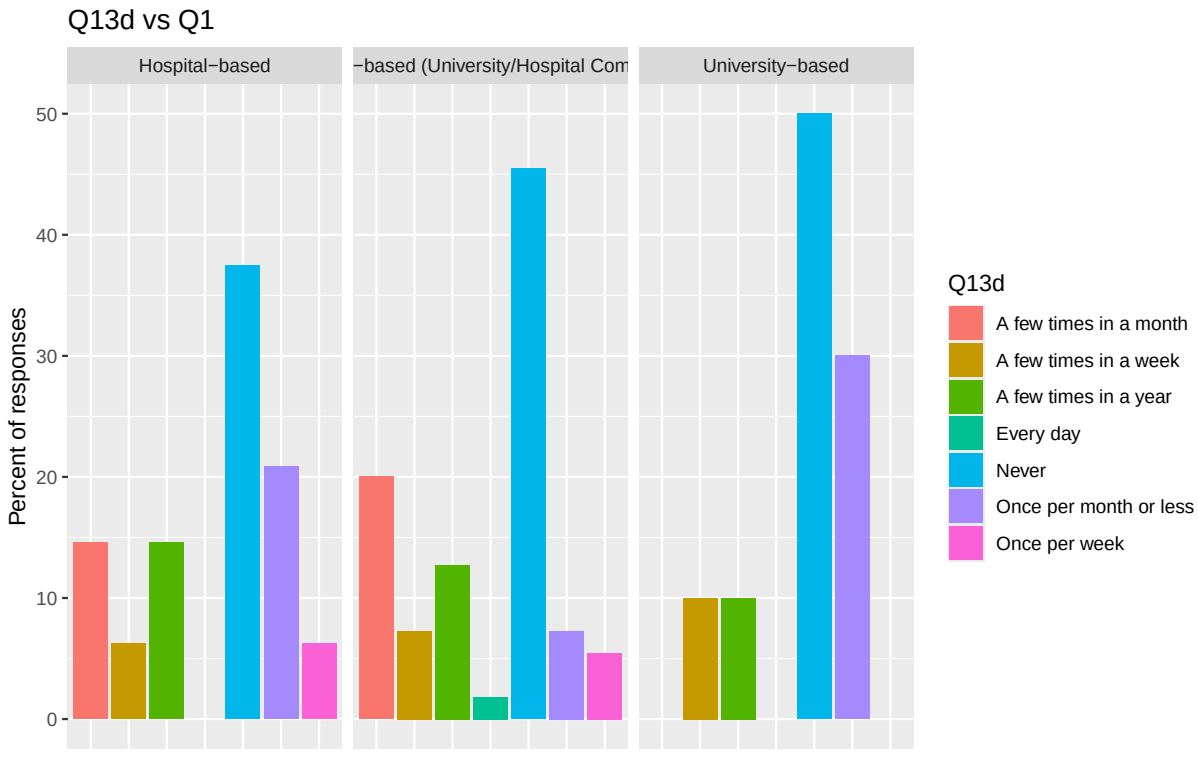
Q13c vs Q2



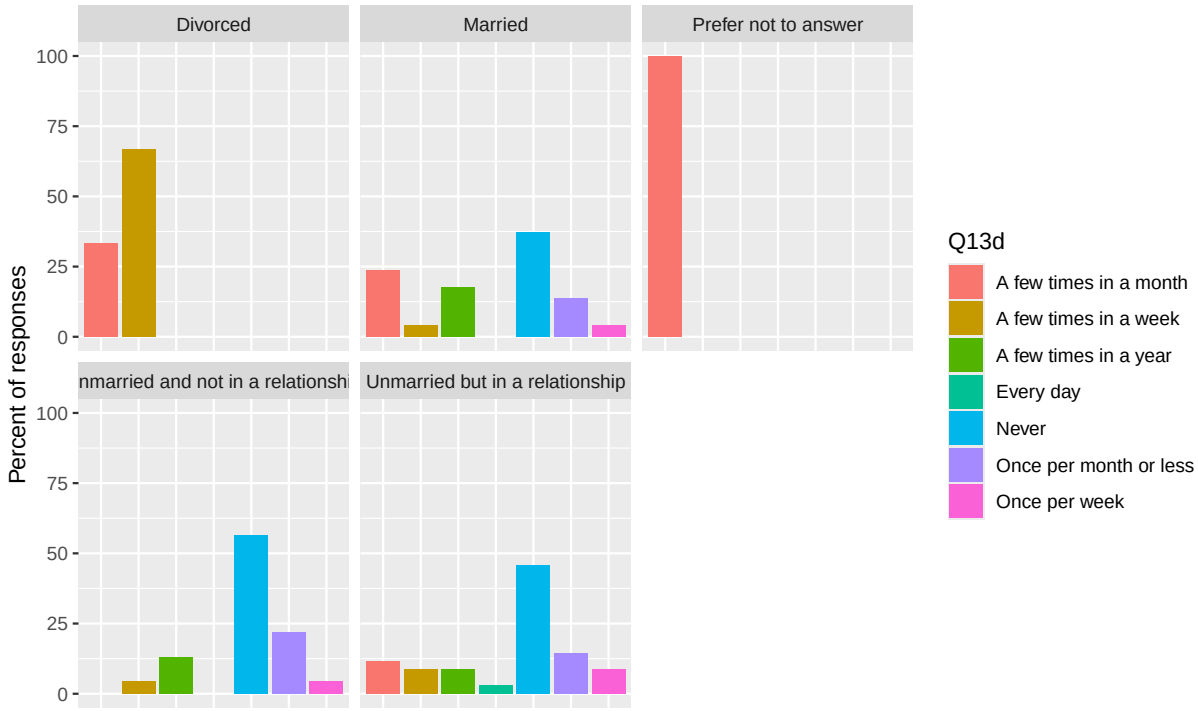

Q13d vs Q10
0 days
1−7 days
16−23 days
24−31 days
8−15 days
Percent of responses
100
75
50
25
0
Q13d
A few times in a month
A few times in a week
A few times in a year
Every day
Never
Once per month or less
Once per week


Q13d vs Q11
40 hours or less
41−50 hours
51−60 hours
61−70 hours
71−80 hours
More than 81 hours
Percent of responses
100
75
50
25
0
Q13d
A few times in a month
A few times in a week
A few times in a year
Every day
Never
Once per month or less
Once per week

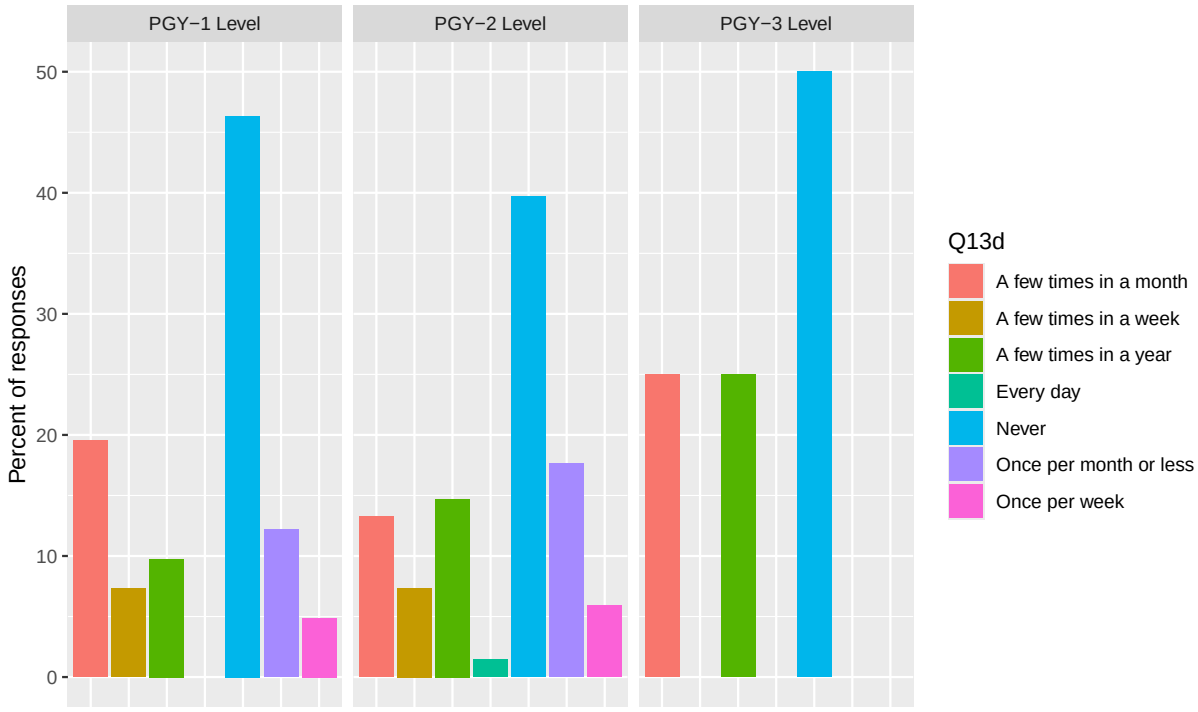

Q13d vs Q12
1 hour or less
2 to 3 hours
4 to 6 hours
7 hours or more
Percent of responses
0
20
40
Q13d
A few times in a month
A few times in a week
A few times in a year
Every day
Never
Once per month or less
Once per week

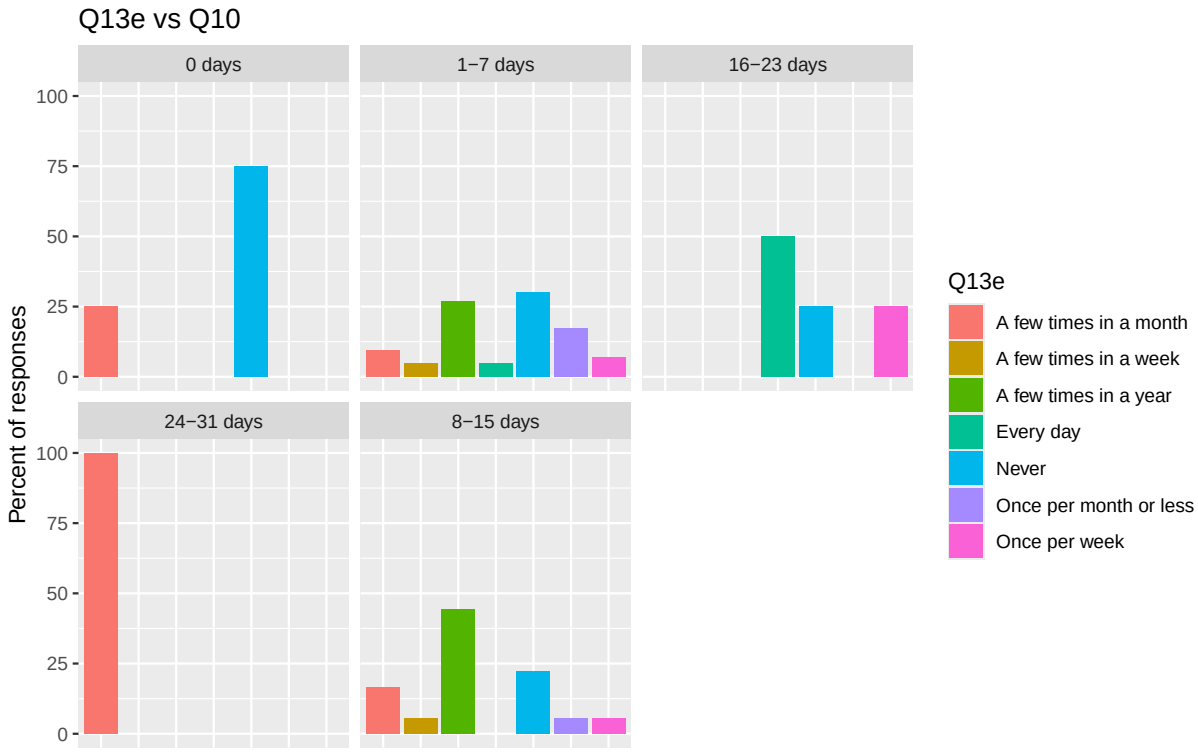

Q13d vs Q1
Hospital−based
−based (University/Hospital Com
University−based
Percent of responses
0
10
20
30
40
50
Q13d
A few times in a month
A few times in a week
A few times in a year
Every day
Never
Once per month or less
Once per week

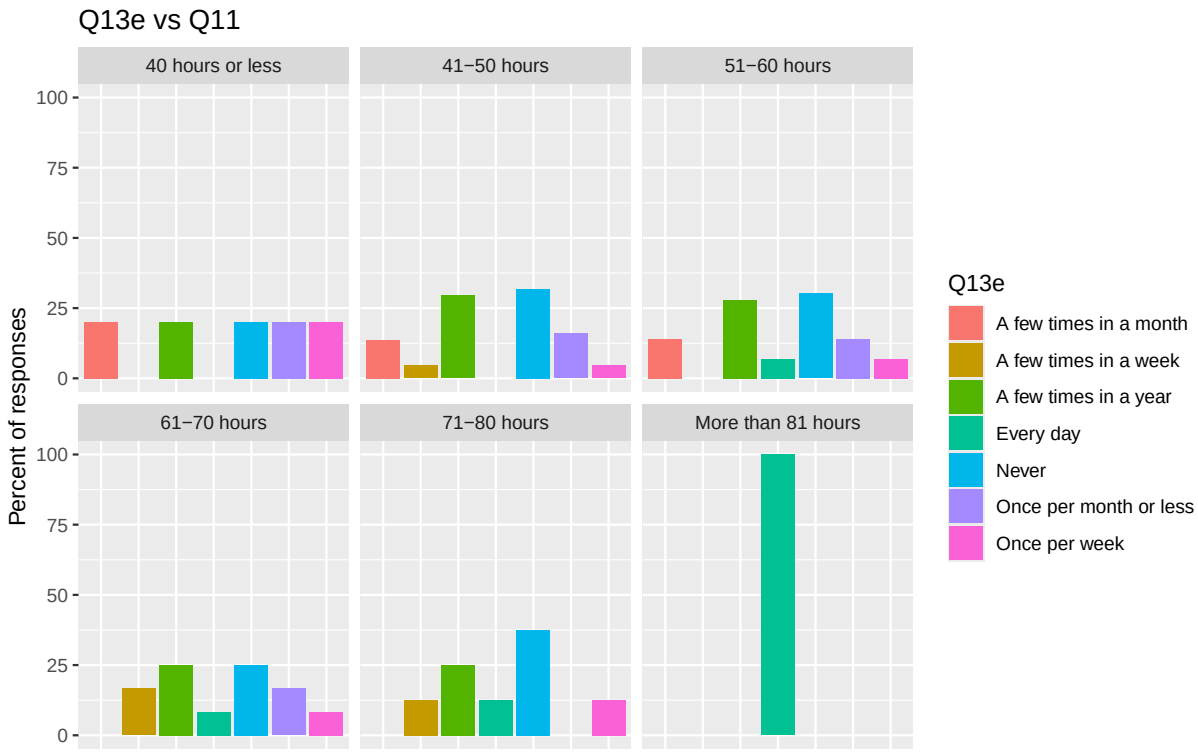
Q13d vs Q5

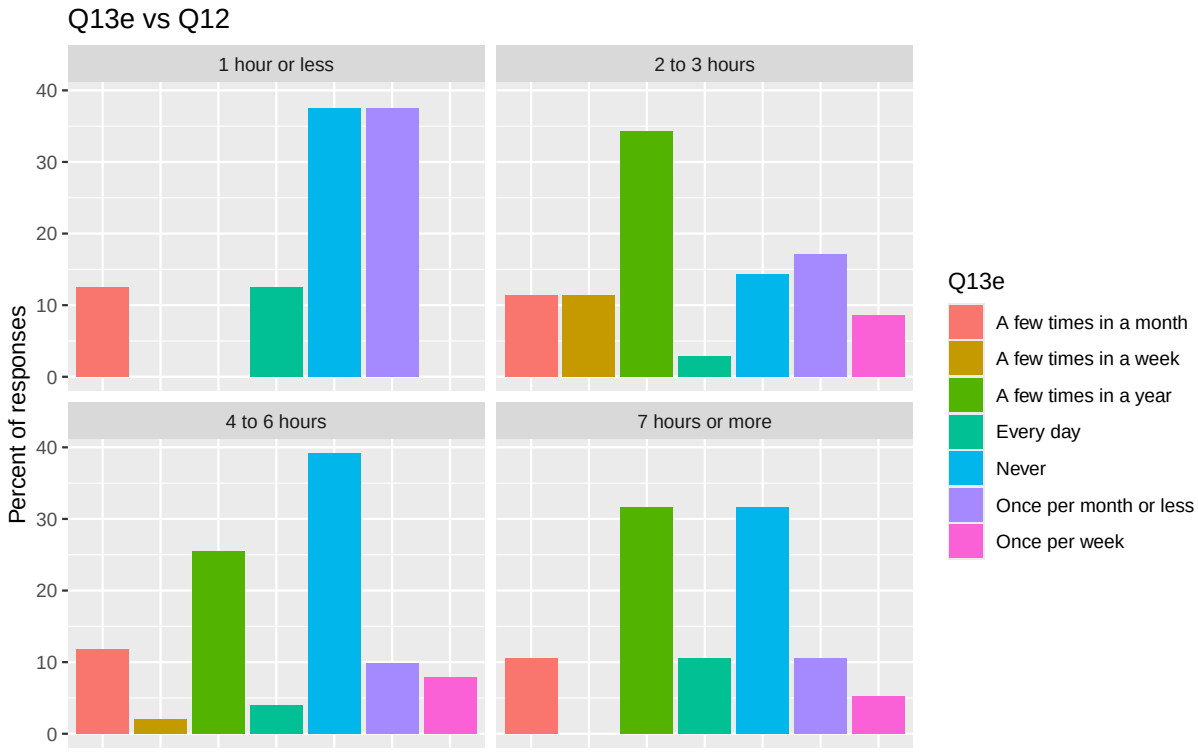


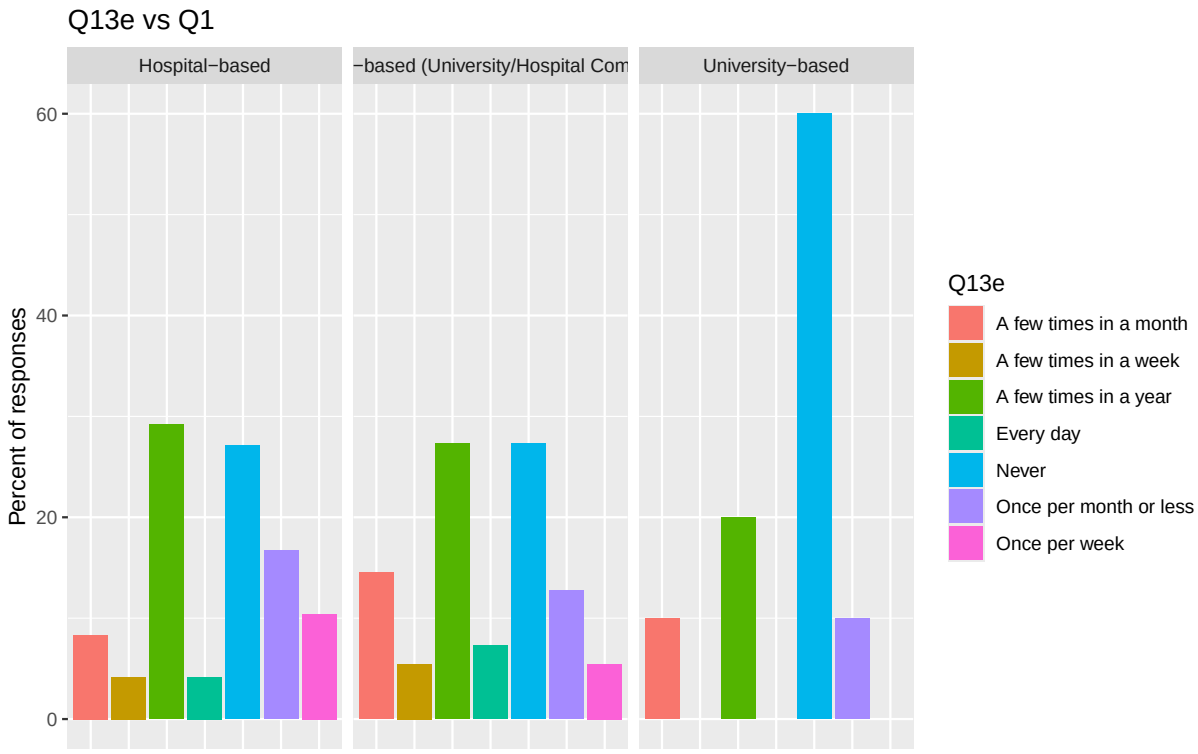
Q13d vs Q2

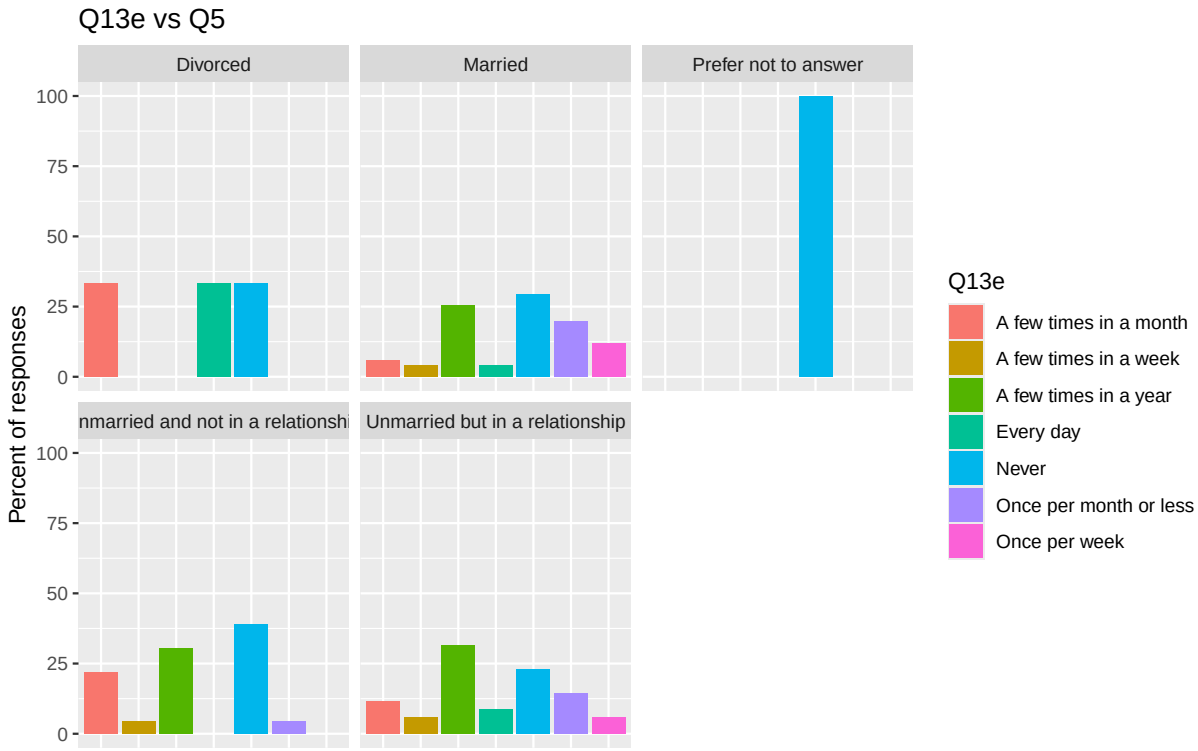


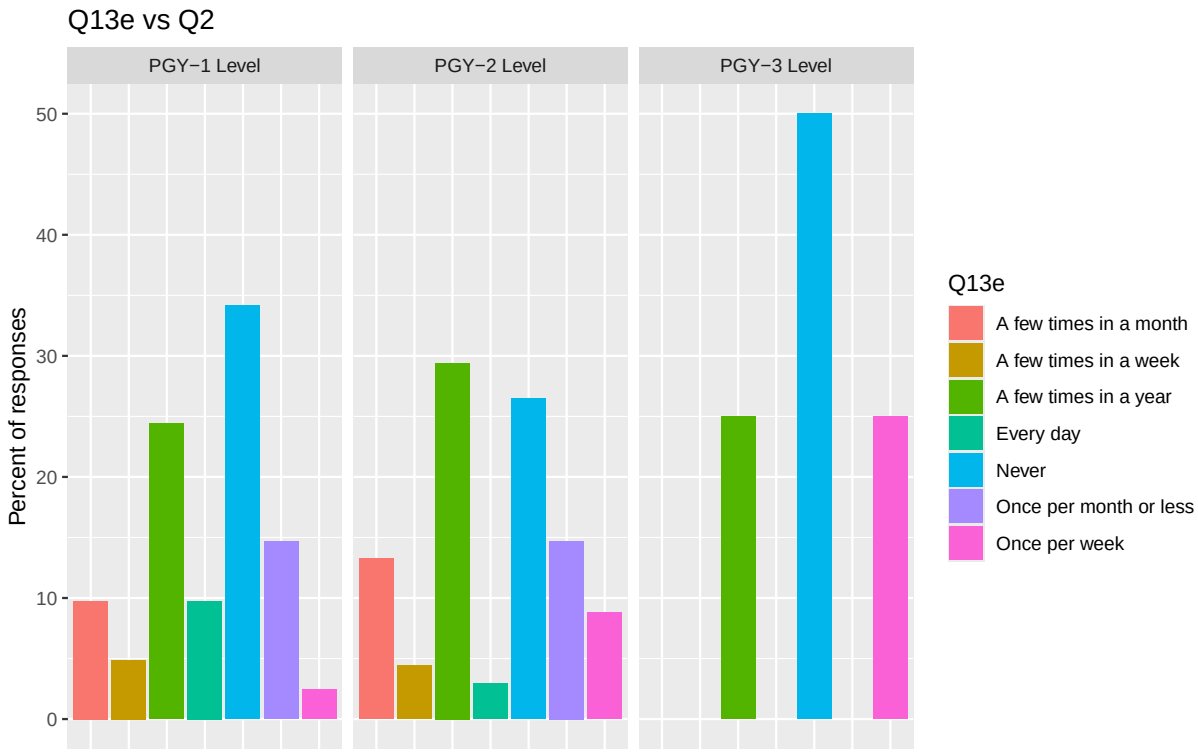

Q13e vs Q10
0 days
1−7 days
16−23 days
24−31 days
8−15 days
Percent of responses
100
75
50
25
0
Q13e
A few times in a month
A few times in a week
A few times in a year
Every day
Never
Once per month or less
Once per week

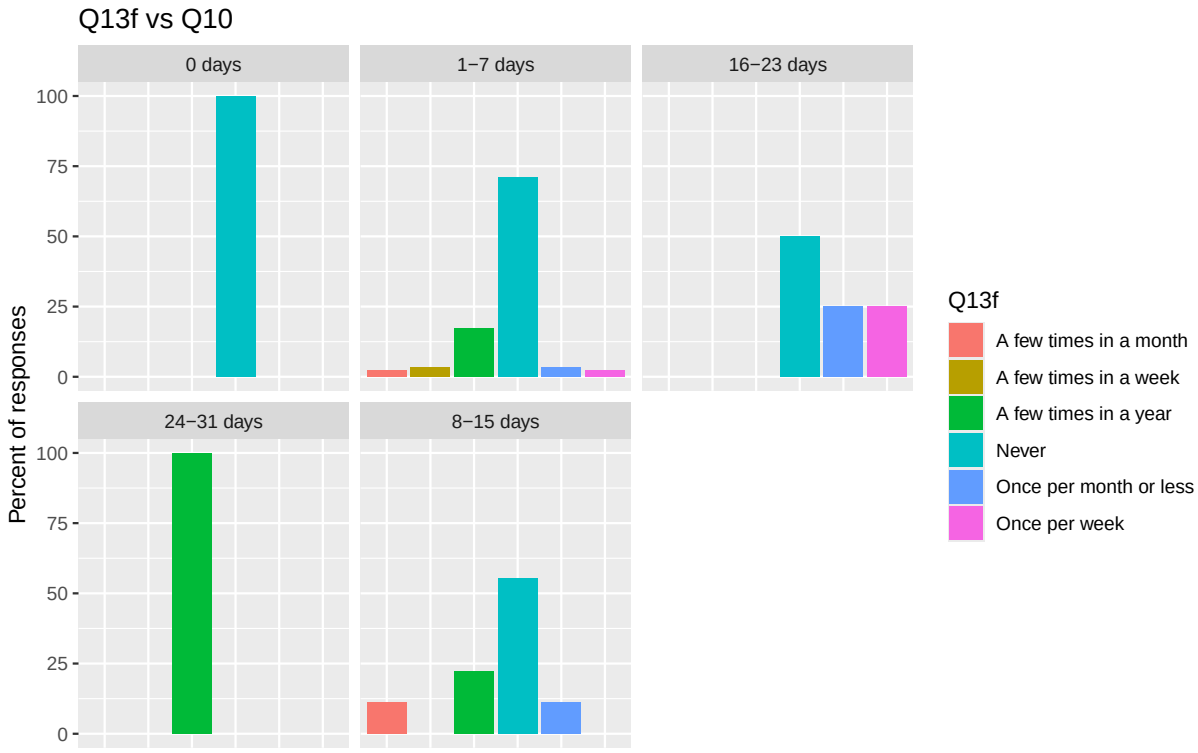

Q13e vs Q11
40 hours or less
41−50 hours
51−60 hours
61−70 hours
71−80 hours
More than 81 hours
Percent of responses
100
75
50
25
0
Q13e
A few times in a month
A few times in a week
A few times in a year
Every day
Never
Once per month or less
Once per week

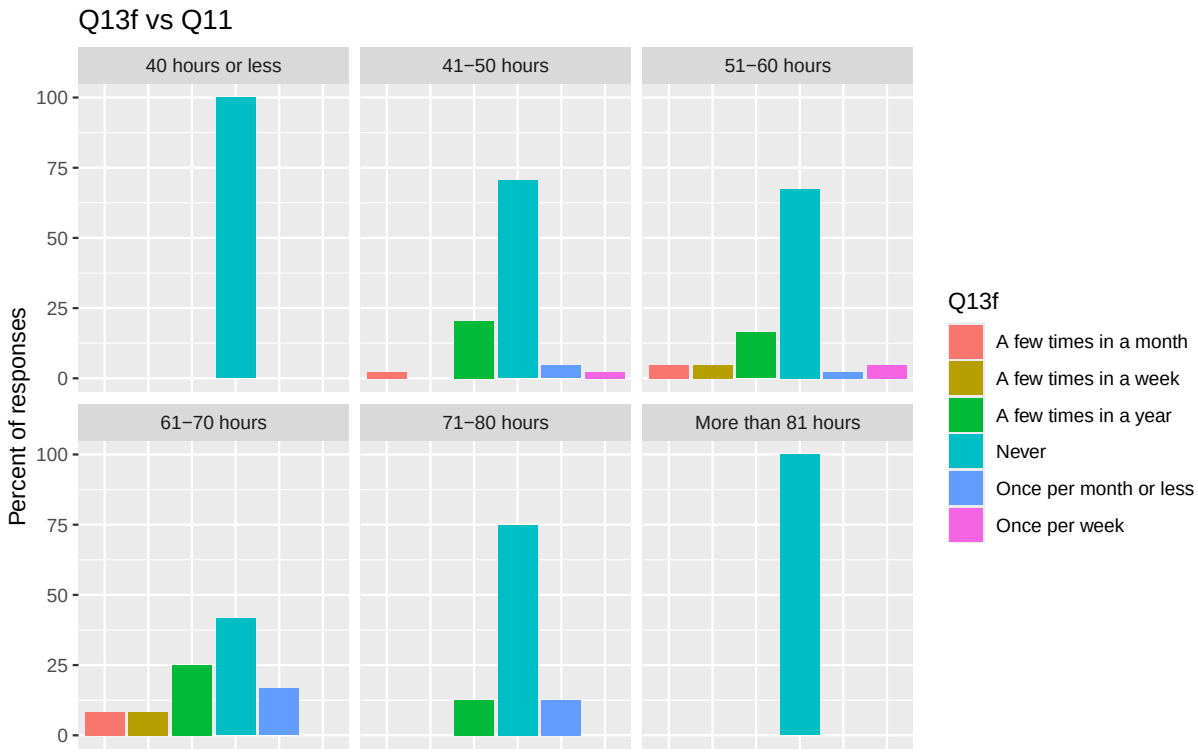

Q13e vs Q12
1 hour or less
2 to 3 hours
4 to 6 hours
7 hours or more
Percent of responses
0
10
20
30
40
Q13e
A few times in a month
A few times in a week
A few times in a year
Every day
Never
Once per month or less
Once per week

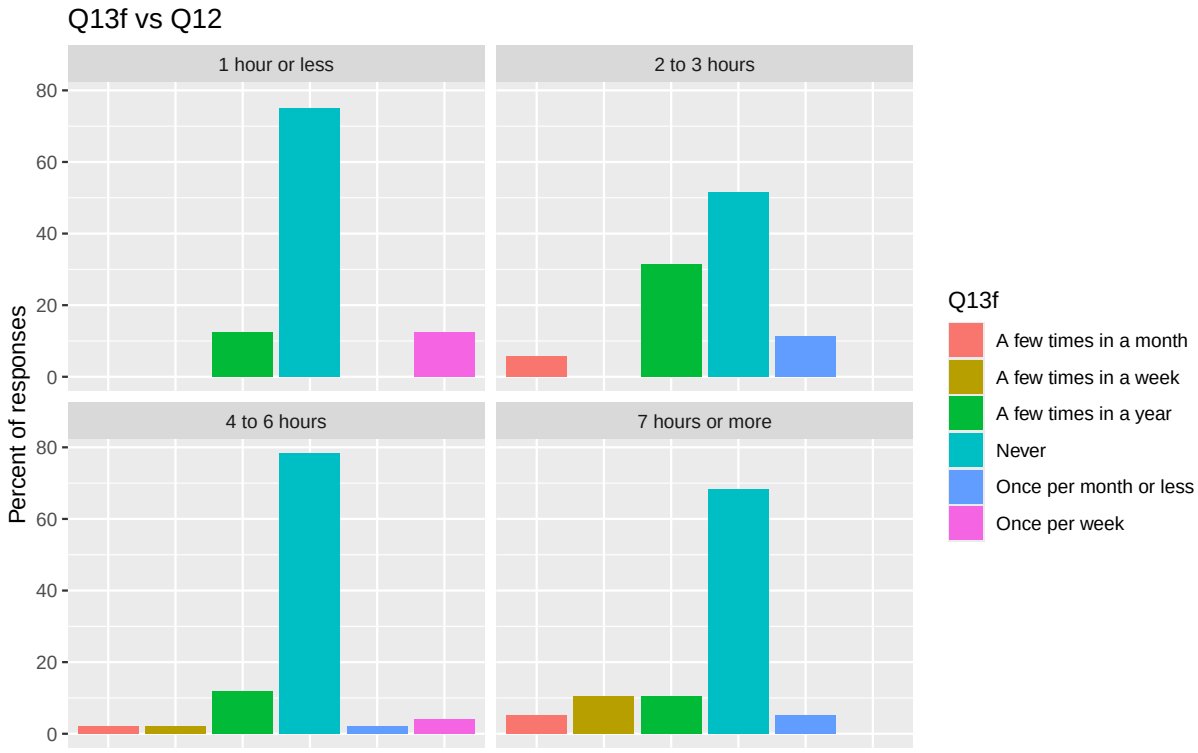

Q13e vs Q1
Hospital−based
−based (University/Hospital Com
University−based
Percent of responses
0
20
40
60
Q13e
A few times in a month
A few times in a week
A few times in a year
Every day
Never
Once per month or less
Once per week

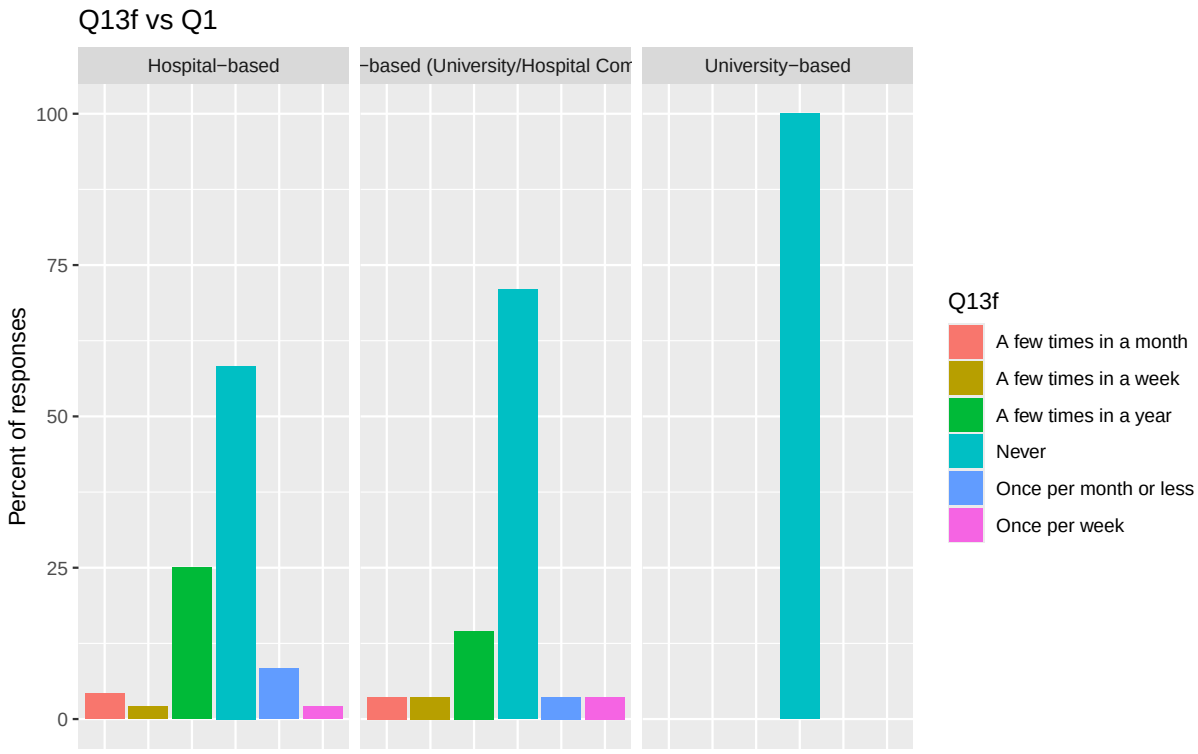

Q13e vs Q5
Divorced
Married
Prefer not to answer
nmarried and not in a relationshi
Unmarried but in a relationship
Percent of responses
100
75
50
25
0
Q13e
A few times in a month
A few times in a week
A few times in a year
Every day
Never
Once per month or less
Once per week

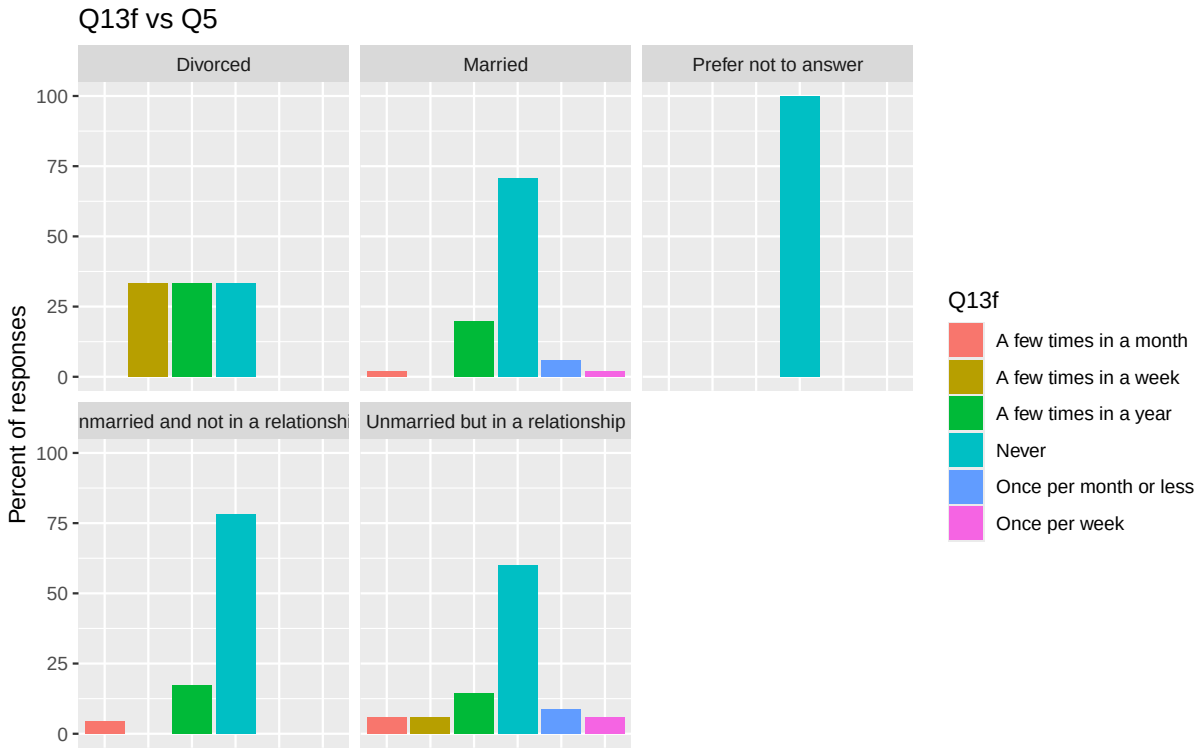

Q13e vs Q2
PGY−1 Level
PGY−2 Level
PGY−3 Level
Percent of responses
50
40
30
20
10
0
Q13e
A few times in a month
A few times in a week
A few times in a year
Every day
Never
Once per month or less
Once per week

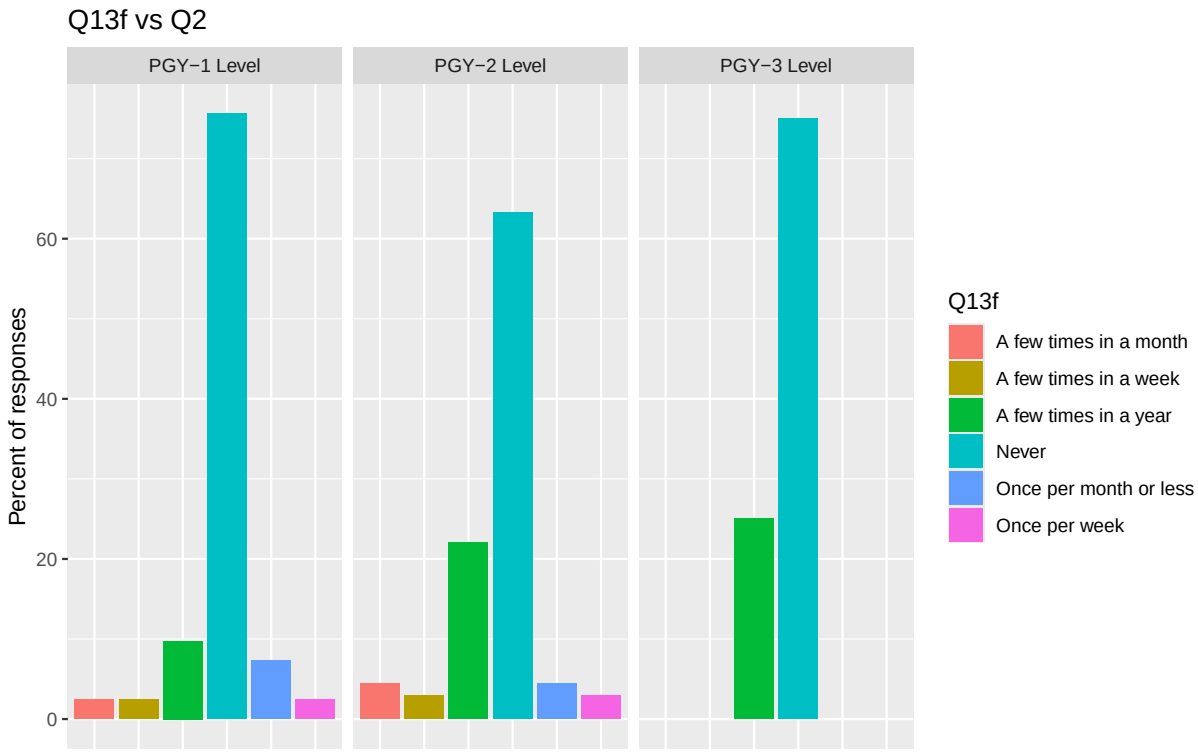

Q13f vs Q10
0 days
1−7 days
16−23 days
24−31 days
8−15 days
Percent of responses
100
75
50
25
0
Q13f
A few times in a month
A few times in a week
A few times in a year
Never
Once per month or less
Once per week

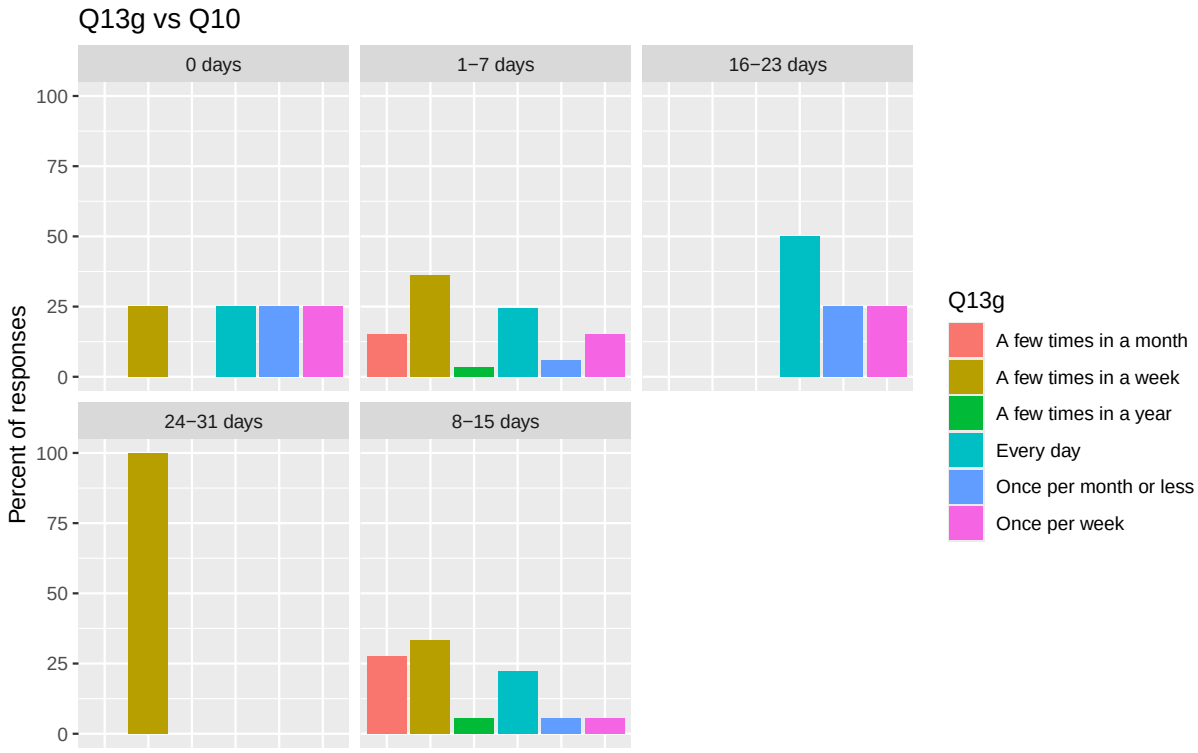

Q13f vs Q11
40 hours or less
41−50 hours
51−60 hours
61−70 hours
71−80 hours
More than 81 hours
Percent of responses
100
75
50
25
0
Q13f
A few times in a month
A few times in a week
A few times in a year
Never
Once per month or less
Once per week

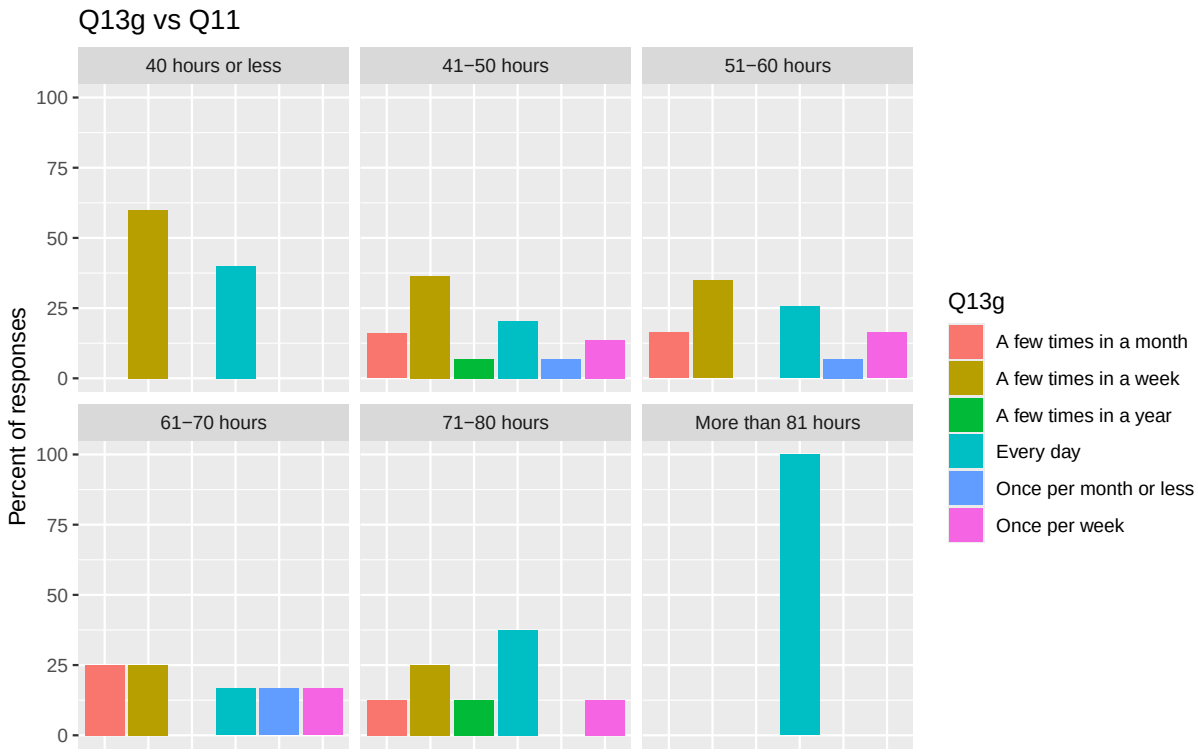

Q13f vs Q12
1 hour or less
2 to 3 hours
4 to 6 hours
7 hours or more
Percent of responses
0
20
40
60
80
Q13f
A few times in a month
A few times in a week
A few times in a year
Never
Once per month or less
Once per week

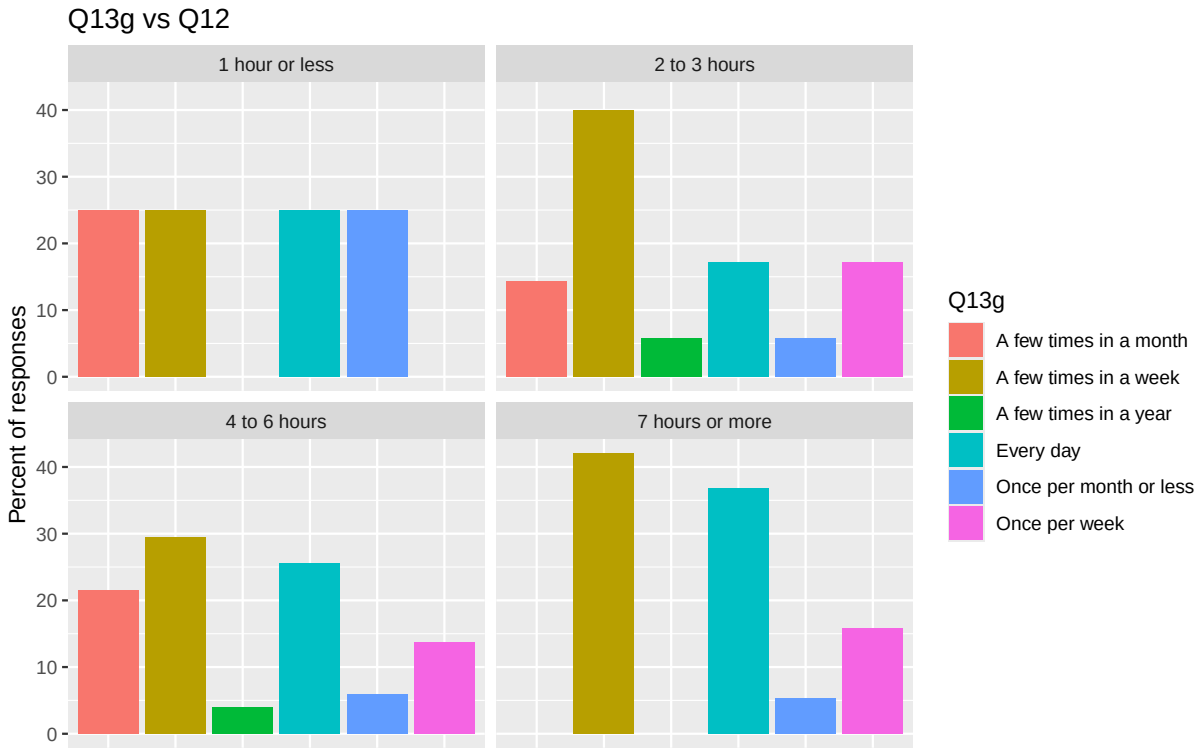

Q13f vs Q1
Hospital−based
−based (University/Hospital Com
University−based
Percent of responses
0
25
50
75
100
Q13f
A few times in a month
A few times in a week
A few times in a year
Never
Once per month or less
Once per week

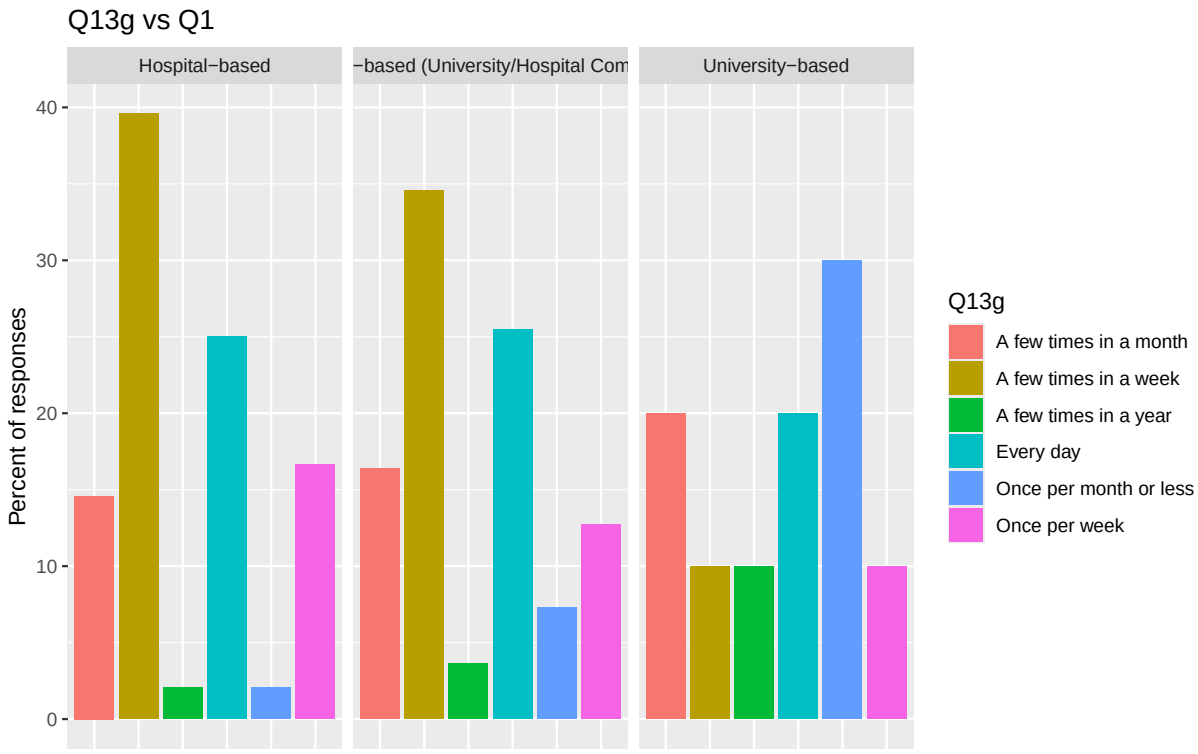

Q13f vs Q5
Divorced
Married
Prefer not to answer
nmarried and not in a relationshi
Unmarried but in a relationship
Percent of responses
100
75
50
25
0
Q13f
A few times in a month
A few times in a week
A few times in a year
Never
Once per month or less
Once per week


Q13f vs Q2
PGY−1 Level
PGY−2 Level
PGY−3 Level
Percent of responses
60
40
20
0
Q13f
A few times in a month
A few times in a week
A few times in a year
Never
Once per month or less
Once per week

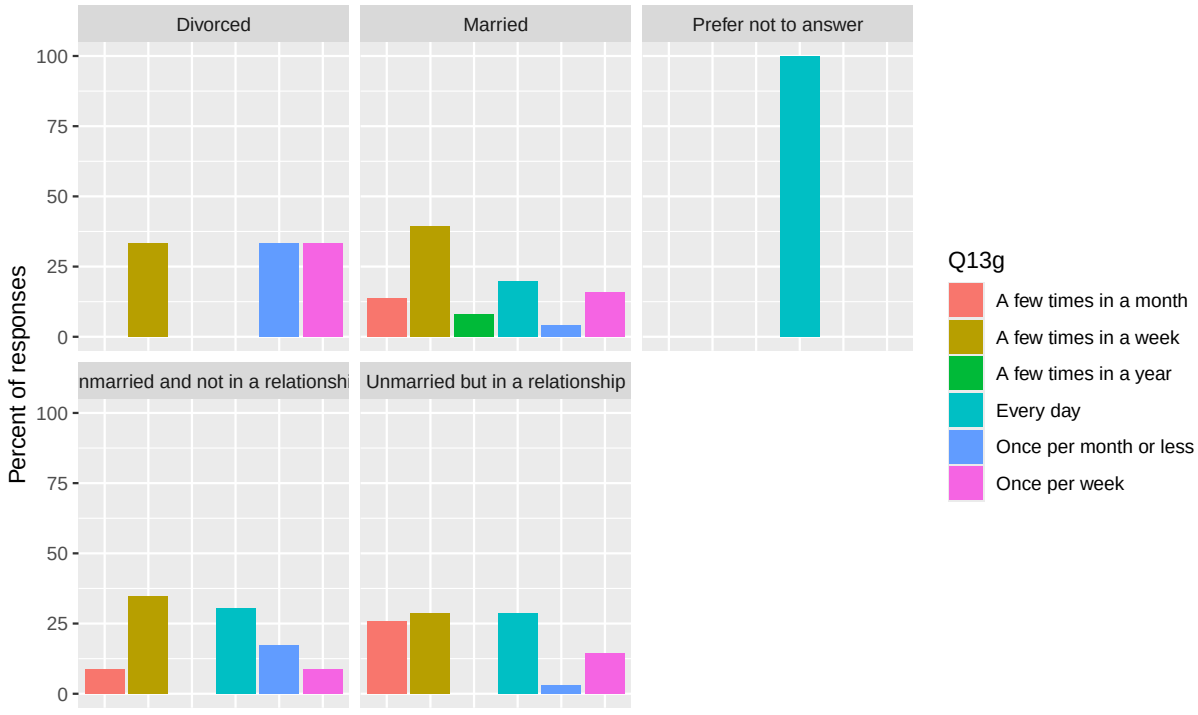

Q13g vs Q10
0 days
1−7 days
16−23 days
24−31 days
8−15 days
Percent of responses
100
75
50
25
0
Q13g
A few times in a month
A few times in a week
A few times in a year
Every day
Once per month or less
Once per week


Q13g vs Q11
40 hours or less
41−50 hours
51−60 hours
61−70 hours
71−80 hours
More than 81 hours
Percent of responses
100
75
50
25
0
Q13g
A few times in a month
A few times in a week
A few times in a year
Every day
Once per month or less
Once per week

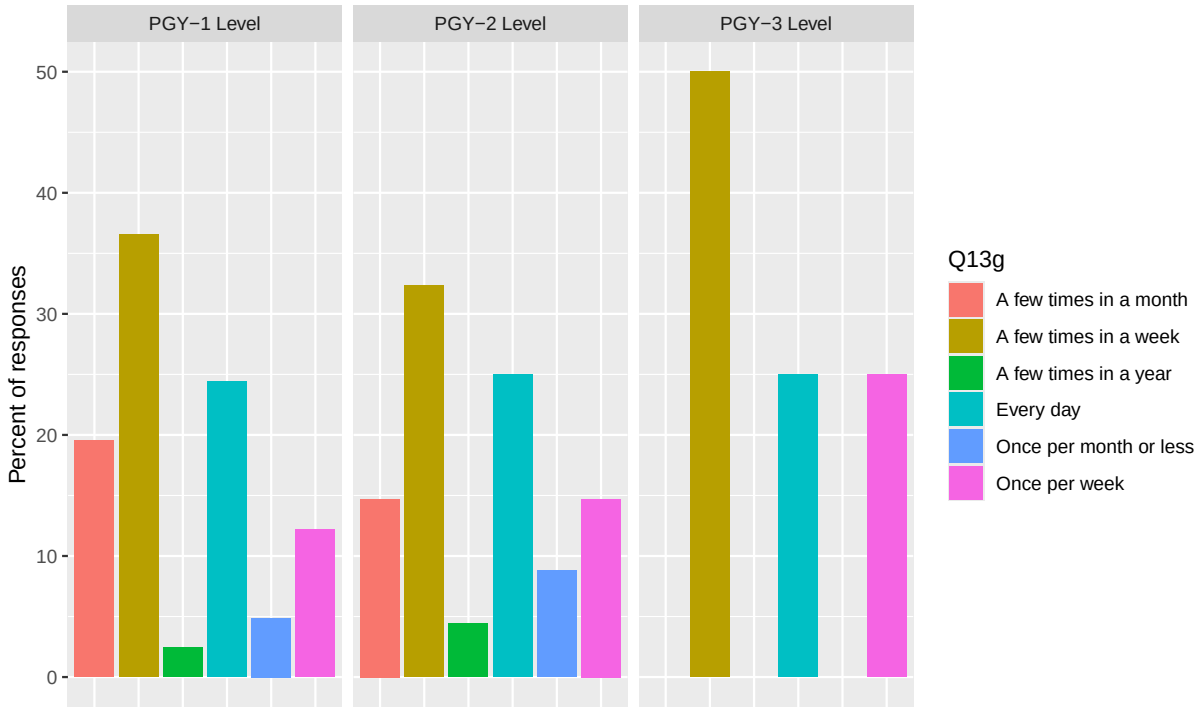

Q13g vs Q12
1 hour or less
2 to 3 hours
4 to 6 hours
7 hours or more
Percent of responses
0
10
20
30
40
Q13g
A few times in a month
A few times in a week
A few times in a year
Every day
Once per month or less
Once per week

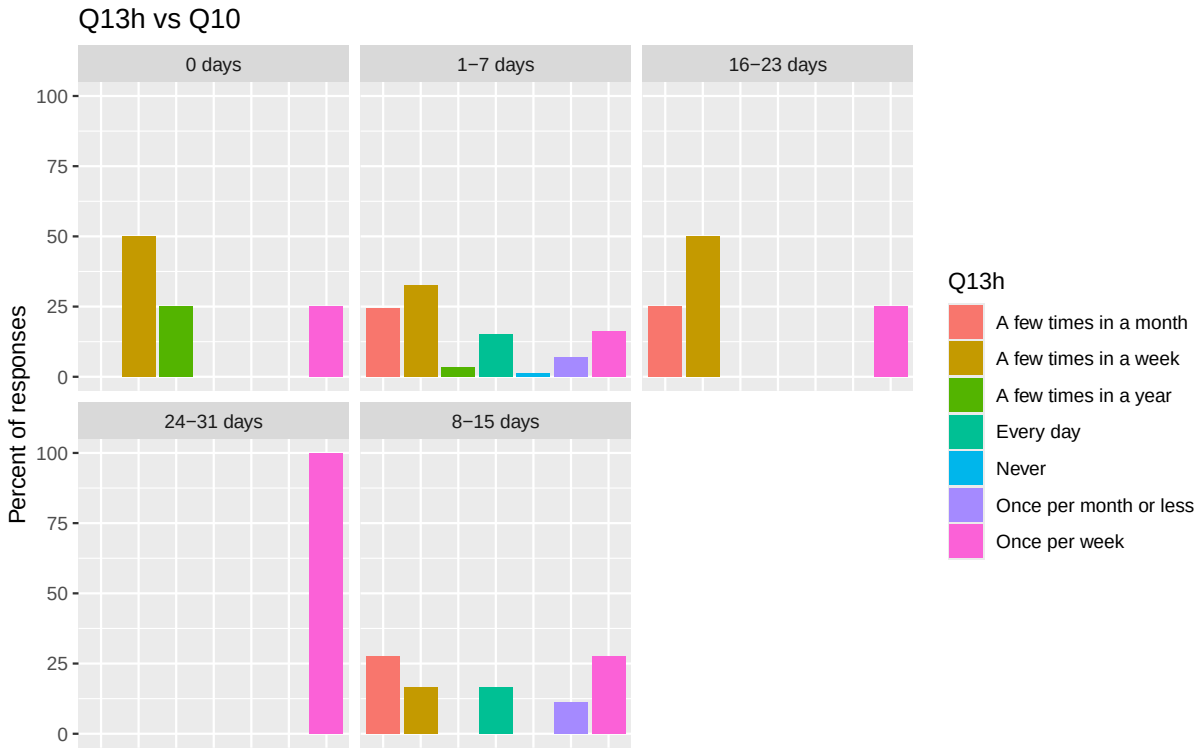

Q13g vs Q1
Hospital−based
−based (University/Hospital Com
University−based
Percent of responses
0
10
20
30
40
Q13g
A few times in a month
A few times in a week
A few times in a year
Every day
Once per month or less
Once per week

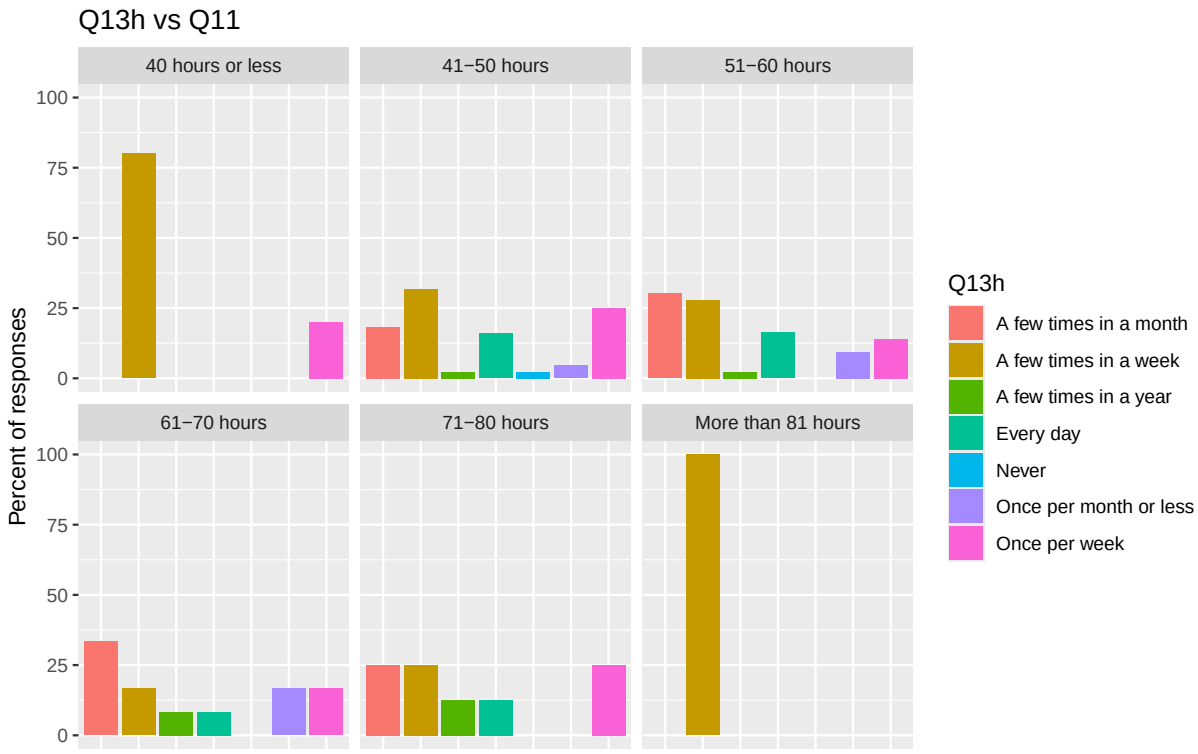
Q13g vs Q5

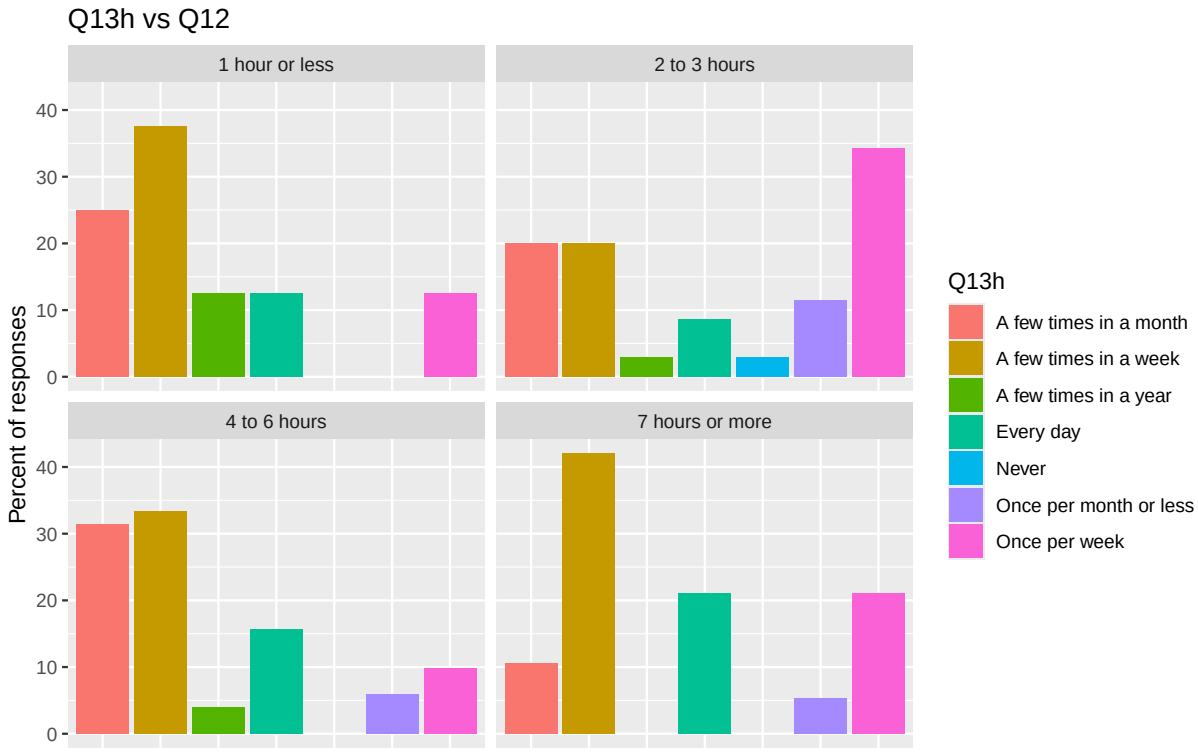


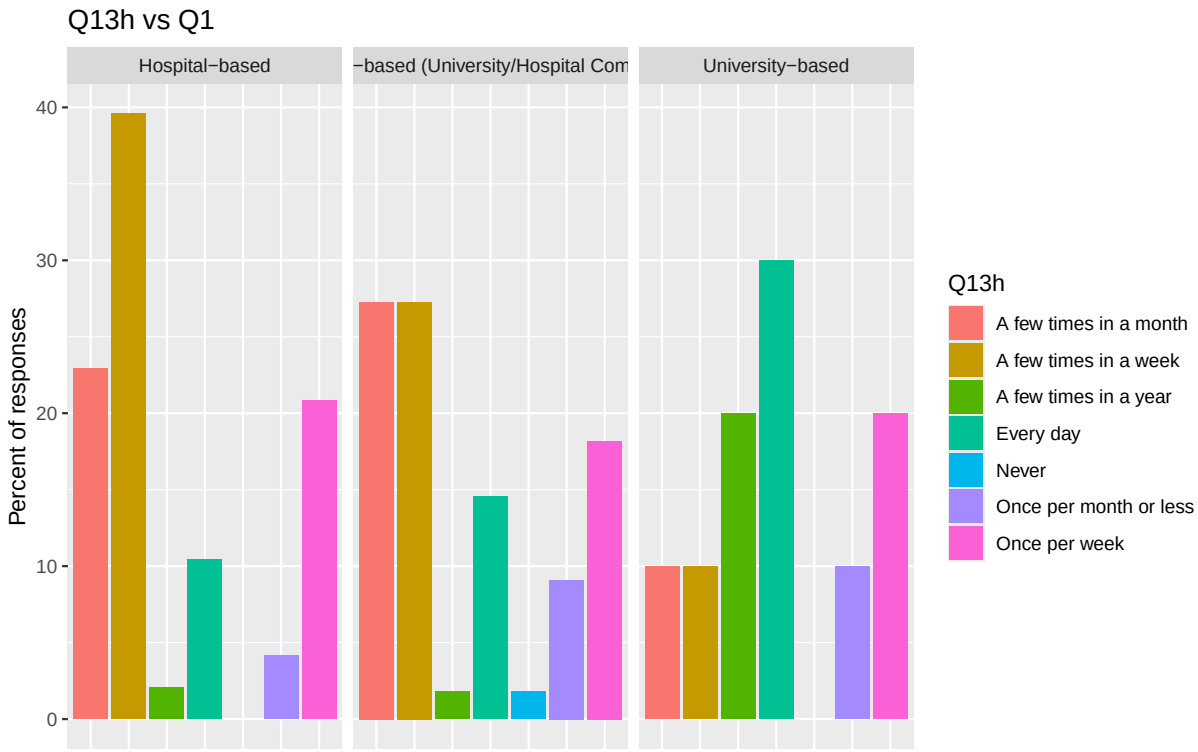
Q13g vs Q2



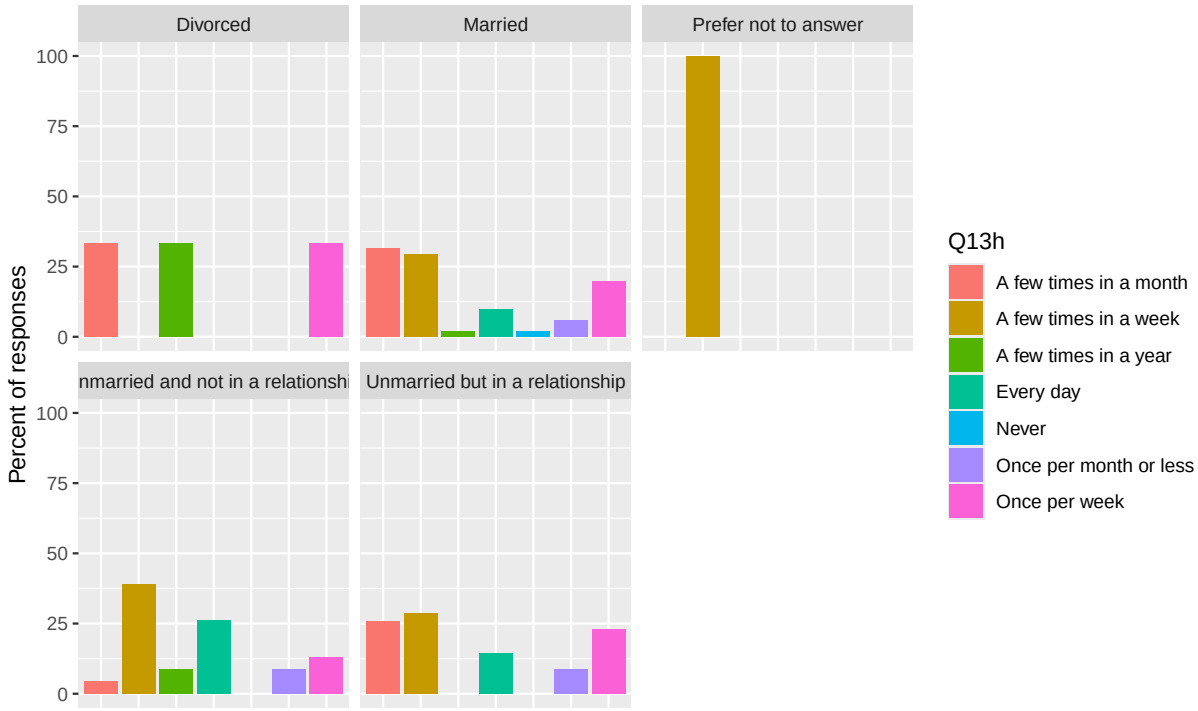

Q13h vs Q10
0 days
1−7 days
16−23 days
24−31 days
8−15 days
Percent of responses
100
75
50
25
0
Q13h
A few times in a month
A few times in a week
A few times in a year
Every day
Never
Once per month or less
Once per week


Q13h vs Q11
40 hours or less
41−50 hours
51−60 hours
61−70 hours
71−80 hours
More than 81 hours
Percent of responses
100
75
50
25
0
Q13h
A few times in a month
A few times in a week
A few times in a year
Every day
Never
Once per month or less
Once per week

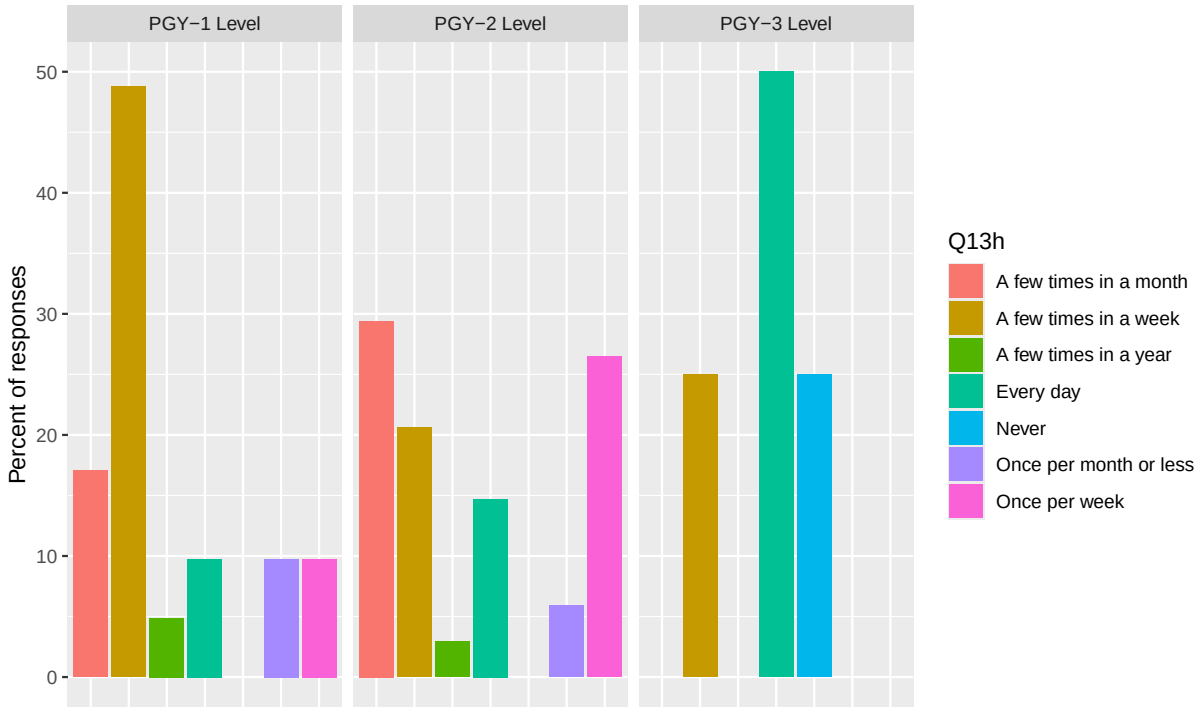

Q13h vs Q12
1 hour or less
2 to 3 hours
4 to 6 hours
7 hours or more
Percent of responses
0
10
20
30
40
Q13h
A few times in a month
A few times in a week
A few times in a year
Every day
Never
Once per month or less
Once per week

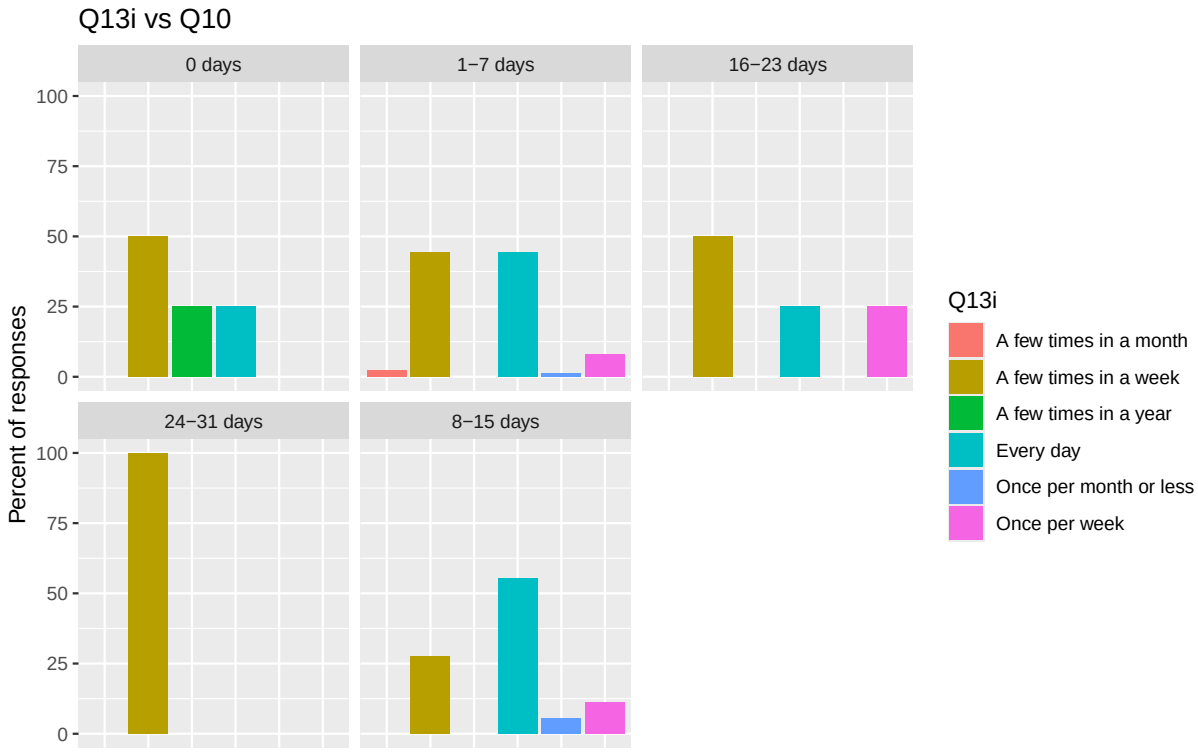

Q13h vs Q1
Hospital−based
−based (University/Hospital Com
University−based
Percent of responses
0
10
20
30
40
Q13h
A few times in a month
A few times in a week
A few times in a year
Every day
Never
Once per month or less
Once per week

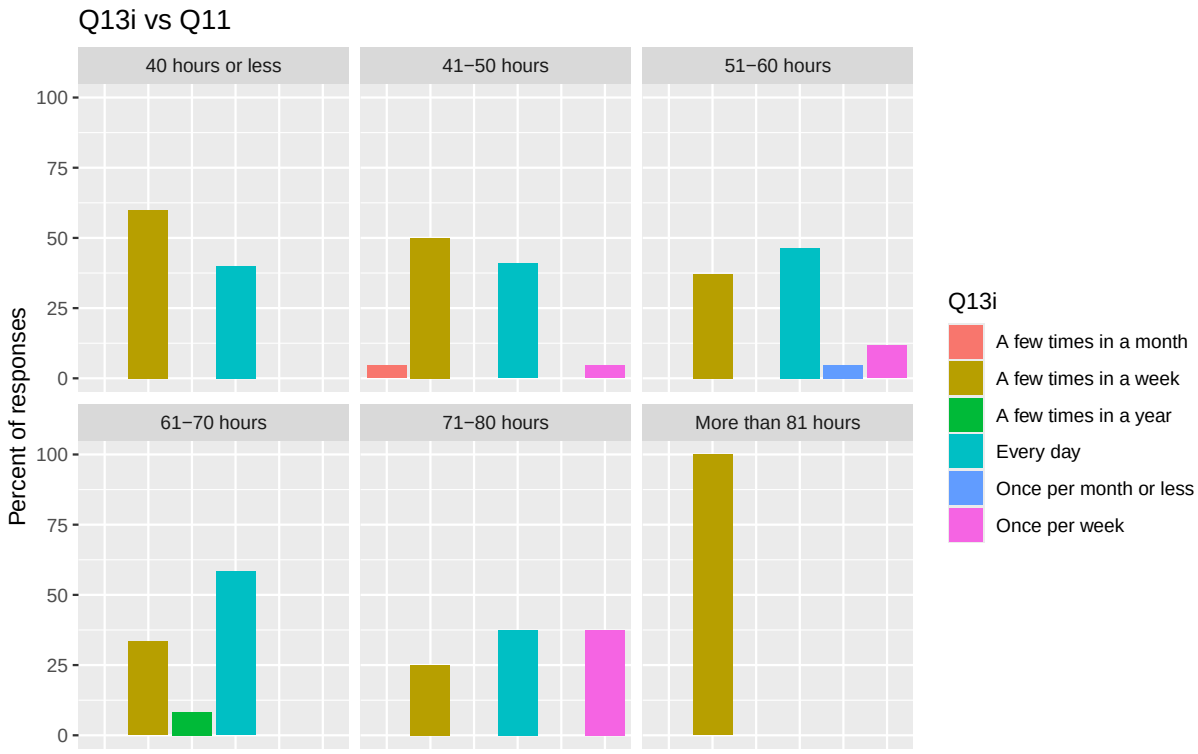
Q13h vs Q5

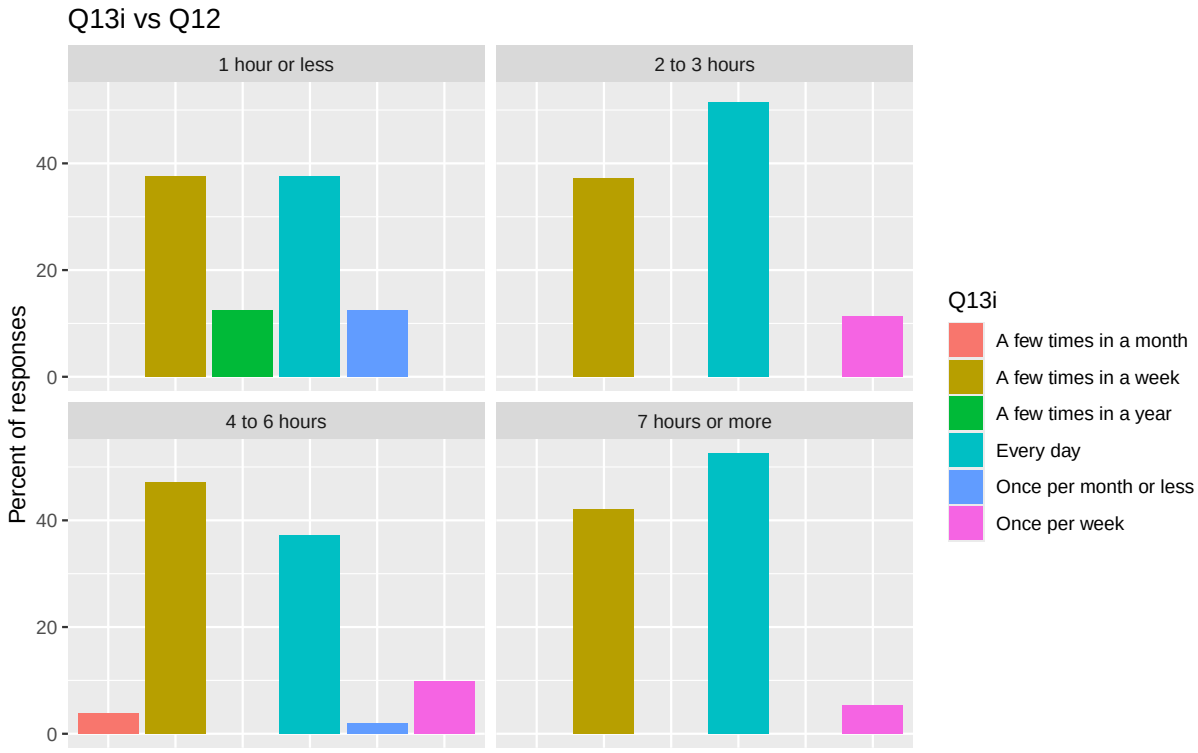


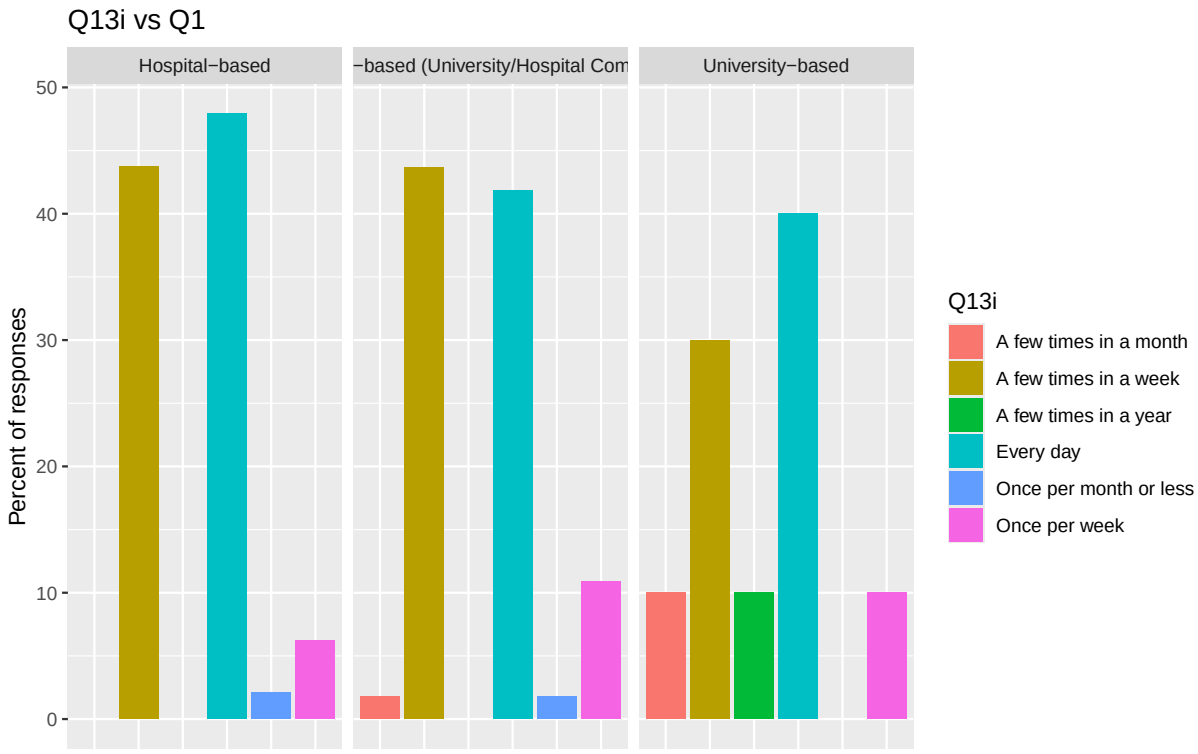
Q13h vs Q2



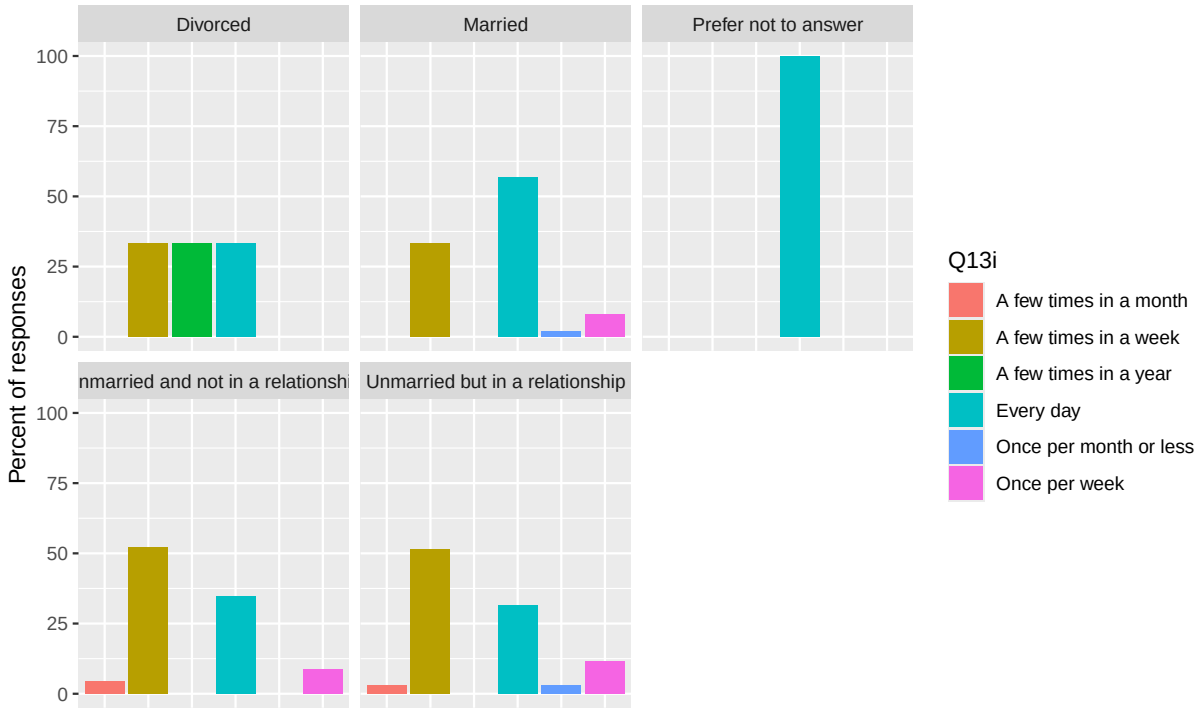

Q13i vs Q10
0 days
1−7 days
16−23 days
24−31 days
8−15 days
Percent of responses
100
75
50
25
0
Q13i
A few times in a month
A few times in a week
A few times in a year
Every day
Once per month or less
Once per week


Q13i vs Q11
40 hours or less
41−50 hours
51−60 hours
61−70 hours
71−80 hours
More than 81 hours
Percent of responses
100
75
50
25
0
Q13i
A few times in a month
A few times in a week
A few times in a year
Every day
Once per month or less
Once per week

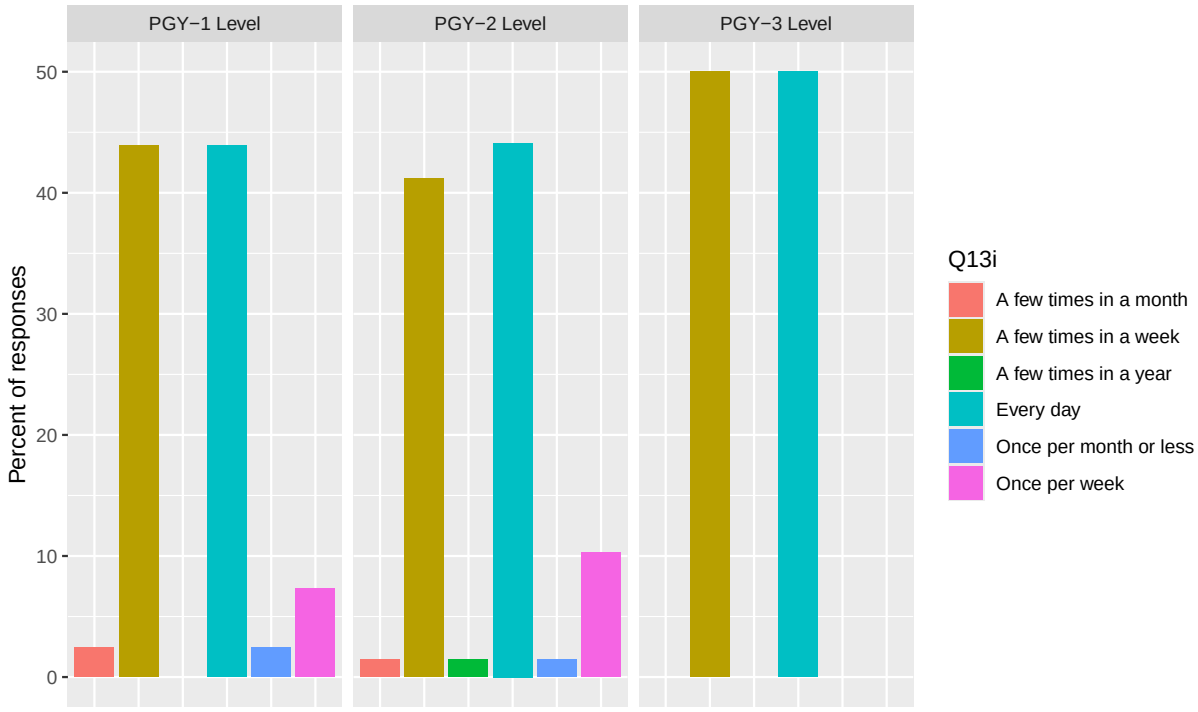

Q13i vs Q12
1 hour or less
2 to 3 hours
4 to 6 hours
7 hours or more
Percent of responses
0
20
40
Q13i
A few times in a month
A few times in a week
A few times in a year
Every day
Once per month or less
Once per week

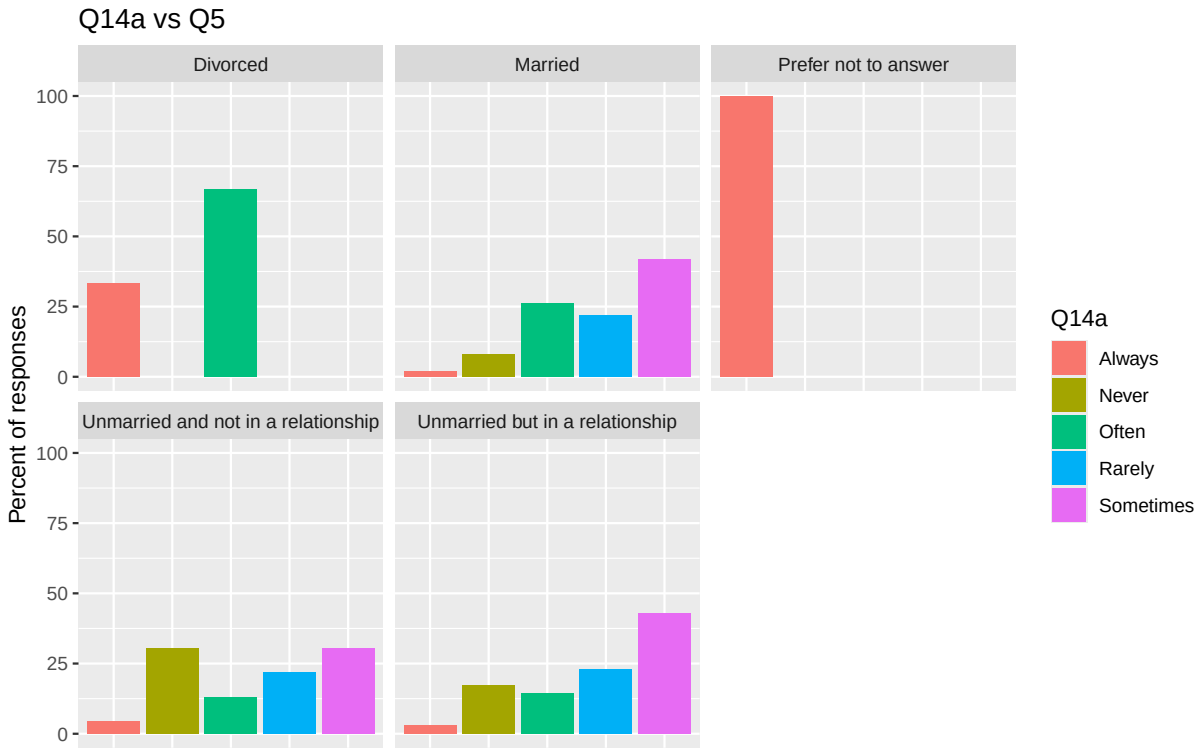

Q13i vs Q1
Hospital−based
−based (University/Hospital Com
University−based
Percent of responses
0
10
20
30
40
50
Q13i
A few times in a month
A few times in a week
A few times in a year
Every day
Once per month or less
Once per week

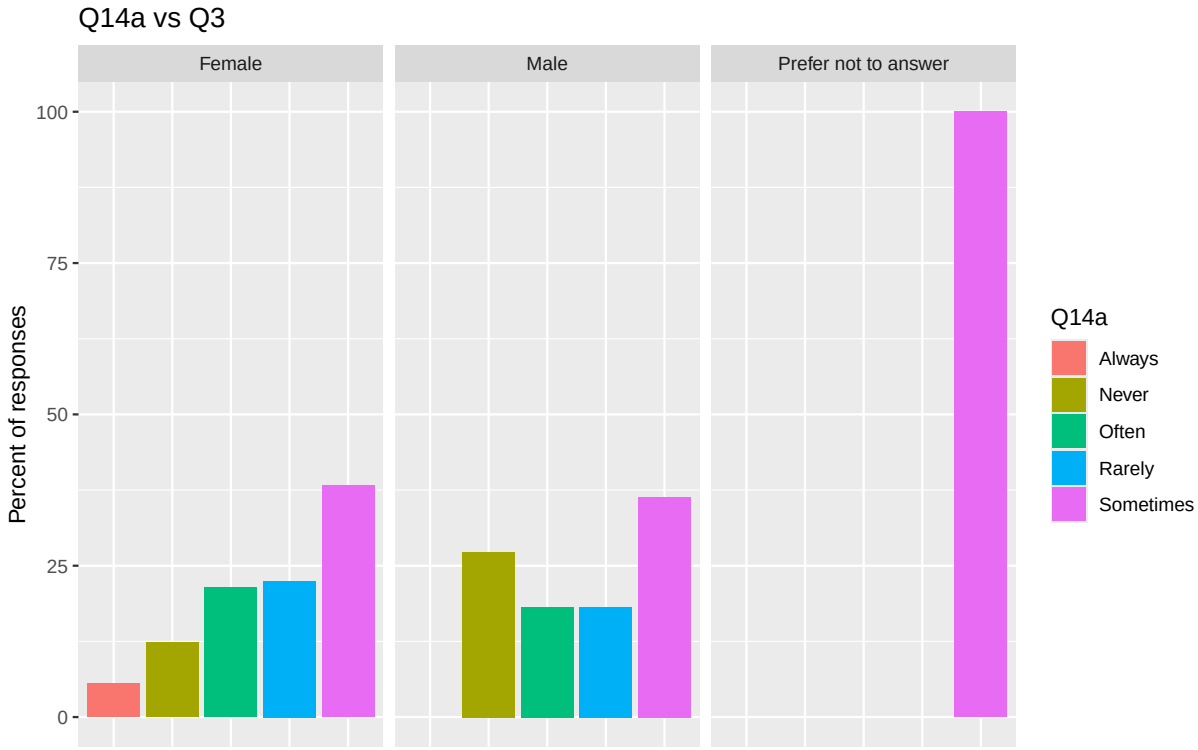
Q13i vs Q5

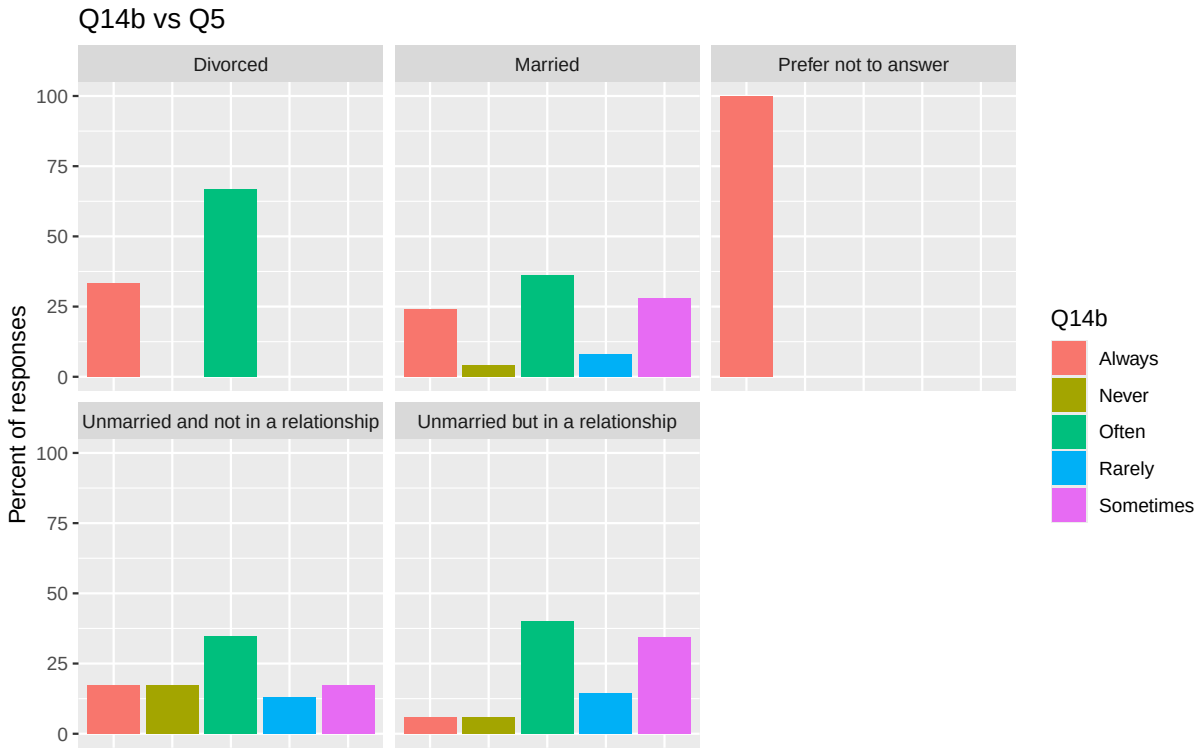


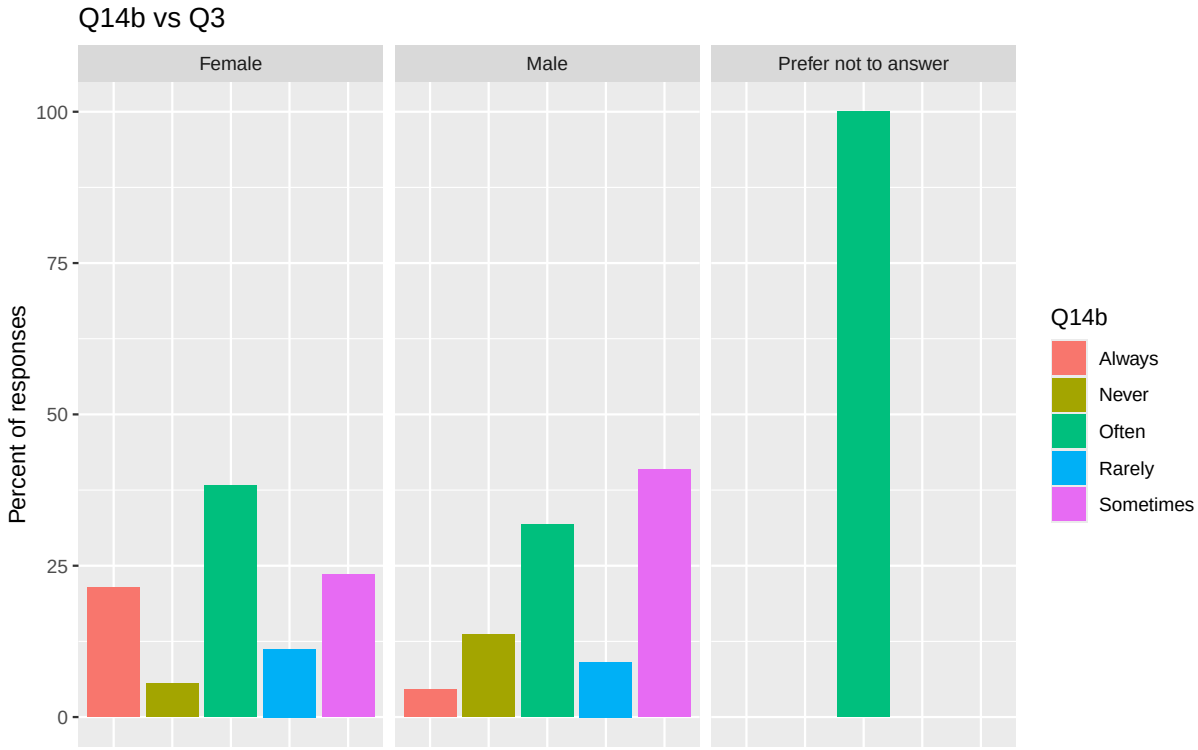
Q13i vs Q2

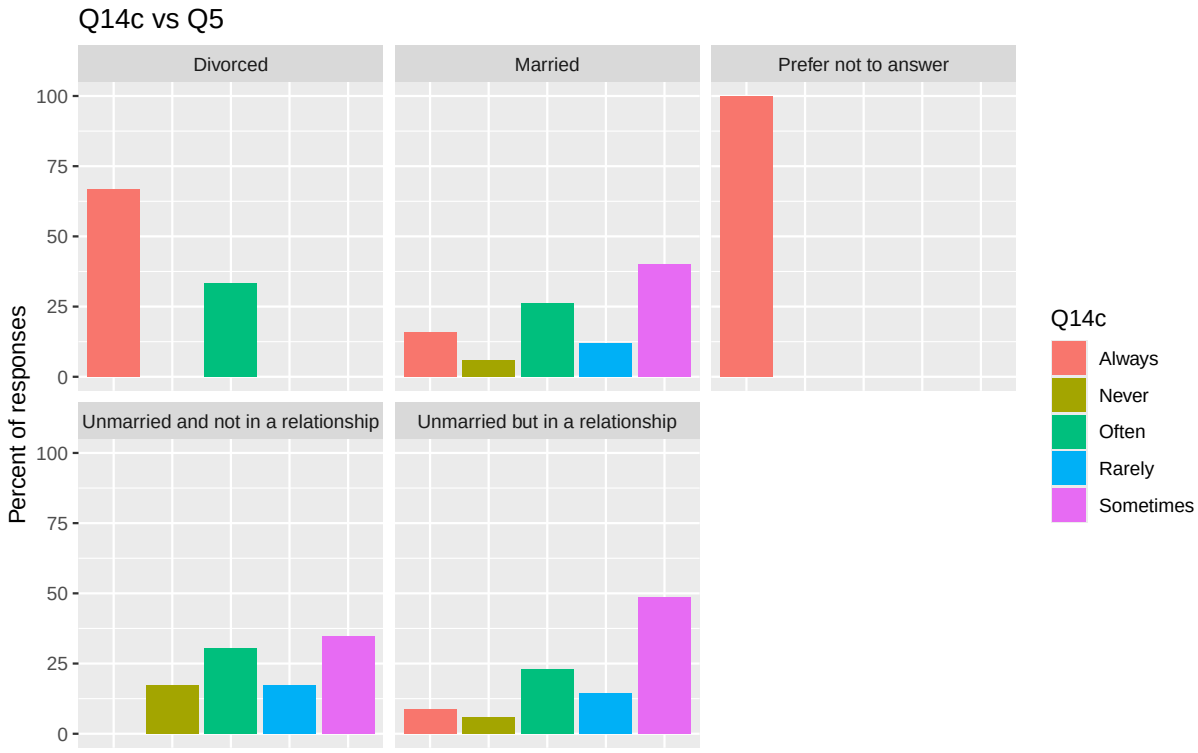


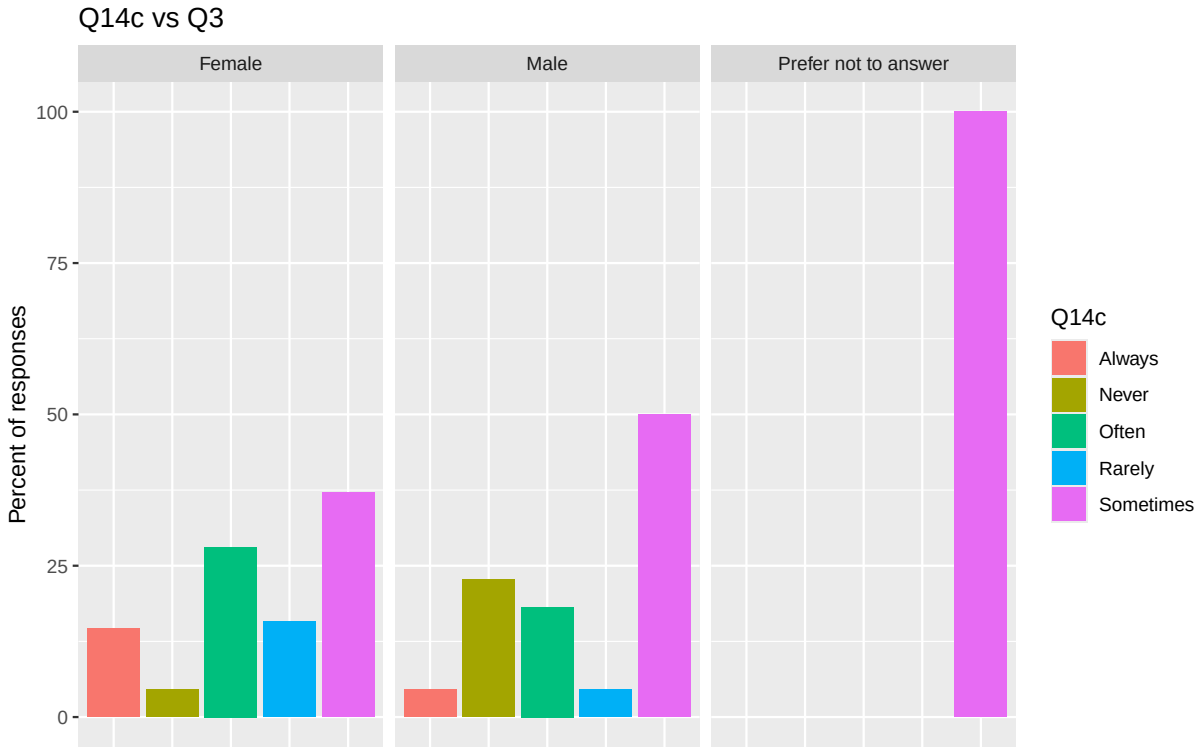

Q14a vs Q5
Divorced
Married
Prefer not to answer
Unmarried and not in a relationship
Unmarried but in a relationship
Percent of responses
100
75
50
25
0
Q14a
Always
Never
Often
Rarely
Sometimes

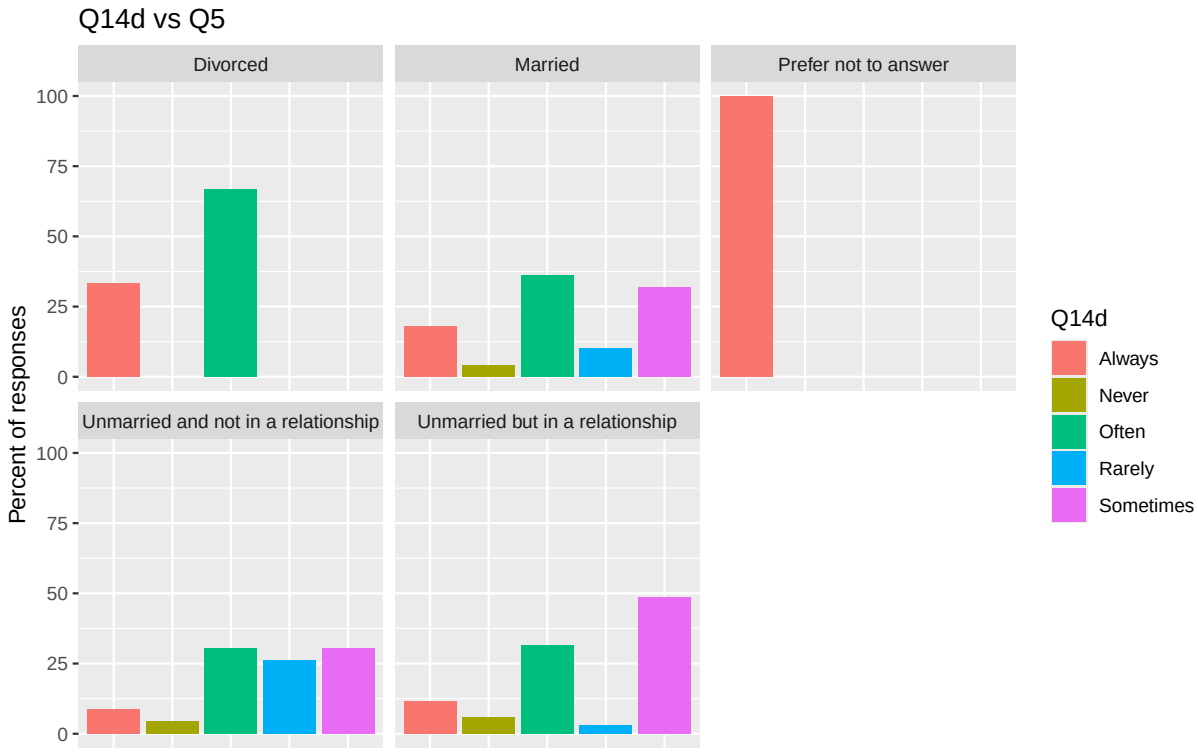

Q14a vs Q3
Female
Male
Prefer not to answer
Percent of responses
100
75
50
25
0
Q14a
Always
Never
Often
Rarely
Sometimes

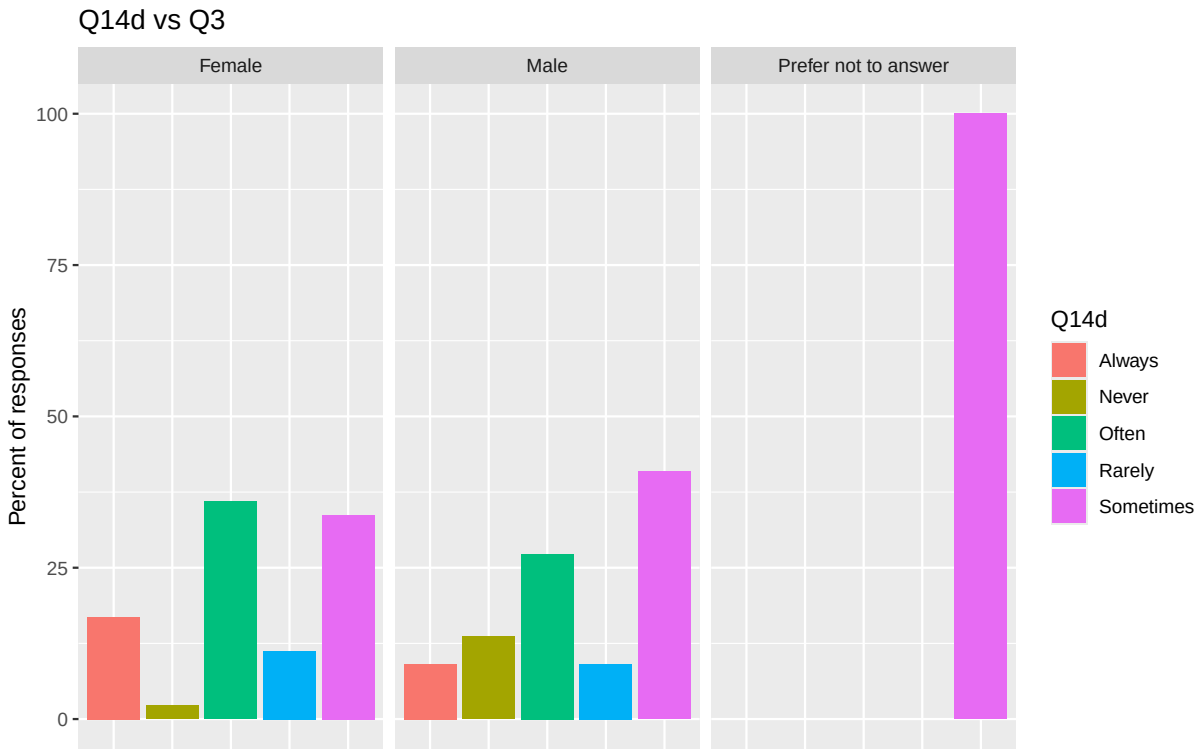

Q14b vs Q5
Divorced
Married
Prefer not to answer
Unmarried and not in a relationship
Unmarried but in a relationship
Percent of responses
100
75
50
25
0
Q14b
Always
Never
Often
Rarely
Sometimes

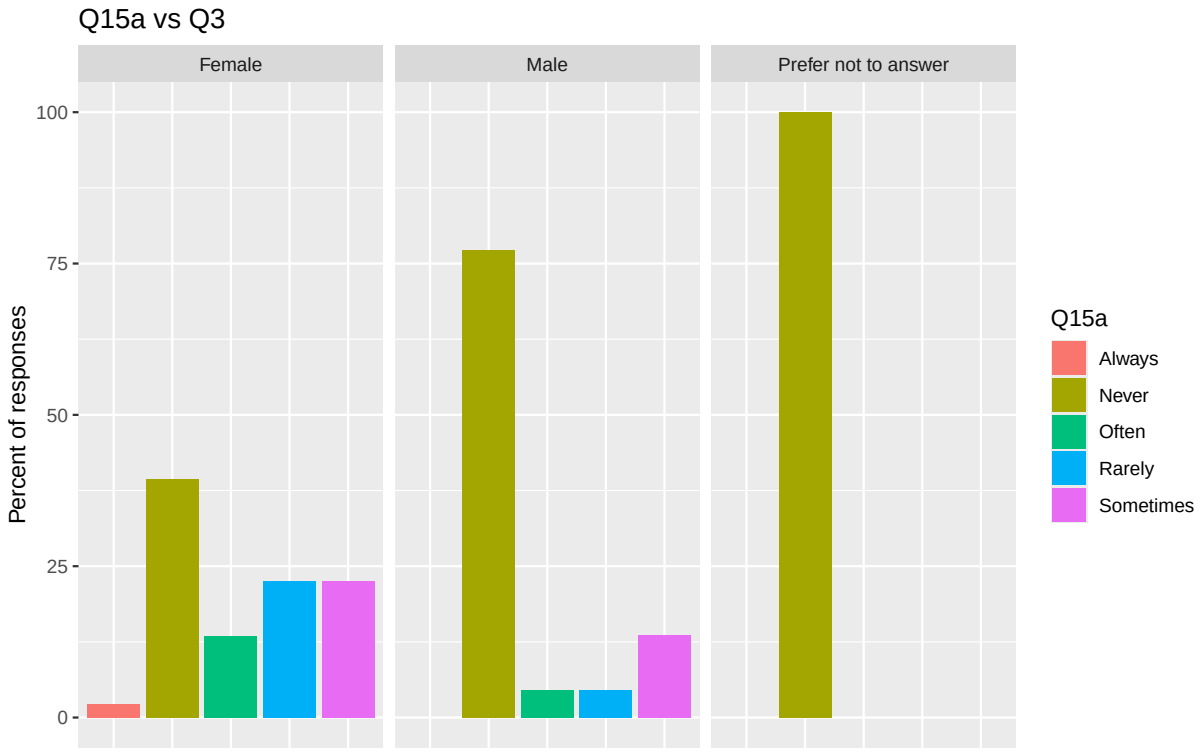

Q14b vs Q3
Female
Male
Prefer not to answer
Percent of responses
100
75
50
25
0
Q14b
Always
Never
Often
Rarely
Sometimes

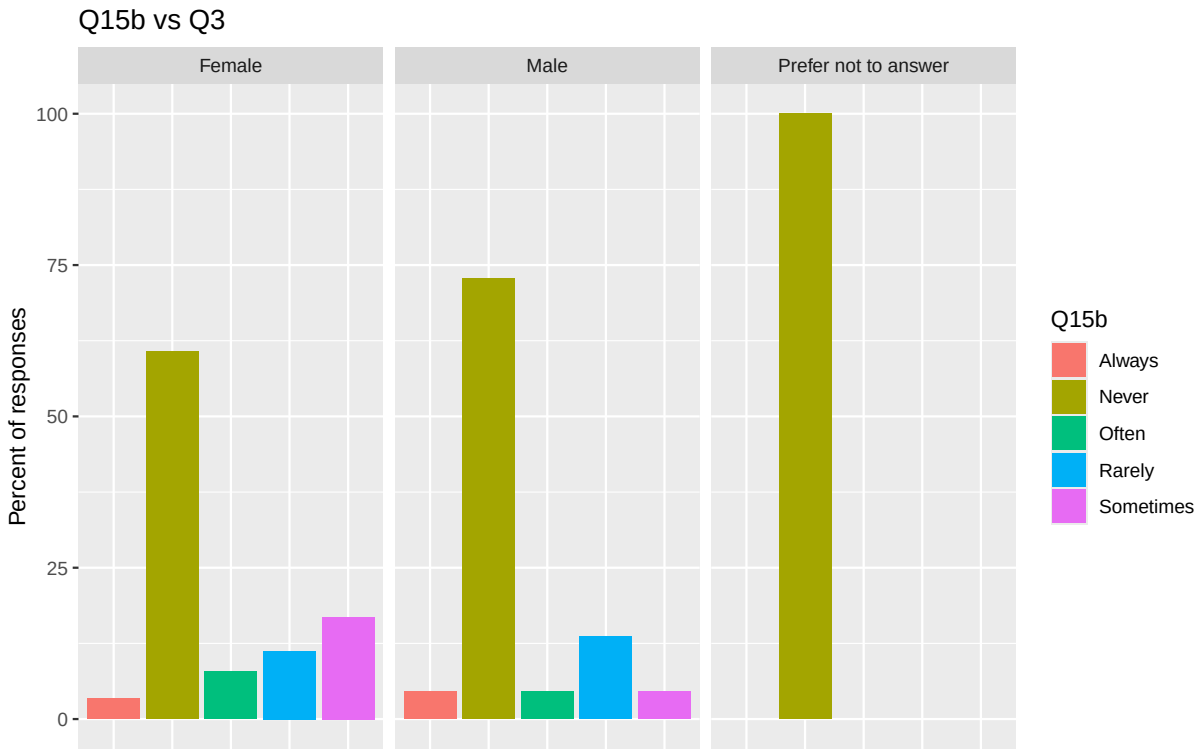

Q14c vs Q5
Divorced
Married
Prefer not to answer
Unmarried and not in a relationship
Unmarried but in a relationship
Percent of responses
100
75
50
25
0
Q14c
Always
Never
Often
Rarely
Sometimes

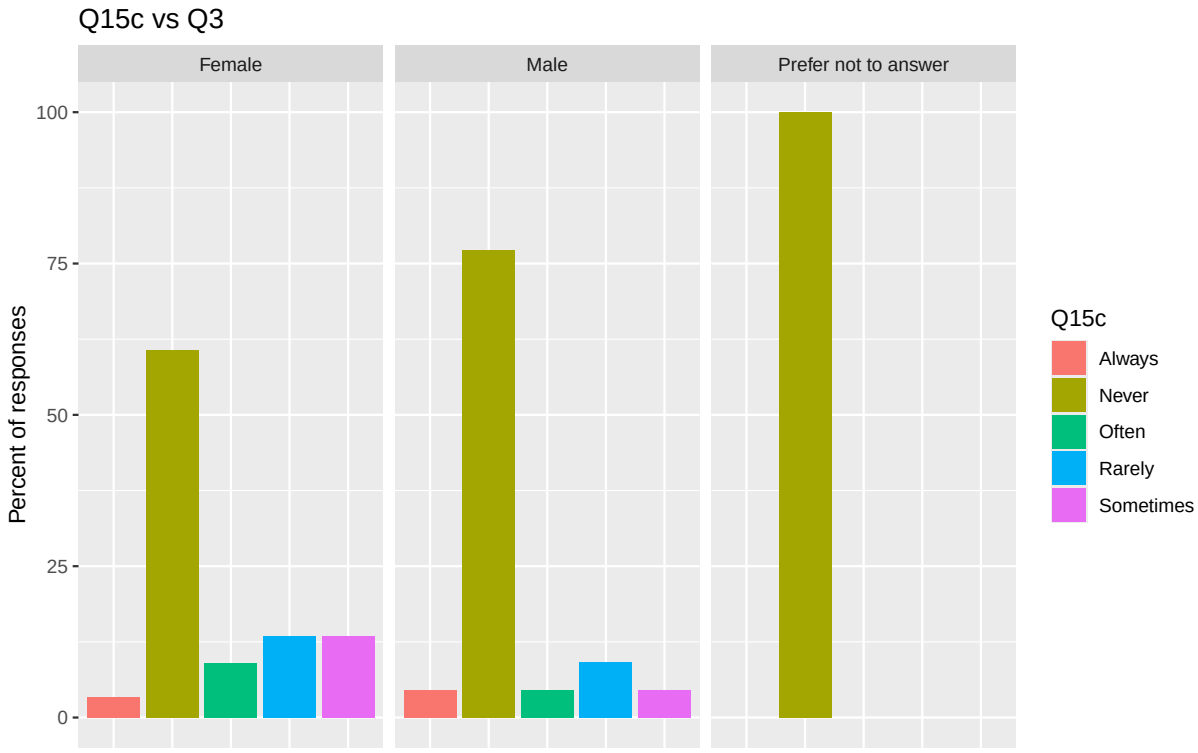

Q14c vs Q3
Female
Male
Prefer not to answer
Percent of responses
100
75
50
25
0
Q14c
Always
Never
Often
Rarely
Sometimes

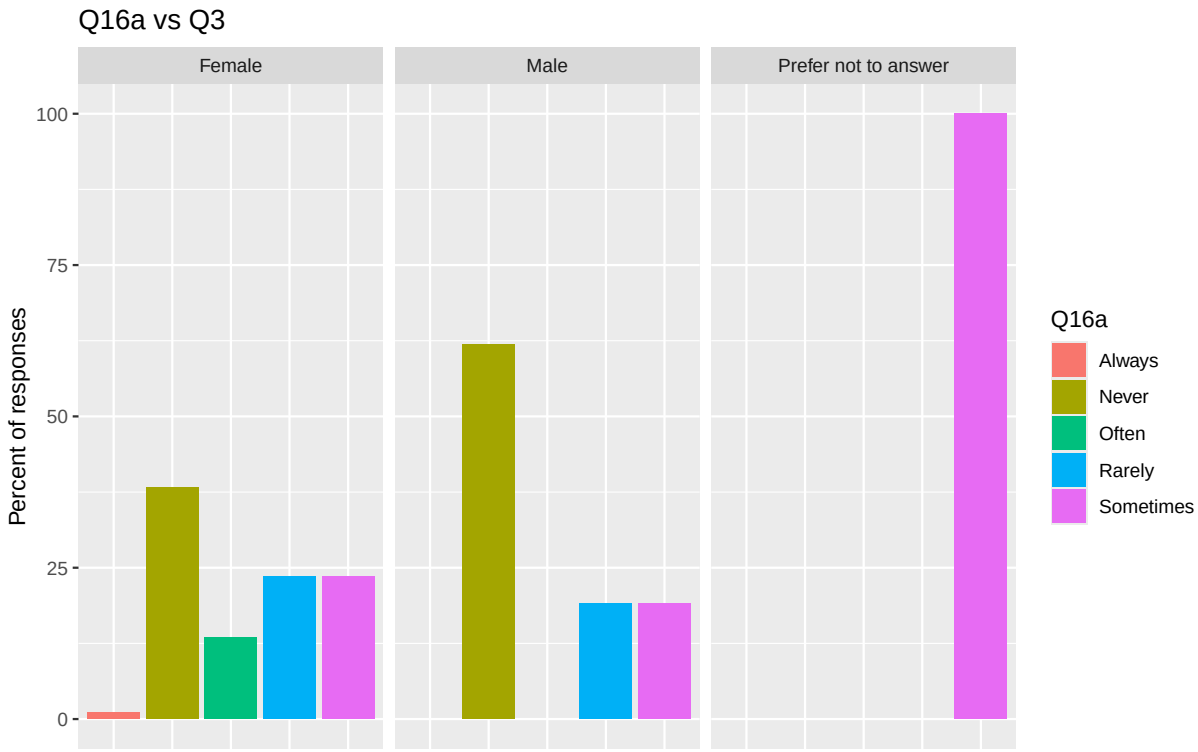

Q14d vs Q5
Divorced
Married
Prefer not to answer
Unmarried and not in a relationship
Unmarried but in a relationship
Percent of responses
100
75
50
25
0
Q14d
Always
Never
Often
Rarely
Sometimes

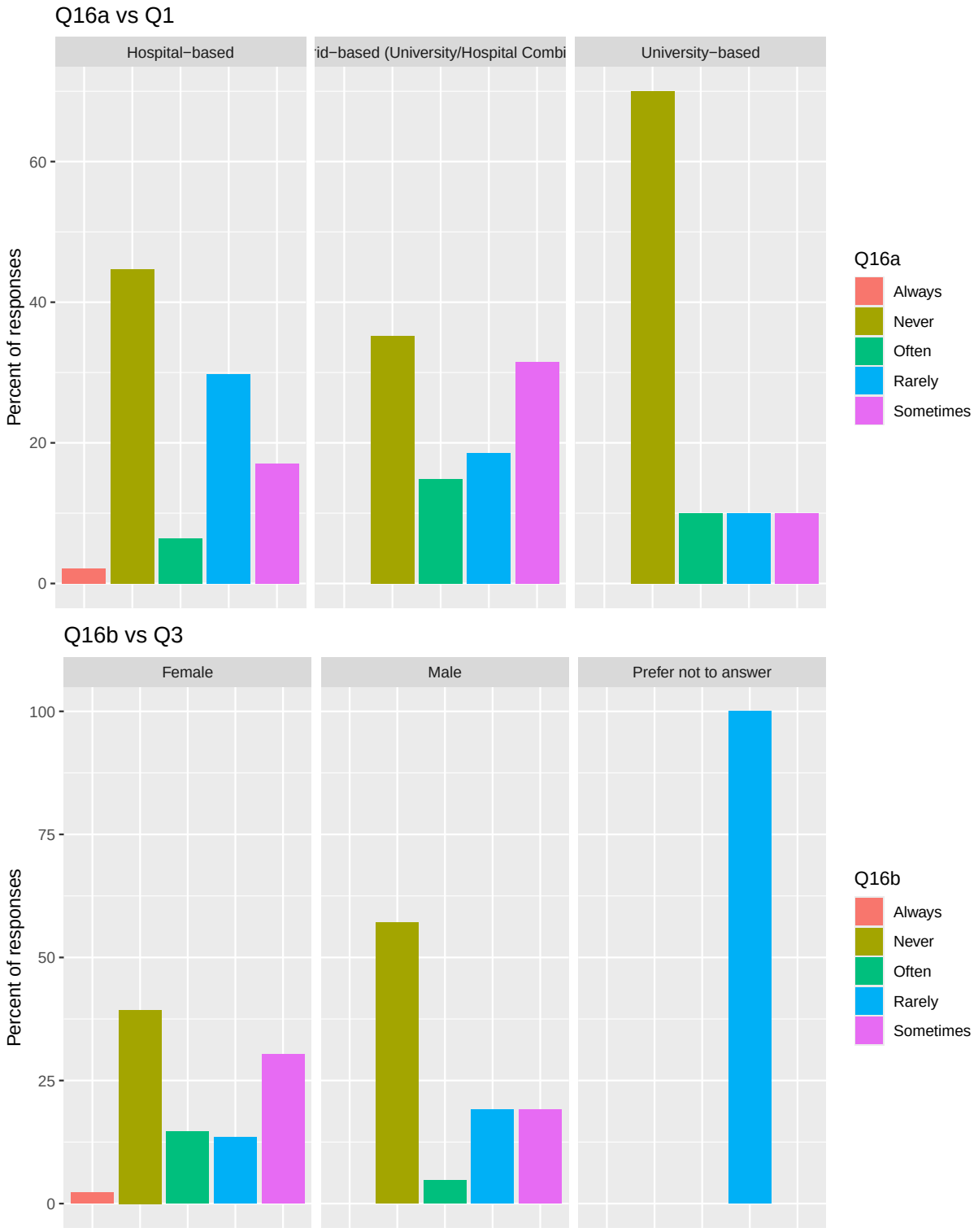

Q14d vs Q3
Female
Male
Prefer not to answer
Percent of responses
100
75
50
25
0
Q14d
Always
Never
Often
Rarely
Sometimes

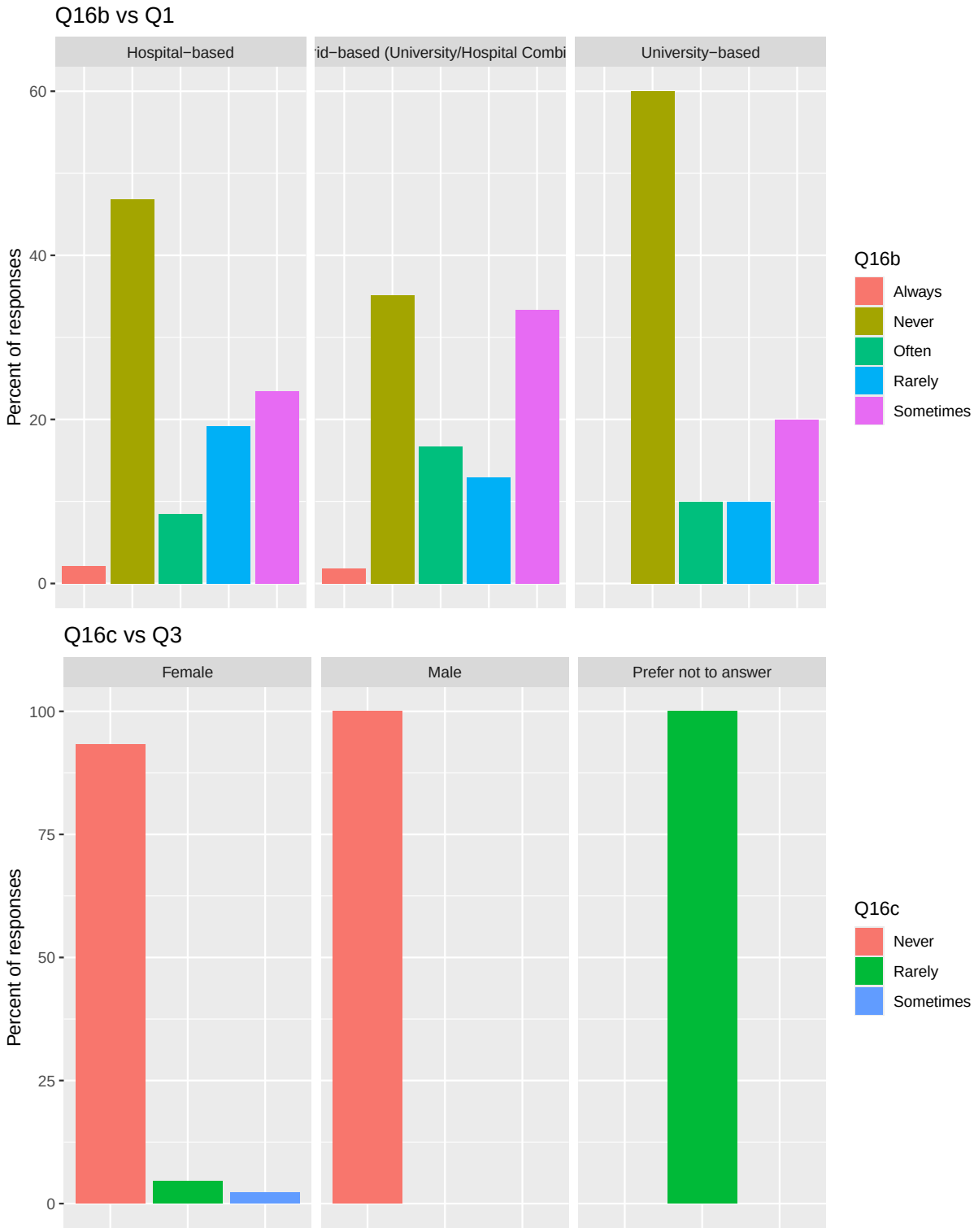

Q15a vs Q3
Female
Male
Prefer not to answer
Percent of responses
100
75
50
25
0
Q15a
Always
Never
Often
Rarely
Sometimes

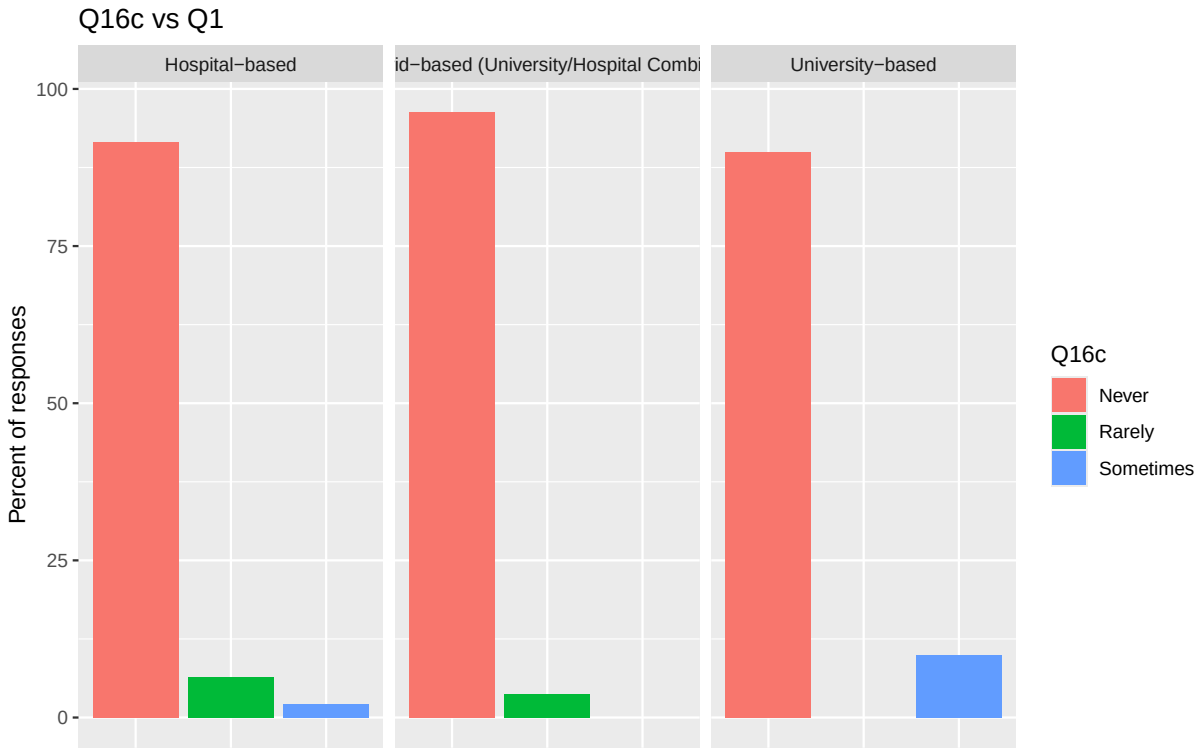

Q15b vs Q3
Female
Male
Prefer not to answer
Percent of responses
100
75
50
25
0
Q15b
Always
Never
Often
Rarely
Sometimes

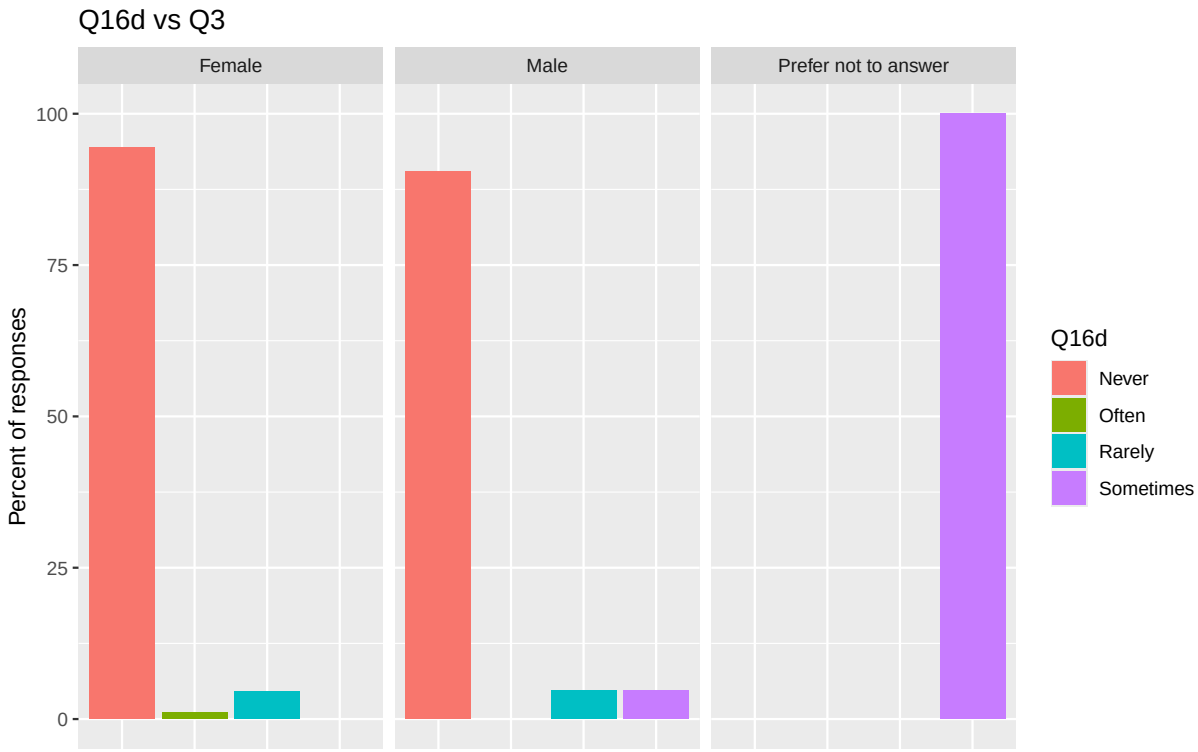

Q15c vs Q3
Female
Male
Prefer not to answer
Percent of responses
100
75
50
25
0
Q15c
Always
Never
Often
Rarely
Sometimes

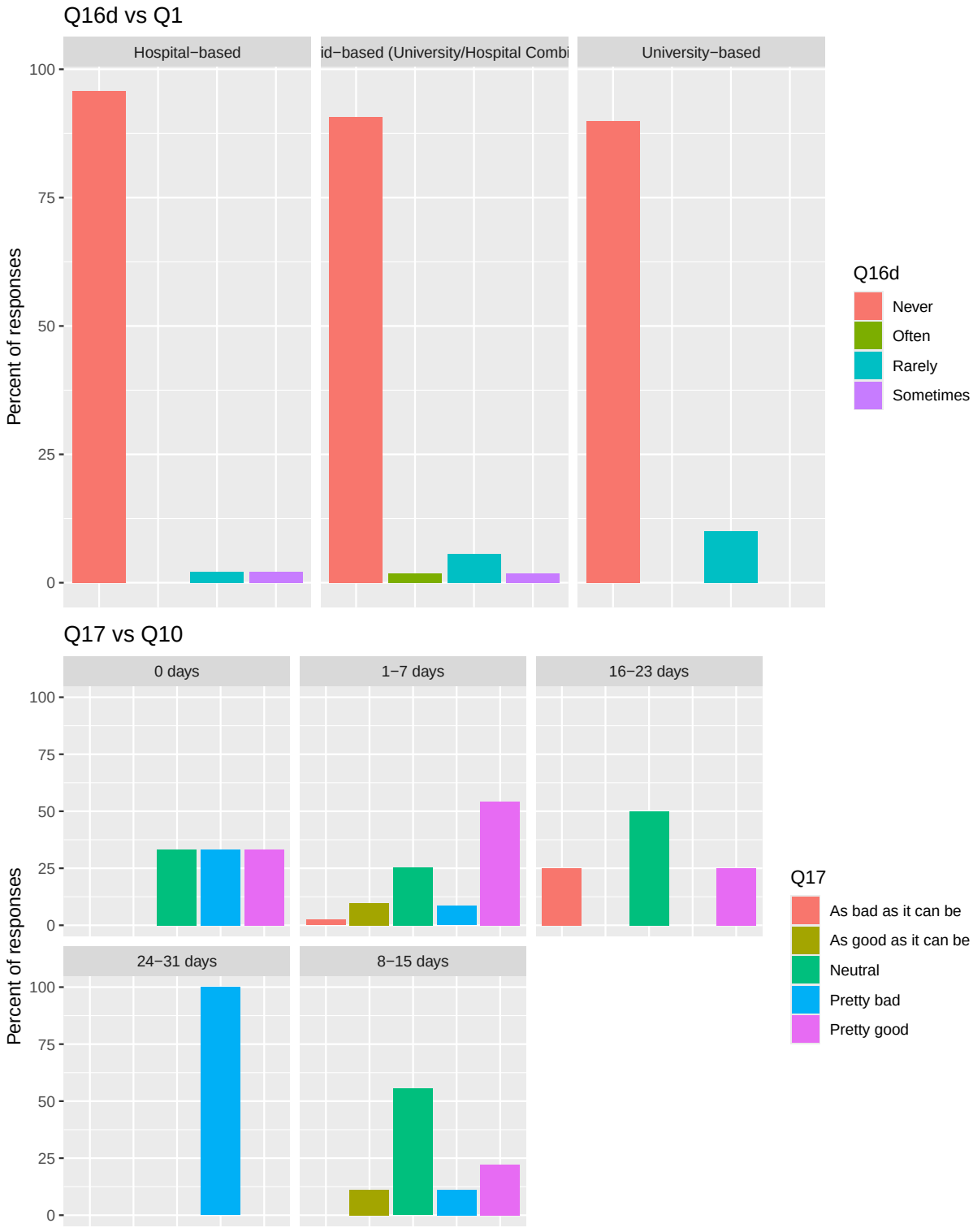

Q16a vs Q3
Female
Male
Prefer not to answer
Percent of responses
100
75
50
25
0
Q16a
Always
Never
Often
Rarely
Sometimes

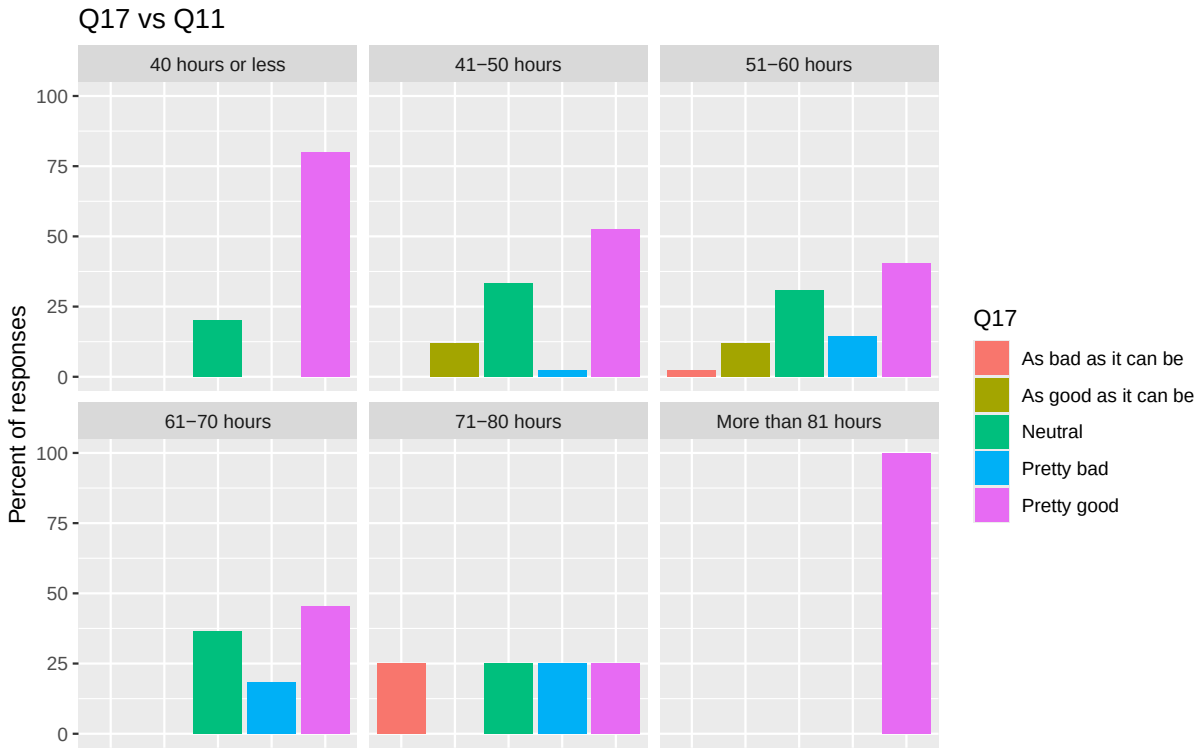

Q16a vs Q1
Hospital−based
id−based (University/Hospital Combi
University−based
Percent of responses
0
20
40
60
Q16a
Always
Never
Often
Rarely
Sometimes
Q16b vs Q3
Female
Male
Prefer not to answer
Percent of responses
0
25
50
75
100
Q16b
Always
Never
Often
Rarely
Sometimes

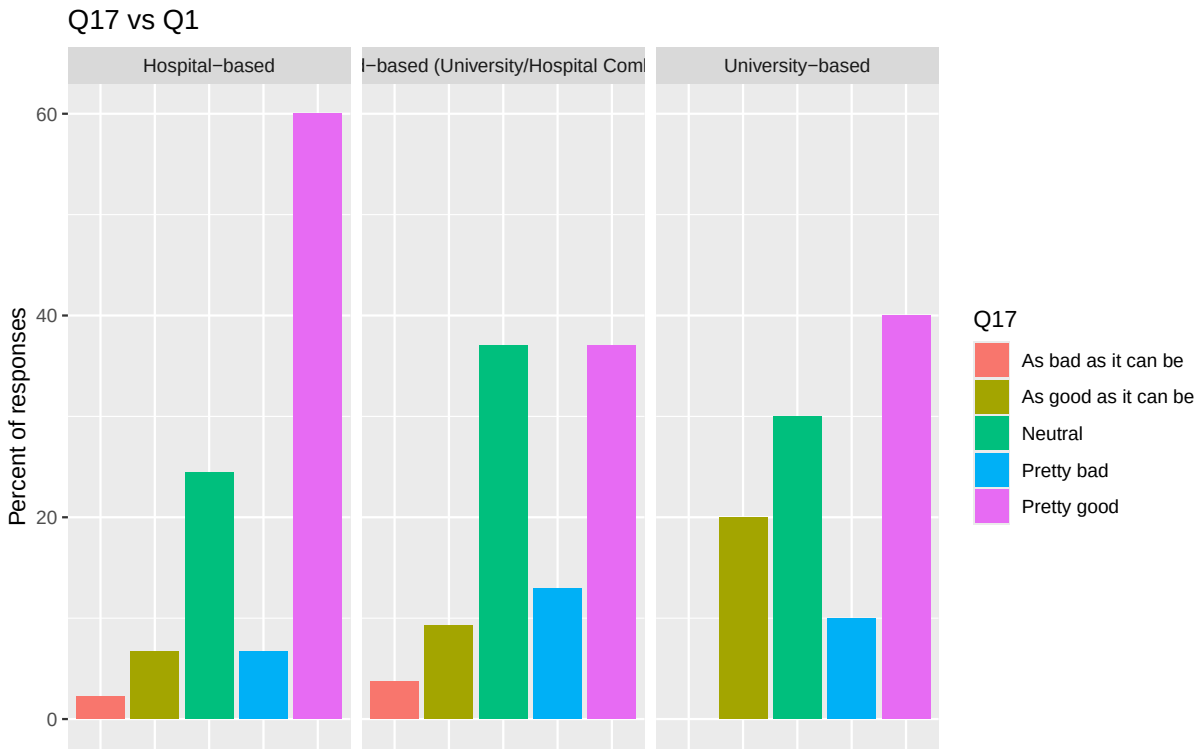

Q16b vs Q1
Hospital−based
id−based (University/Hospital Combi
University−based
Percent of responses
60
40
20
0
Q16b
Always
Never
Often
Rarely
Sometimes
Q16c vs Q3
Female
Male
Prefer not to answer
Percent of responses
100
75
50
25
0
Q16c
Never
Rarely
Sometimes

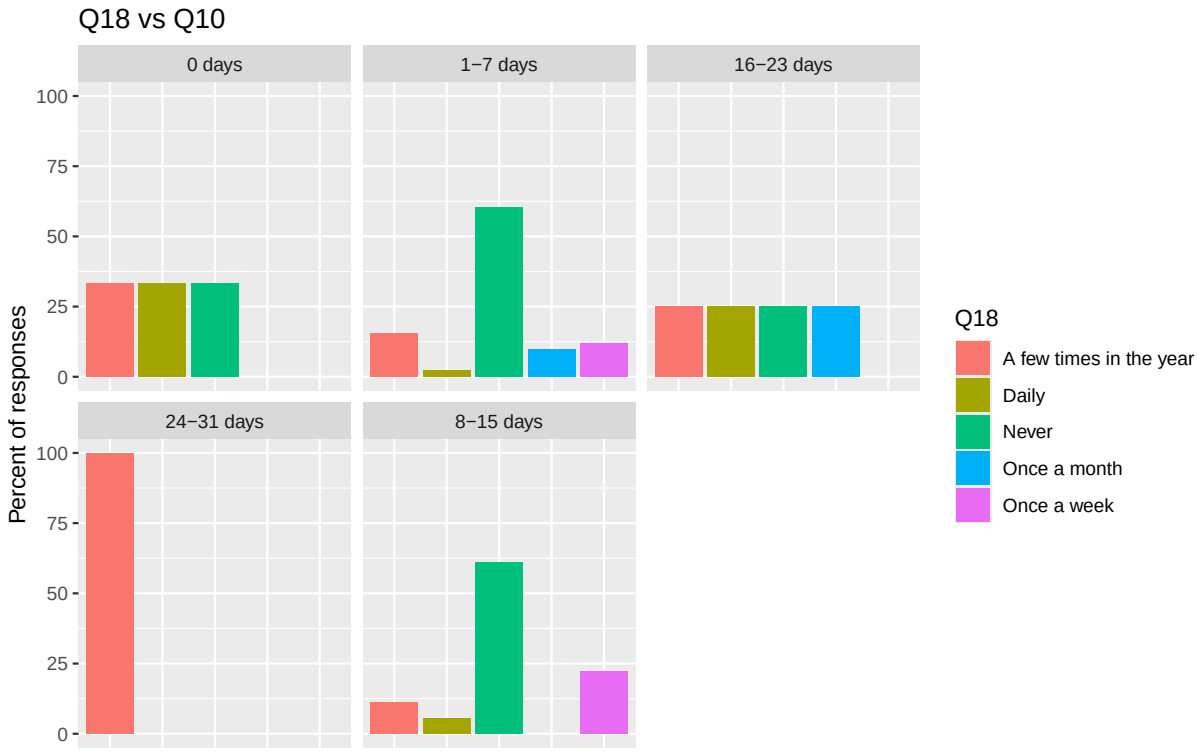

Q16c vs Q1
Hospital−based
id−based (University/Hospital Combi
University−based
Percent of responses
100
75
50
25
0
Q16c
Never
Rarely
Sometimes

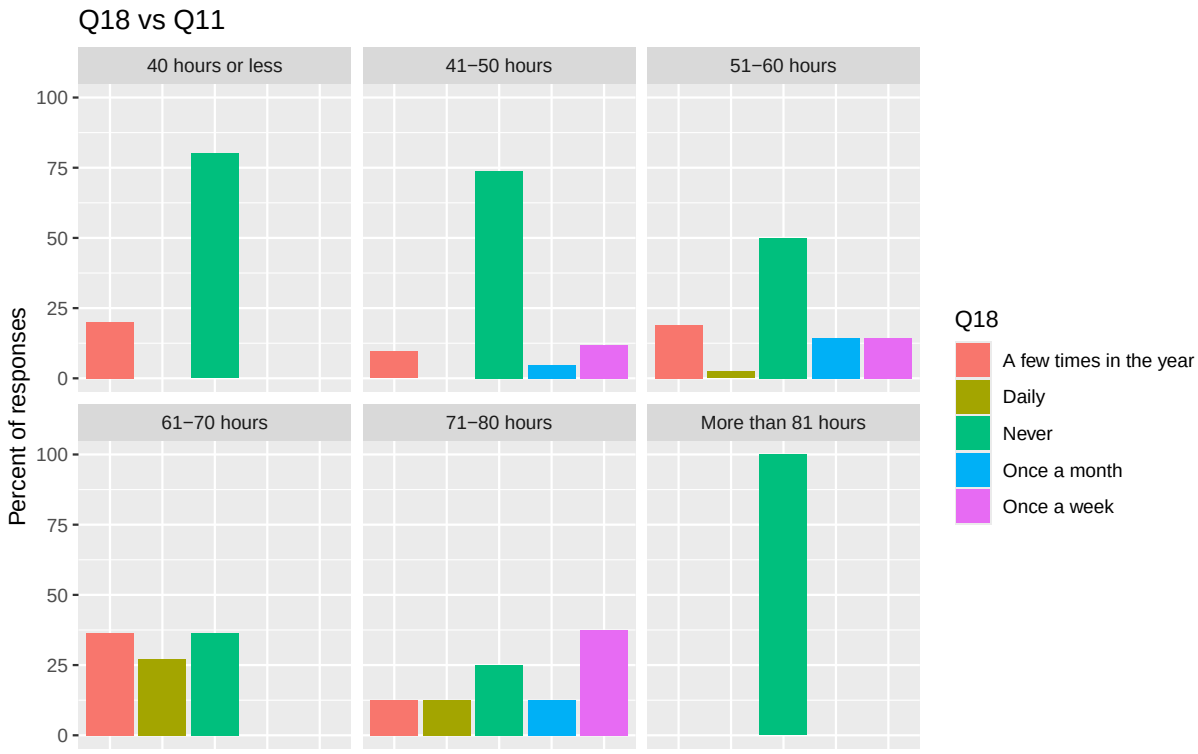

Q16d vs Q3
Female
Male
Prefer not to answer
Percent of responses
100
75
50
25
0
Q16d
Never
Often
Rarely
Sometimes

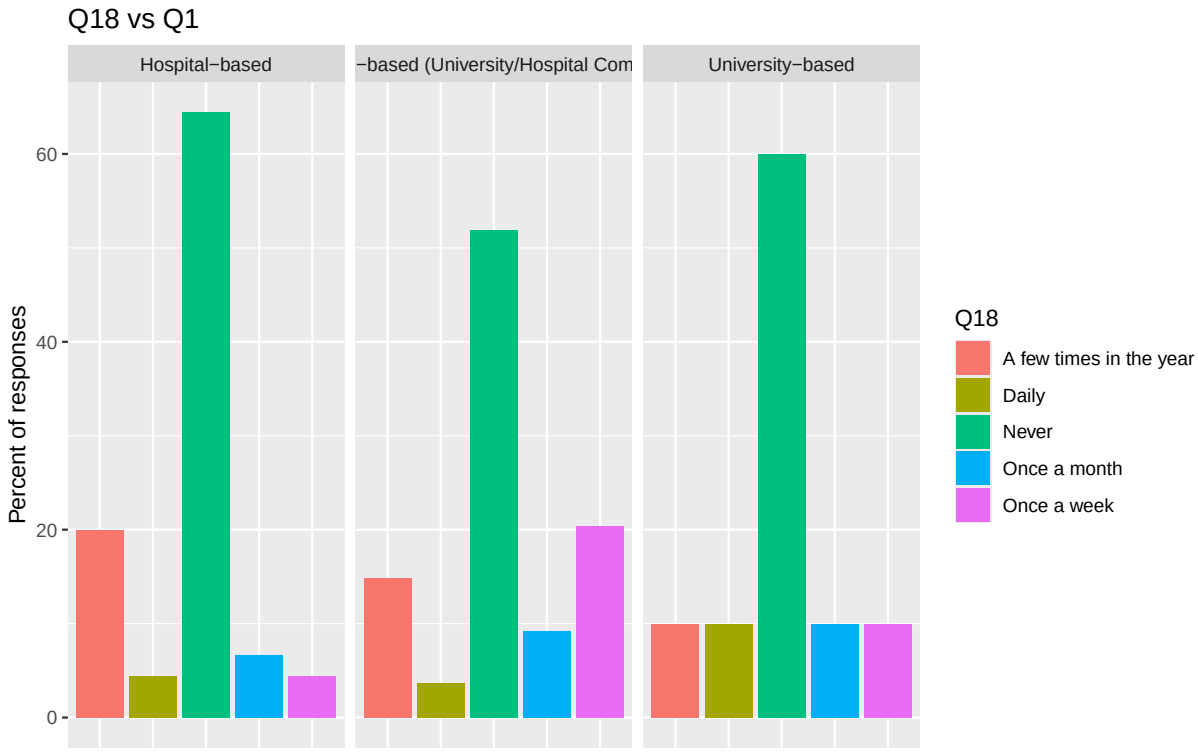

Q16d vs Q1
Hospital−based
id−based (University/Hospital Combi
University−based
Percent of responses
100
75
50
25
0
Q16d
Never
Often
Rarely
Sometimes
Q17 vs Q10
0 days
1−7 days
16−23 days
24−31 days
8−15 days
Q17
As bad as it can be
As good as it can be
Neutral
Pretty bad
Pretty good

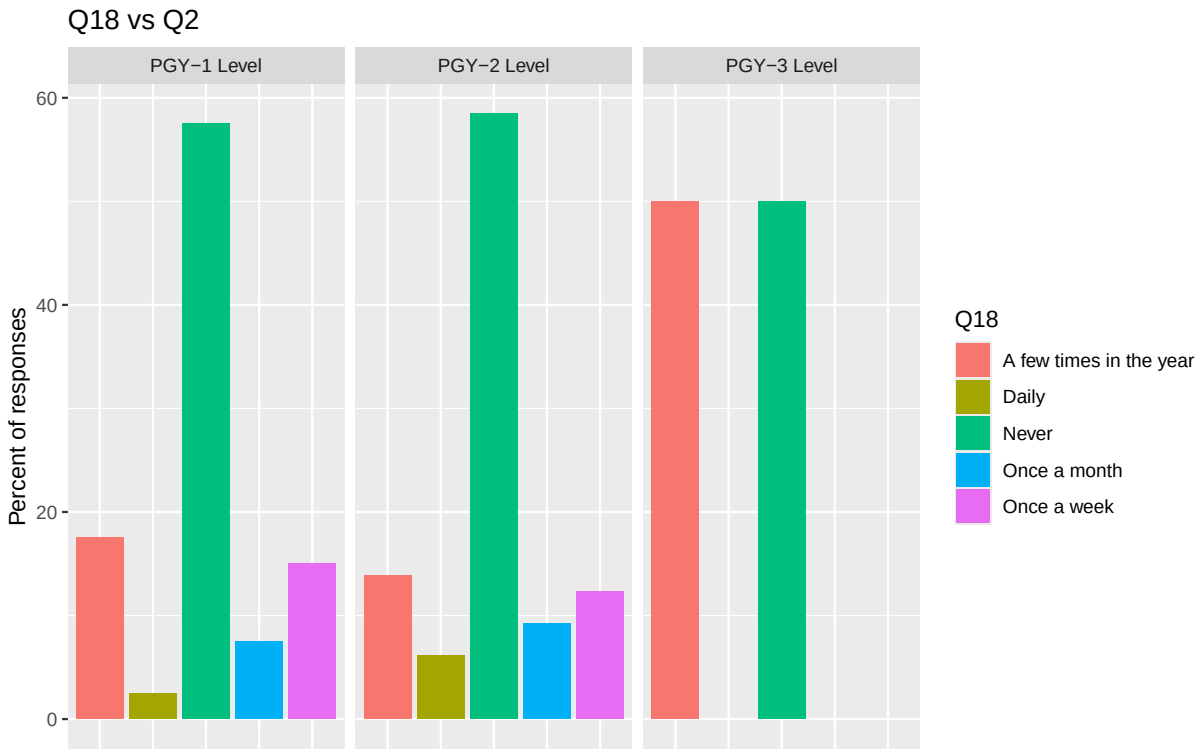

Q17 vs Q11
40 hours or less
41−50 hours
51−60 hours
61−70 hours
71−80 hours
More than 81 hours
Percent of responses
100
75
50
25
0
Q17
As bad as it can be
As good as it can be
Neutral
Pretty bad
Pretty good

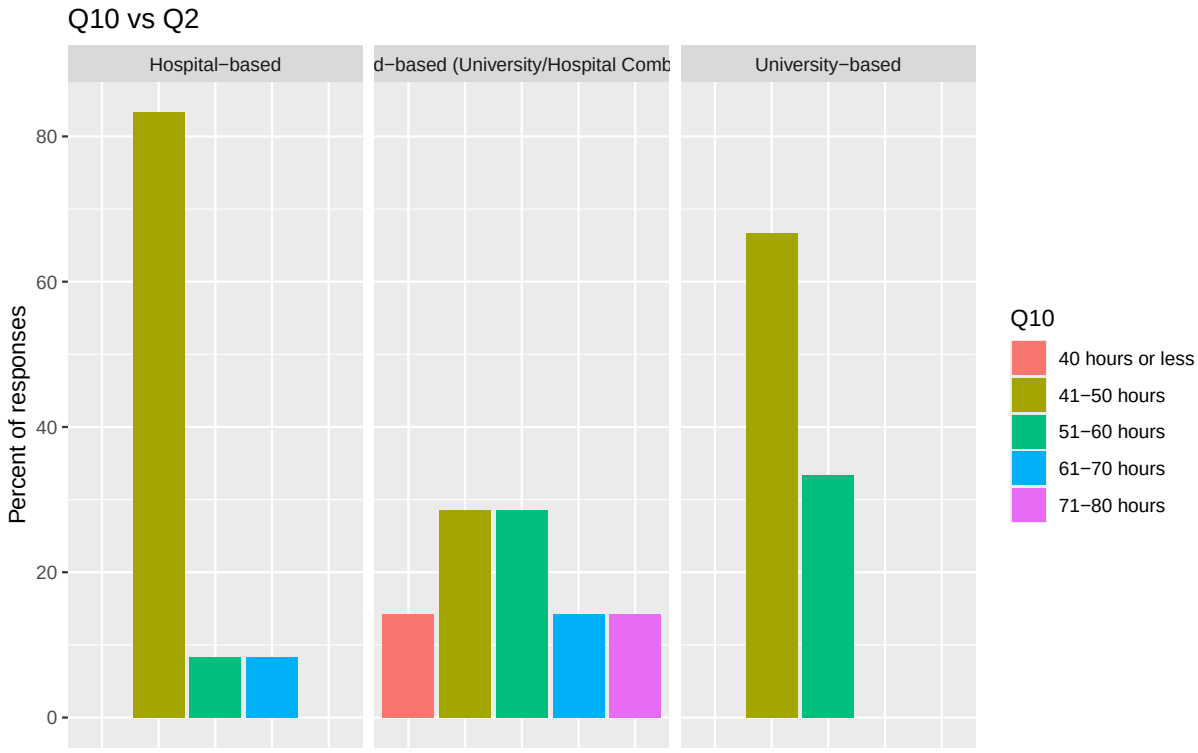

Q17 vs Q1
Hospital−based
I−based (University/Hospital Com
University−based
Percent of responses
60
40
20
0
Q17
As bad as it can be
As good as it can be
Neutral
Pretty bad
Pretty good

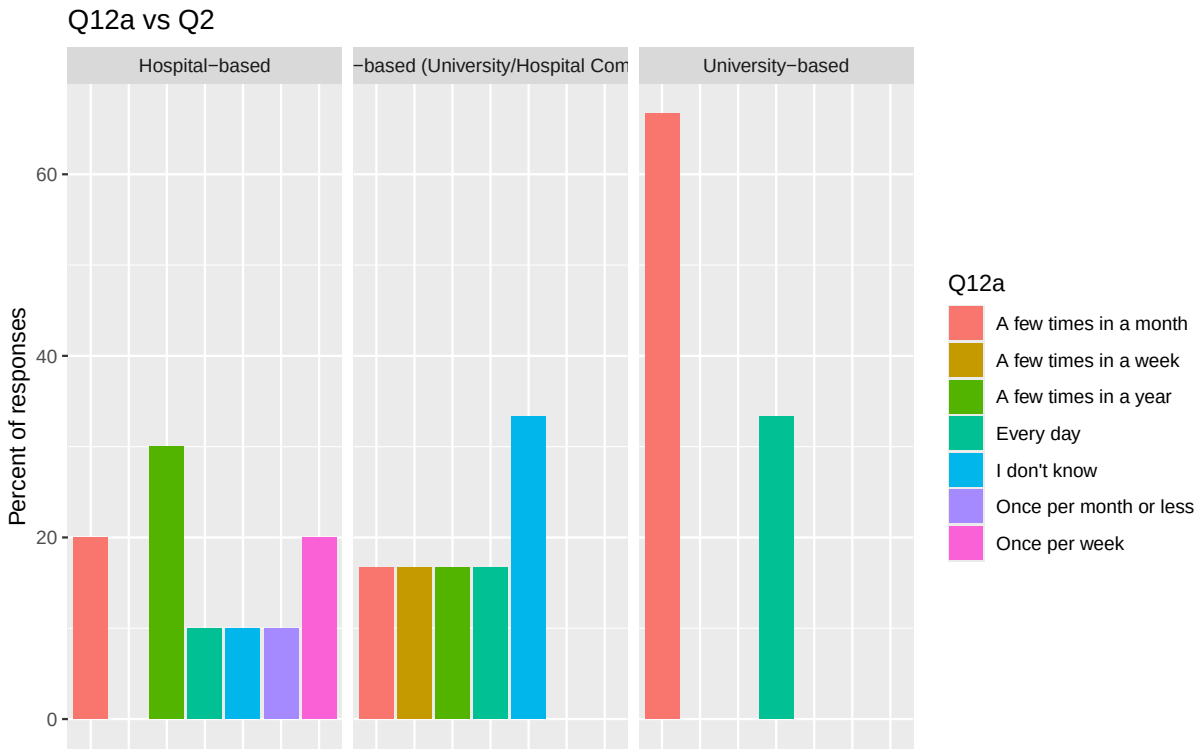

Q18 vs Q10
0 days
1−7 days
16−23 days
24−31 days
8−15 days
Percent of responses
100
75
50
25
0
Q18
A few times in the year
Daily
Never
Once a month
Once a week

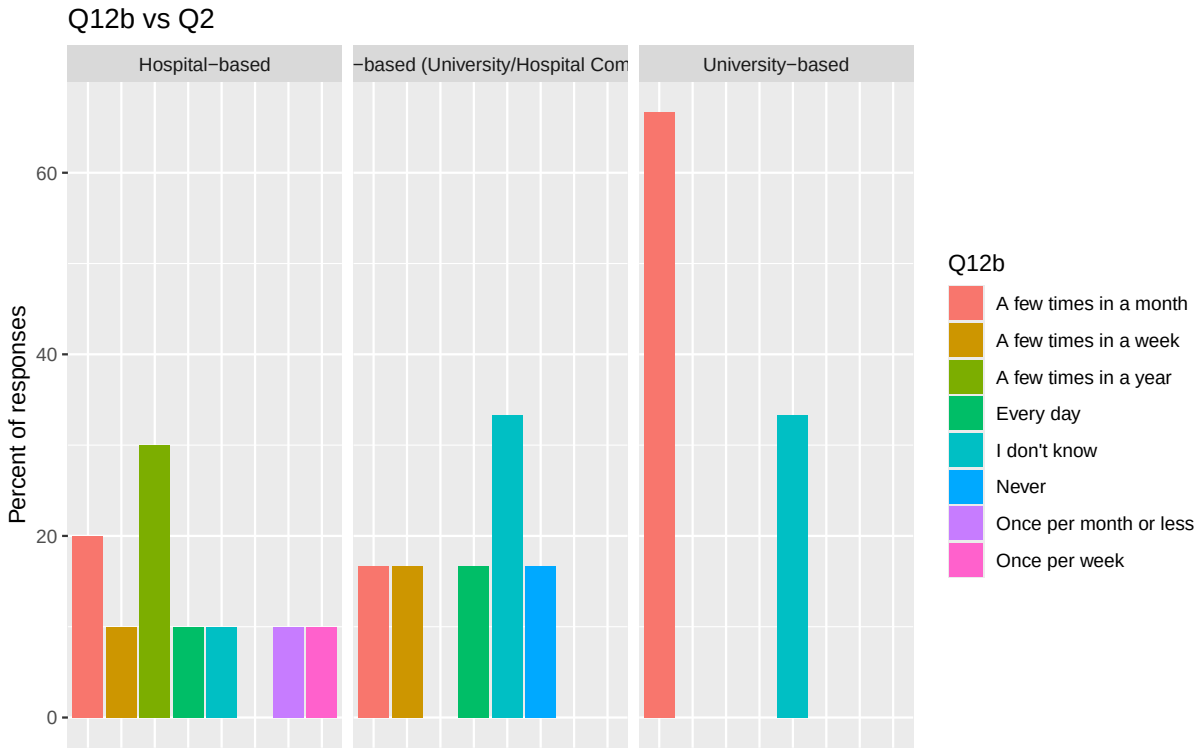

Q18 vs Q11
40 hours or less
41−50 hours
51−60 hours
61−70 hours
71−80 hours
More than 81 hours
Percent of responses
100
75
50
25
0
Q18
A few times in the year
Daily
Never
Once a month
Once a week

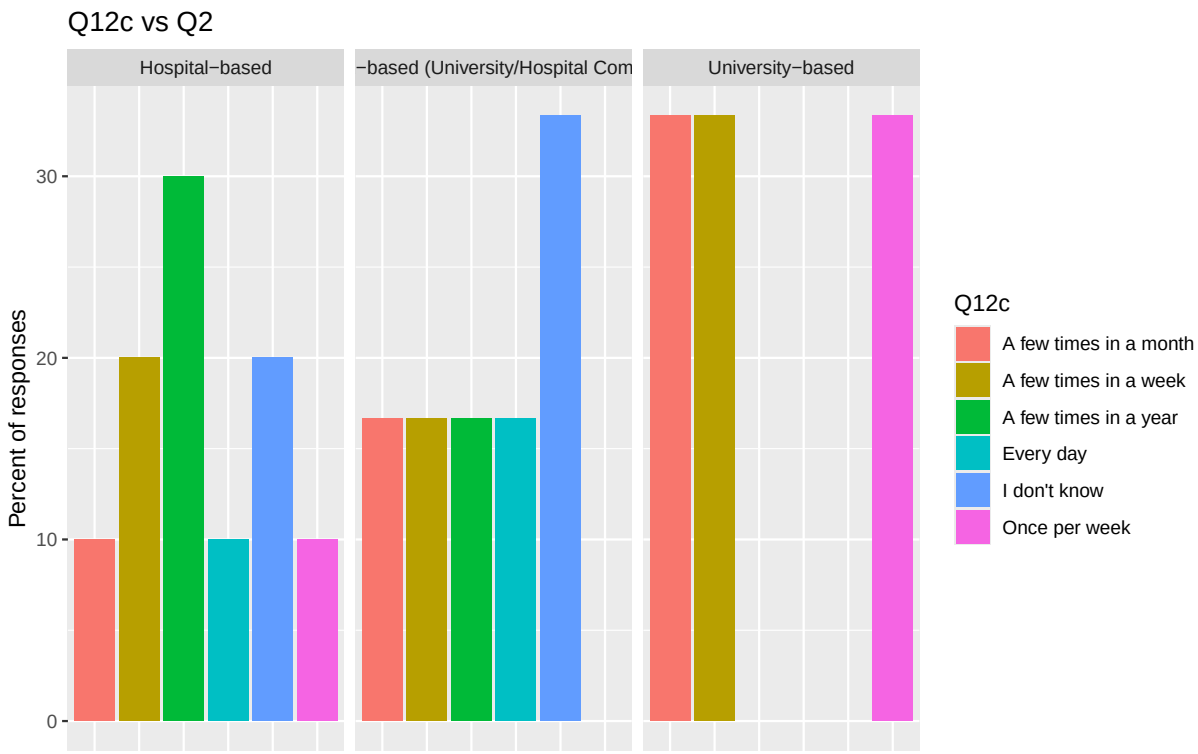

Q18 vs Q1
Hospital−based
−based (University/Hospital Com
University−based
Percent of responses
0
20
40
60
Q18
A few times in the year
Daily
Never
Once a month
Once a week

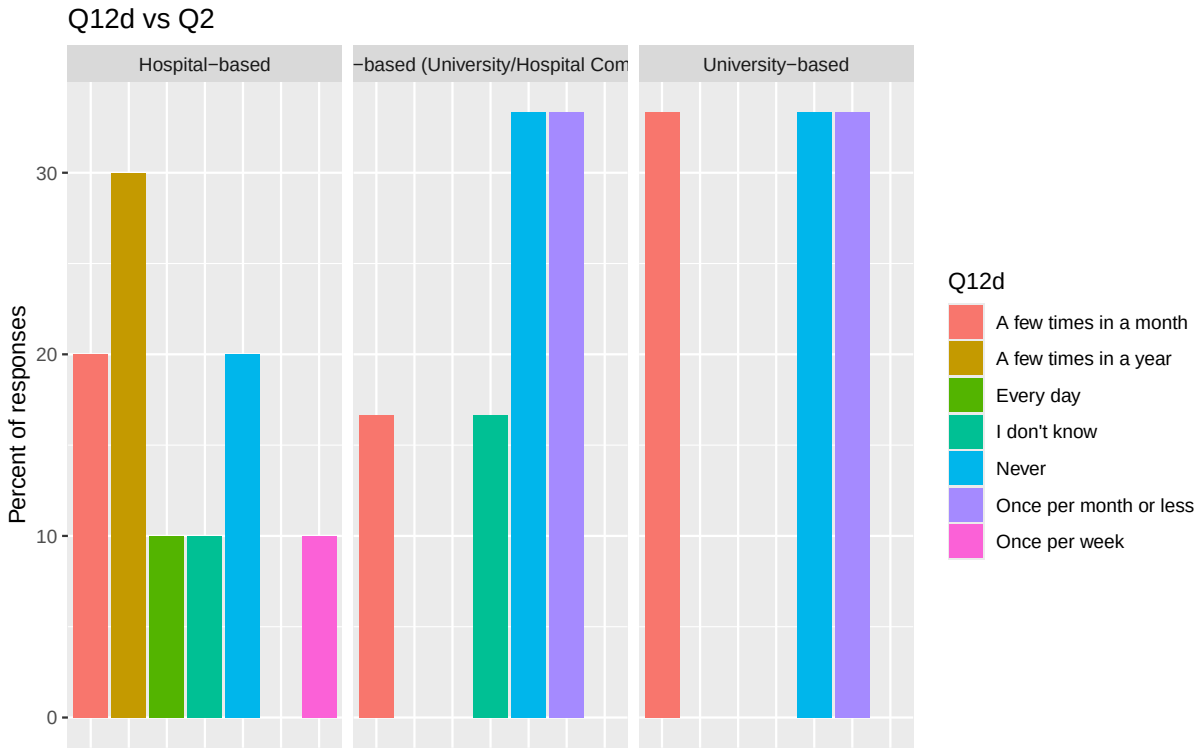

Q18 vs Q2
PGY−1 Level
PGY−2 Level
PGY−3 Level
Percent of responses
0
20
40
60
Q18
A few times in the year
Daily
Never
Once a month
Once a week

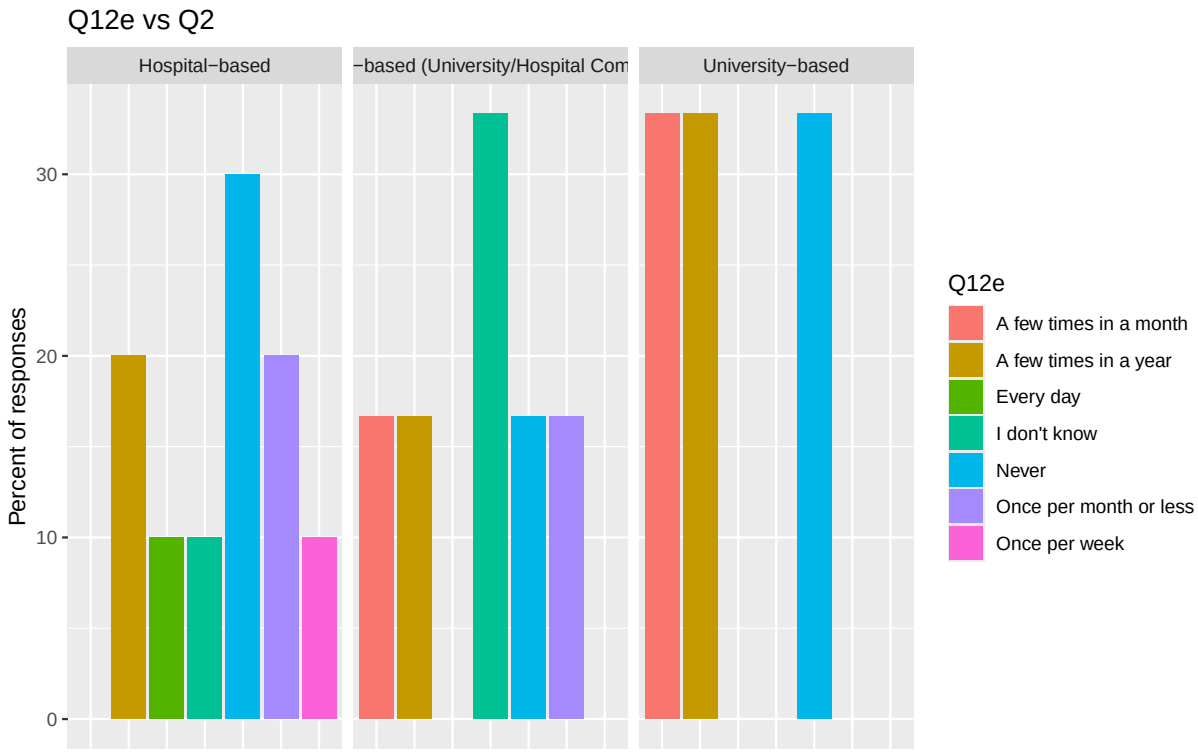

Q10 vs Q2
Hospital−based
d−based (University/Hospital Comb
University−based
Percent of responses
0
20
40
60
80
Q10
40 hours or less
41−50 hours
51−60 hours
61−70 hours
71−80 hours

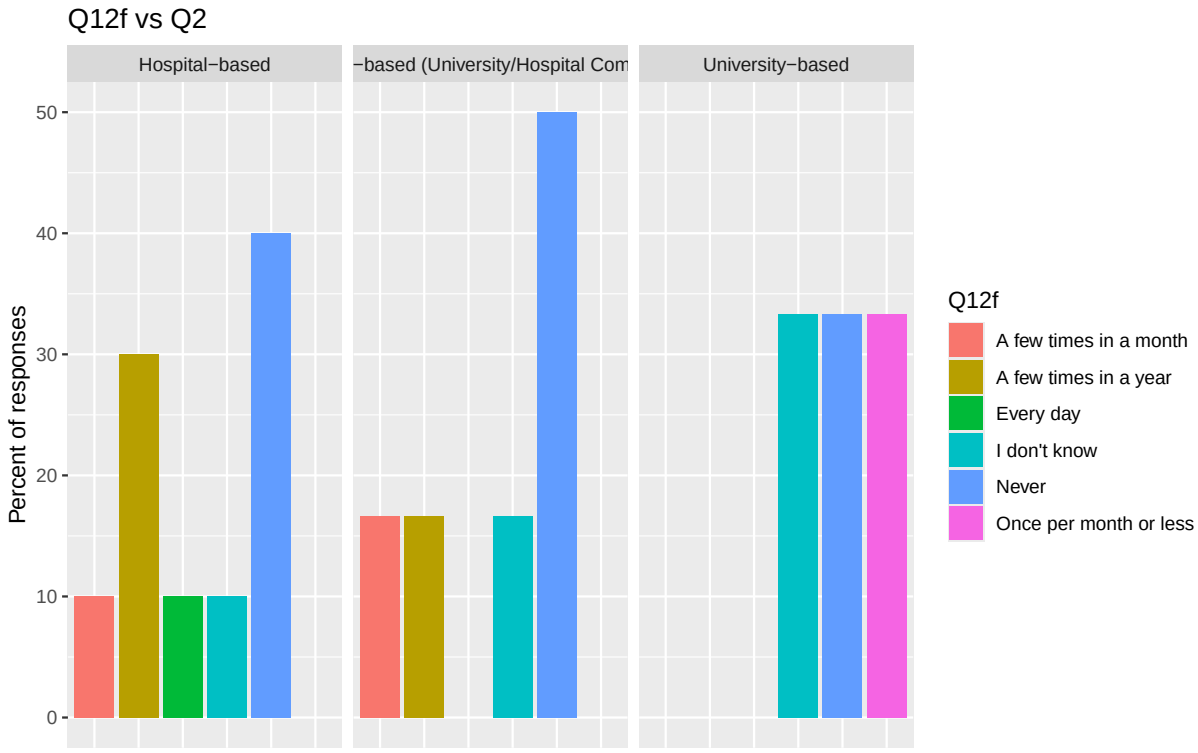

Q12a vs Q2
Hospital−based
−based (University/Hospital Com
University−based
Percent of responses
0
20
40
60
Q12a
A few times in a month
A few times in a week
A few times in a year
Every day
I don't know
Once per month or less
Once per week

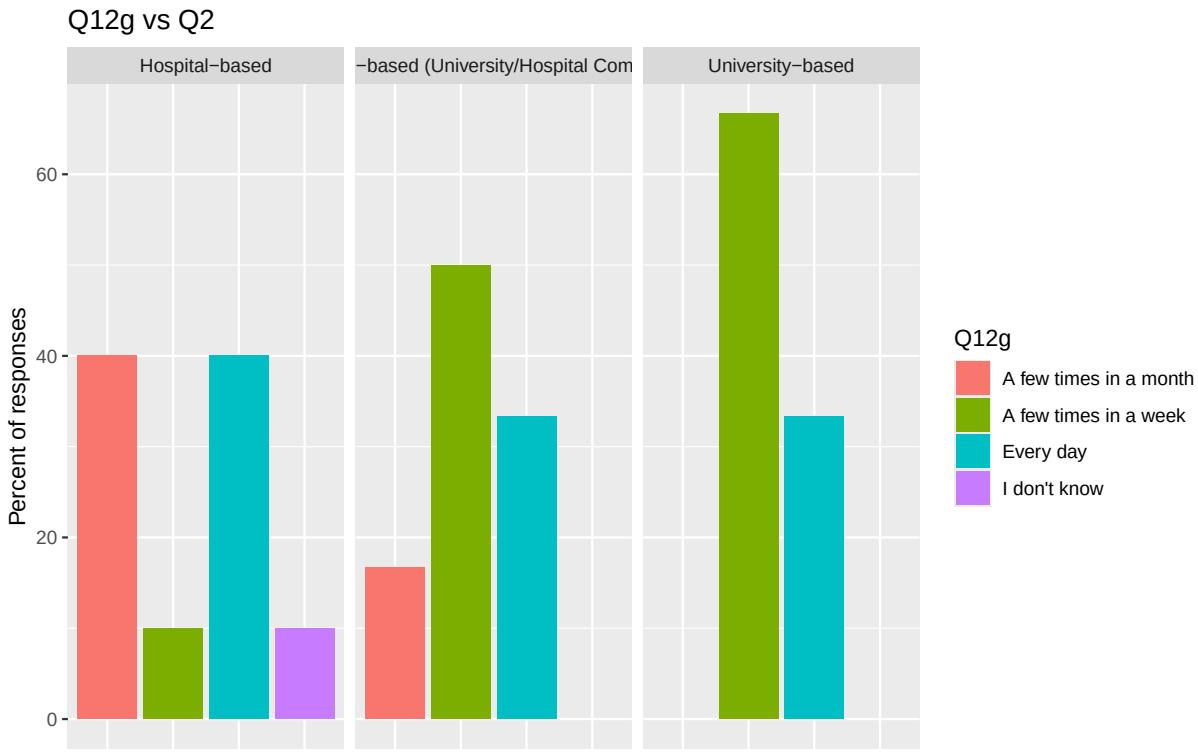

Q12b vs Q2
Hospital−based
−based (University/Hospital Com
University−based
Percent of responses
60
40
20
0
Q12b
A few times in a month
A few times in a week
A few times in a year
Every day
I don't know
Never
Once per month or less
Once per week

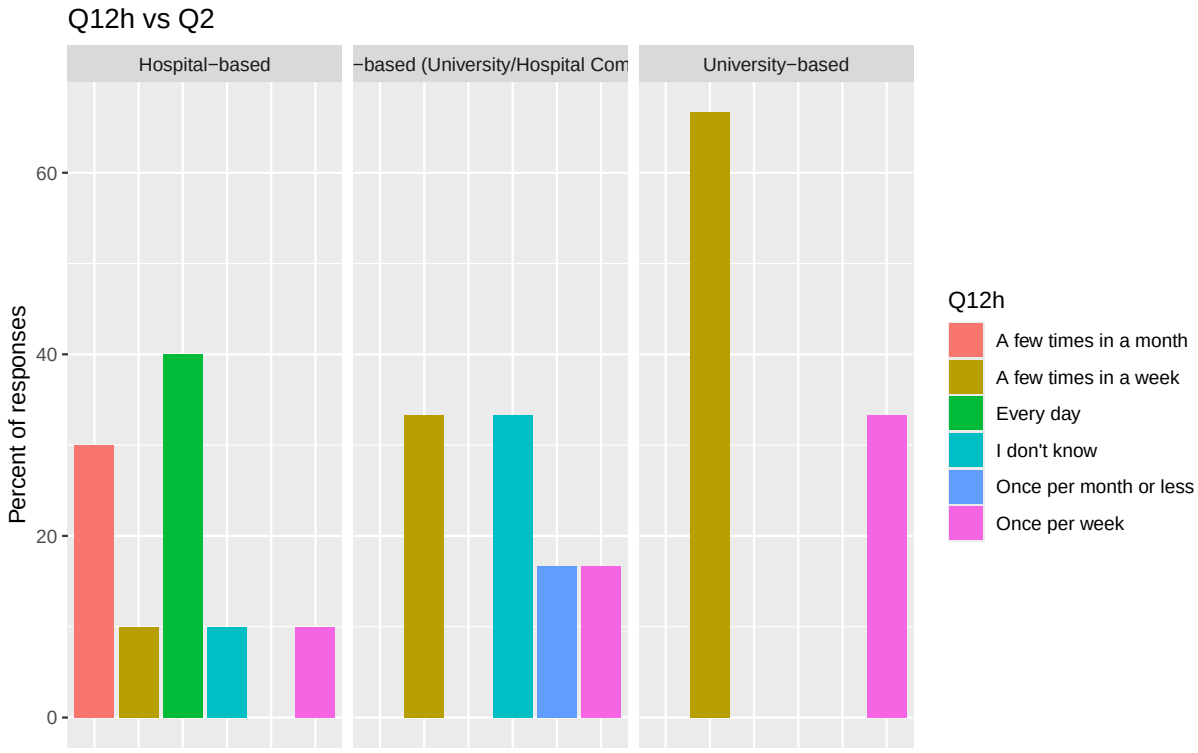

Q12c vs Q2
Hospital−based
−based (University/Hospital Com
University−based
Percent of responses
30
20
10
0
Q12c
A few times in a month
A few times in a week
A few times in a year
Every day
I don't know
Once per week

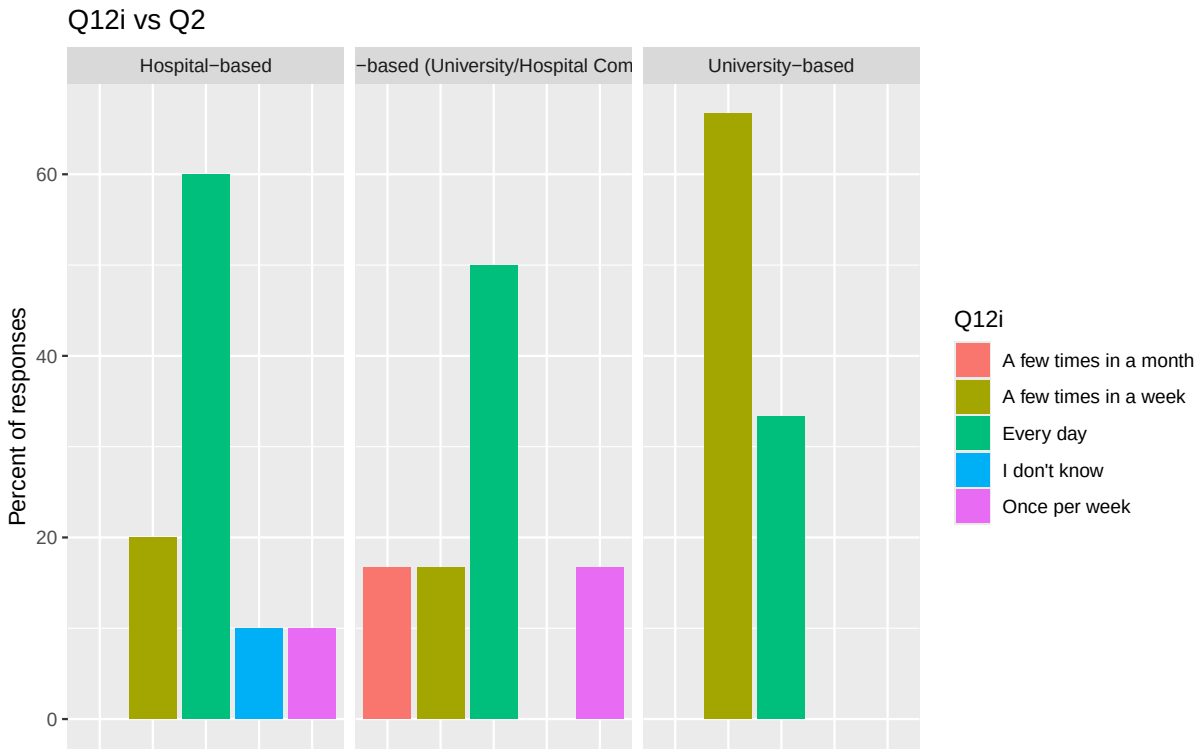

Q12d vs Q2
Hospital−based
−based (University/Hospital Com
University−based
Percent of responses
30
20
10
0
Q12d
A few times in a month
A few times in a year
Every day
I don't know
Never
Once per month or less
Once per week

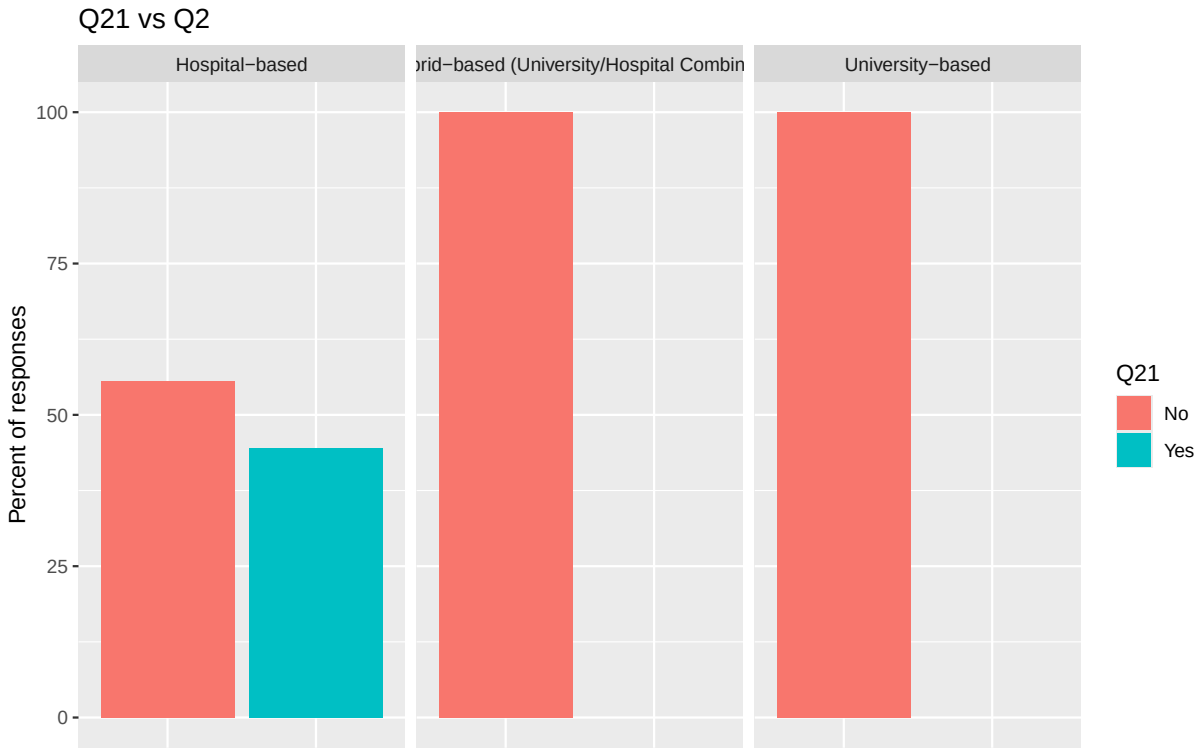

Q12e vs Q2
Hospital−based
−based (University/Hospital Com
University−based
Percent of responses
30
20
10
0
Q12e
A few times in a month
A few times in a year
Every day
I don't know
Never
Once per month or less
Once per week

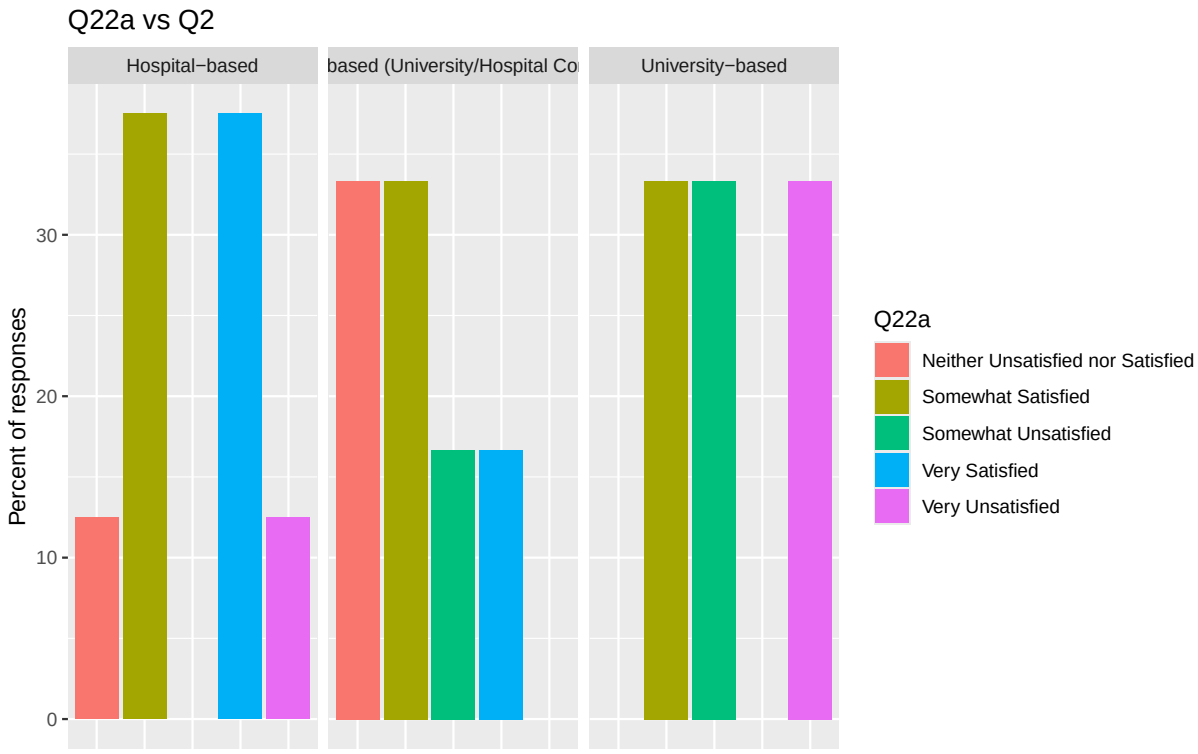

Q12f vs Q2
Hospital−based
−based (University/Hospital Com
University−based
Percent of responses
50
40
30
20
10
0
Q12f
A few times in a month
A few times in a year
Every day
I don't know
Never
Once per month or less

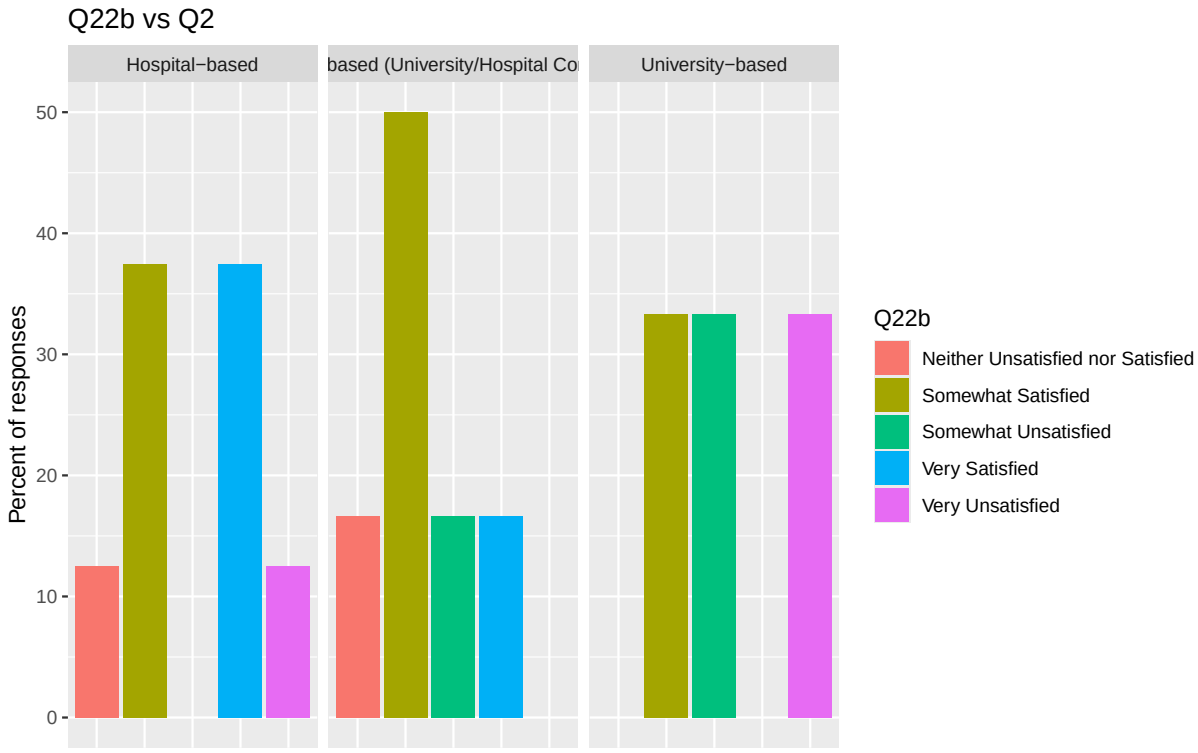

Q12g vs Q2
Hospital−based
−based (University/Hospital Com
University−based
Percent of responses
60
40
20
0
Q12g
A few times in a month
A few times in a week
Every day
I don't know

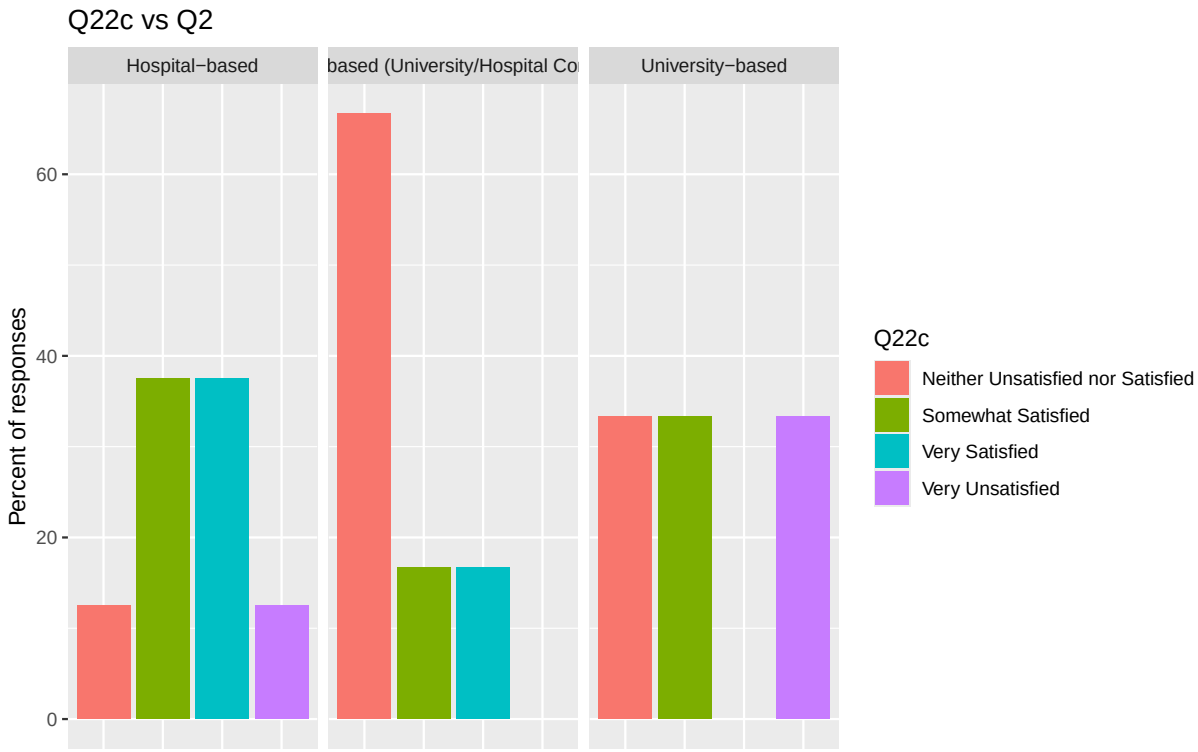

Q12h vs Q2
Hospital−based
−based (University/Hospital Com
University−based
Percent of responses
0
20
40
60
Q12h
A few times in a month
A few times in a week
Every day
I don't know
Once per month or less
Once per week

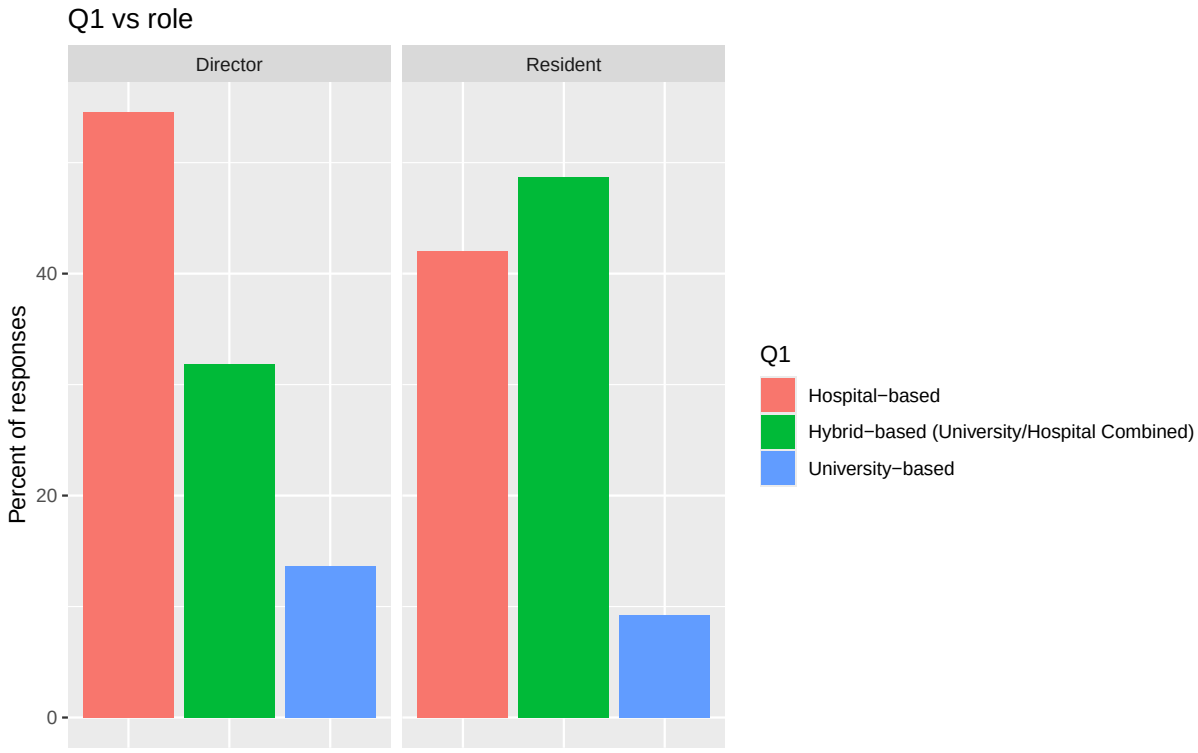

Q12i vs Q2
Hospital−based
−based (University/Hospital Com
University−based
Percent of responses
0
20
40
60
Q12i
A few times in a month
A few times in a week
Every day
I don't know
Once per week

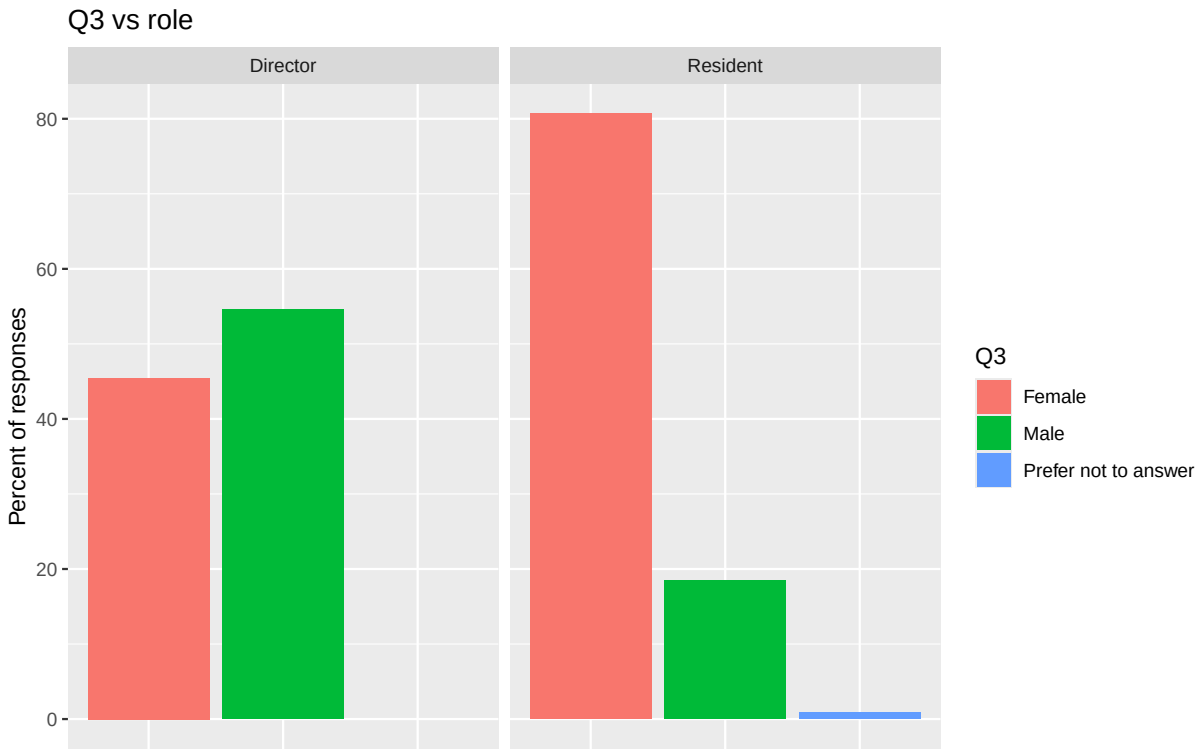

Q21 vs Q2
Hospital−based
rid−based (University/Hospital Combin
University−based
100
75
50
25
0
Percent of responses
Q21
No
Yes


Q22a vs Q2
Hospital−based
based (University/Hospital Co
University−based
30
20
10
0
Percent of responses
Q22a
Neither Unsatisfied nor Satisfied
Somewhat Satisfied
Somewhat Unsatisfied
Very Satisfied
Very Unsatisfied


Q22b vs Q2
Hospital−based
based (University/Hospital Co
University−based
Percent of responses
0
10
20
30
40
50
Q22b
Neither Unsatisfied nor Satisfied
Somewhat Satisfied
Somewhat Unsatisfied
Very Satisfied
Very Unsatisfied


Q22c vs Q2
Hospital−based
based (University/Hospital Co
University−based
Percent of responses
0
20
40
60
Q22c
Neither Unsatisfied nor Satisfied
Somewhat Satisfied
Very Satisfied
Very Unsatisfied


Q1 vs role
Director
Resident
Percent of responses
0
20
40
Q1
Hospital−based
Hybrid−based (University/Hospital Combined)
University−based


Q3 vs role
Director
Resident
Percent of responses
0
20
40
60
80
Q3
Female
Male
Prefer not to answer

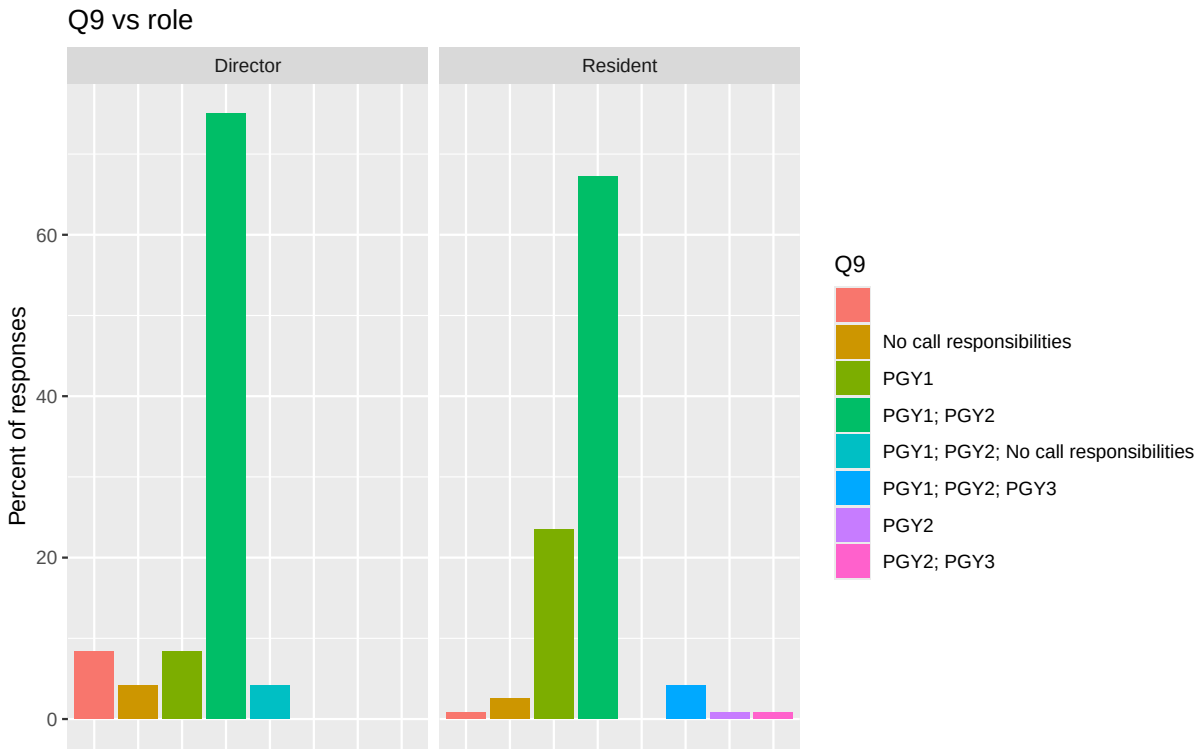
ebraska, North Dakota, Ohio, South Dakota, Wisconsin, and the Canadian provinces of Manitoba and Ontario)

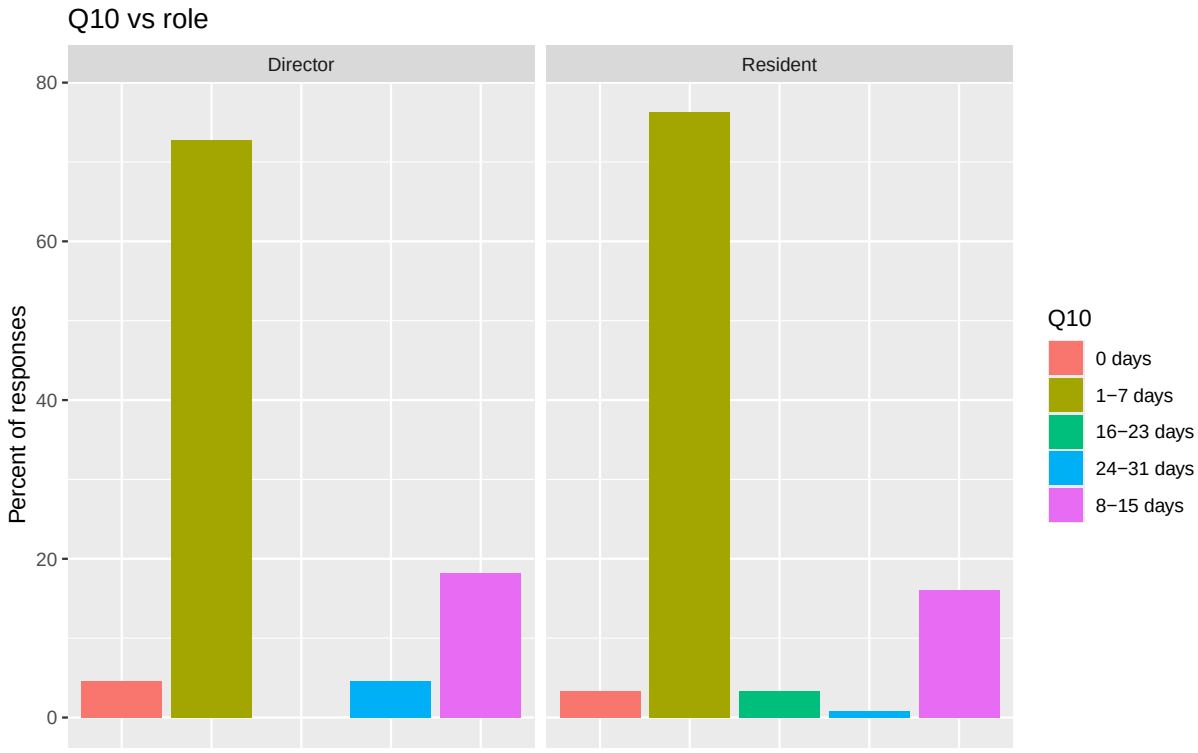
ıe, Maryland, Massachusetts, New Hampshire, New Jersey, New York, Pennsylvania, Rhode Island, Vermont, and the Canadian provinces of New

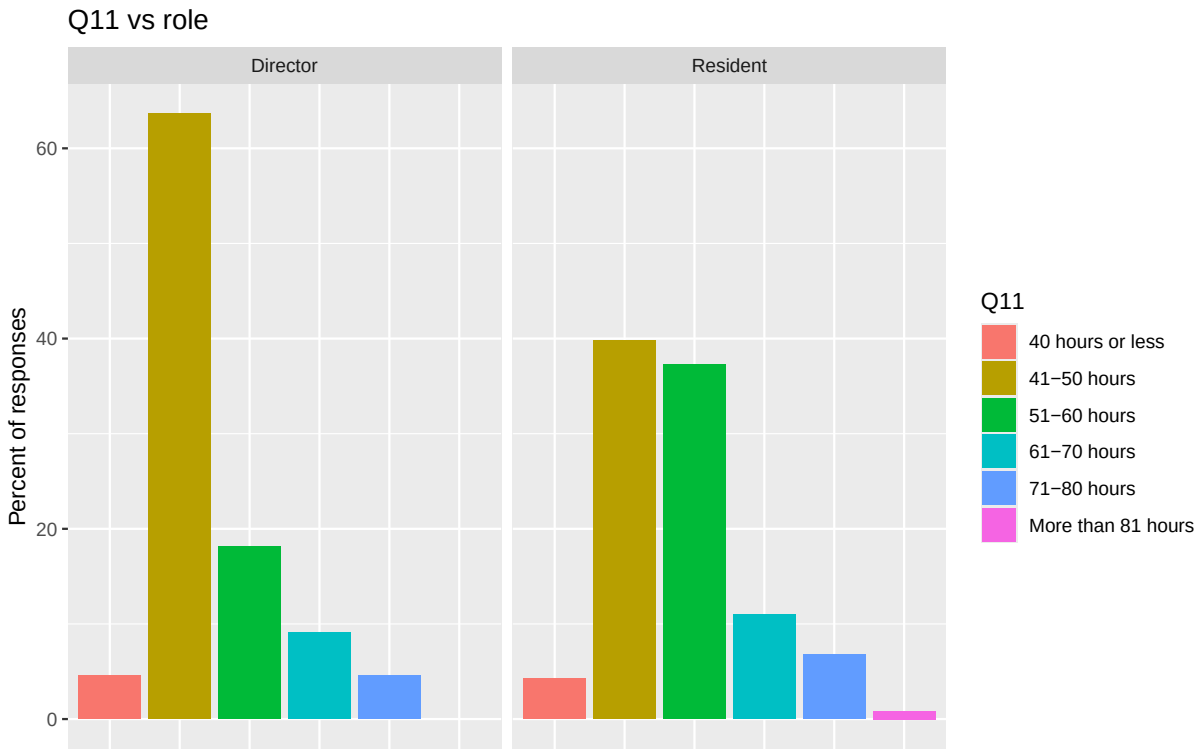
i, North Carolina, South Carolina, Tennessee, Virginia, West Virginia, the Commonwealth of Puerto Rico, and the U.S. Virgin Islands)

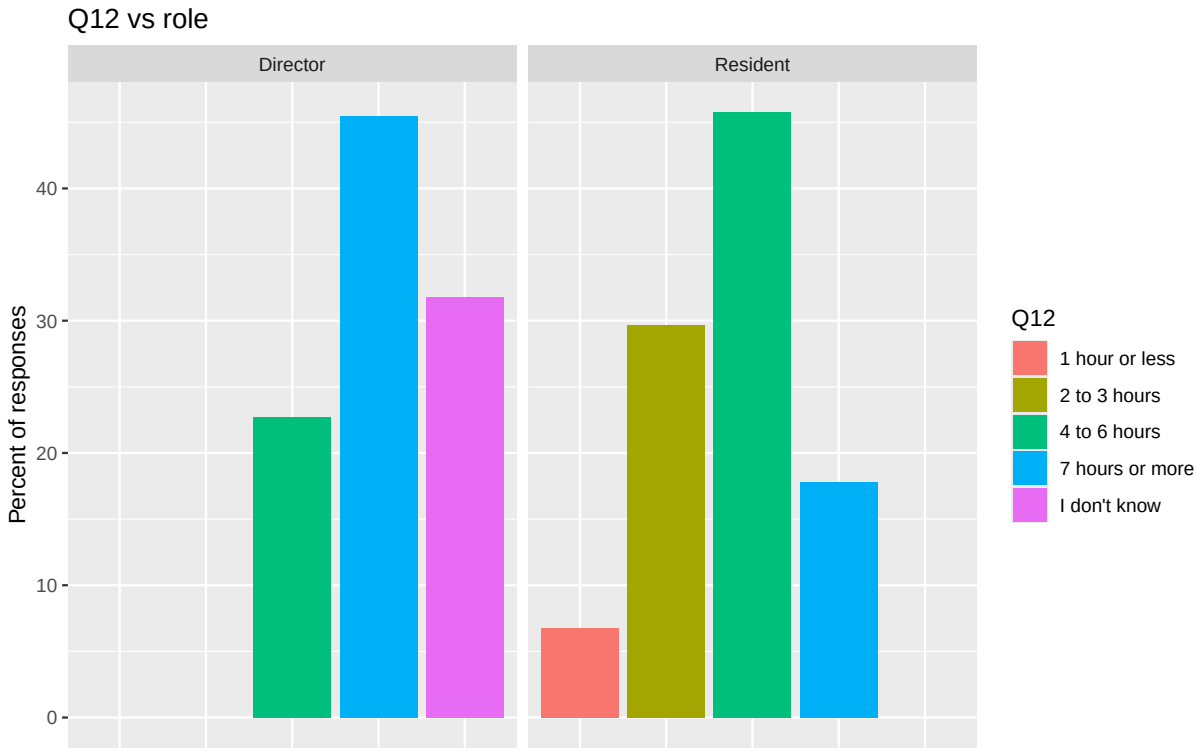
sas, Louisiana, Texas and Mexico

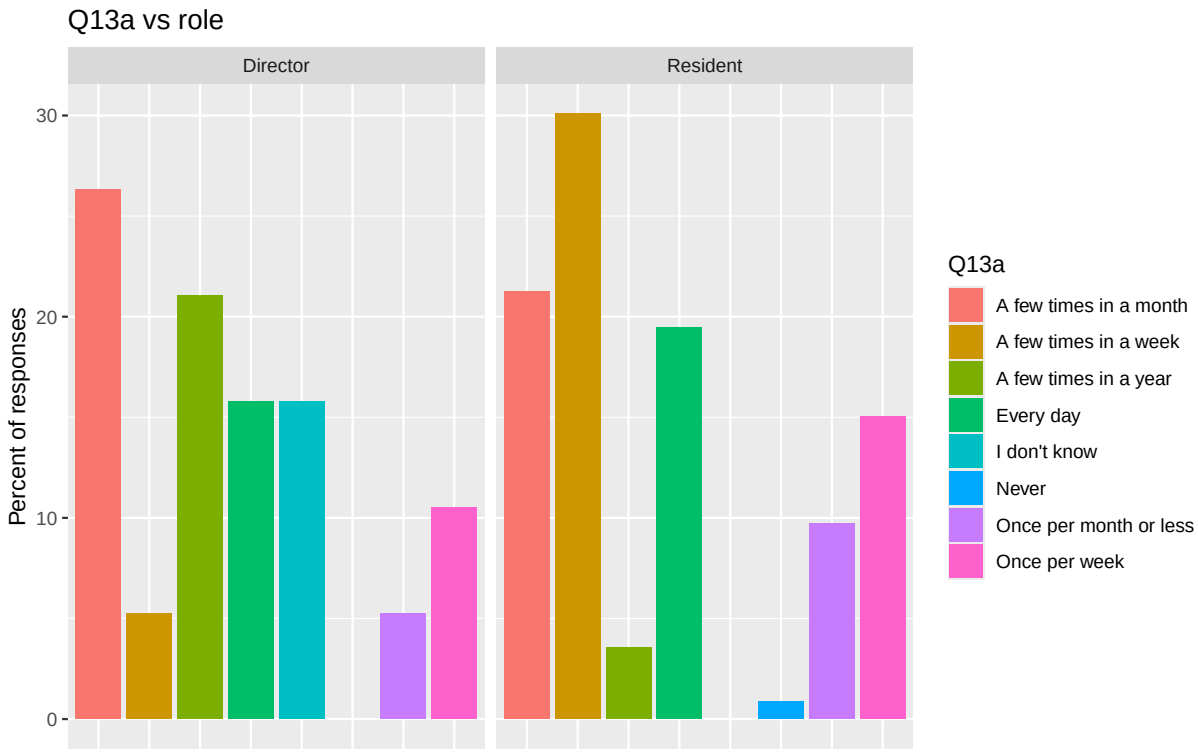
Nevada, Oregon, Utah, Washington, Wyoming, and the Canadian provinces of Saskatchewan, Alberta, British Columbia, Northwest Territories, Nu

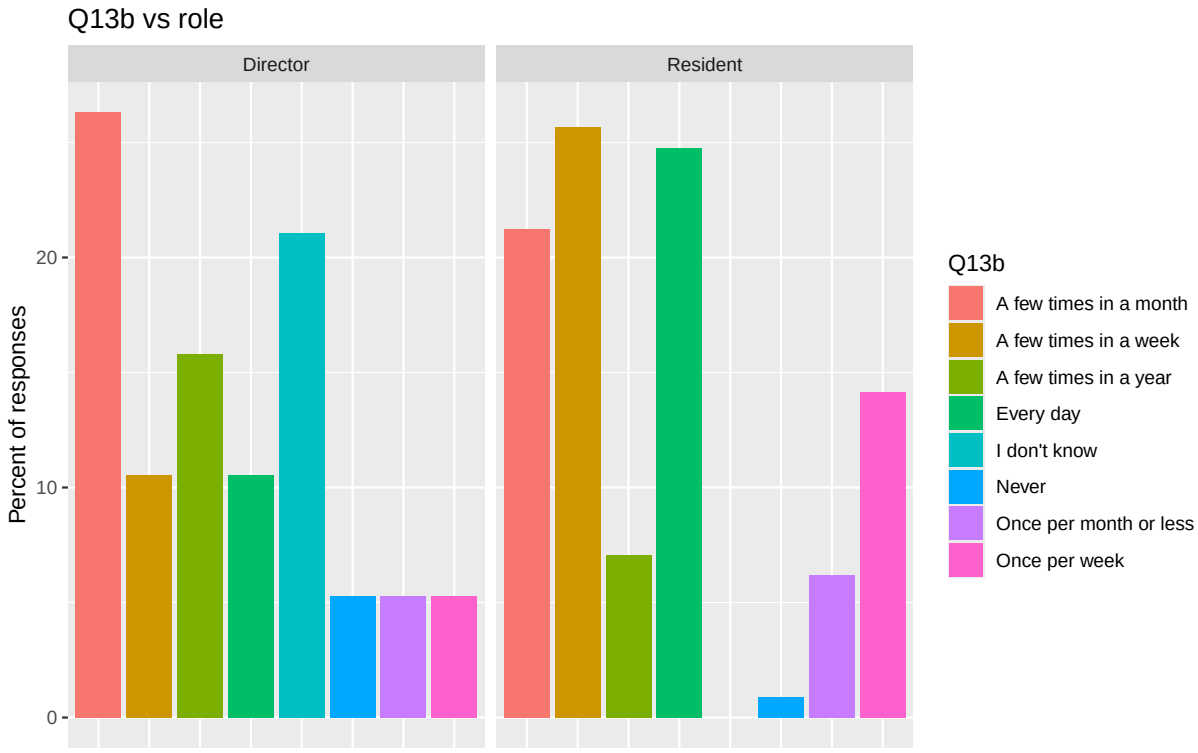


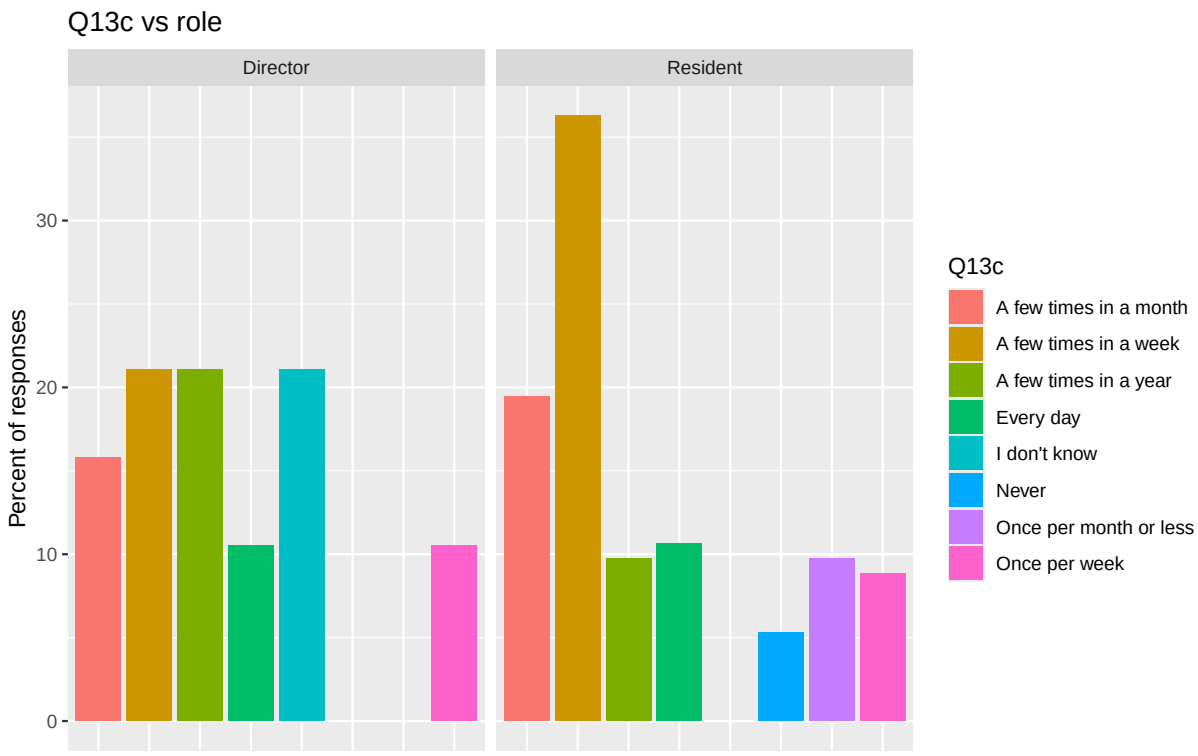

Q10 vs role
Director
Resident
Percent of responses
80
60
40
20
0
Q10
0 days
1−7 days
16−23 days
24−31 days
8−15 days

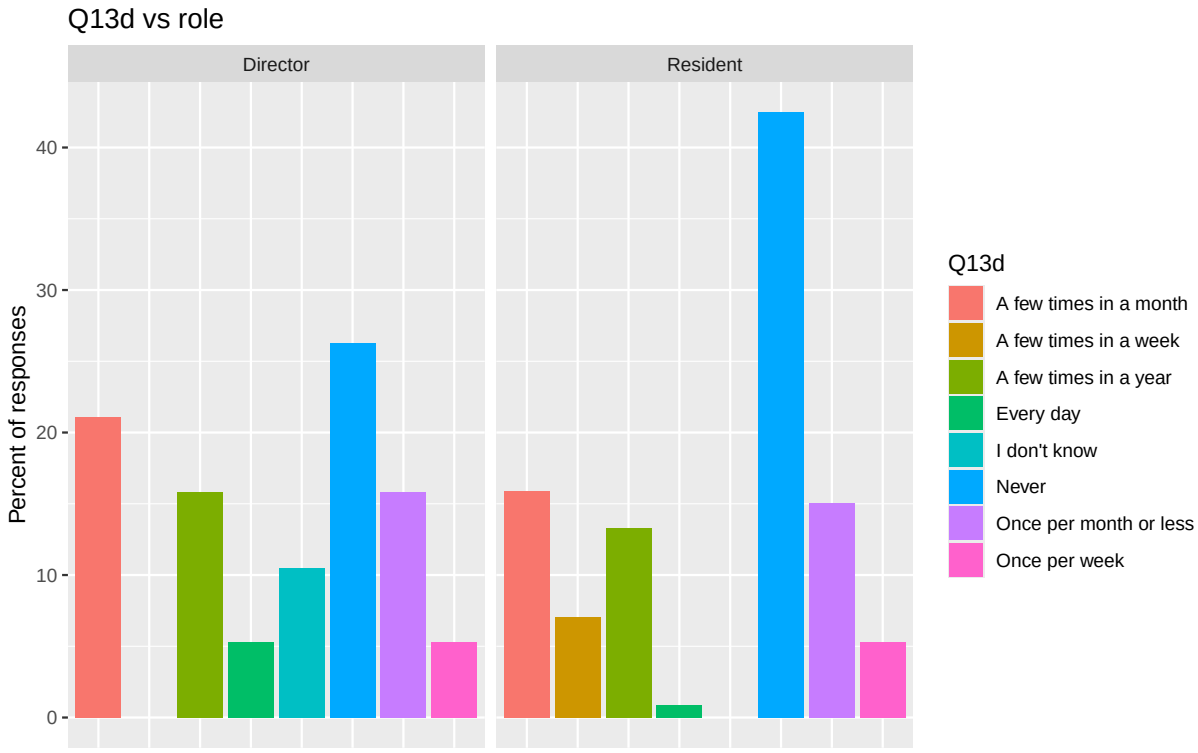

Q11 vs role
Director
Resident
Percent of responses
60
40
20
0
Q11
40 hours or less
41−50 hours
51−60 hours
61−70 hours
71−80 hours
More than 81 hours

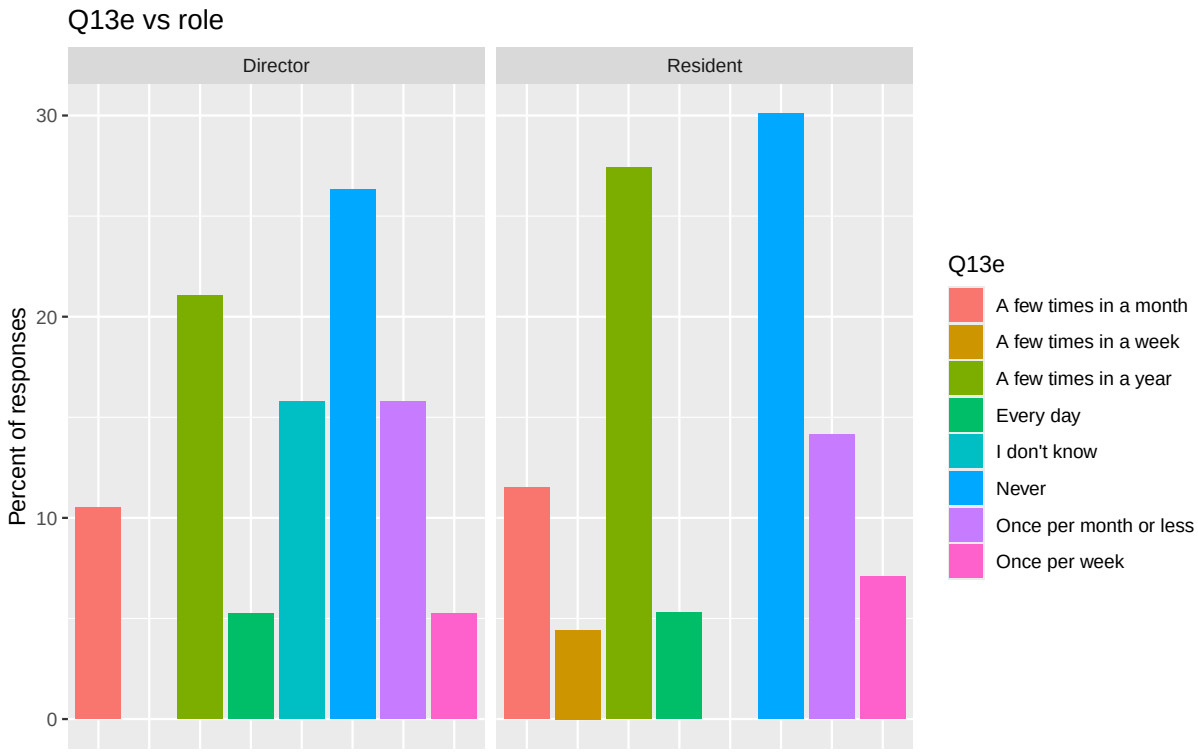

Q12 vs role
Director
Resident
Percent of responses
0
10
20
30
40
Q12
1 hour or less
2 to 3 hours
4 to 6 hours
7 hours or more
I don't know

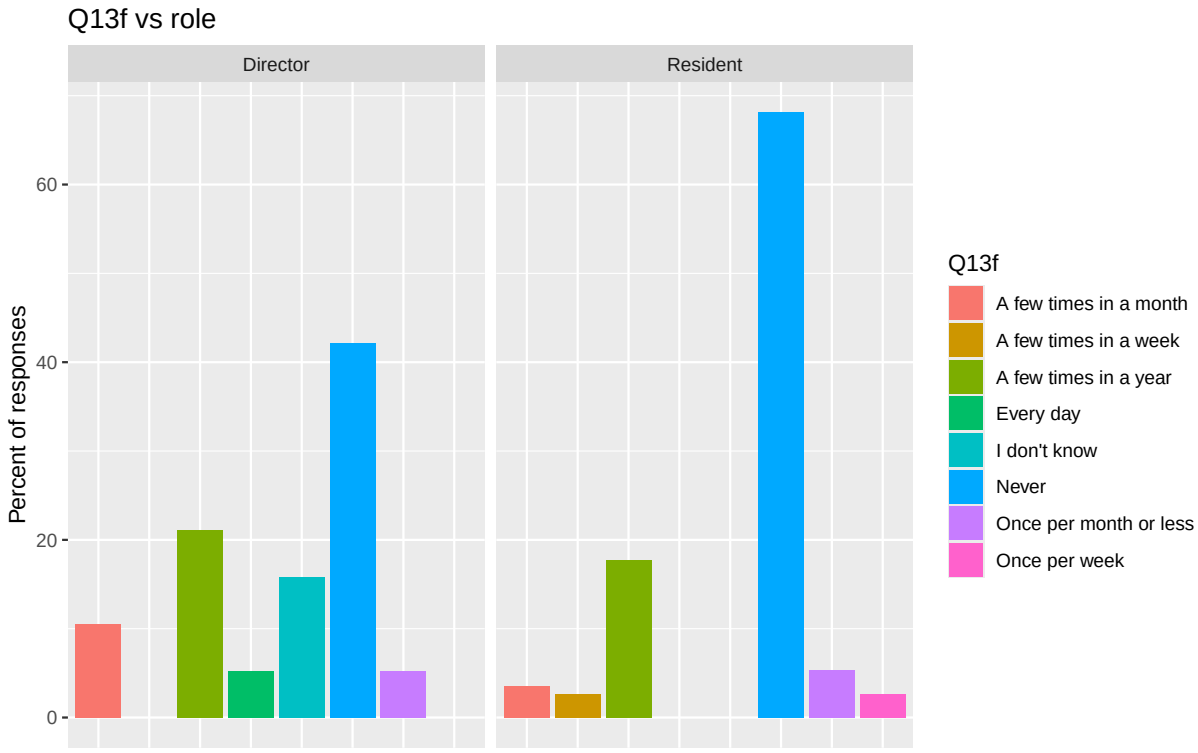

Q13a vs role
Director
Resident
Percent of responses
0
10
20
30
Q13a
A few times in a month
A few times in a week
A few times in a year
Every day
I don't know
Never
Once per month or less
Once per week

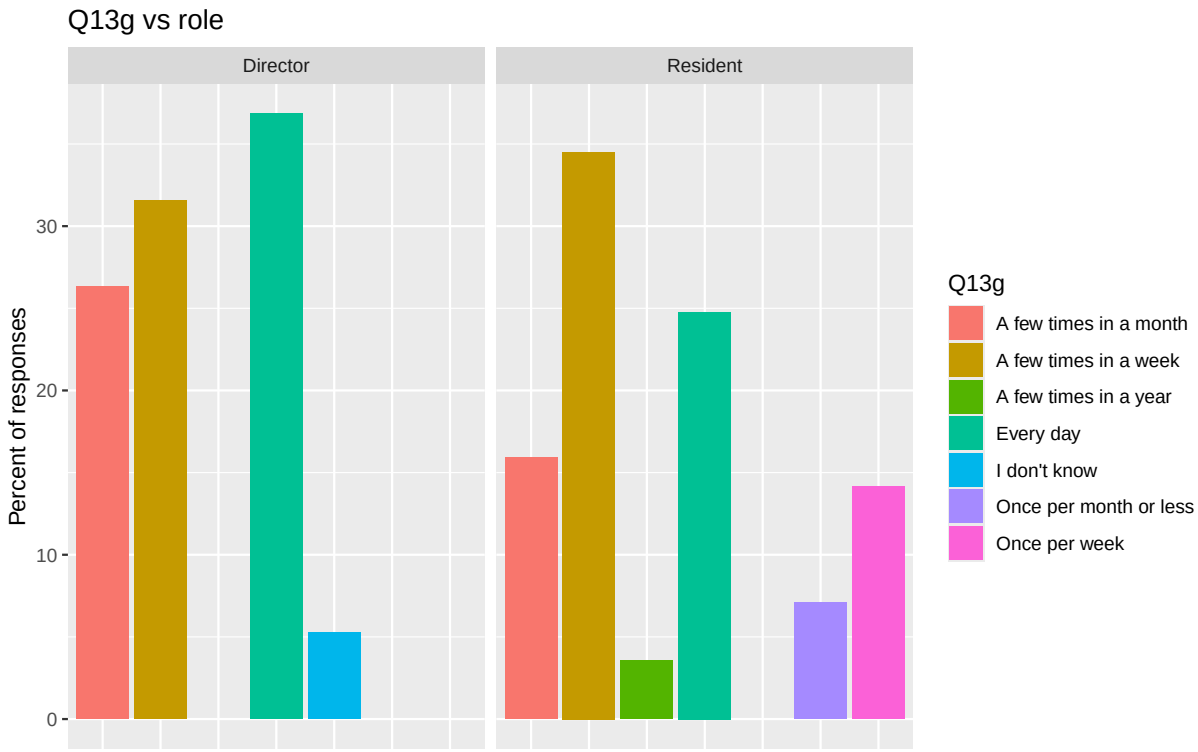

Q13b vs role
Director
Resident
Percent of responses
20
10
0
Q13b
A few times in a month
A few times in a week
A few times in a year
Every day
I don't know
Never
Once per month or less
Once per week

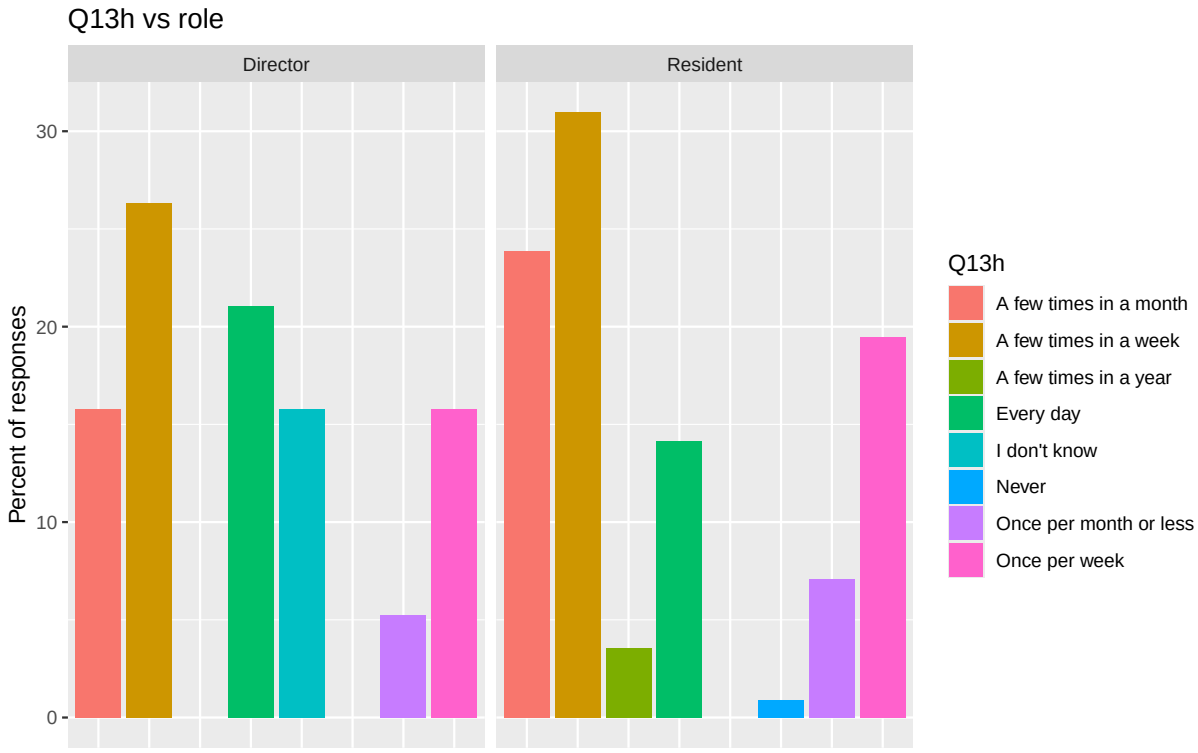

Q13c vs role
Director
Resident
Percent of responses
30
20
10
0
Q13c
A few times in a month
A few times in a week
A few times in a year
Every day
I don't know
Never
Once per month or less
Once per week

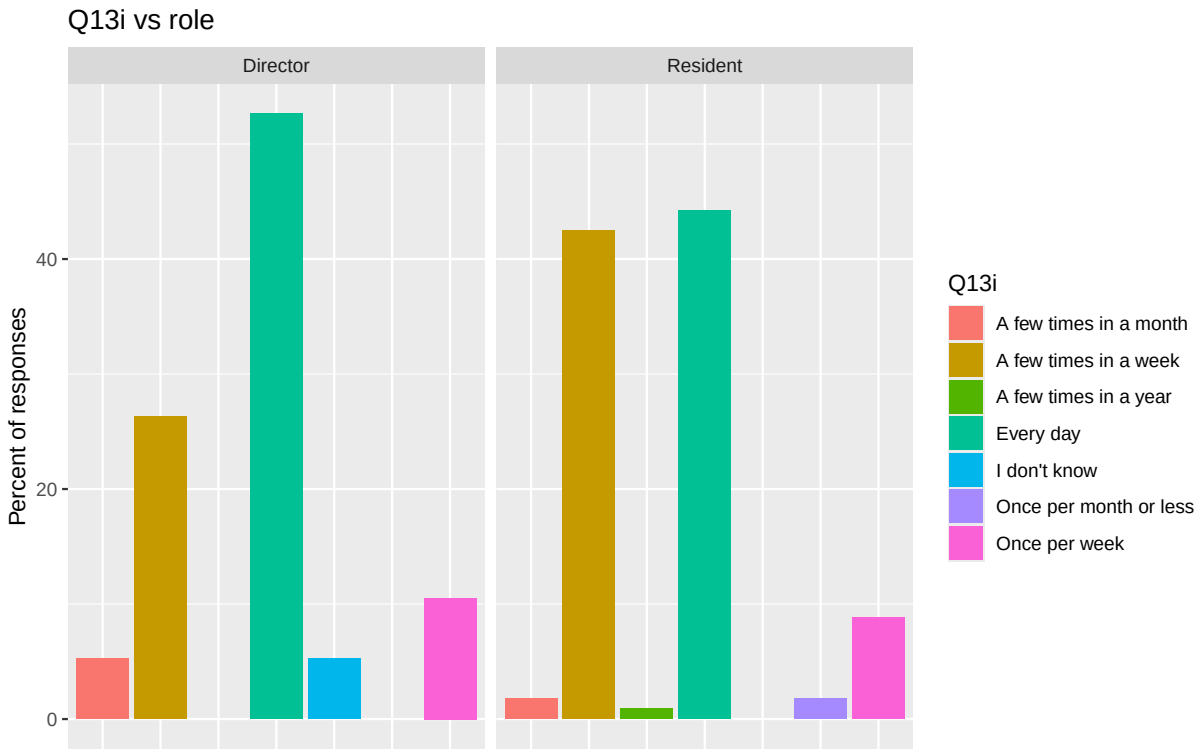

Q13d vs role
Director
Resident
Percent of responses
0
10
20
30
40
Q13d
A few times in a month
A few times in a week
A few times in a year
Every day
I don't know
Never
Once per month or less
Once per week

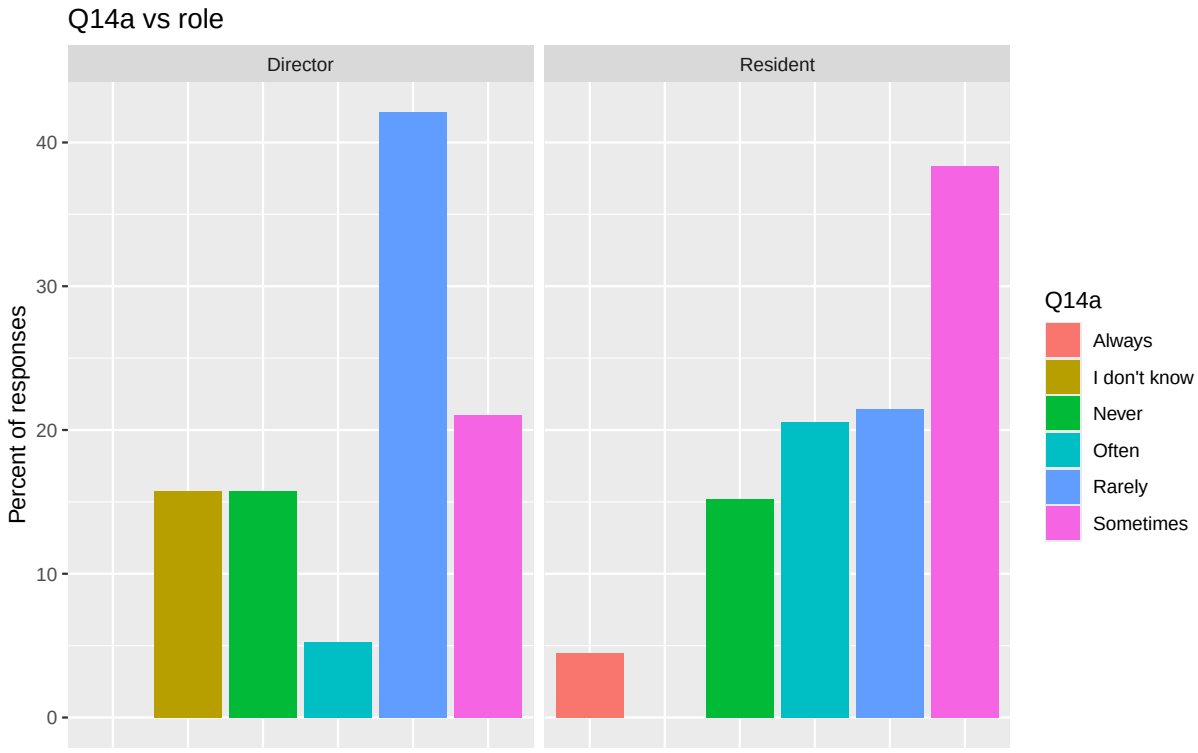

Q13e vs role
Director
Resident
Percent of responses
0
10
20
30
Q13e
A few times in a month
A few times in a week
A few times in a year
Every day
I don't know
Never
Once per month or less
Once per week

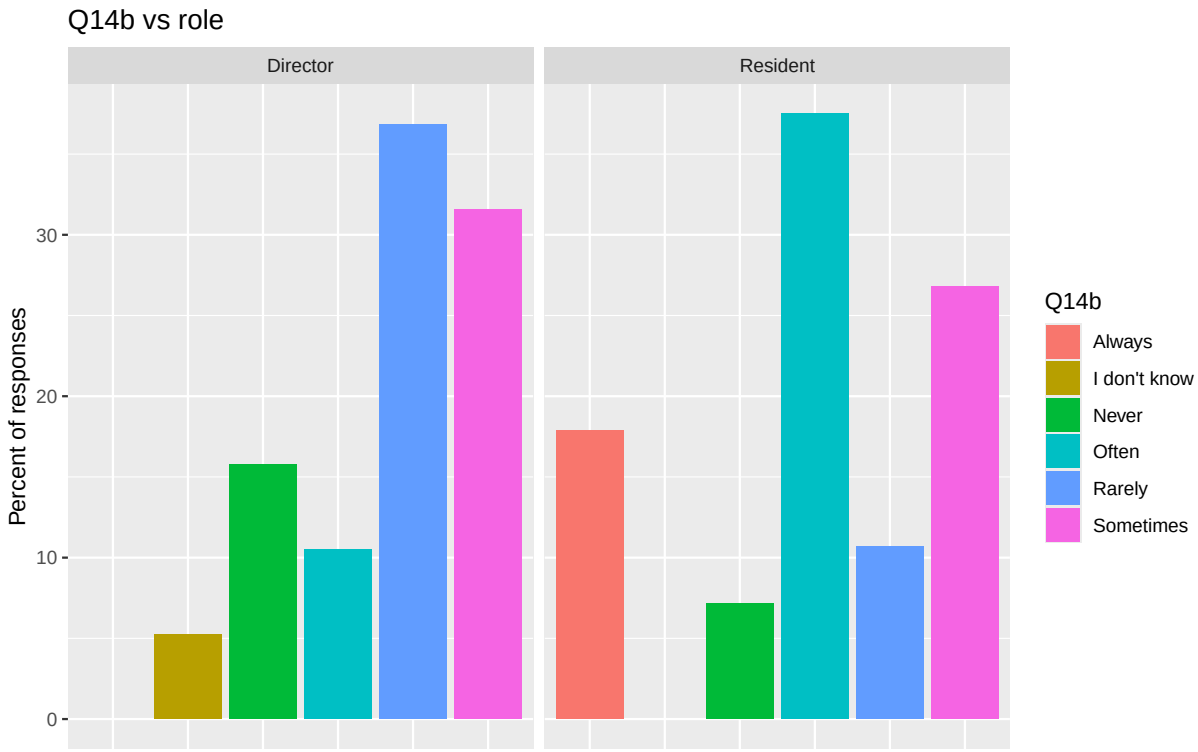

Q13f vs role
Director
Resident
Percent of responses
0
20
40
60
Q13f
A few times in a month
A few times in a week
A few times in a year
Every day
I don't know
Never
Once per month or less
Once per week

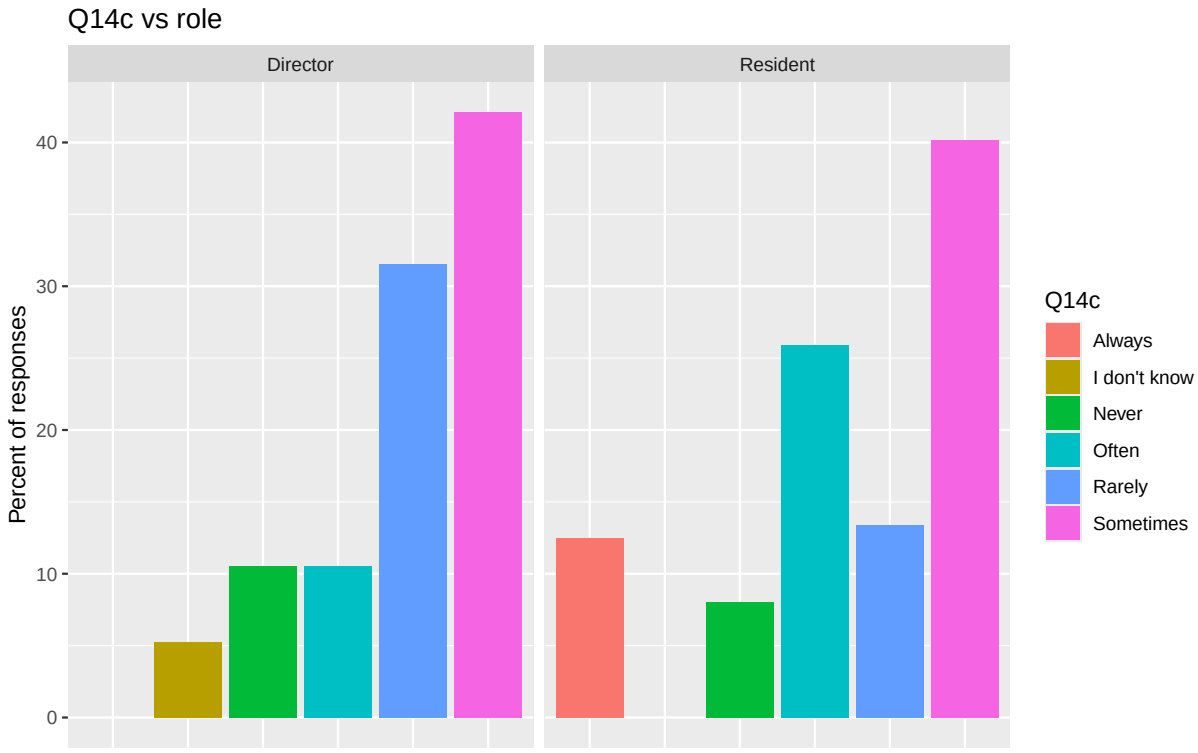

Q13g vs role
Director
Resident
Percent of responses
0
10
20
30
Q13g
A few times in a month
A few times in a week
A few times in a year
Every day
I don't know
Once per month or less
Once per week

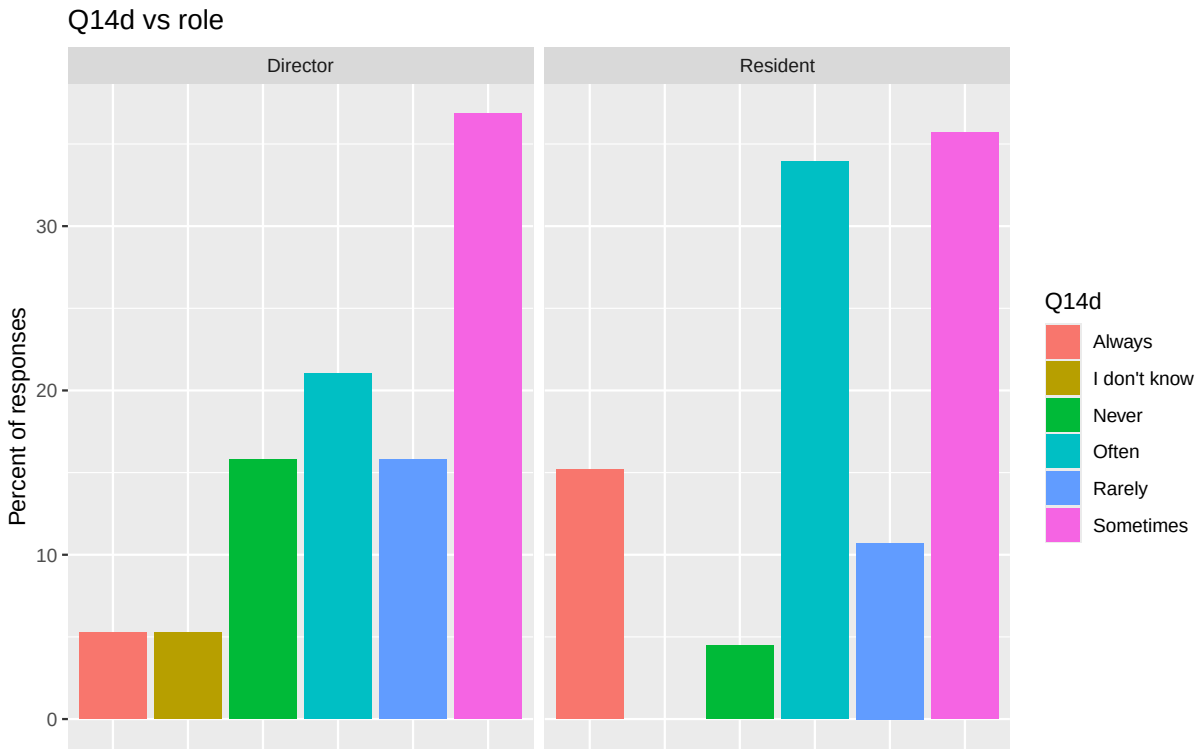

Q13h vs role
Director
Resident
Percent of responses
30
20
10
0
Q13h
A few times in a month
A few times in a week
A few times in a year
Every day
I don't know
Never
Once per month or less
Once per week

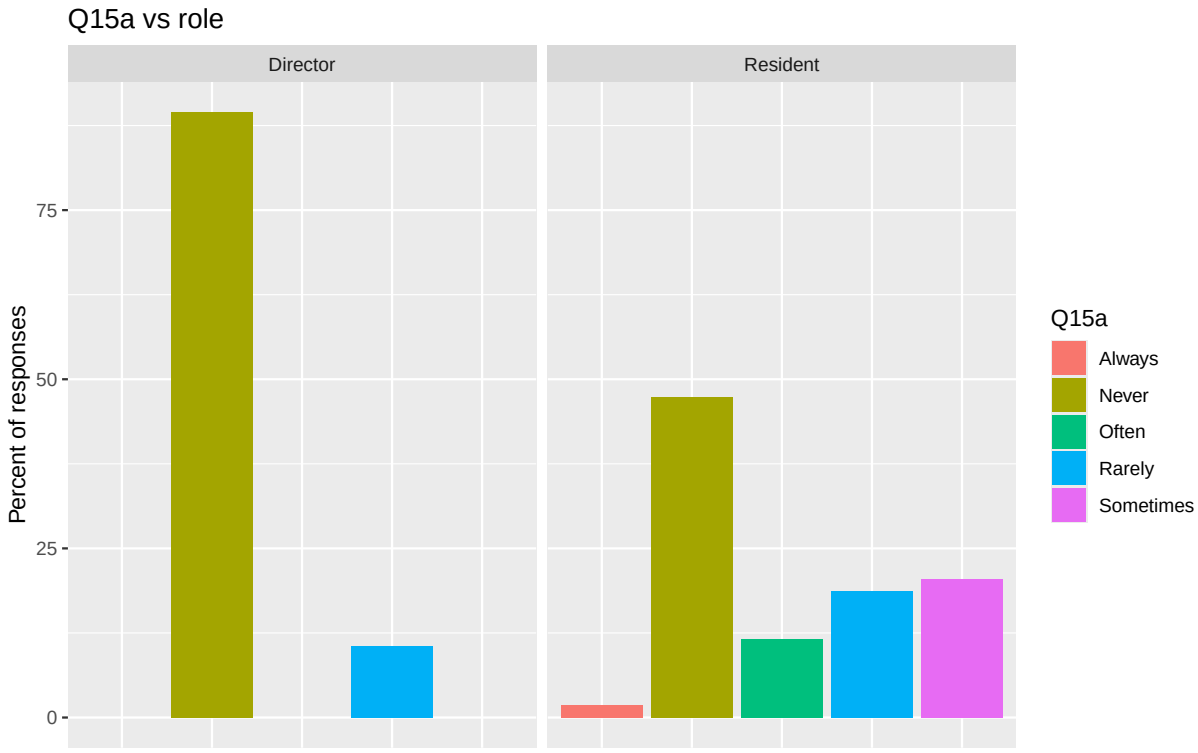

Q13i vs role
Director
Resident
Percent of responses
40
20
0
Q13i
A few times in a month
A few times in a week
A few times in a year
Every day
I don't know
Once per month or less
Once per week

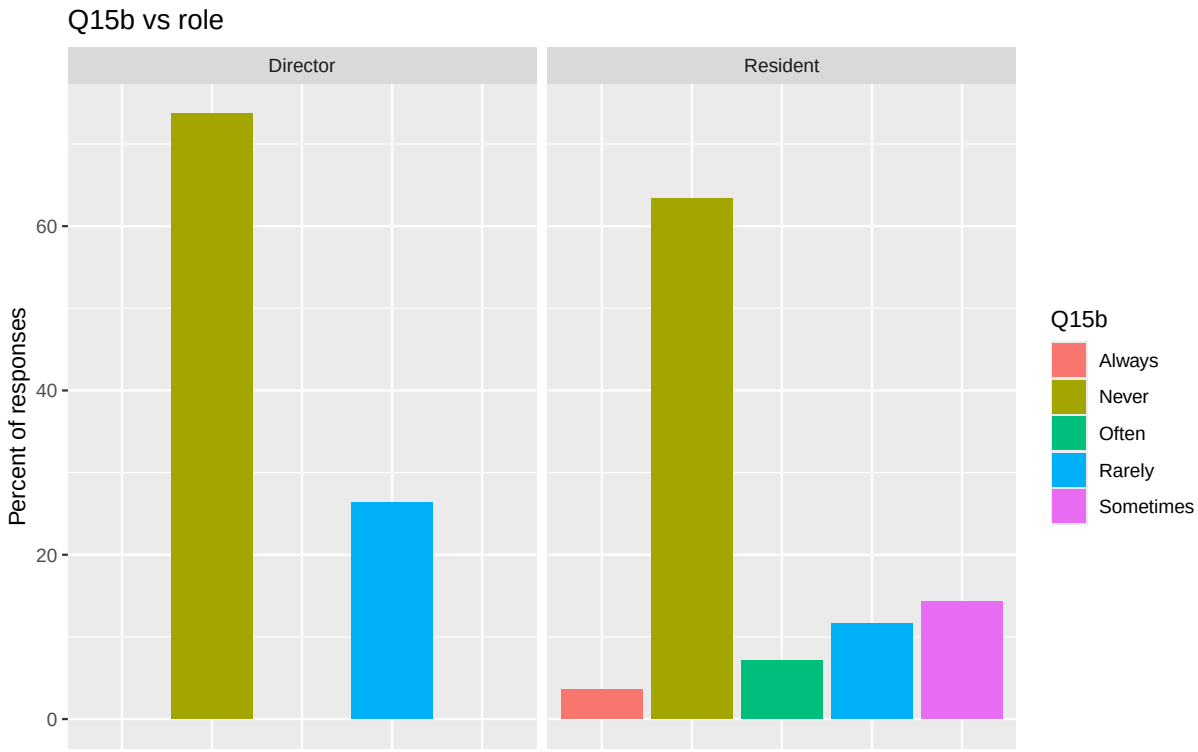

Q14a vs role
Director
Resident
Percent of responses
0
10
20
30
40
Q14a
Always
I don't know
Never
Often
Rarely
Sometimes

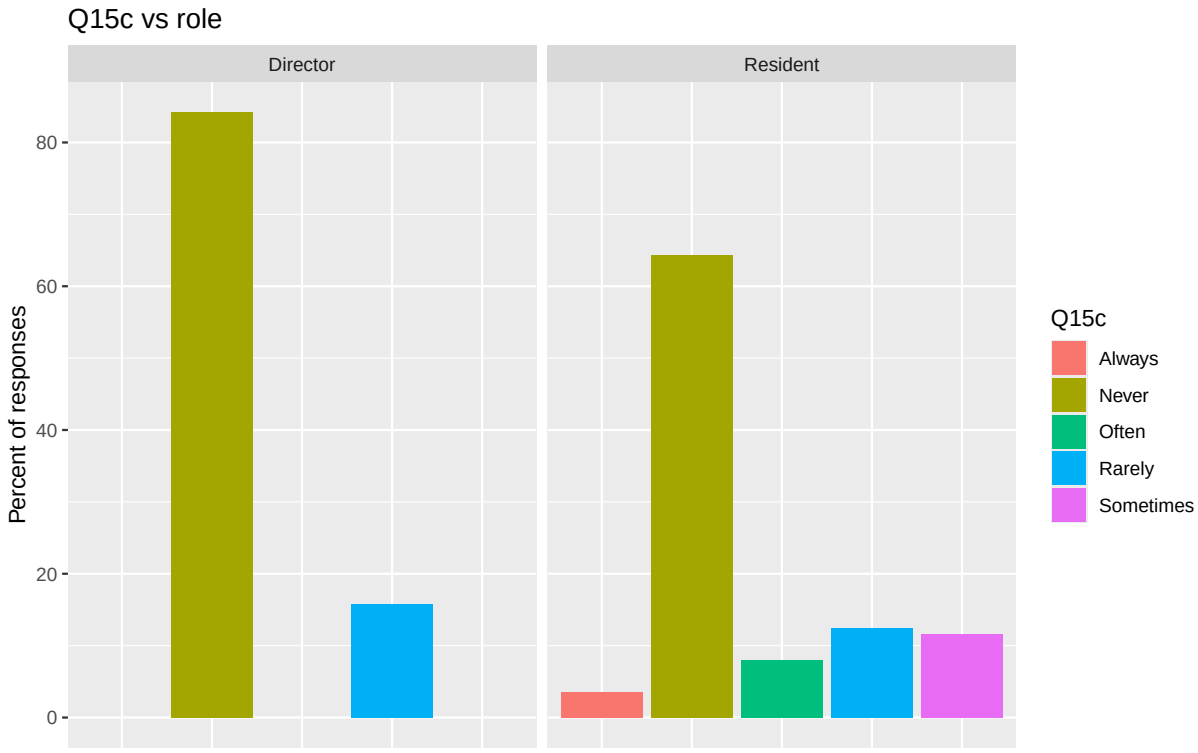

Q14b vs role
Director
Resident
Percent of responses
0
10
20
30
Q14b
Always
I don't know
Never
Often
Rarely
Sometimes

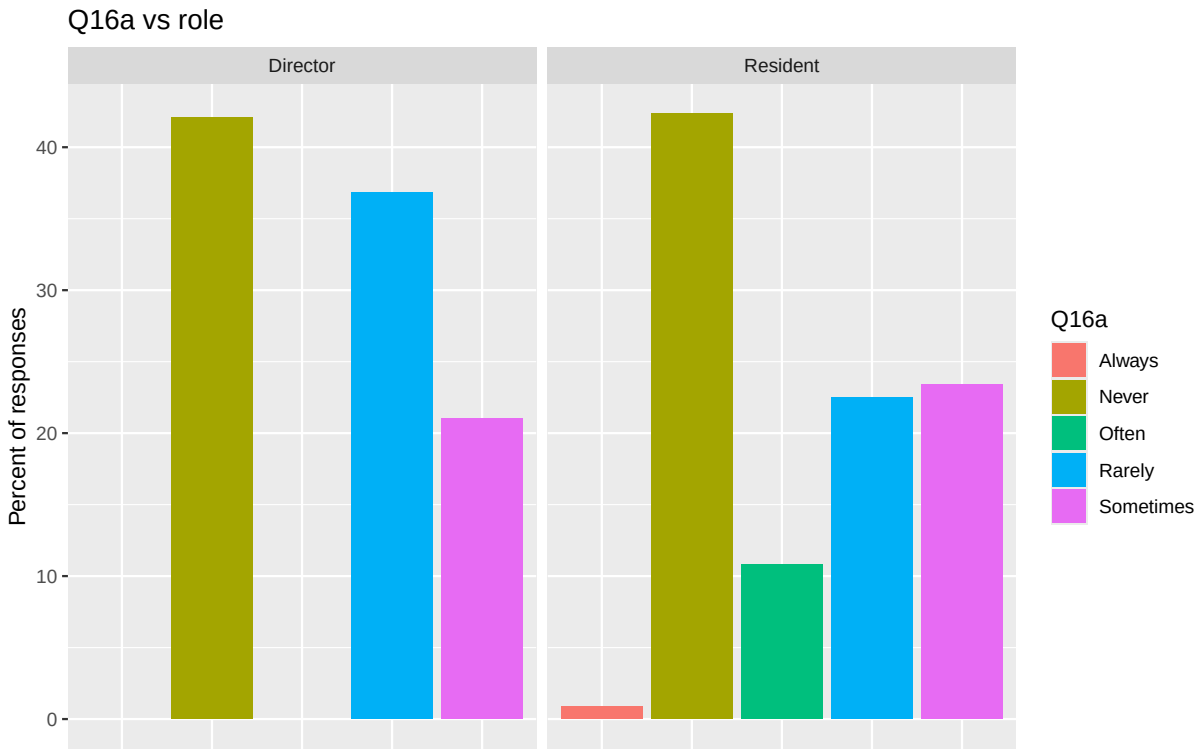

Q14c vs role
Director
Resident
Percent of responses
0
10
20
30
40
Q14c
Always
I don't know
Never
Often
Rarely
Sometimes

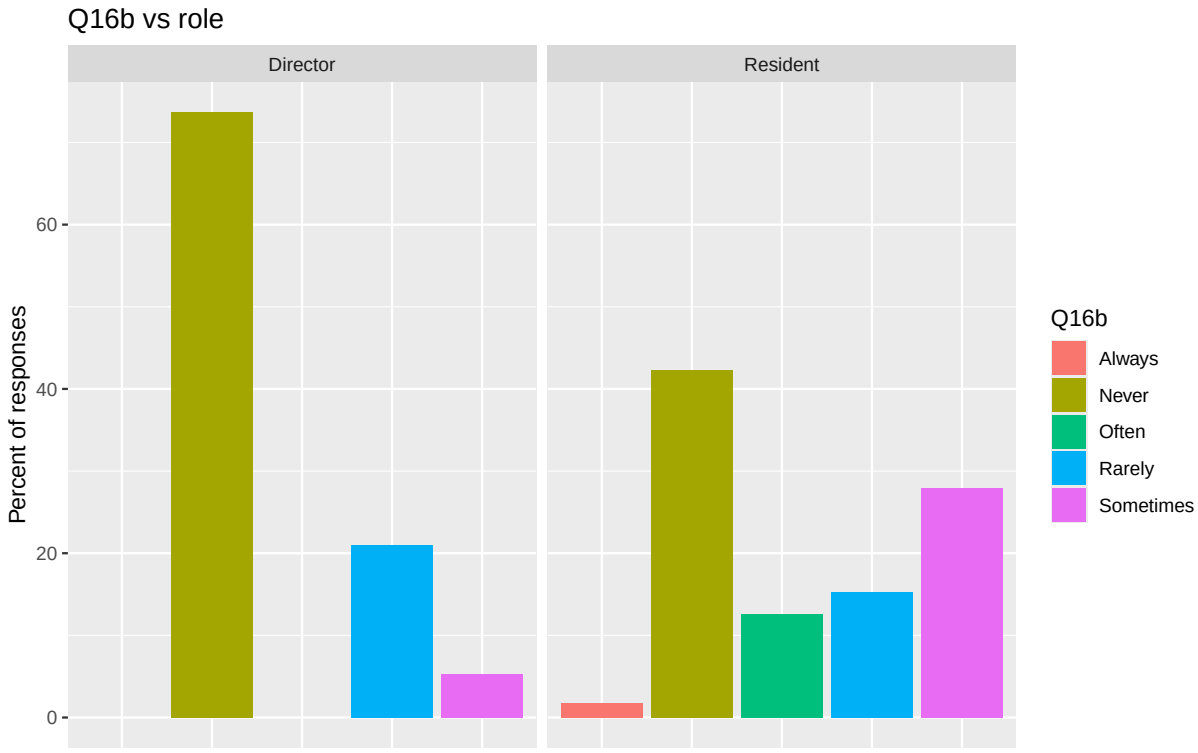

Q14d vs role
Director
Resident
Percent of responses
0
10
20
30
Q14d
Always
I don't know
Never
Often
Rarely
Sometimes

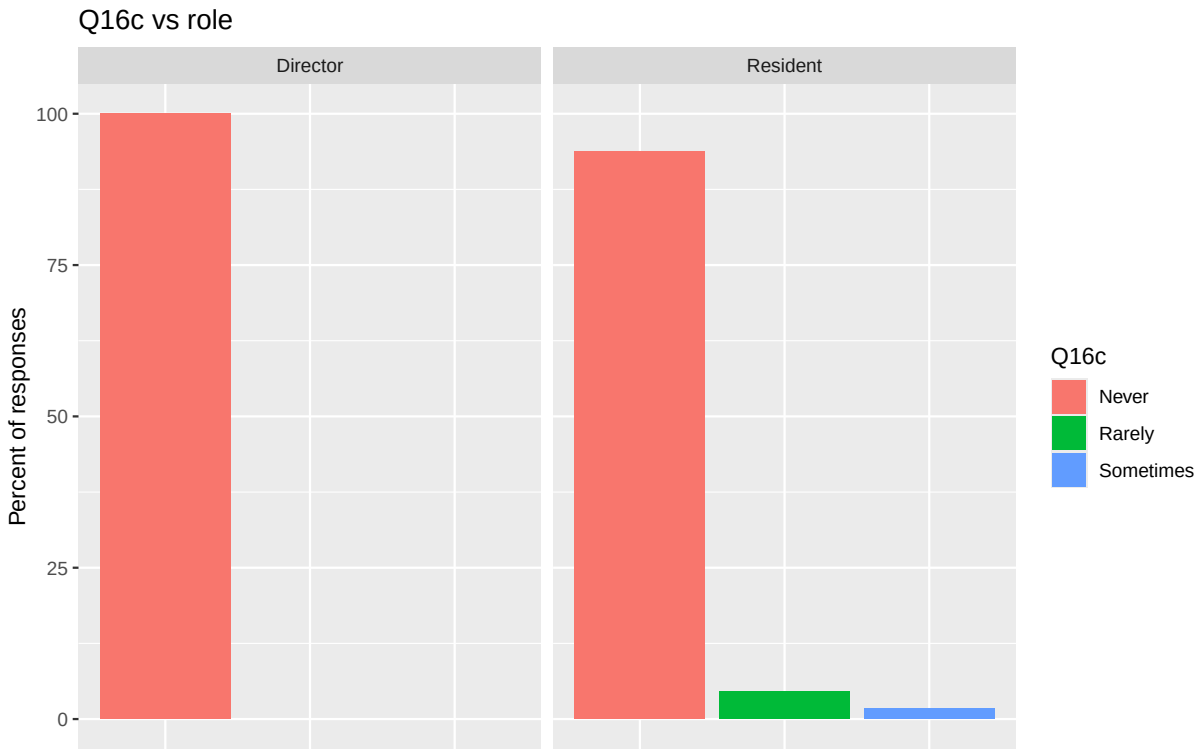

Q15a vs role
Director
Resident
Percent of responses
75
50
25
0
Q15a
Always
Never
Often
Rarely
Sometimes

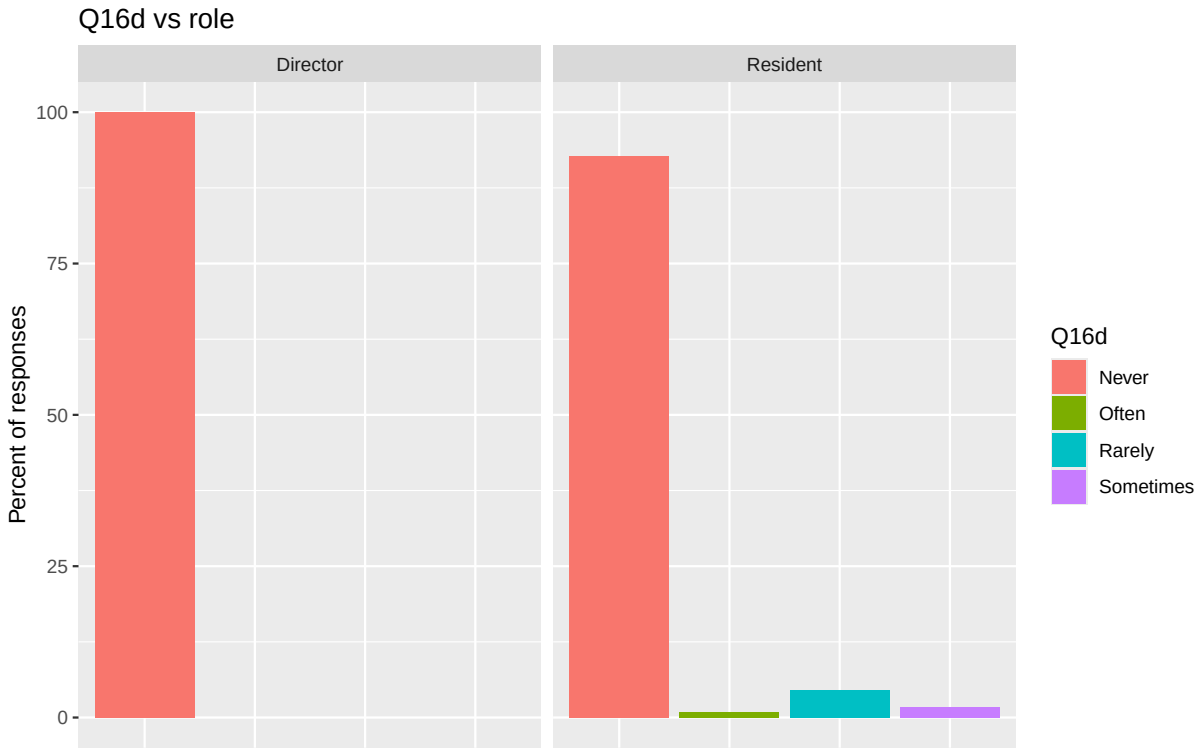

Q15b vs role
Director
Resident
Percent of responses
60
40
20
0
Q15b
Always
Never
Often
Rarely
Sometimes

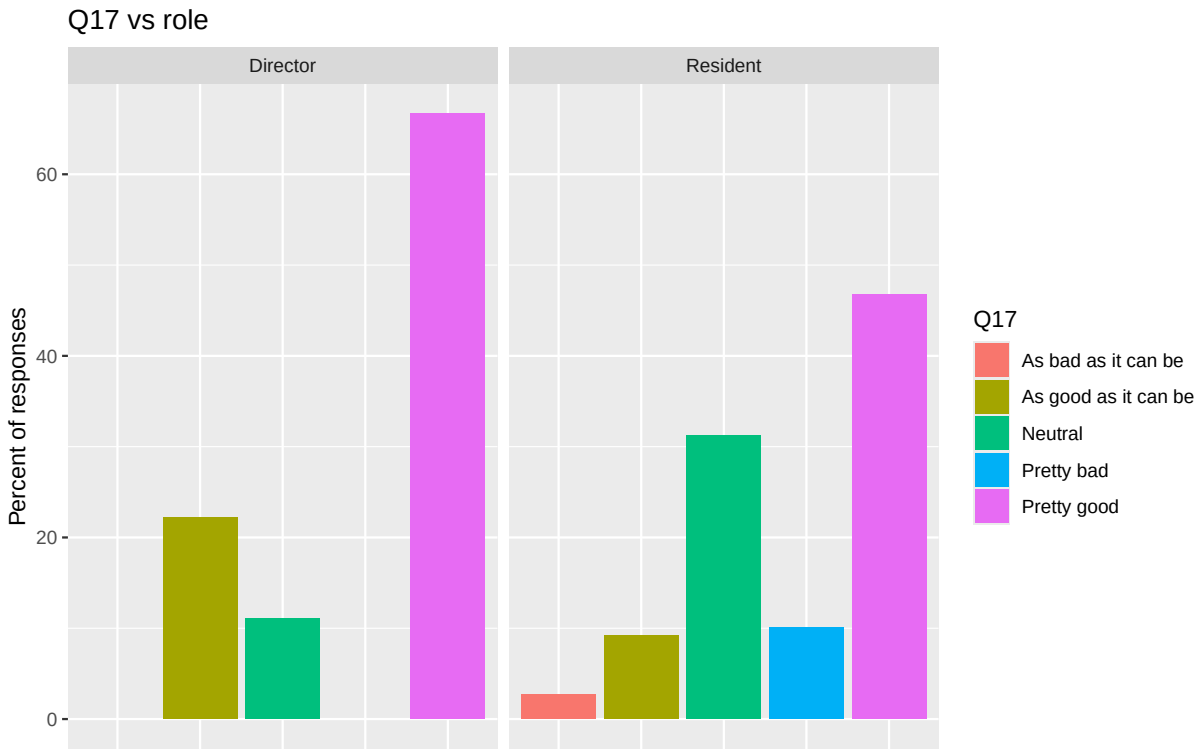

Q15c vs role
Director
Resident
Percent of responses
0
20
40
60
80
Q15c
Always
Never
Often
Rarely
Sometimes

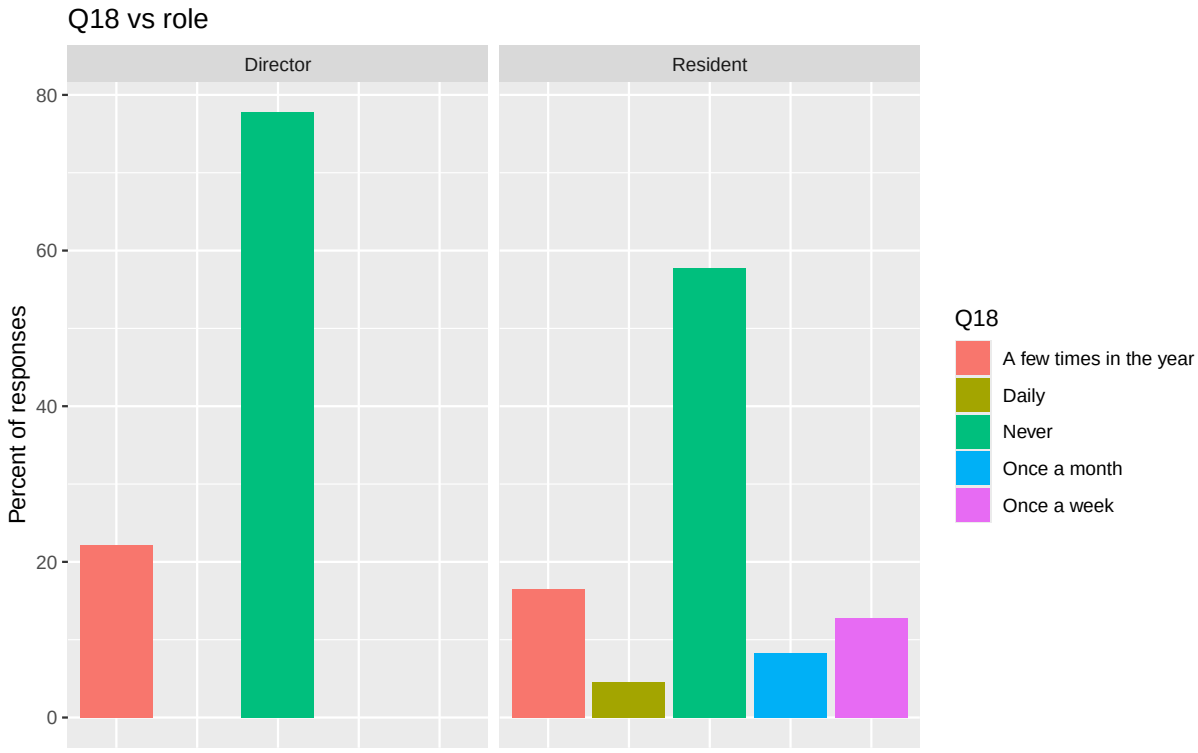

Q16a vs role
Director
Resident
Percent of responses
0
10
20
30
40
Q16a
Always
Never
Often
Rarely
Sometimes

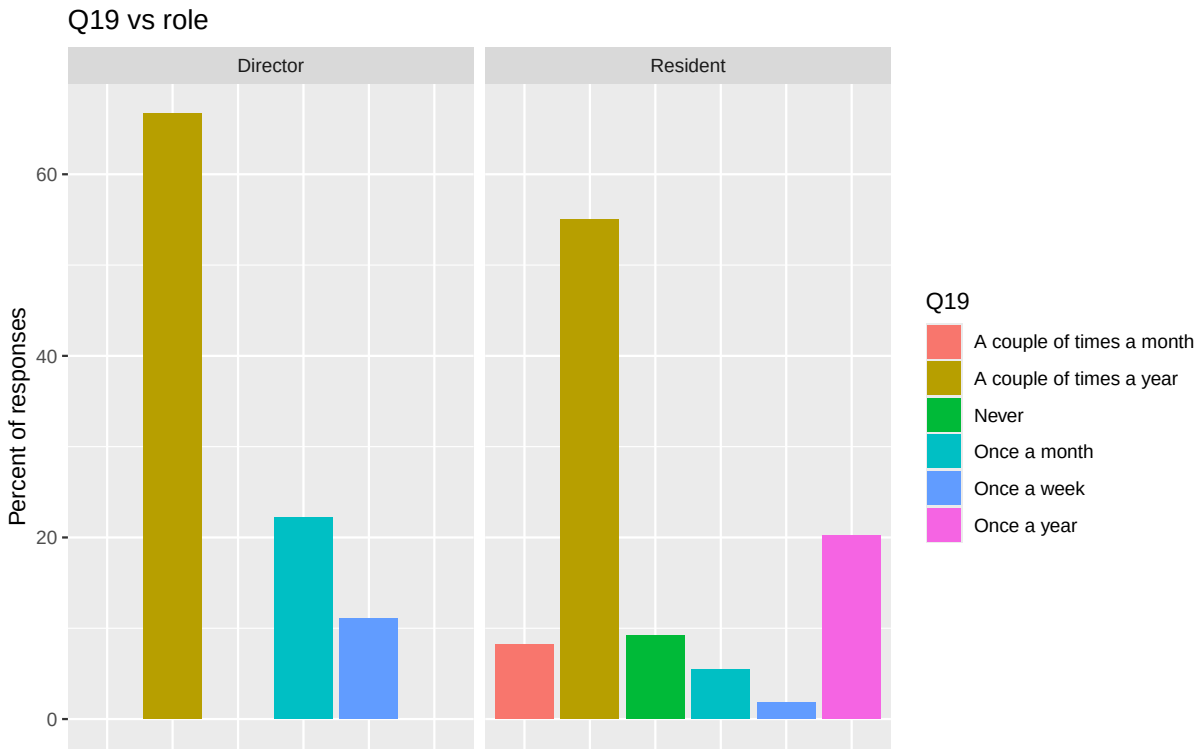

Q16b vs role
Director
Resident
Percent of responses
0
20
40
60
Q16b
Always
Never
Often
Rarely
Sometimes

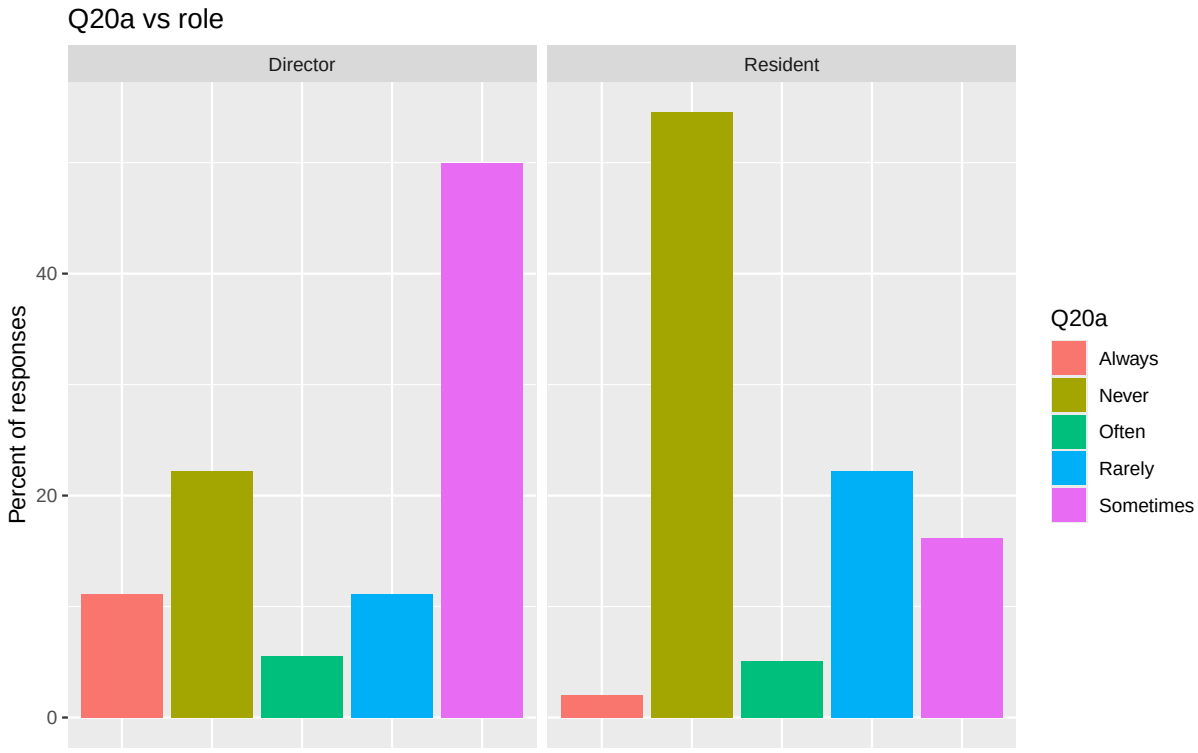

Q16c vs role
Director
Resident
Percent of responses
0
25
50
75
100
Q16c
Never
Rarely
Sometimes

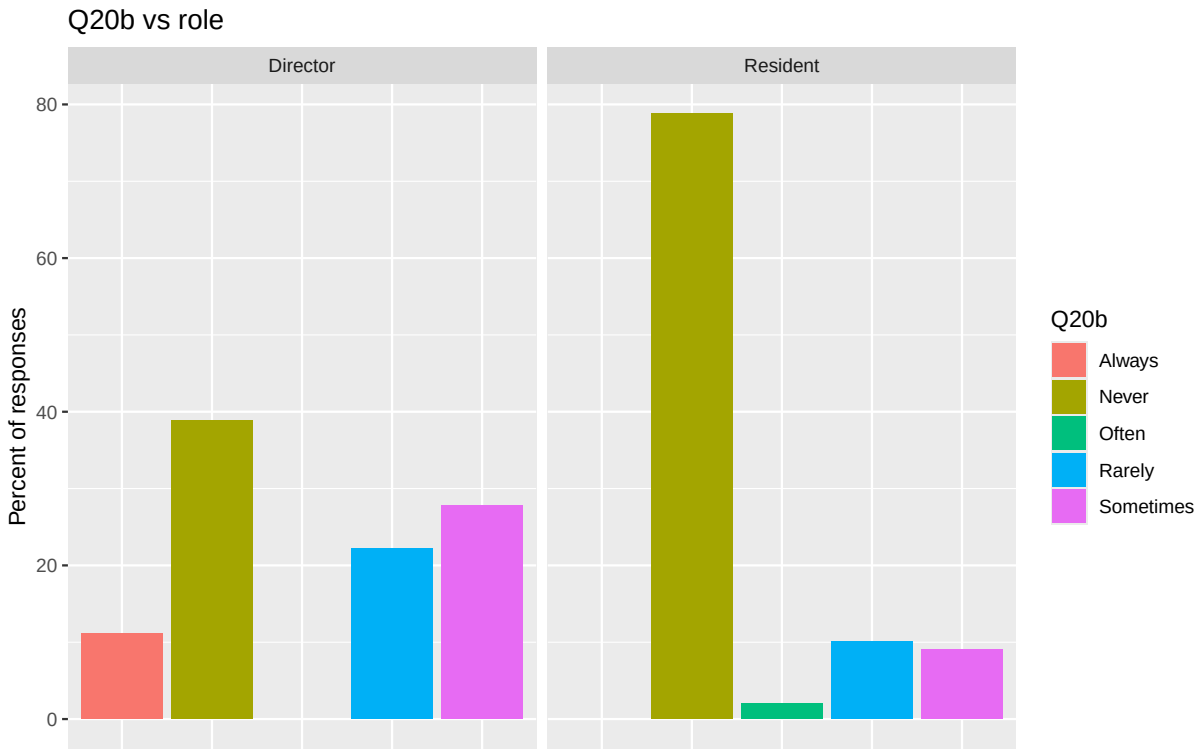

Q16d vs role
Director
Resident
100
75
50
25
0
Percent of responses
Q16d
Never
Often
Rarely
Sometimes

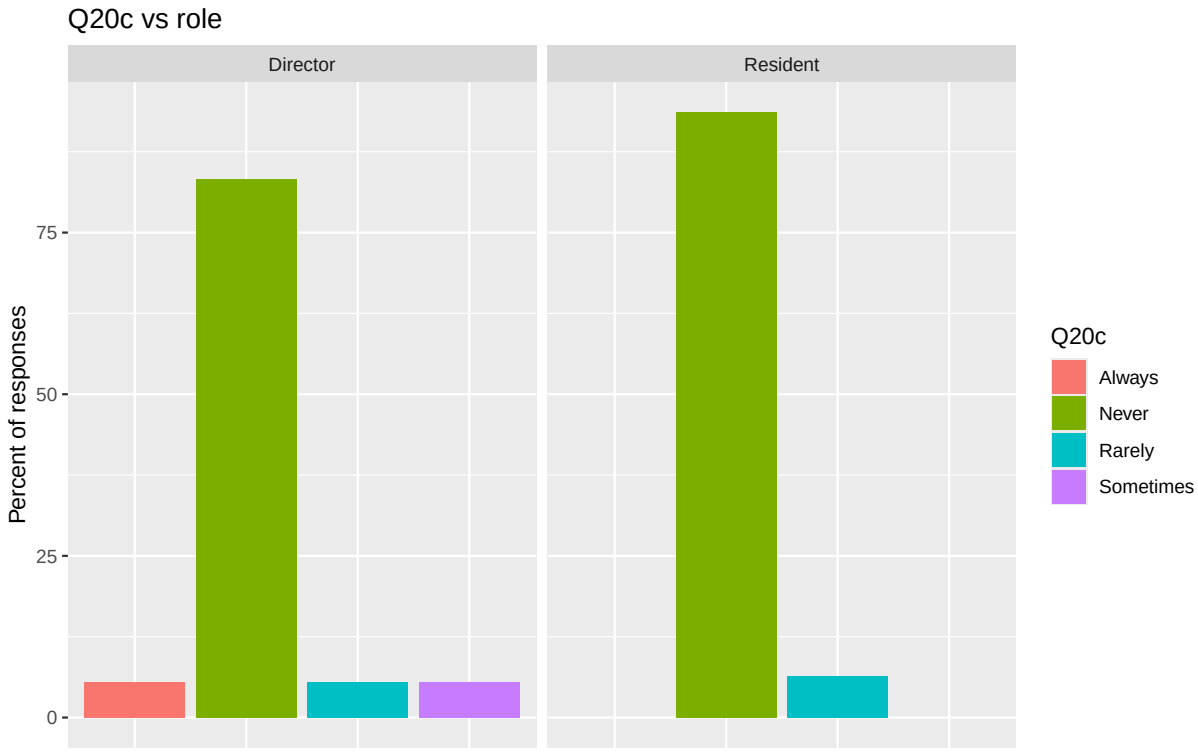

Q17 vs role
Director
Resident
60
40
20
0
Percent of responses
Q17
As bad as it can be
As good as it can be
Neutral
Pretty bad
Pretty good

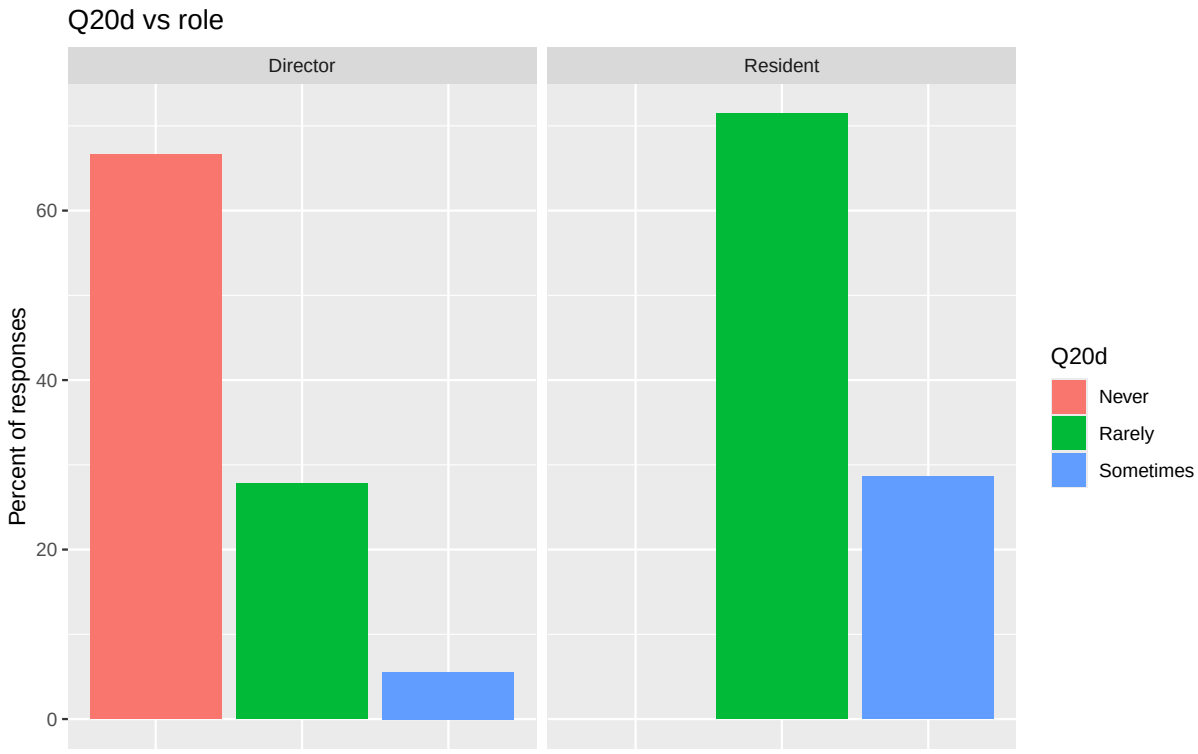

Q18 vs role
Director
Resident
Percent of responses
80
60
40
20
0
Q18
A few times in the year
Daily
Never
Once a month
Once a week

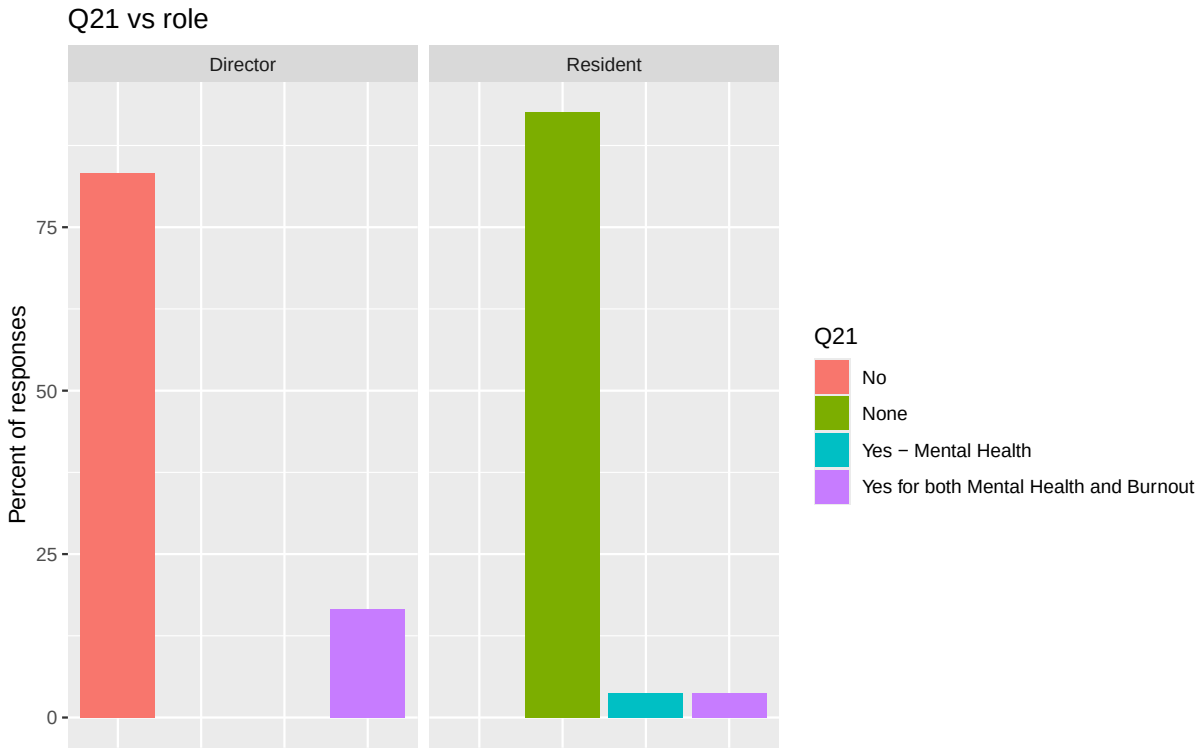

Q19 vs role
Director
Resident
Percent of responses
60
40
20
0
Q19
A couple of times a month
A couple of times a year
Never
Once a month
Once a week
Once a year

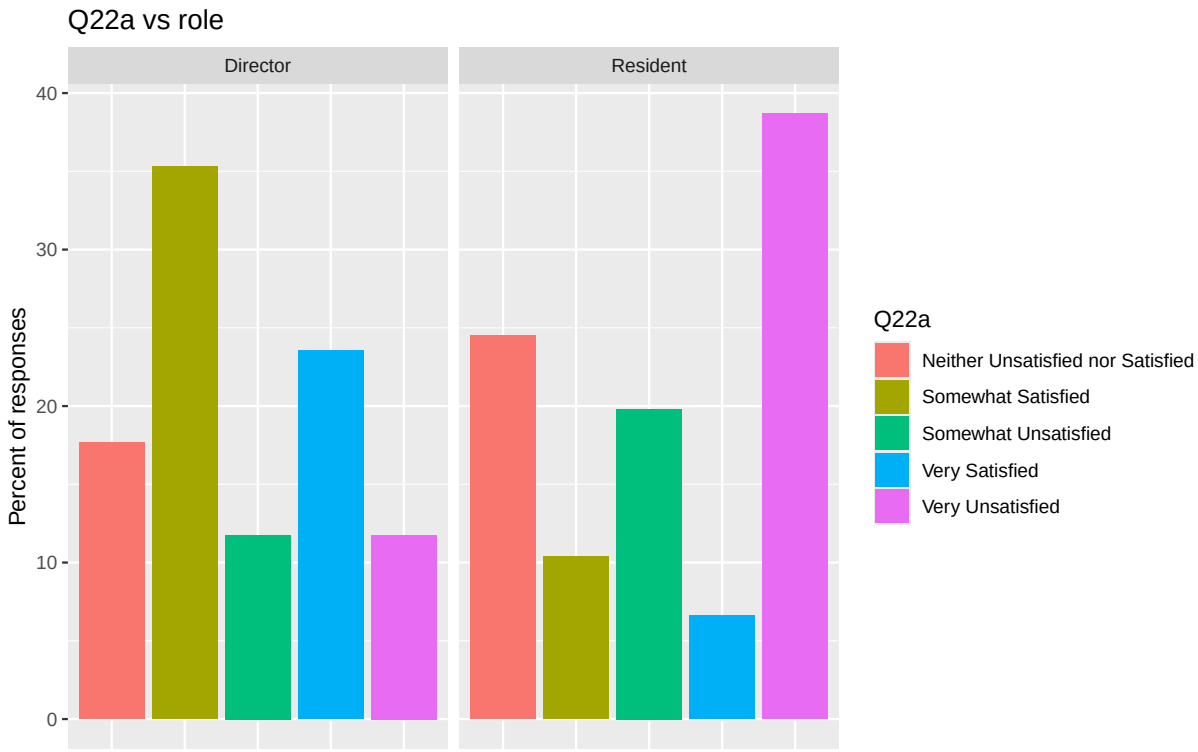

Q20a vs role
Director
Resident
Percent of responses
0
20
40
Q20a
Always
Never
Often
Rarely
Sometimes

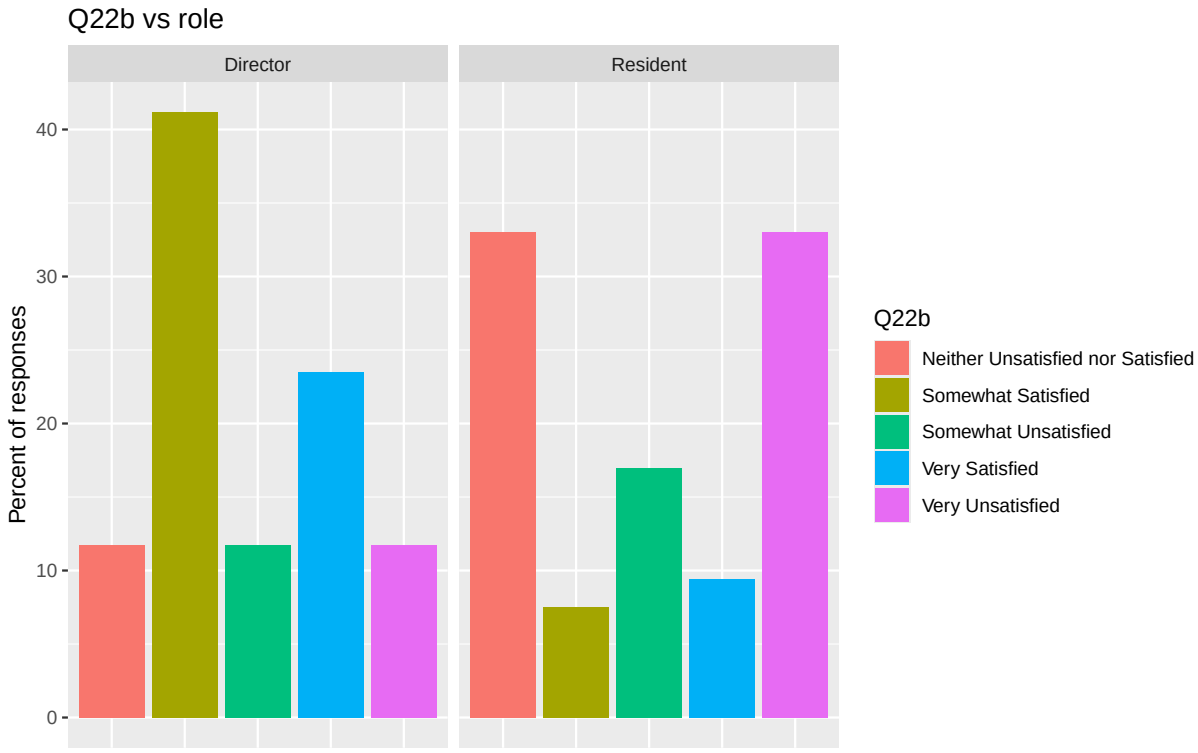

Q20b vs role
Director
Resident
Percent of responses
0
20
40
60
80
Q20b
Always
Never
Often
Rarely
Sometimes

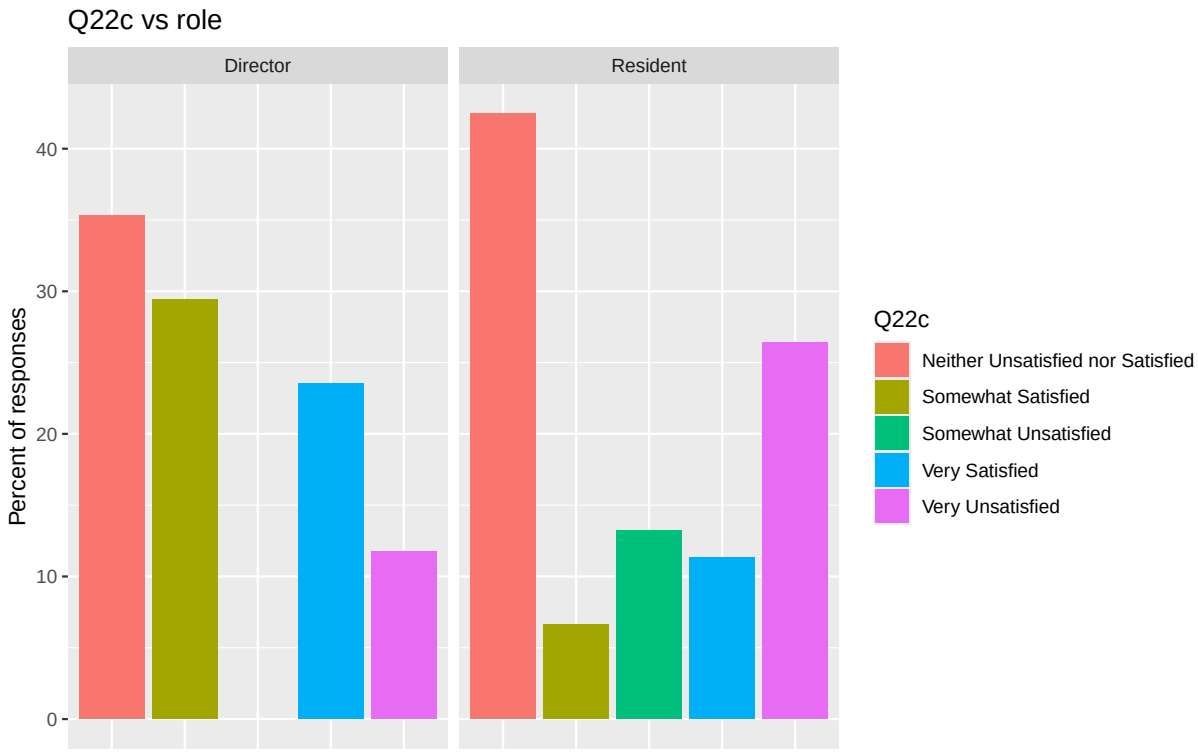

Q20c vs role
Director
Resident
Percent of responses
75
50
25
0
Q20c
Always
Never
Rarely
Sometimes

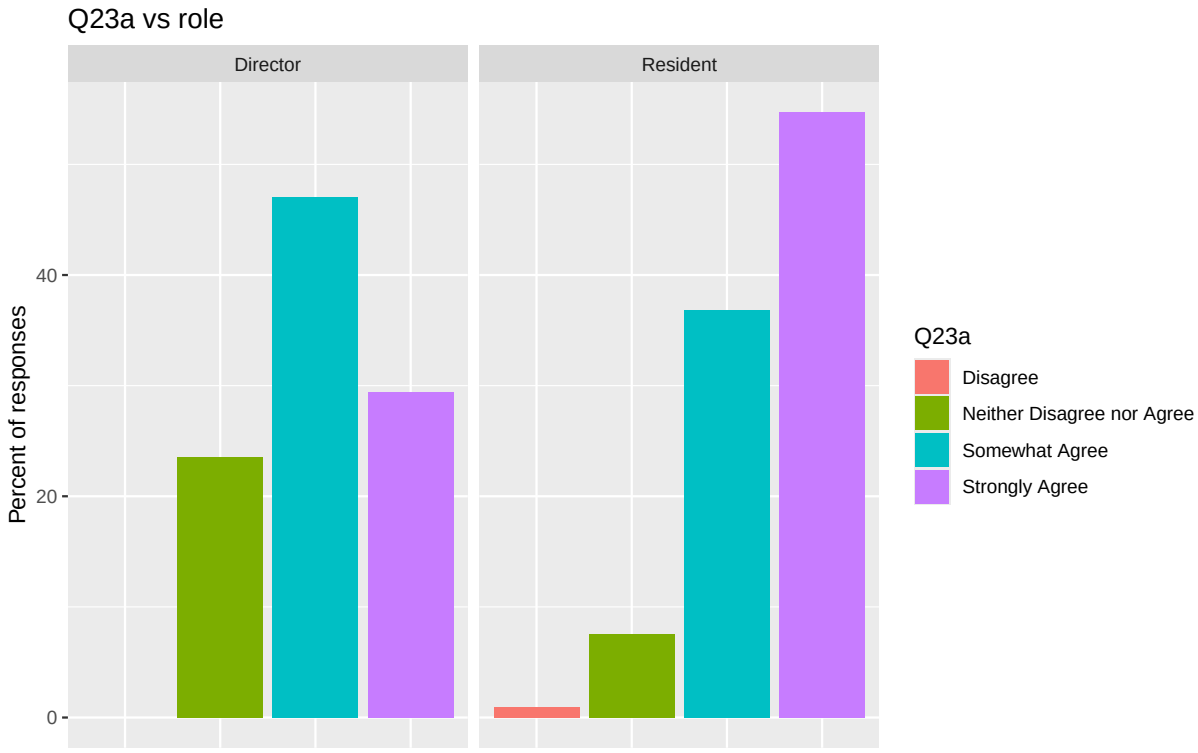

Q20d vs role
Director
Resident
Percent of responses
60
40
20
0
Q20d
Never
Rarely
Sometimes

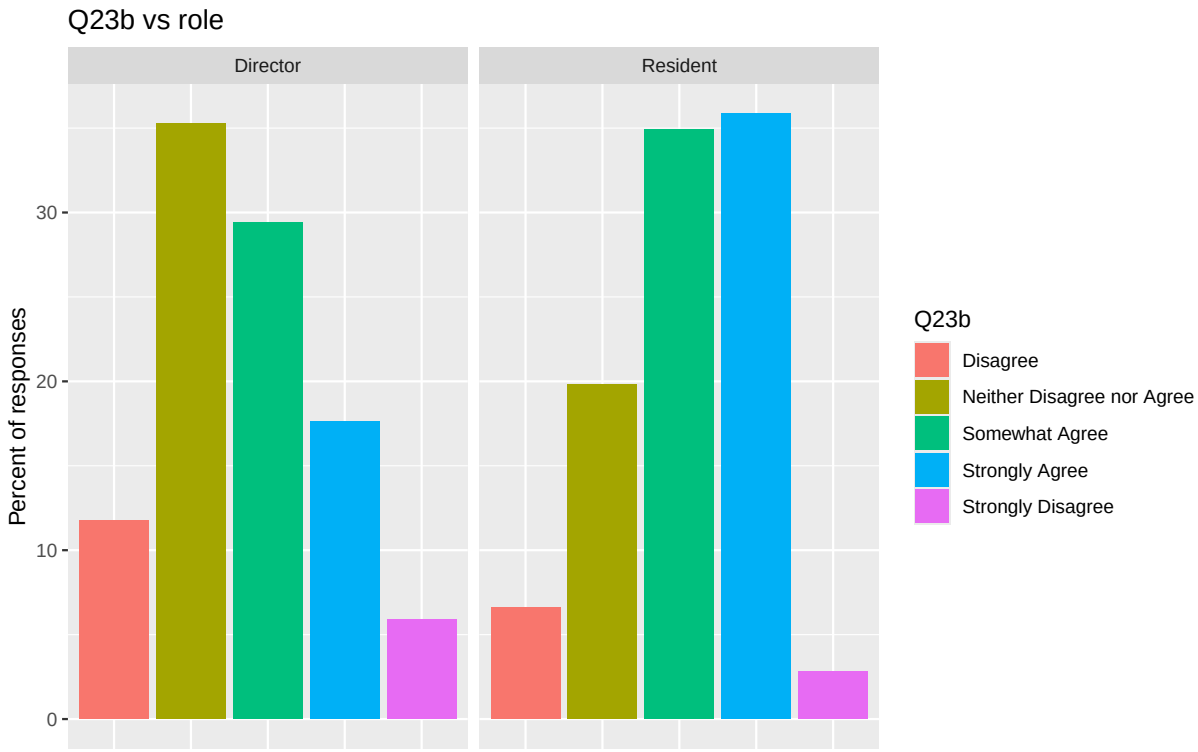

Q21 vs role
Director
Resident
Percent of responses
0
25
50
75
Q21
No
None
Yes − Mental Health
Yes for both Mental Health and Burnout

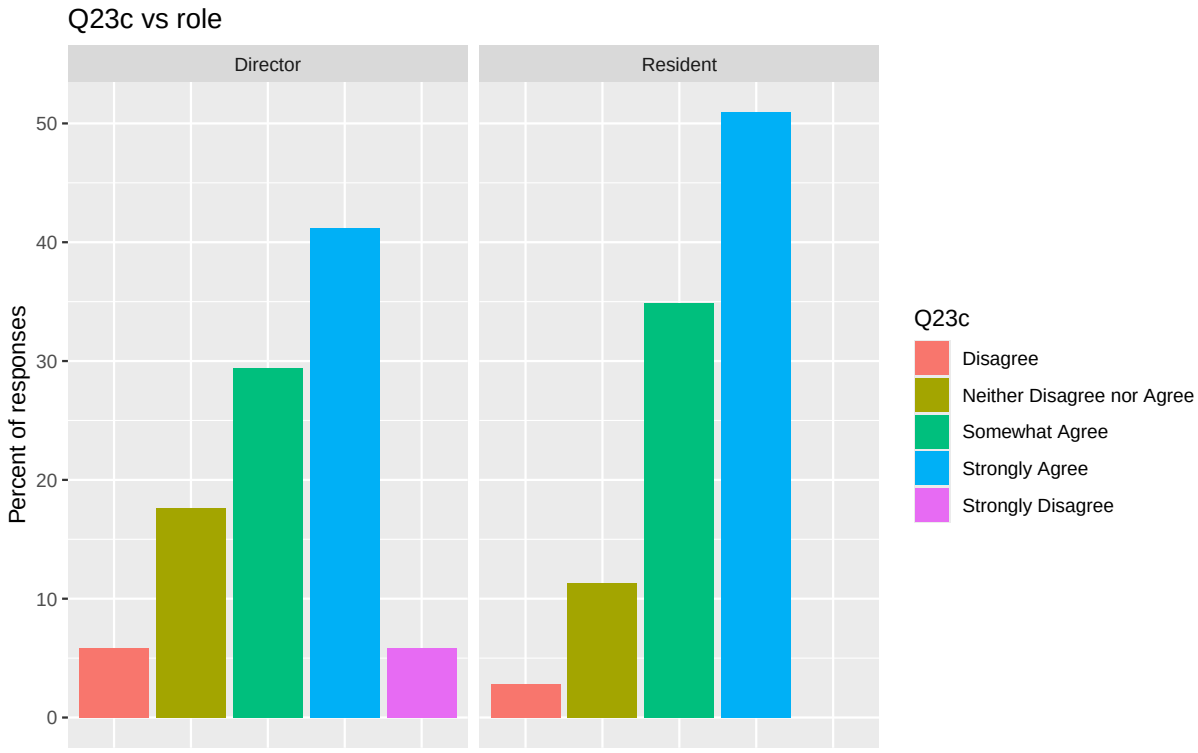

Q22a vs role
Director
Resident
Percent of responses
0
10
20
30
40
Q22a
Neither Unsatisfied nor Satisfied
Somewhat Satisfied
Somewhat Unsatisfied
Very Satisfied
Very Unsatisfied

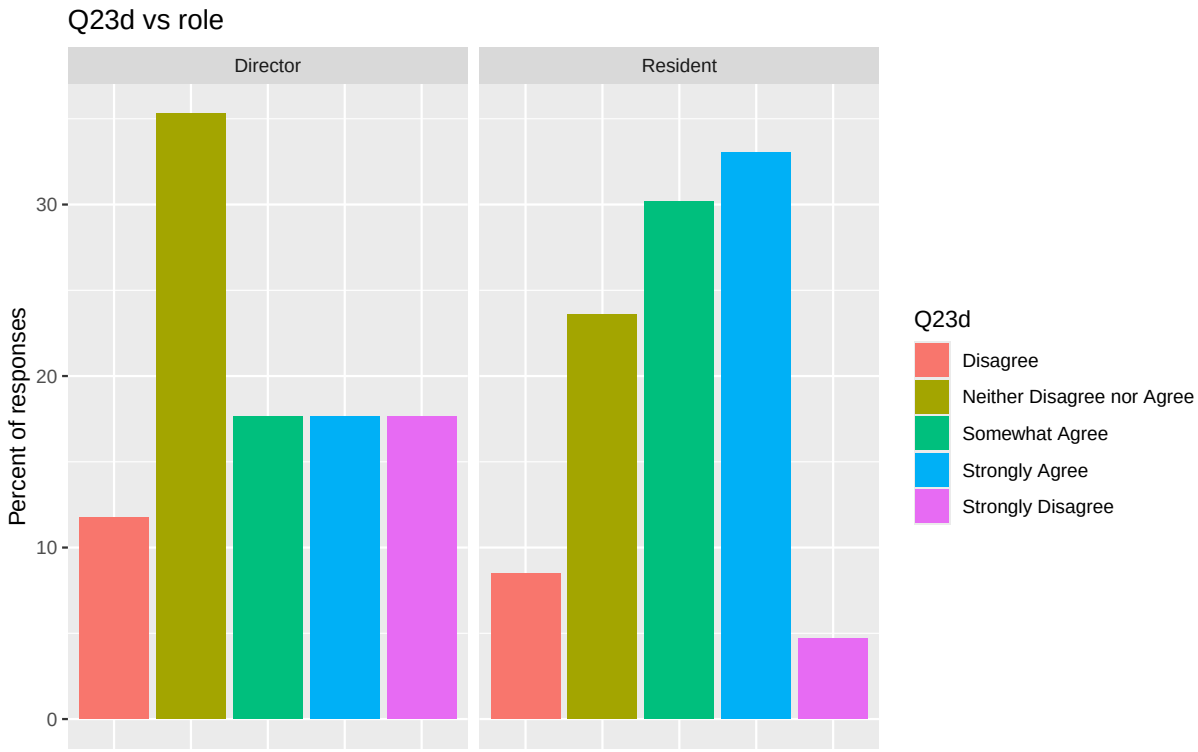

Q22b vs role
Director
Resident
Percent of responses
0
10
20
30
40
Q22b
Neither Unsatisfied nor Satisfied
Somewhat Satisfied
Somewhat Unsatisfied
Very Satisfied
Very Unsatisfied

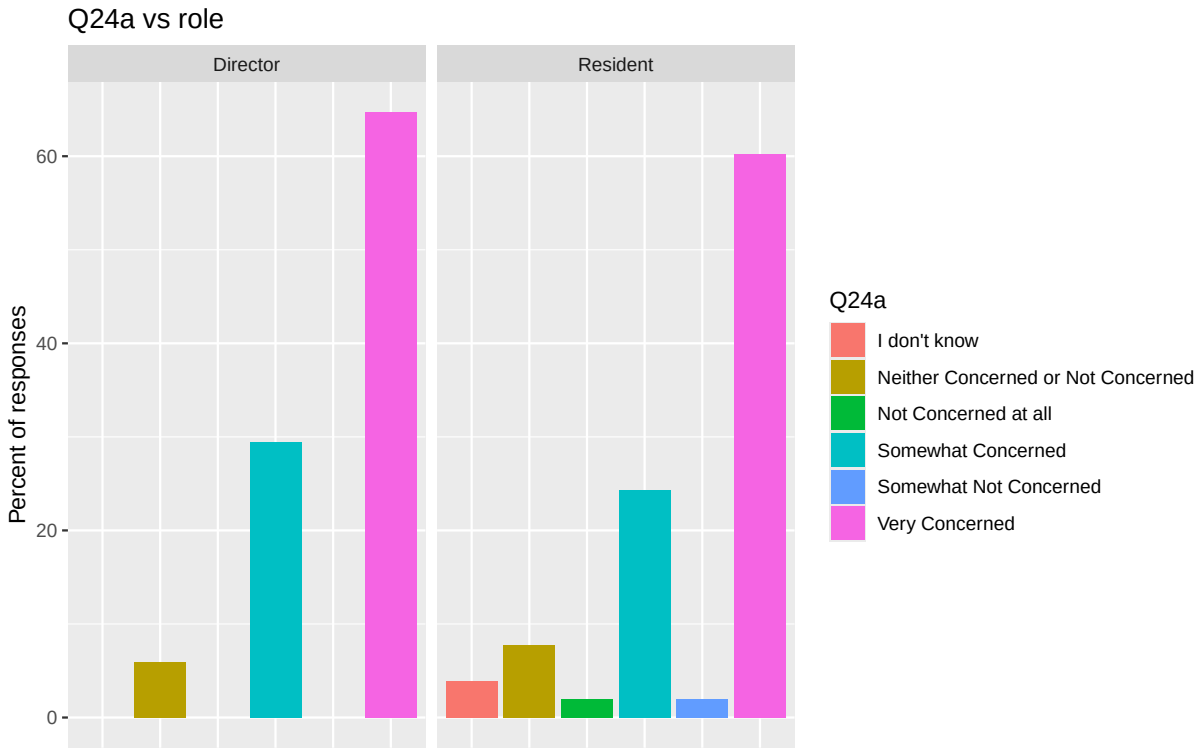

Q22c vs role
Director
Resident
Percent of responses
0
10
20
30
40
Q22c
Neither Unsatisfied nor Satisfied
Somewhat Satisfied
Somewhat Unsatisfied
Very Satisfied
Very Unsatisfied

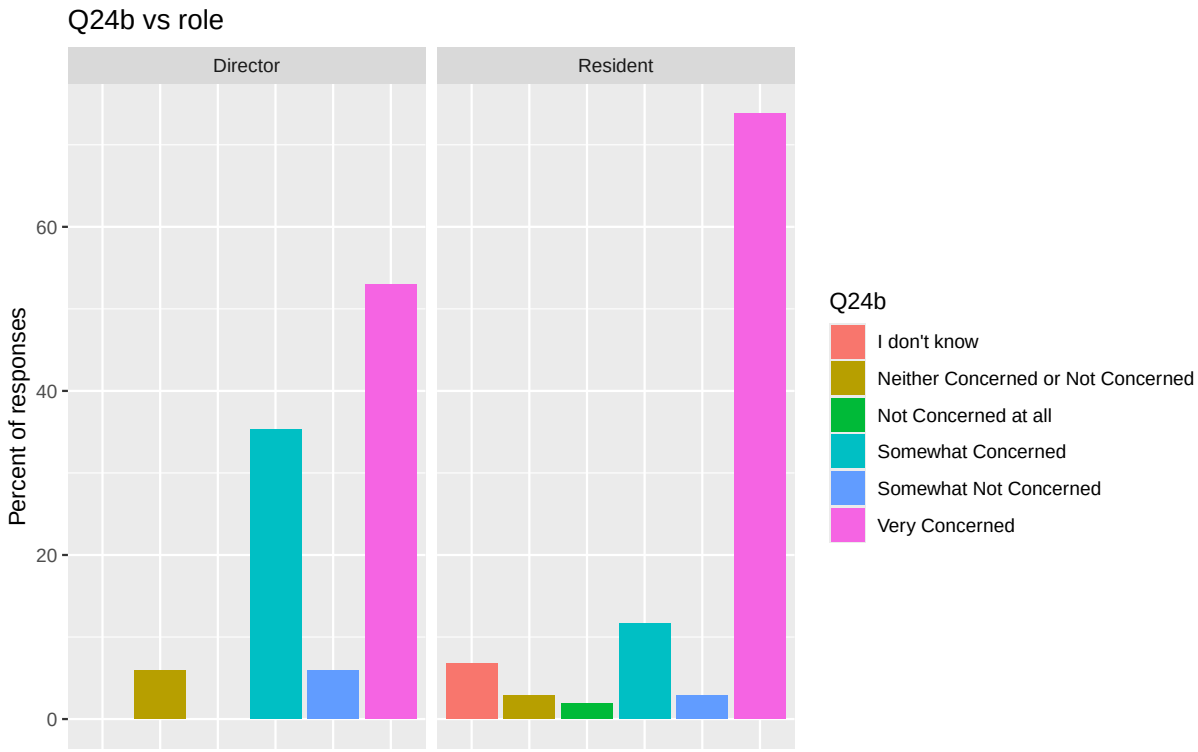

Q23a vs role
Director
Resident
Percent of responses
40
20
0
Q23a
Disagree
Neither Disagree nor Agree
Somewhat Agree
Strongly Agree

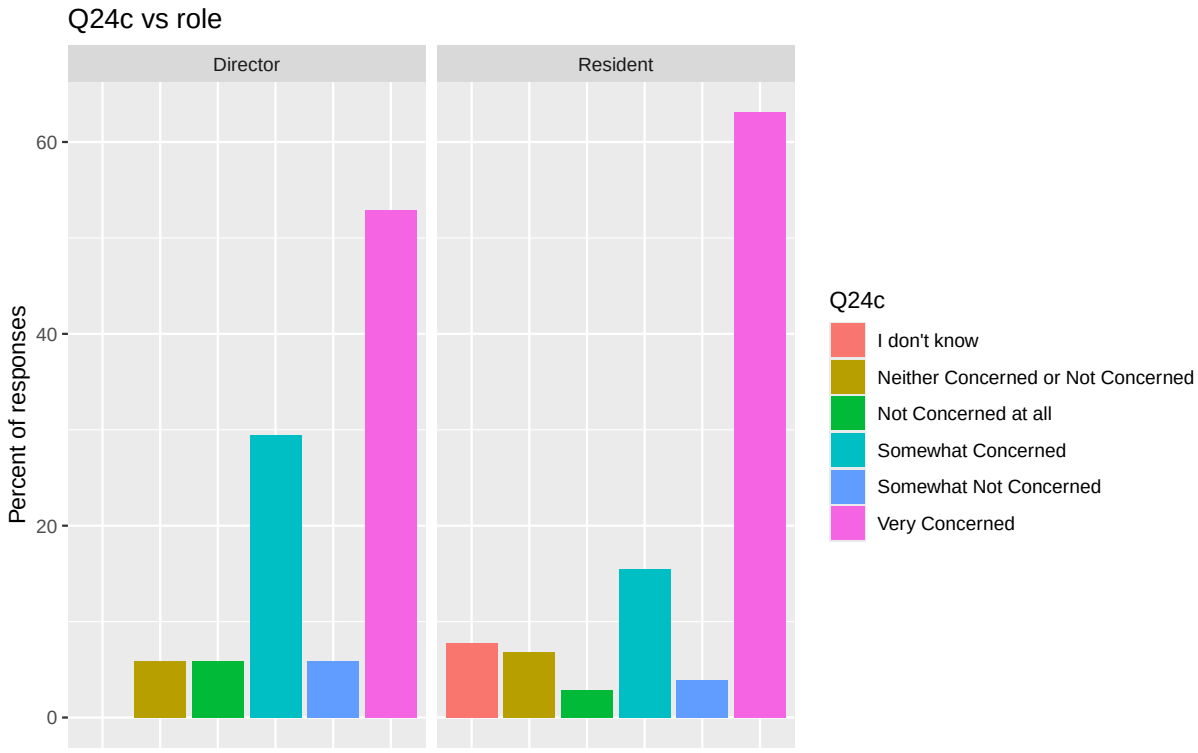

Q23b vs role
Director
Resident
Percent of responses
30
20
10
0
Q23b
Disagree
Neither Disagree nor Agree
Somewhat Agree
Strongly Agree
Strongly Disagree

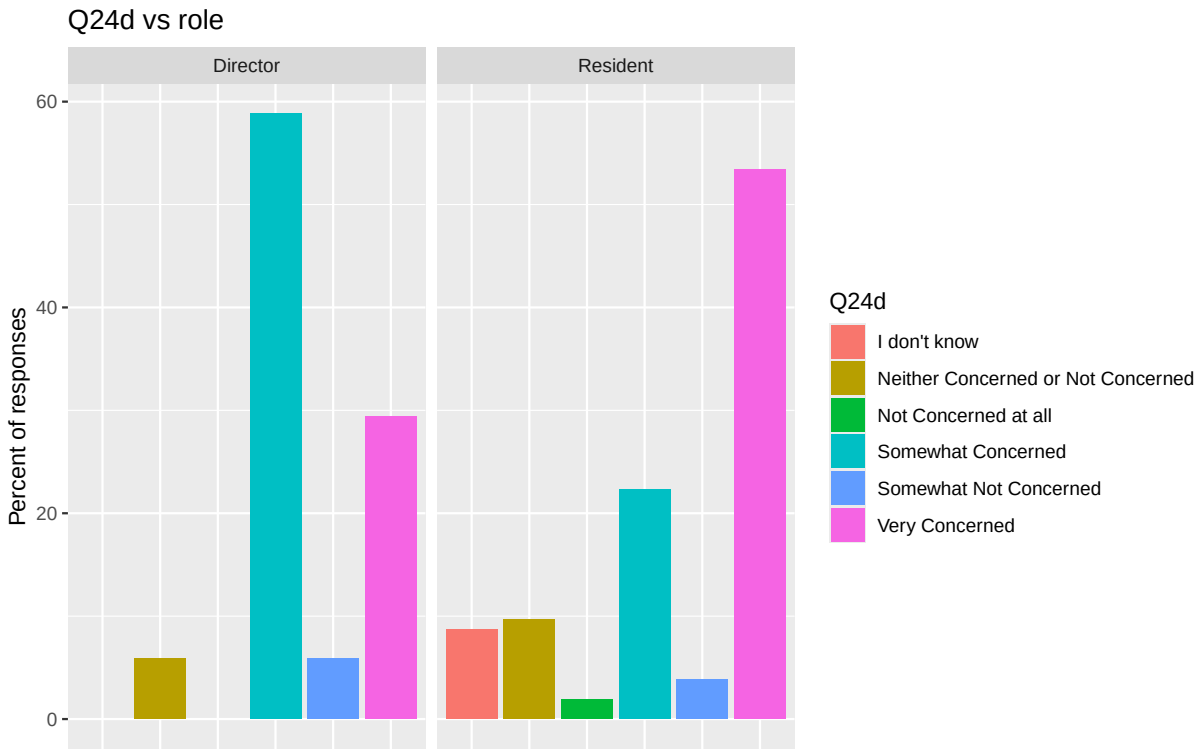

Q23c vs role
Director
Resident
Percent of responses
0
10
20
30
40
50
Q23c
Disagree
Neither Disagree nor Agree
Somewhat Agree
Strongly Agree
Strongly Disagree

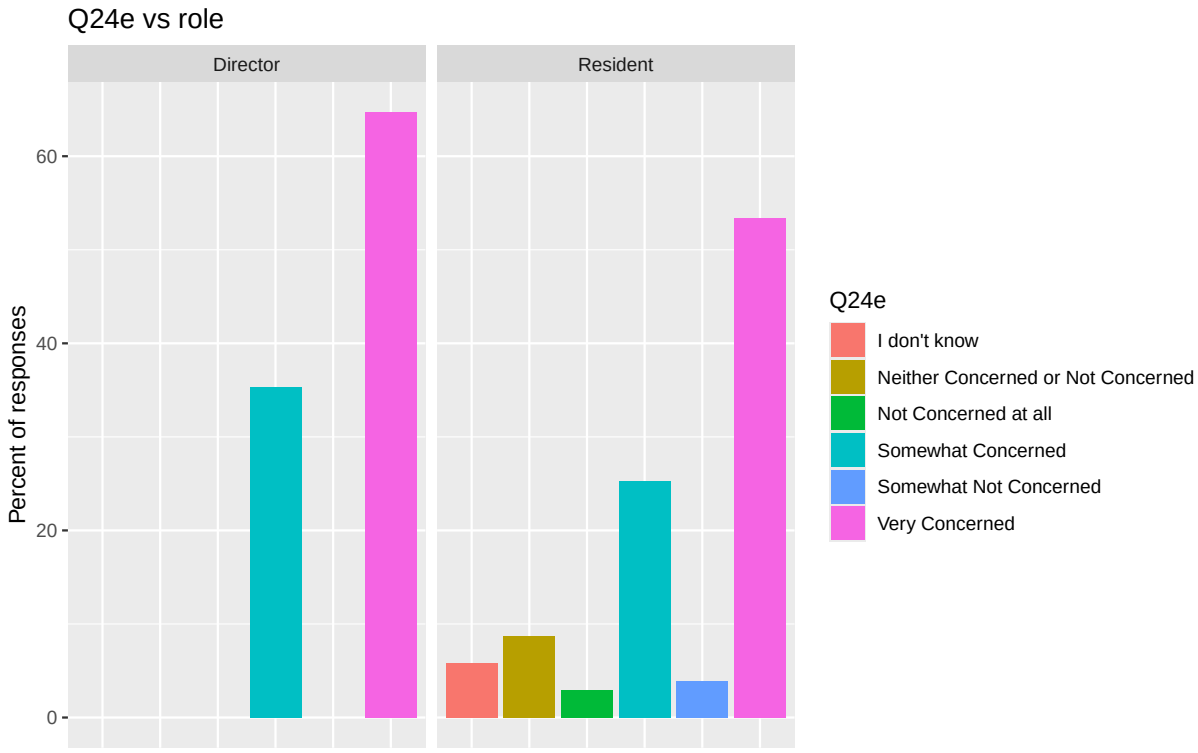

Q23d vs role
Director
Resident
Percent of responses
0
10
20
30
Q23d
Disagree
Neither Disagree nor Agree
Somewhat Agree
Strongly Agree
Strongly Disagree

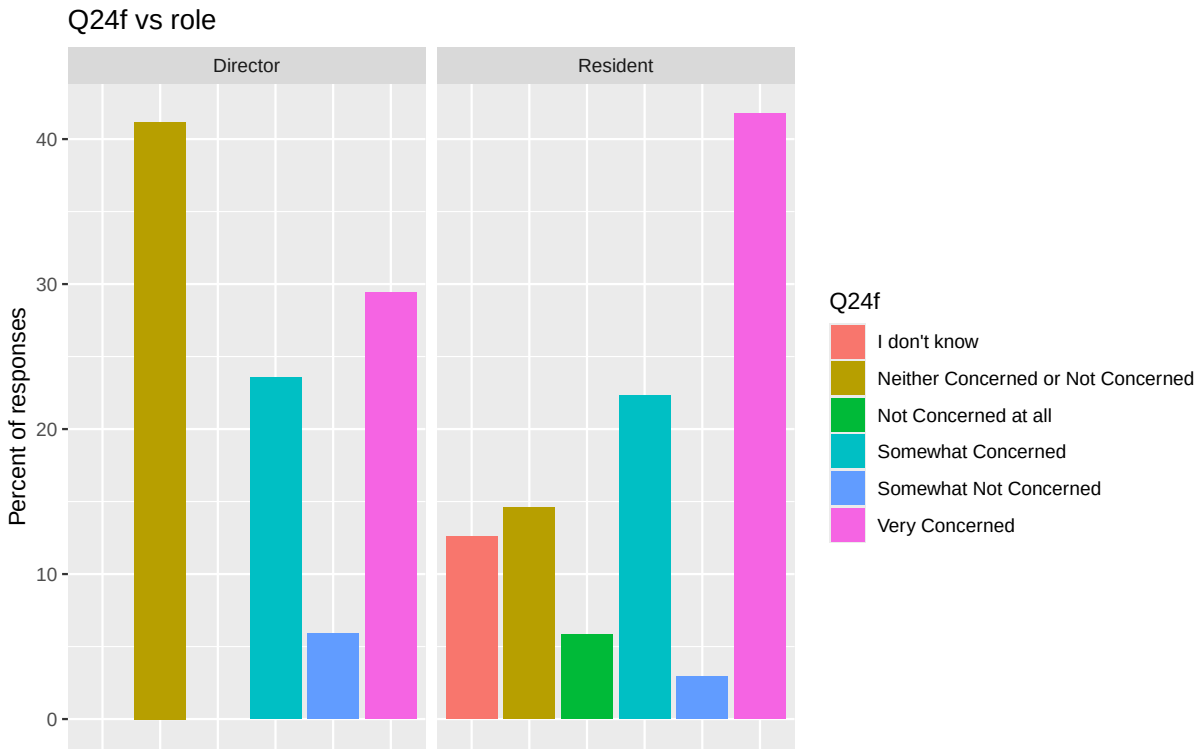

Q24a vs role
Director
Resident
Percent of responses
0
20
40
60
Q24a
I don't know
Neither Concerned or Not Concerned
Not Concerned at all
Somewhat Concerned
Somewhat Not Concerned
Very Concerned

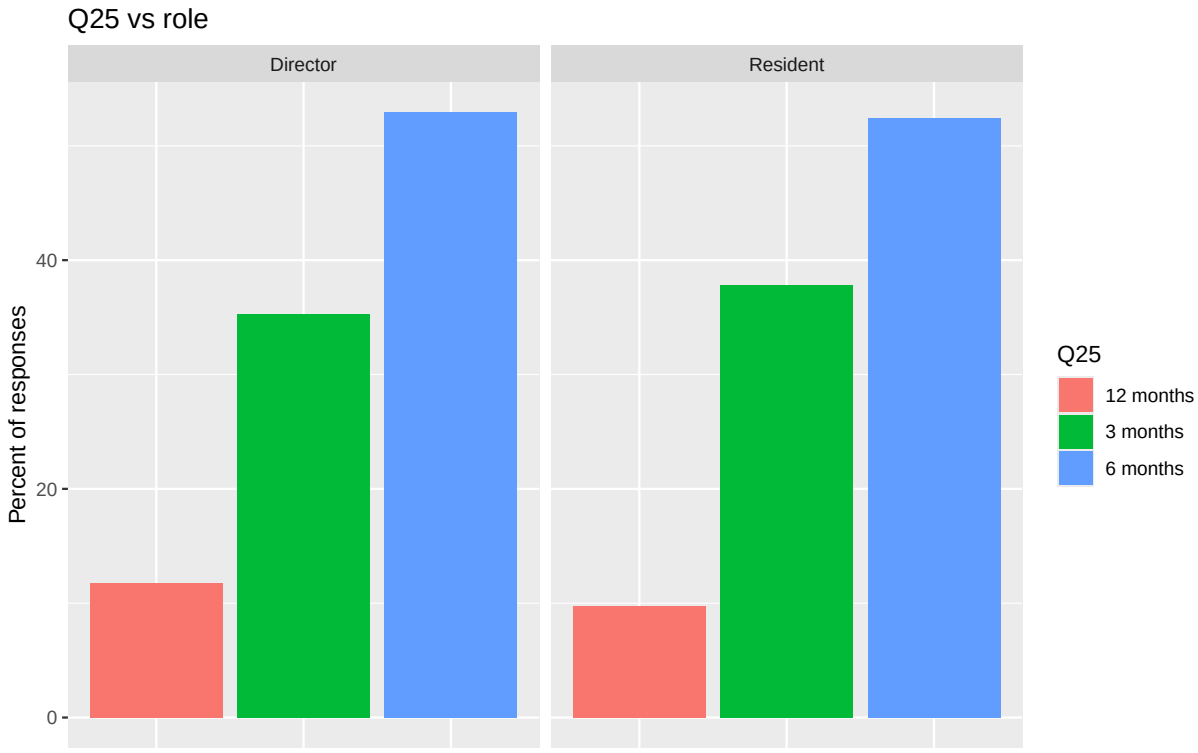

Q24b vs role
Director
Resident
Percent of responses
0
20
40
60
Q24b
I don't know
Neither Concerned or Not Concerned
Not Concerned at all
Somewhat Concerned
Somewhat Not Concerned
Very Concerned

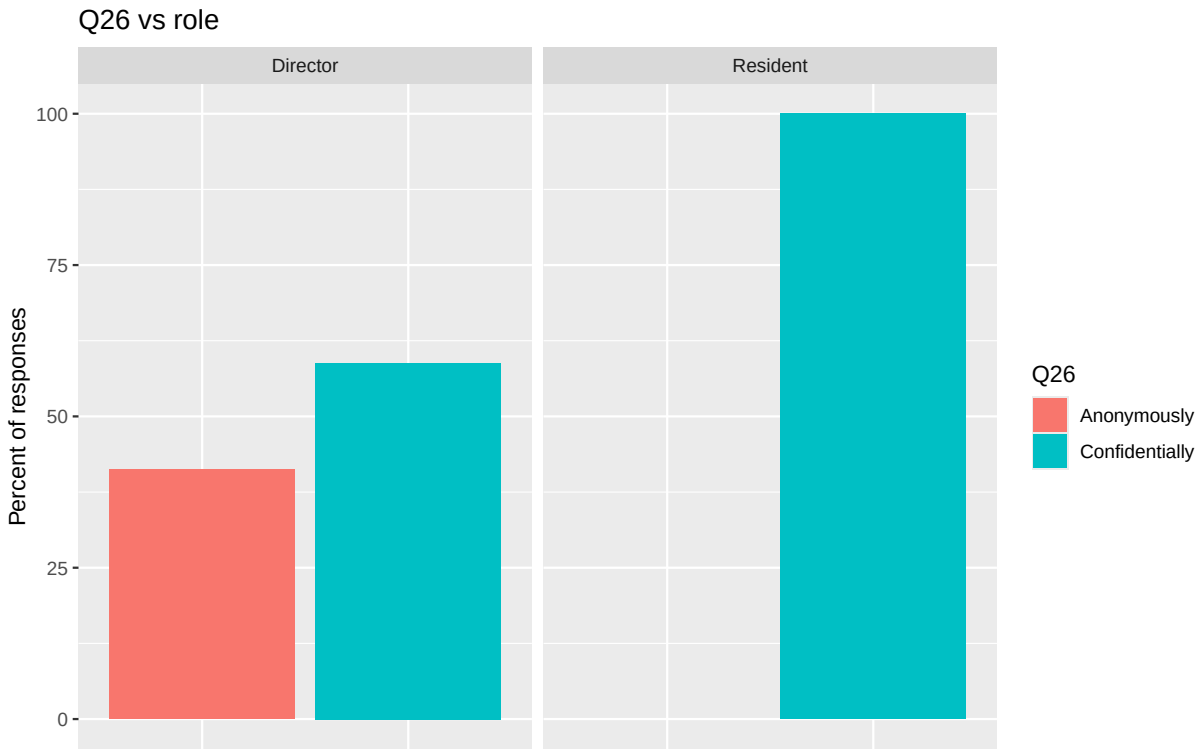

Q24c vs role
Director
Resident
Percent of responses
0
20
40
60
Q24c
I don't know
Neither Concerned or Not Concerned
Not Concerned at all
Somewhat Concerned
Somewhat Not Concerned
Very Concerned


Q24d vs role
Director
Resident
Percent of responses
0
20
40
60
Q24d
I don't know
Neither Concerned or Not Concerned
Not Concerned at all
Somewhat Concerned
Somewhat Not Concerned
Very Concerned


Q24e vs role
Director
Resident
Percent of responses
0
20
40
60
Q24e
I don't know
Neither Concerned or Not Concerned
Not Concerned at all
Somewhat Concerned
Somewhat Not Concerned
Very Concerned


Q24f vs role
Director
Resident
Percent of responses
0
10
20
30
40
Q24f
I don't know
Neither Concerned or Not Concerned
Not Concerned at all
Somewhat Concerned
Somewhat Not Concerned
Very Concerned


Q25 vs role
Director
Resident
Percent of responses
40
20
0
Q25
12 months
3 months
6 months


Q26 vs role
Director
Resident
Percent of responses
100
75
50
25
0
Q26
Anonymously
Confidentially



Appendix B: Code

```
library(tidyverse)
library(zoo)
library(stringr)
library(gt)

if (Sys.info()[["user"]] == "gregorymatthews"){
  headers_raw <- read_csv("/Users/gregorymatthews/Dentist Survey Levine/Falchook/Resident Exp
} else {
  headers_raw <- read_csv("/Users/jdumanski24/Desktop/CDSC/Levine/Resident Expanded(in).csv"
}
questions <- as.character(headers_raw[1,])

answer_choices <- as.character(headers_raw[2,])

questions_filled <- na.locf(ifelse(questions == "" | is.na(questions), NA, questions))

clean_headers <- paste(questions_filled, answer_choices, sep = "_")
```

```r
if (Sys.info()[["user"]] == "gregorymatthews"){
  f_res_raw <- read_csv("/Users/gregorymatthews/Dentist Survey Levine/Falchook/Resident Expa
} else {
  f_res_raw <- read_csv("/Users/jdumanski24/Desktop/CDSC/Levine/Resident Expanded(in).csv", 
}

f_res <- f_res_raw %>%
  select(c(10:ncol(f_res_raw)))

f_res <- f_res %>%
  mutate(
    across(
      everything(),
      ~ na_if(as.character(.), "")
    )
  )

f_res_q <- f_res %>%
  mutate(Q1 = do.call(coalesce, pick(1:3))) %>%
  mutate(Q2 = do.call(coalesce, pick(4:6))) %>%
  mutate(Q3 = do.call(coalesce, pick(7:9))) %>%
  mutate(Q4 = do.call(coalesce, pick(10:10))) %>%
  mutate(Q5 = do.call(coalesce, pick(11:15))) %>%
  mutate(Q6 = do.call(coalesce, pick(16:17))) %>%
  mutate(Q7 = do.call(coalesce, pick(18:18))) %>%
  mutate(Q8 = do.call(coalesce, pick(19:23))) %>%
  #mutate(Q9 = do.call(coalesce, pick(24:27))) %>%
  mutate(Q10 = do.call(coalesce, pick(28:32))) %>%
  mutate(Q11 = do.call(coalesce, pick(33:38))) %>%
  mutate(Q12 = do.call(coalesce, pick(39:42))) %>%
  mutate(Q13a = do.call(coalesce, pick(43:49))) %>%
  mutate(Q13b = do.call(coalesce, pick(50:56))) %>%
  mutate(Q13c = do.call(coalesce, pick(57:63))) %>%
  mutate(Q13d = do.call(coalesce, pick(64:70))) %>%
  mutate(Q13e = do.call(coalesce, pick(71:77))) %>%
  mutate(Q13f = do.call(coalesce, pick(78:84))) %>%
  mutate(Q13g = do.call(coalesce, pick(85:91))) %>%
  mutate(Q13h = do.call(coalesce, pick(92:98))) %>%
  mutate(Q13i = do.call(coalesce, pick(99:105))) %>%
  mutate(Q14a = do.call(coalesce, pick(106:110))) %>%
  mutate(Q14b = do.call(coalesce, pick(111:115))) %>%
  mutate(Q14c = do.call(coalesce, pick(116:120))) %>%
```

```
  mutate(Q14d = do.call(coalesce, pick(121:125))) %>%
  mutate(Q15a = do.call(coalesce, pick(126:130))) %>%
  mutate(Q15b = do.call(coalesce, pick(131:135))) %>%
  mutate(Q15c = do.call(coalesce, pick(136:140))) %>%
  mutate(Q16a = do.call(coalesce, pick(141:145))) %>%
  mutate(Q16b = do.call(coalesce, pick(146:150))) %>%
  mutate(Q16c = do.call(coalesce, pick(151:155))) %>%
  mutate(Q16d = do.call(coalesce, pick(156:160))) %>%
  mutate(Q17 = do.call(coalesce, pick(161:165))) %>%
  mutate(Q18 = do.call(coalesce, pick(166:170))) %>%
  mutate(Q19 = do.call(coalesce, pick(171:177))) %>%
  mutate(Q20a = do.call(coalesce, pick(178:182))) %>%
  mutate(Q20b = do.call(coalesce, pick(183:187))) %>%
  mutate(Q20c = do.call(coalesce, pick(188:192))) %>%
  mutate(Q20d = do.call(coalesce, pick(193:196))) %>%
  mutate(Q21 = do.call(coalesce, pick(197:200))) %>%
  mutate(Q22a = do.call(coalesce, pick(201:205))) %>%
  mutate(Q22b = do.call(coalesce, pick(206:210))) %>%
  mutate(Q22c = do.call(coalesce, pick(211:215))) %>%
  mutate(Q23a = do.call(coalesce, pick(216:220))) %>%
  mutate(Q23b = do.call(coalesce, pick(221:225))) %>%
  mutate(Q23c = do.call(coalesce, pick(226:230))) %>%
  mutate(Q23d = do.call(coalesce, pick(231:235))) %>%
  mutate(Q24a = do.call(coalesce, pick(236:241))) %>%
  mutate(Q24b = do.call(coalesce, pick(242:247))) %>%
  mutate(Q24c = do.call(coalesce, pick(248:253))) %>%
  mutate(Q24d = do.call(coalesce, pick(254:259))) %>%
  mutate(Q24e = do.call(coalesce, pick(260:265))) %>%
  mutate(Q24f = do.call(coalesce, pick(266:271))) %>%
  mutate(Q25 = do.call(coalesce, pick(272:275))) %>%
  mutate(Q26 = do.call(coalesce, pick(276:278))) %>%
  #mutate(Q27 = do.call(coalesce, pick(279:289))) %>%
  mutate(Q28 = do.call(coalesce, pick(290:290)))

f_res_coalesced_partial <- f_res_q %>%
  select(291:298, 24:27, 299:344, 279:289, 345)

f_res_coalesced <- f_res_coalesced_partial %>%
  mutate(Q9 = pmap_chr(select(., 9:12), ~ paste(na.omit(c(...)), collapse = "; "))) %>%
  mutate(Q27 = pmap_chr(select(., 59:69), ~ paste(na.omit(c(...)), collapse = "; "))) %>%
  select(1:8, 71, 13:58, 72, 70)
```

```r
f_res_coalesced$role <- "Resident"
f_res_coalesced <- f_res_coalesced %>%
  mutate(Q26 = case_when(
  Q26 == "Confidentially (one-on-one meetings)" ~ "Confidentially",
  Q26 == "Anonymously (online questionnaire)" ~ "Anonymously"
  ))

if (Sys.info()[["user"]] == "gregorymatthews"){
  dir_headers_raw <- read_csv("/Users/gregorymatthews/Dentist Survey Levine/Falchook/PD expa
} else {
  dir_headers_raw <- read_csv("/Users/jdumanski24/Desktop/CDSC/Levine/PD expanded(in).csv", 
}

dir_questions <- as.character(dir_headers_raw[1,])

dir_answer_choices <- as.character(dir_headers_raw[2,])

dir_questions_filled <- na.locf(ifelse(dir_questions == "" | is.na(dir_questions), NA, dir_q

dir_clean_headers <- paste(dir_questions_filled, dir_answer_choices, sep = "_")

if (Sys.info()[["user"]] == "gregorymatthews"){
  f_dir_raw <- read_csv("/Users/gregorymatthews/Dentist Survey Levine/Falchook/PD expanded.cs
    f_dir_raw <- read_csv("/Users/jdumanski24/Desktop/CDSC/Levine/PD expanded(in).csv", skip
  }

f_dir <- f_dir_raw %>%
  select(c(10:ncol(f_dir_raw)))
#view(f_dir)

f_dir <- f_dir %>%
  mutate(
    across(
      everything(),
      ~ na_if(as.character(.), "")
    )
  )

f_dir_q <- f_dir %>%
  mutate(Q1 = do.call(coalesce, pick(1:2))) %>%
  mutate(Q2 = do.call(coalesce, pick(3:5))) %>%
  mutate(Q3 = do.call(coalesce, pick(6:6))) %>%
```

```
  mutate(Q4 = do.call(coalesce, pick(7:9))) %>%
  mutate(Q5 = do.call(coalesce, pick(10:10))) %>%
  mutate(Q6 = do.call(coalesce, pick(11:11))) %>%
  mutate(Q7 = do.call(coalesce, pick(12:16))) %>%
  #mutate(Q8 = do.call(coalesce, pick(17:20))) %>%
  mutate(Q9 = do.call(coalesce, pick(21:25))) %>%
  mutate(Q10 = do.call(coalesce, pick(26:32))) %>%
  mutate(Q11 = do.call(coalesce, pick(33:37))) %>%
  mutate(Q12a = do.call(coalesce, pick(38:45))) %>%
  mutate(Q12b = do.call(coalesce, pick(46:53))) %>%
  mutate(Q12c = do.call(coalesce, pick(54:61))) %>%
  mutate(Q12d = do.call(coalesce, pick(62:69))) %>%
  mutate(Q12e = do.call(coalesce, pick(70:77))) %>%
  mutate(Q12f = do.call(coalesce, pick(78:85))) %>%
  mutate(Q12g = do.call(coalesce, pick(86:93))) %>%
  mutate(Q12h = do.call(coalesce, pick(94:101))) %>%
  mutate(Q12i = do.call(coalesce, pick(102:109))) %>%
  mutate(Q13a = do.call(coalesce, pick(110:115))) %>%
  mutate(Q13b = do.call(coalesce, pick(116:121))) %>%
  mutate(Q13c = do.call(coalesce, pick(122:127))) %>%
  mutate(Q13d = do.call(coalesce, pick(128:133))) %>%
  mutate(Q14a = do.call(coalesce, pick(134:138))) %>%
  mutate(Q14b = do.call(coalesce, pick(139:143))) %>%
  mutate(Q14c = do.call(coalesce, pick(144:148))) %>%
  mutate(Q15a = do.call(coalesce, pick(149:153))) %>%
  mutate(Q15b = do.call(coalesce, pick(154:158))) %>%
  mutate(Q15c = do.call(coalesce, pick(159:163))) %>%
  mutate(Q15d = do.call(coalesce, pick(164:168))) %>%
  mutate(Q16 = do.call(coalesce, pick(169:173))) %>%
  mutate(Q17 = do.call(coalesce, pick(174:178))) %>%
  mutate(Q18 = do.call(coalesce, pick(179:184))) %>%
  mutate(Q19a = do.call(coalesce, pick(185:189))) %>%
  mutate(Q19b = do.call(coalesce, pick(190:194))) %>%
  mutate(Q19c = do.call(coalesce, pick(195:199))) %>%
  mutate(Q19d = do.call(coalesce, pick(200:204))) %>%
  mutate(Q20 = do.call(coalesce, pick(205:208))) %>%
  mutate(Q21 = do.call(coalesce, pick(209:210))) %>%
  mutate(Q22a = do.call(coalesce, pick(211:215))) %>%
  mutate(Q22b = do.call(coalesce, pick(216:220))) %>%
  mutate(Q22c = do.call(coalesce, pick(221:225))) %>%
  mutate(Q23a = do.call(coalesce, pick(226:230))) %>%
  mutate(Q23b = do.call(coalesce, pick(231:235))) %>%
```

```
  mutate(Q23c = do.call(coalesce, pick(236:240))) %>%
  mutate(Q23d = do.call(coalesce, pick(241:245))) %>%
  mutate(Q24a = do.call(coalesce, pick(246:250))) %>%
  mutate(Q24b = do.call(coalesce, pick(251:255))) %>%
  mutate(Q24c = do.call(coalesce, pick(256:260))) %>%
  mutate(Q24d = do.call(coalesce, pick(261:265))) %>%
  mutate(Q24e = do.call(coalesce, pick(266:270))) %>%
  mutate(Q24f = do.call(coalesce, pick(271:275))) %>%
  mutate(Q25 = do.call(coalesce, pick(276:278))) %>%
  mutate(Q26 = do.call(coalesce, pick(279:281))) %>%
  #mutate(Q27 = do.call(coalesce, pick(282:292))) %>%
  mutate(Q28 = do.call(coalesce, pick(293:293)))

f_dir_coalesced_partial <- f_dir_q %>%
  select(294:300, 17:20, 301:347, 282:292, 348)

f_dir_coalesced <- f_dir_coalesced_partial %>%
  mutate(Q8 = pmap_chr(select(., 8:11), ~ paste(na.omit(c(...)), collapse = "; "))) %>%
  mutate(Q27 = pmap_chr(select(., 59:69), ~ paste(na.omit(c(...)), collapse = "; "))) %>%
  select(1:7, 71, 12:58, 72, 70)
f_dir_coalesced$role <- "Director"
f_dir_coalesced <- f_dir_coalesced %>%
  mutate(Q26 = case_when(
  Q26 == "Confidentially (one-on-one meetings)" ~ "Confidentially",
  Q26 == "Anonymously (online questionnaire)" ~ "Anonymously"
  ))

question1s <- c("Q11", rep("Q13a", 6), rep("Q13b", 6), rep("Q13c", 6), rep("Q13d", 6), rep("
question2s <- c("Q1", rep(c("Q10", "Q11", "Q12", "Q1", "Q5", "Q2"), 9), rep(c("Q5", "Q3"), 4

question_dic_res <- data.frame(question1s, question2s)
# when trying to cross reference one question (question1) by another grouping variable (quest
validation <- function(question1, question2, data){
  df <- data %>%
    transmute(
      response1 = .data[[question1]],
      response2 = .data[[question2]]
    ) %>%
    drop_na()

  comparison_res <- df %>%
    group_by(response1, response2) %>%
```

```
    summarize(counts = n(), .groups = "drop") %>%
    group_by(response2) %>%
    mutate(percent = counts / sum(counts) * 100) %>%
    ungroup()

  table_comparison <- table(df$response2, df$response1)
#table_hours
  test <- suppressWarnings(chisq.test(table_comparison))

  if (mean(test$expected <= 5) < 0.8 | any(test$expected < 1)){
    fish_test <- fisher.test(table_comparison, simulate.p.value = TRUE)
    if (fish_test$p.value <= 0.05){
     significant <- TRUE
   }
   else {
      significant <- FALSE
   }
    return(c("question1", "question2", round(fish_test$p.value, digits = 4), significant, "fi
  }
  else{
    if (test$p.value <= 0.05){
     significant <- TRUE
   }
   else {
      significant <- FALSE
   }
    }
  return(c("question1", "question2", round(test$p.value, digits = 4), significant, "chisq"))
}


res_results <- data.frame(
  Question1 = character(),
  Question2 = character(),
  P_value = numeric(),
  Significant = logical(),
  test = character()
)

for (i in seq_len(nrow(question_dic_res))) {

  res_results[i, ] <- validation(
```

```
    question1 = question_dic_res$question1s[i],
    question2 = question_dic_res$question2s[i],
    data = f_res_coalesced
  )
}

res_results <- res_results %>%
  mutate(
    Adj_P = round(p.adjust(P_value, method = "BH"), digits = 4),
    Sig_Adj = Adj_P <= 0.05,
    Question1 = question_dic_res$question1s,
    Question2 = question_dic_res$question2s
  ) %>% arrange(Adj_P)
gt(res_results)

question1s_dir <- c("Q10", "Q12a", "Q12b", "Q12c", "Q12d", "Q12e", "Q12f", "Q12g", "Q12h", "
question2s_dir <- c(rep("Q2",14))

question_dic_dir <- data.frame(question1s_dir, question2s_dir)
dir_results <- data.frame(
  Question1 = character(),
  Question2 = character(),
  P_value = numeric(),
  Significant = logical(),
  test = character()
)

for (i in seq_len(nrow(question_dic_dir))) {
  dir_results[i, ] <- validation(
    question1 = question_dic_dir$question1s_dir[i],
    question2 = question_dic_dir$question2s_dir[i],
    data = f_dir_coalesced
  )
}

dir_results <- dir_results %>%
  mutate(
    Adj_P = round(p.adjust(P_value, method = "BH"), digits = 4),
    Sig_Adj = Adj_P <= 0.05,
    Question1 = question_dic_dir$question1s,
    Question2 = question_dic_dir$question2s
  ) %>% arrange(Adj_P)
```

```r
gt(dir_results)
common_qs_res <- c("Q1", "Q3", "Q8", "Q9", "Q10", "Q11", "Q12", "Q13a", "Q13b","Q13c","Q13d"
"Q16a", "Q16b", "Q16c", "Q16d", "Q17",  "Q18",  "Q19",  "Q20a", "Q20b", "Q20c", "Q20d", "Q21
common_qs_dir <- c("Q2", "Q4", "Q7", "Q8", "Q9", "Q10", "Q11", "Q12a", "Q12b","Q12c","Q12d",
"Q15a", "Q15b", "Q15c", "Q15d", "Q16",  "Q17",  "Q18",  "Q19a", "Q19b", "Q19c", "Q19d", "Q20

full_dic <- data.frame(common_qs_res, common_qs_dir)

res_common <- f_res_coalesced %>%
  select(all_of(common_qs_res))

dir_common <- f_dir_coalesced %>%
  select(all_of(common_qs_dir)) %>%
  setNames(common_qs_res)
res_dir <- bind_rows(
  res_common,
  dir_common
)
res_dir_results <- data.frame(
  Question1 = character(),
  Question2 = character(),
  P_value = numeric(),
  Significant = logical(),
  test = character()
)

for (i in seq_len(ncol(res_dir)-1)) {

  res_dir_results[i, ] <- validation(
    question1 = colnames(res_dir)[i],
    question2 = colnames(res_dir)[52],
    data = res_dir
  )
}

res_dir_results <- res_dir_results %>%
  mutate(
    Adj_P = round(p.adjust(P_value, method = "BH"), digits = 4),
    Sig_Adj = Adj_P <= 0.05,
    Question1 = colnames(res_dir)[1:51],
    Question2 = "role"
  ) %>% arrange(Adj_P)
```

```r
gt(res_dir_results)
stack_bar <- function(q_reference, q_group, data){
  bar_df <- data %>%
    group_by(.data[[q_reference]], .data[[q_group]]) %>%
    summarize(counts = n(), .groups = "drop") %>%
    drop_na() %>%
    group_by(.data[[q_group]]) %>%
    mutate(percent = counts / sum(counts) * 100) %>%
    ungroup()

  p <- ggplot(bar_df, aes(x = .data[[q_reference]], y = percent, fill = .data[[q_reference]]
    geom_col() +
    facet_wrap(vars(.data[[q_group]])) +
    labs(
      title = paste(q_reference, "vs", q_group, sep = " "),
      x = NULL,
      y = "Percent of responses"
    ) +
    theme(
      axis.text.x = element_blank(),
      axis.ticks.x = element_blank()
    )
  return(p)
}

plots_res <- pmap(question_dic_res,
              function(question1s, question2s){
                stack_bar(
                  q_reference = question1s,
                  q_group = question2s,
                  data = f_res_coalesced
                )
              })
walk(plots_res, print)

plots_dir <- pmap(question_dic_dir,
                  function(question1s_dir, question2s_dir){
                    stack_bar(
                      q_reference = question1s_dir,
                      q_group = question2s_dir,
                      data = f_dir_coalesced
                    )
```

```
                  })
walk(plots_dir, print)

question_dic_res_dir <- data.frame(question1s_res_dir = colnames(res_dir)
                                   , question2s_res_dir = rep("role", ncol(res_dir)))
question_dic_res_dir <- question_dic_res_dir %>%
  filter(question1s_res_dir != "role")
plots_comp <- pmap(question_dic_res_dir,
                  function(question1s_res_dir, question2s_res_dir){
                    stack_bar(
                      q_reference = question1s_res_dir,
                      q_group = question2s_res_dir,
                      data = res_dir
                    )
                  })
walk(plots_comp, print)
```